Data Sheet

S6D0170

Preliminary

240*RGB X 432*DOT 1-CHIP DRIVER IC WITH INTERNAL GRAM
FOR 262,144-COLORS TFT-LCD

Dec. 04. 2006

Ver. 0.80

System LSI Division
Semiconductor Business
SAMSUNG ELECTRONICS CO., LTD.
(http://www.samsung.com/Products/Semiconductor/DisplayDriverIC)

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INTRODUCTION

The S6D0170 is a single-chip solution for TFT-LCD panel. Source drivers with built-in memory, gate driver and power generation circuits are integrated into one chip. The S6D0170 can display up to 240-RGB x 432-dot graphics on 260k-color TFT-LCD panel.

The S6D0170 has embedded Graphics RAM (GRAM) to store 240-RGB x 432-dot 260k-color image, internal boosters which generates LCD driving voltages, voltage-divided potentiometer and the voltage follower circuit for LCD driver.

The S6D0170 also supports 18-/16-/9-/8-bit high-speed bus interface to enable efficient data transfer to the GRAM.

The motion picture area can be designated in GRAM by window function. The specified window area can be updated selectively so that motion picture can be displayed simultaneously independent of still picture area.

The S6D0170 has various functions for optimizing power consumption in the LCD system: operation at low voltage (1.5V), register-controlled power-save mode, low-current mode, partial display mode and so on.

The S6D0170 is suitable for any medium-sized or small portable mobile solution requiring long-term driving capabilities, such as digital cellular phones supporting a web browser, bi-directional pagers and small PDAs, etc.

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FEATURES

240-RGBx432-dot TFT-LCD display controller/driver IC for 262,144 colors (720 channel-source drivers)

18-/16-/9-/8-bit high-speed parallel bus interface (80- and 68- system) and serial peripheral interface (SPI)

18-/16-/6-bit RGB interface

Writing to a window-ram address area by using a window-address function

Various color-display control functions

- 262,144 colors can be displayed at the same time (including RGB separated gamma adjust)
- Vertical scroll display function in raster-row units

Internal ram capacity: 240 x 18 x 432 = 1,866,240 bits

Low-power operation supports:

- Power-save mode: standby mode, deep standby mode
- Partial display mode in any position
- Maximum 7-times step-up circuit for generating LCD drive voltage
- Voltage followers to decrease direct current flow in the LCD drive system
- Charge sharing function for the switching performance of step-up circuits and operational amplifiers

1-raster row inversion drive (Reverse the polarity of driving voltage in every selected raster row)

Internal oscillation circuit and external hardware reset

Structure for TFT-display retention volume (only for Cst)

Alternating functions for TFT-LCD counter-electrode power

- Line alternating drive of VCOM.

Internal power supply circuit

- Step-up circuit: from 5 to 7 times positive-polarity, 3 to 5 times negative-polarity
- Adjustment of VCOM amplitude: internal 64-level digital potentiometer

Operating voltage

• Apply voltage
- VDD3 to VSS = 1.65 to 3.3 v logic interface level
- VCI to VSS = 2.4 to 3.3 v power supply voltage

• Generated voltage
- For the source driver: AVDD to VSS = 3.5 to 6v (power supply for LCD drive circuits)
  GVDD to VSS = 2.5 to 5v (reference voltage for grayscale generator)
- For the gate level shifter: VGH to VSS = +8.75 to +19.25, VGL to VSS = -13.75 to -5.25 v
  VGH to VGL = 14.0 to 30.0 v
- For the step-up circuit: VCI1 to VSS = 1.75 to 3v (refer to instruction description)
- For the TFT-LCD counter electrode: VCOMH to VSS = 2.5 to 5v
  VCOML to VSS = -VCI1 + 0.5v to VSS
  VCOM amplitude (max) = 6.0v

Released package type

- S6D0170 is released COG type package format only.

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BLOCK DIAGRAM

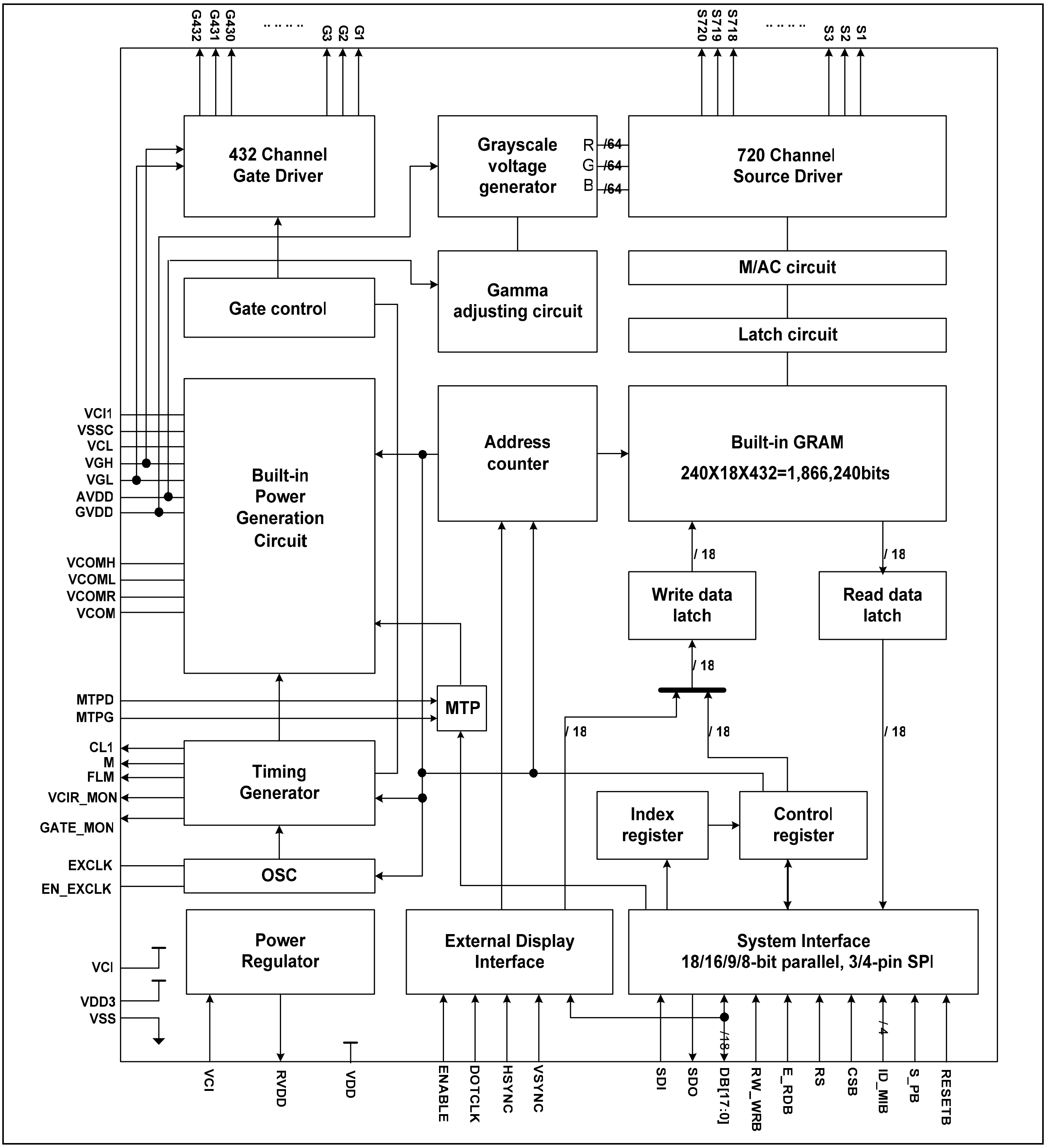

Figure 1. S6D0170 Block Diagram

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PAD CONFIGURATION

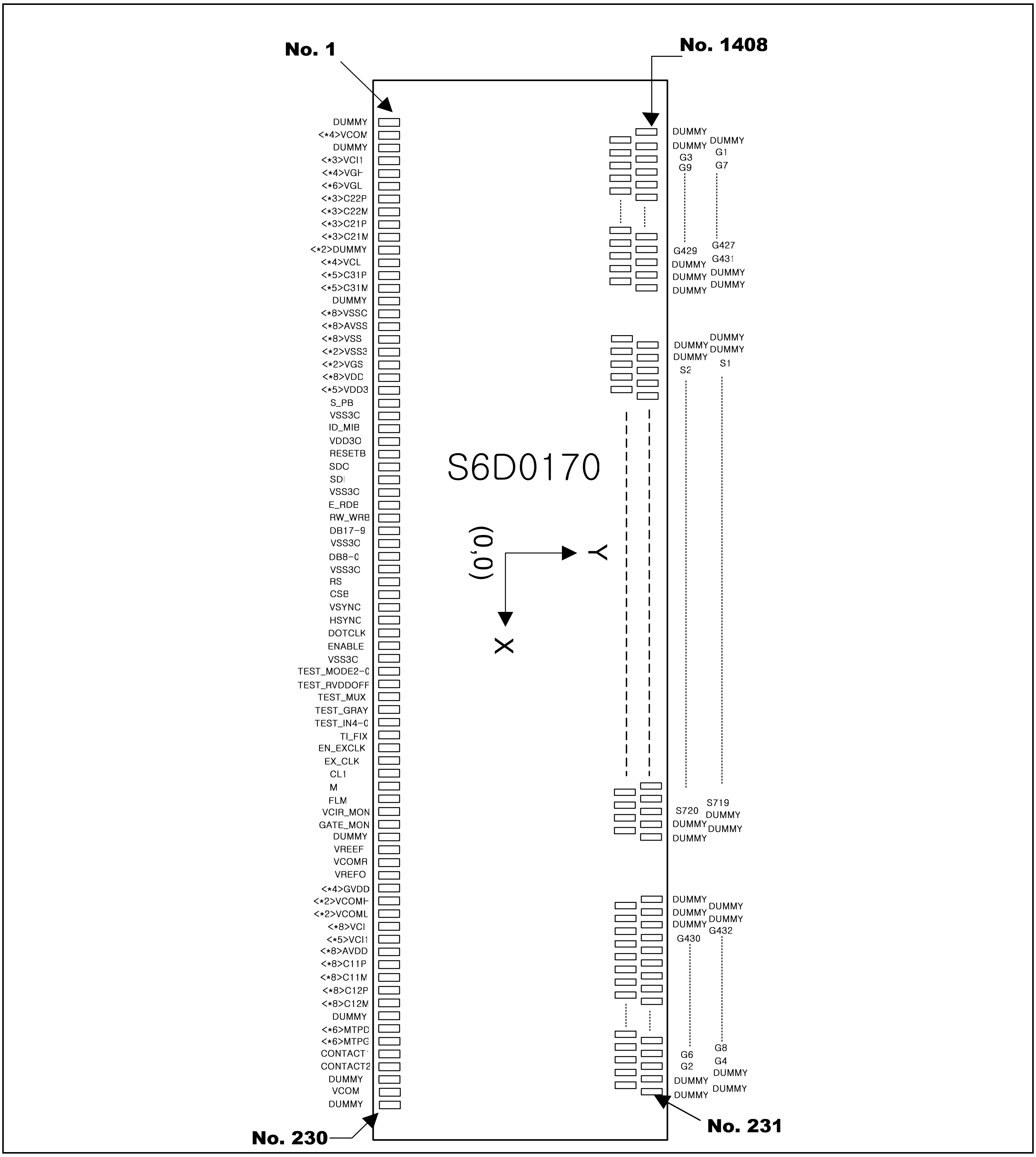

Figure 2. Pad Configuration

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Table 1. S6D0170 Pad Dimensions

Items	Pad name.	Size		Unit
		X	Y	
Chip size[1)]	-	21,190	1,437	um
Bumped Pad Size	Input Pads	60	91	
	Output Pads	19	110	
Bumped Pad Height	All Pads	15 (typ.) ±3		

NOTES:
[1)] Scribe lane included in this chip size (Scribe lane: 100um)

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ALIGN KEY CONFIGURATION AND COORDINATE

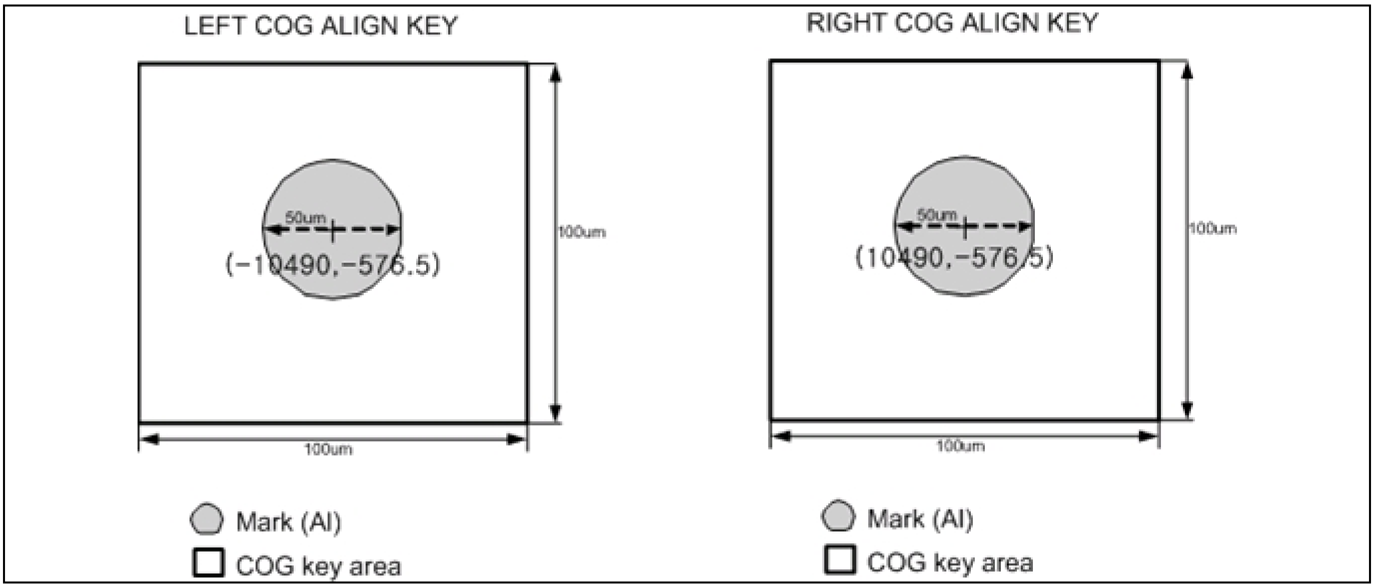

Figure 3. COG Align Key1

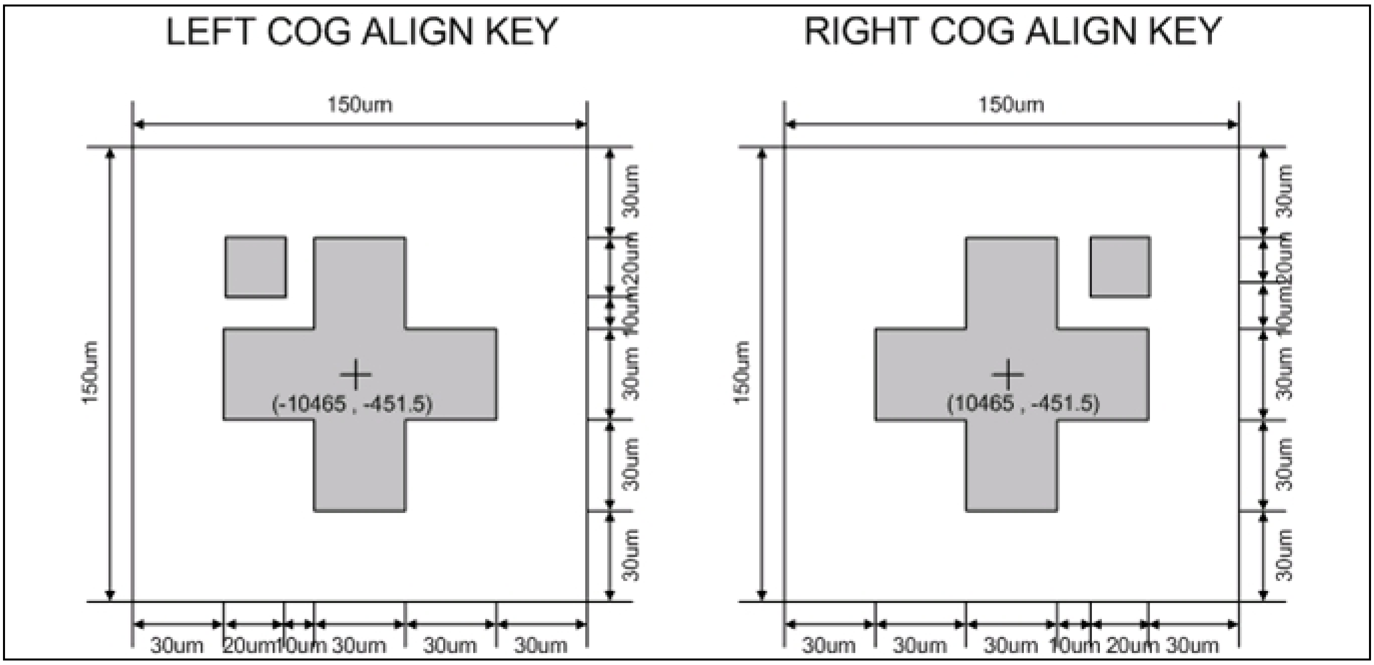

Figure 4. COG Align Key2

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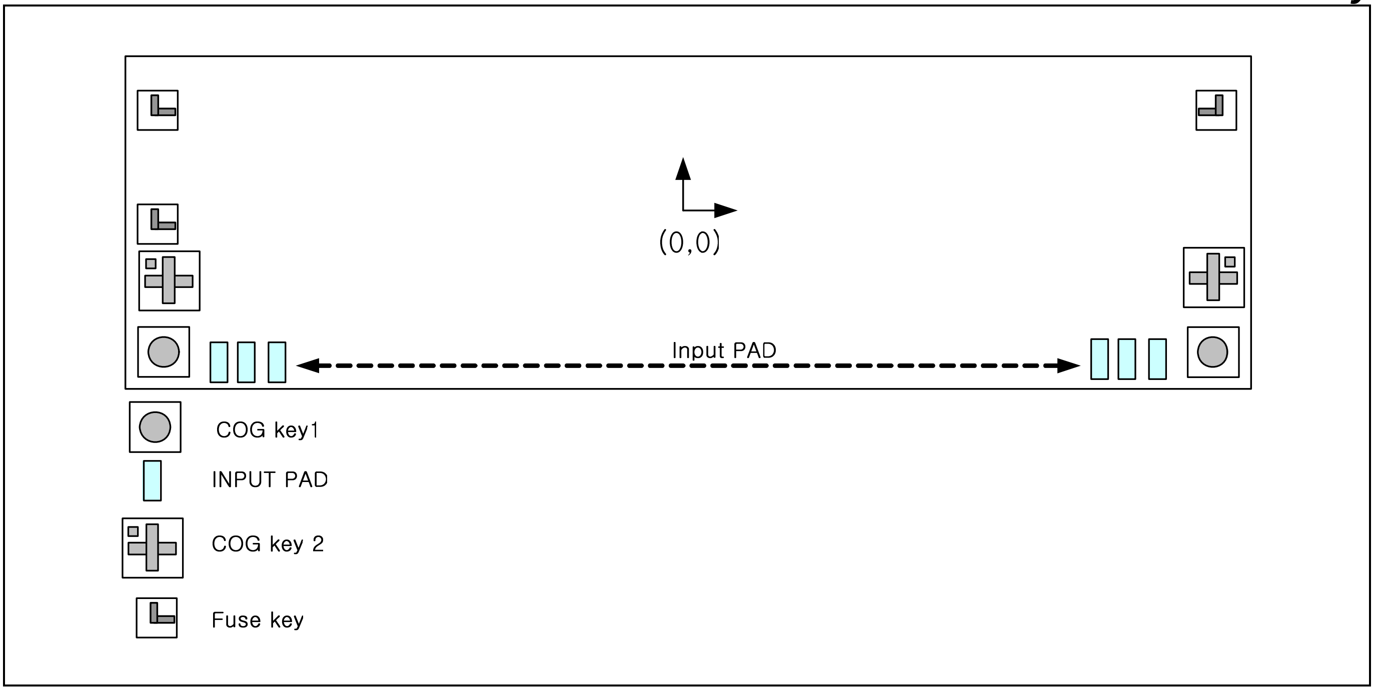

Figure 5. Align Key Configuration

NOTES:
1. Gold bumped height: 15 ± 3 um
2. Wafer thickness: 300 um (typ)

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PAD CENTER COORDINATES

Table 2. Pad Center Coordinates

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
1	DUMMY	−10305	−616	51	VSSC	−5805	−616	101	DB16	−1305	−616
2	VCOM	−10215	−616	52	VSSC	−5715	−616	102	DB15	−1215	−616
3	VCOM	−10125	−616	53	VSSC	−5625	−616	103	DB14	−1125	−616
4	VCOM	−10035	−616	54	VSSC	−5535	−616	104	DB13	−1035	−616
5	VCOM	−9945	−616	55	VSSC	−5445	−616	105	DB12	−945	−616
6	DUMMY	−9855	−616	56	VSSC	−5355	−616	106	DB11	−855	−616
7	VCI1	−9765	−616	57	AVSS	−5265	−616	107	DB10	−765	−616
8	VCI1	−9675	−616	58	AVSS	−5175	−616	108	DB9	−675	−616
9	VCI1	−9585	−616	59	AVSS	−5085	−616	109	VSS3O	−585	−616
10	VGH	−9495	−616	60	AVSS	−4995	−616	110	DB8	−495	−616
11	VGH	−9405	−616	61	AVSS	−4905	−616	111	DB7	−405	−616
12	VGH	−9315	−616	62	AVSS	−4815	−616	112	DB6	−315	−616
13	VGH	−9225	−616	63	AVSS	−4725	−616	113	DB5	−225	−616
14	VGL	−9135	−616	64	AVSS	−4635	−616	114	DB4	−135	−616
15	VGL	−9045	−616	65	VSS	−4545	−616	115	DB3	−45	−616
16	VGL	−8955	−616	66	VSS	−4455	−616	116	DB2	45	−616
17	VGL	−8865	−616	67	VSS	−4365	−616	117	DB1	135	−616
18	VGL	−8775	−616	68	VSS	−4275	−616	118	DB0	225	−616
19	VGL	−8685	−616	69	VSS	−4185	−616	119	VSS3O	315	−616
20	C22P	−8595	−616	70	VSS	−4095	−616	120	RS	405	−616
21	C22P	−8505	−616	71	VSS	−4005	−616	121	CSB	495	−616
22	C22P	−8415	−616	72	VSS	−3915	−616	122	VSYNC	585	−616
23	C22M	−8325	−616	73	VSS3	−3825	−616	123	HSYNC	675	−616
24	C22M	−8235	−616	74	VSS3	−3735	−616	124	DOTCLK	765	−616
25	C22M	−8145	−616	75	VGS	−3645	−616	125	ENABLE	855	−616
26	C21P	−8055	−616	76	VGS	−3555	−616	126	VSS3O	945	−616
27	C21P	−7965	−616	77	VDD	−3465	−616	127	TEST_MODE2	1035	−616
28	C21P	−7875	−616	78	VDD	−3375	−616	128	TEST_MODE1	1125	−616
29	C21M	−7785	−616	79	VDD	−3285	−616	129	TEST_MODE0	1215	−616
30	C21M	−7695	−616	80	VDD	−3195	−616	130	TEST_RVDDOFF	1305	−616
31	C21M	−7605	−616	81	VDD	−3105	−616	131	TEST_MUX	1395	−616
32	DUMMY	−7515	−616	82	VDD	−3015	−616	132	TEST_GRAY	1485	−616
33	DUMMY	−7425	−616	83	VDD	−2925	−616	133	TEST_IN4	1575	−616
34	VCL	−7335	−616	84	VDD	−2835	−616	134	TEST_IN3	1665	−616
35	VCL	−7245	−616	85	VDD3	−2745	−616	135	TEST_IN2	1755	−616
36	VCL	−7155	−616	86	VDD3	−2655	−616	136	TEST_IN1	1845	−616
37	VCL	−7065	−616	87	VDD3	−2565	−616	137	TEST_IN0	1935	−616
38	C31P	−6975	−616	88	VDD3	−2475	−616	138	TI_FIX	2025	−616
39	C31P	−6885	−616	89	VDD3	−2385	−616	139	EN_EXCLK	2115	−616
40	C31P	−6795	−616	90	S_PB	−2295	−616	140	EXCLK	2205	−616
41	C31P	−6705	−616	91	VSS3O	−2205	−616	141	CL1	2295	−616
42	C31P	−6615	−616	92	ID_MIB	−2115	−616	142	M	2385	−616
43	C31M	−6525	−616	93	VDD3O	−2025	−616	143	FLM	2475	−616
44	C31M	−6435	−616	94	RESETB	−1935	−616	144	VCIR_MON	2565	−616
45	C31M	−6345	−616	95	SDO	−1845	−616	145	GATE_MON	2655	−616
46	C31M	−6255	−616	96	SDI	−1755	−616	146	DUMMY	2745	−616
47	C31M	−6165	−616	97	VSS3O	−1665	−616	147	VREF	2835	−616
48	DUMMY	−6075	−616	98	E_RDB	−1575	−616	148	VCOMR	2925	−616
49	VSSC	−5985	−616	99	RW_WRB	−1485	−616	149	GVDD	3015	−616
50	VSSC	−5895	−616	100	DB17	−1395	−616	150	GVDD	3105	−616

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Table 3. Pad Center Coordinates (continued)

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
151	GVDD	3195	−616	201	C12P	7695	−616	251	G<34>	10113.5	606.5
152	GVDD	3285	−616	202	C12M	7785	−616	252	G<36>	10096.5	461.5
153	VCOMH	3375	−616	203	C12M	7875	−616	253	G<38>	10079.5	606.5
154	VCOMH	3465	−616	204	C12M	7965	−616	254	G<40>	10062.5	461.5
155	VCOML	3555	−616	205	C12M	8055	−616	255	G<42>	10045.5	606.5
156	VCOML	3645	−616	206	C12M	8145	−616	256	G<44>	10028.5	461.5
157	VCI	3735	−616	207	C12M	8235	−616	257	G<46>	10011.5	606.5
158	VCI	3825	−616	208	C12M	8325	−616	258	G<48>	9994.5	461.5
159	VCI	3915	−616	209	C12M	8415	−616	259	G<50>	9977.5	606.5
160	VCI	4005	−616	210	DUMMY	8505	−616	260	G<52>	9960.5	461.5
161	VCI	4095	−616	211	MTPD	8595	−616	261	G<54>	9943.5	606.5
162	VCI	4185	−616	212	MTPD	8685	−616	262	G<56>	9926.5	461.5
163	VCI	4275	−616	213	MTPD	8775	−616	263	G<58>	9909.5	606.5
164	VCI	4365	−616	214	MTPD	8865	−616	264	G<60>	9892.5	461.5
165	VCI1	4455	−616	215	MTPD	8955	−616	265	G<62>	9875.5	606.5
166	VCI1	4545	−616	216	MTPD	9045	−616	266	G<64>	9858.5	461.5
167	VCI1	4635	−616	217	MTPG	9135	−616	267	G<66>	9841.5	606.5
168	VCI1	4725	−616	218	MTPG	9225	−616	268	G<68>	9824.5	461.5
169	VCI1	4815	−616	219	MTPG	9315	−616	269	G<70>	9807.5	606.5
170	AVDD	4905	−616	220	MTPG	9405	−616	270	G<72>	9790.5	461.5
171	AVDD	4995	−616	221	MTPG	9495	−616	271	G<74>	9773.5	606.5
172	AVDD	5085	−616	222	MTPG	9585	−616	272	G<76>	9756.5	461.5
173	AVDD	5175	−616	223	CONTACT1	9675	−616	273	G<78>	9739.5	606.5
174	AVDD	5265	−616	224	CONTACT2	9765	−616	274	G<80>	9722.5	461.5
175	AVDD	5355	−616	225	DUMMY	9855	−616	275	G<82>	9705.5	606.5
176	AVDD	5445	−616	226	VCOM	9945	−616	276	G<84>	9688.5	461.5
177	AVDD	5535	−616	227	VCOM	10035	−616	277	G<86>	9671.5	606.5
178	C11P	5625	−616	228	VCOM	10125	−616	278	G<88>	9654.5	461.5
179	C11P	5715	−616	229	VCOM	10215	−616	279	G<90>	9637.5	606.5
180	C11P	5805	−616	230	DUMMY	10305	−616	280	G<92>	9620.5	461.5
181	C11P	5895	−616	231	DUMMY	10453.5	606.5	281	G<94>	9603.5	606.5
182	C11P	5985	−616	232	DUMMY	10436.5	461.5	282	G<96>	9586.5	461.5
183	C11P	6075	−616	233	DUMMY	10419.5	606.5	283	G<98>	9569.5	606.5
184	C11P	6165	−616	234	DUMMY	10402.5	461.5	284	G<100>	9552.5	461.5
185	C11P	6255	−616	235	G<2>	10385.5	606.5	285	G<102>	9535.5	606.5
186	C11M	6345	−616	236	G<4>	10368.5	461.5	286	G<104>	9518.5	461.5
187	C11M	6435	−616	237	G<6>	10351.5	606.5	287	G<106>	9501.5	606.5
188	C11M	6525	−616	238	G<8>	10334.5	461.5	288	G<108>	9484.5	461.5
189	C11M	6615	−616	239	G<10>	10317.5	606.5	289	G<110>	9467.5	606.5
190	C11M	6705	−616	240	G<12>	10300.5	461.5	290	G<112>	9450.5	461.5
191	C11M	6795	−616	241	G<14>	10283.5	606.5	291	G<114>	9433.5	606.5
192	C11M	6885	−616	242	G<16>	10266.5	461.5	292	G<116>	9416.5	461.5
193	C11M	6975	−616	243	G<18>	10249.5	606.5	293	G<118>	9399.5	606.5
194	C12P	7065	−616	244	G<20>	10232.5	461.5	294	G<120>	9382.5	461.5
195	C12P	7155	−616	245	G<22>	10215.5	606.5	295	G<122>	9365.5	606.5
196	C12P	7245	−616	246	G<24>	10198.5	461.5	296	G<124>	9348.5	461.5
197	C12P	7335	−616	247	G<26>	10181.5	606.5	297	G<126>	9331.5	606.5
198	C12P	7425	−616	248	G<28>	10164.5	461.5	298	G<128>	9314.5	461.5
199	C12P	7515	−616	249	G<30>	10147.5	606.5	299	G<130>	9297.5	606.5
200	C12P	7605	−616	250	G<32>	10130.5	461.5	300	G<132>	9280.5	461.5

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Table 4. Pad Center Coordinates (continued)

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
301	G<134>	9263.5	606.5	351	G<234>	8413.5	606.5	401	G<334>	7563.5	606.5
302	G<136>	9246.5	461.5	352	G<236>	8396.5	461.5	402	G<336>	7546.5	461.5
303	G<138>	9229.5	606.5	353	G<238>	8379.5	606.5	403	G<338>	7529.5	606.5
304	G<140>	9212.5	461.5	354	G<240>	8362.5	461.5	404	G<340>	7512.5	461.5
305	G<142>	9195.5	606.5	355	G<242>	8345.5	606.5	405	G<342>	7495.5	606.5
306	G<144>	9178.5	461.5	356	G<244>	8328.5	461.5	406	G<344>	7478.5	461.5
307	G<146>	9161.5	606.5	357	G<246>	8311.5	606.5	407	G<346>	7461.5	606.5
308	G<148>	9144.5	461.5	358	G<248>	8294.5	461.5	408	G<348>	7444.5	461.5
309	G<150>	9127.5	606.5	359	G<250>	8277.5	606.5	409	G<350>	7427.5	606.5
310	G<152>	9110.5	461.5	360	G<252>	8260.5	461.5	410	G<352>	7410.5	461.5
311	G<154>	9093.5	606.5	361	G<254>	8243.5	606.5	411	G<354>	7393.5	606.5
312	G<156>	9076.5	461.5	362	G<256>	8226.5	461.5	412	G<356>	7376.5	461.5
313	G<158>	9059.5	606.5	363	G<258>	8209.5	606.5	413	G<358>	7359.5	606.5
314	G<160>	9042.5	461.5	364	G<260>	8192.5	461.5	414	G<360>	7342.5	461.5
315	G<162>	9025.5	606.5	365	G<262>	8175.5	606.5	415	G<362>	7325.5	606.5
316	G<164>	9008.5	461.5	366	G<264>	8158.5	461.5	416	G<364>	7308.5	461.5
317	G<166>	8991.5	606.5	367	G<266>	8141.5	606.5	417	G<366>	7291.5	606.5
318	G<168>	8974.5	461.5	368	G<268>	8124.5	461.5	418	G<368>	7274.5	461.5
319	G<170>	8957.5	606.5	369	G<270>	8107.5	606.5	419	G<370>	7257.5	606.5
320	G<172>	8940.5	461.5	370	G<272>	8090.5	461.5	420	G<372>	7240.5	461.5
321	G<174>	8923.5	606.5	371	G<274>	8073.5	606.5	421	G<374>	7223.5	606.5
322	G<176>	8906.5	461.5	372	G<276>	8056.5	461.5	422	G<376>	7206.5	461.5
323	G<178>	8889.5	606.5	373	G<278>	8039.5	606.5	423	G<378>	7189.5	606.5
324	G<180>	8872.5	461.5	374	G<280>	8022.5	461.5	424	G<380>	7172.5	461.5
325	G<182>	8855.5	606.5	375	G<282>	8005.5	606.5	425	G<382>	7155.5	606.5
326	G<184>	8838.5	461.5	376	G<284>	7988.5	461.5	426	G<384>	7138.5	461.5
327	G<186>	8821.5	606.5	377	G<286>	7971.5	606.5	427	G<386>	7121.5	606.5
328	G<188>	8804.5	461.5	378	G<288>	7954.5	461.5	428	G<388>	7104.5	461.5
329	G<190>	8787.5	606.5	379	G<290>	7937.5	606.5	429	G<390>	7087.5	606.5
330	G<192>	8770.5	461.5	380	G<292>	7920.5	461.5	430	G<392>	7070.5	461.5
331	G<194>	8753.5	606.5	381	G<294>	7903.5	606.5	431	G<394>	7053.5	606.5
332	G<196>	8736.5	461.5	382	G<296>	7886.5	461.5	432	G<396>	7036.5	461.5
333	G<198>	8719.5	606.5	383	G<298>	7869.5	606.5	433	G<398>	7019.5	606.5
334	G<200>	8702.5	461.5	384	G<300>	7852.5	461.5	434	G<400>	7002.5	461.5
335	G<202>	8685.5	606.5	385	G<302>	7835.5	606.5	435	G<402>	6985.5	606.5
336	G<204>	8668.5	461.5	386	G<304>	7818.5	461.5	436	G<404>	6968.5	461.5
337	G<206>	8651.5	606.5	387	G<306>	7801.5	606.5	437	G<406>	6951.5	606.5
338	G<208>	8634.5	461.5	388	G<308>	7784.5	461.5	438	G<408>	6934.5	461.5
339	G<210>	8617.5	606.5	389	G<310>	7767.5	606.5	439	G<410>	6917.5	606.5
340	G<212>	8600.5	461.5	390	G<312>	7750.5	461.5	440	G<412>	6900.5	461.5
341	G<214>	8583.5	606.5	391	G<314>	7733.5	606.5	441	G<414>	6883.5	606.5
342	G<216>	8566.5	461.5	392	G<316>	7716.5	461.5	442	G<416>	6866.5	461.5
343	G<218>	8549.5	606.5	393	G<318>	7699.5	606.5	443	G<418>	6849.5	606.5
344	G<220>	8532.5	461.5	394	G<320>	7682.5	461.5	444	G<420>	6832.5	461.5
345	G<222>	8515.5	606.5	395	G<322>	7665.5	606.5	445	G<422>	6815.5	606.5
346	G<224>	8498.5	461.5	396	G<324>	7648.5	461.5	446	G<424>	6798.5	461.5
347	G<226>	8481.5	606.5	397	G<326>	7631.5	606.5	447	G<426>	6781.5	606.5
348	G<228>	8464.5	461.5	398	G<328>	7614.5	461.5	448	G<428>	6764.5	461.5
349	G<230>	8447.5	606.5	399	G<330>	7597.5	606.5	449	G<430>	6747.5	606.5
350	G<232>	8430.5	461.5	400	G<332>	7580.5	461.5	450	G<432>	6730.5	461.5

Preliminary

Table 5. Pad Center Coordinates (continued)

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
451	DUMMY	6713.5	606.5	501	S<679>	5414.5	461.5	551	S<629>	4564.5	461.5
452	DUMMY	6696.5	461.5	502	S<678>	5397.5	606.5	552	S<628>	4547.5	606.5
453	DUMMY	6679.5	606.5	503	S<677>	5380.5	461.5	553	S<627>	4530.5	461.5
454	DUMMY	6662.5	461.5	504	S<676>	5363.5	606.5	554	S<626>	4513.5	606.5
455	DUMMY	6645.5	606.5	505	S<675>	5346.5	461.5	555	S<625>	4496.5	461.5
456	DUMMY	6179.5	606.5	506	S<674>	5329.5	606.5	556	S<624>	4479.5	606.5
457	DUMMY	6162.5	461.5	507	S<673>	5312.5	461.5	557	S<623>	4462.5	461.5
458	DUMMY	6145.5	606.5	508	S<672>	5295.5	606.5	558	S<622>	4445.5	606.5
459	DUMMY	6128.5	461.5	509	S<671>	5278.5	461.5	559	S<621>	4428.5	461.5
460	S<720>	6111.5	606.5	510	S<670>	5261.5	606.5	560	S<620>	4411.5	606.5
461	S<719>	6094.5	461.5	511	S<669>	5244.5	461.5	561	S<619>	4394.5	461.5
462	S<718>	6077.5	606.5	512	S<668>	5227.5	606.5	562	S<618>	4377.5	606.5
463	S<717>	6060.5	461.5	513	S<667>	5210.5	461.5	563	S<617>	4360.5	461.5
464	S<716>	6043.5	606.5	514	S<666>	5193.5	606.5	564	S<616>	4343.5	606.5
465	S<715>	6026.5	461.5	515	S<665>	5176.5	461.5	565	S<615>	4326.5	461.5
466	S<714>	6009.5	606.5	516	S<664>	5159.5	606.5	566	S<614>	4309.5	606.5
467	S<713>	5992.5	461.5	517	S<663>	5142.5	461.5	567	S<613>	4292.5	461.5
468	S<712>	5975.5	606.5	518	S<662>	5125.5	606.5	568	S<612>	4275.5	606.5
469	S<711>	5958.5	461.5	519	S<661>	5108.5	461.5	569	S<611>	4258.5	461.5
470	S<710>	5941.5	606.5	520	S<660>	5091.5	606.5	570	S<610>	4241.5	606.5
471	S<709>	5924.5	461.5	521	S<659>	5074.5	461.5	571	S<609>	4224.5	461.5
472	S<708>	5907.5	606.5	522	S<658>	5057.5	606.5	572	S<608>	4207.5	606.5
473	S<707>	5890.5	461.5	523	S<657>	5040.5	461.5	573	S<607>	4190.5	461.5
474	S<706>	5873.5	606.5	524	S<656>	5023.5	606.5	574	S<606>	4173.5	606.5
475	S<705>	5856.5	461.5	525	S<655>	5006.5	461.5	575	S<605>	4156.5	461.5
476	S<704>	5839.5	606.5	526	S<654>	4989.5	606.5	576	S<604>	4139.5	606.5
477	S<703>	5822.5	461.5	527	S<653>	4972.5	461.5	577	S<603>	4122.5	461.5
478	S<702>	5805.5	606.5	528	S<652>	4955.5	606.5	578	S<602>	4105.5	606.5
479	S<701>	5788.5	461.5	529	S<651>	4938.5	461.5	579	S<601>	4088.5	461.5
480	S<700>	5771.5	606.5	530	S<650>	4921.5	606.5	580	S<600>	4071.5	606.5
481	S<699>	5754.5	461.5	531	S<649>	4904.5	461.5	581	S<599>	4054.5	461.5
482	S<698>	5737.5	606.5	532	S<648>	4887.5	606.5	582	S<598>	4037.5	606.5
483	S<697>	5720.5	461.5	533	S<647>	4870.5	461.5	583	S<597>	4020.5	461.5
484	S<696>	5703.5	606.5	534	S<646>	4853.5	606.5	584	S<596>	4003.5	606.5
485	S<695>	5686.5	461.5	535	S<645>	4836.5	461.5	585	S<595>	3986.5	461.5
486	S<694>	5669.5	606.5	536	S<644>	4819.5	606.5	586	S<594>	3969.5	606.5
487	S<693>	5652.5	461.5	537	S<643>	4802.5	461.5	587	S<593>	3952.5	461.5
488	S<692>	5635.5	606.5	538	S<642>	4785.5	606.5	588	S<592>	3935.5	606.5
489	S<691>	5618.5	461.5	539	S<641>	4768.5	461.5	589	S<591>	3918.5	461.5
490	S<690>	5601.5	606.5	540	S<640>	4751.5	606.5	590	S<590>	3901.5	606.5
491	S<689>	5584.5	461.5	541	S<639>	4734.5	461.5	591	S<589>	3884.5	461.5
492	S<688>	5567.5	606.5	542	S<638>	4717.5	606.5	592	S<588>	3867.5	606.5
493	S<687>	5550.5	461.5	543	S<637>	4700.5	461.5	593	S<587>	3850.5	461.5
494	S<686>	5533.5	606.5	544	S<636>	4683.5	606.5	594	S<586>	3833.5	606.5
495	S<685>	5516.5	461.5	545	S<635>	4666.5	461.5	595	S<585>	3816.5	461.5
496	S<684>	5499.5	606.5	546	S<634>	4649.5	606.5	596	S<584>	3799.5	606.5
497	S<683>	5482.5	461.5	547	S<633>	4632.5	461.5	597	S<583>	3782.5	461.5
498	S<682>	5465.5	606.5	548	S<632>	4615.5	606.5	598	S<582>	3765.5	606.5
499	S<681>	5448.5	461.5	549	S<631>	4598.5	461.5	599	S<581>	3748.5	461.5
500	S<680>	5431.5	606.5	550	S<630>	4581.5	606.5	600	S<580>	3731.5	606.5

Preliminary

Table 6. Pad Center Coordinates (continued)

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
601	S<579>	3714.5	461.5	651	S<529>	2864.5	461.5	701	S<479>	2014.5	461.5
602	S<578>	3697.5	606.5	652	S<528>	2847.5	606.5	702	S<478>	1997.5	606.5
603	S<577>	3680.5	461.5	653	S<527>	2830.5	461.5	703	S<477>	1980.5	461.5
604	S<576>	3663.5	606.5	654	S<526>	2813.5	606.5	704	S<476>	1963.5	606.5
605	S<575>	3646.5	461.5	655	S<525>	2796.5	461.5	705	S<475>	1946.5	461.5
606	S<574>	3629.5	606.5	656	S<524>	2779.5	606.5	706	S<474>	1929.5	606.5
607	S<573>	3612.5	461.5	657	S<523>	2762.5	461.5	707	S<473>	1912.5	461.5
608	S<572>	3595.5	606.5	658	S<522>	2745.5	606.5	708	S<472>	1895.5	606.5
609	S<571>	3578.5	461.5	659	S<521>	2728.5	461.5	709	S<471>	1878.5	461.5
610	S<570>	3561.5	606.5	660	S<520>	2711.5	606.5	710	S<470>	1861.5	606.5
611	S<569>	3544.5	461.5	661	S<519>	2694.5	461.5	711	S<469>	1844.5	461.5
612	S<568>	3527.5	606.5	662	S<518>	2677.5	606.5	712	S<468>	1827.5	606.5
613	S<567>	3510.5	461.5	663	S<517>	2660.5	461.5	713	S<467>	1810.5	461.5
614	S<566>	3493.5	606.5	664	S<516>	2643.5	606.5	714	S<466>	1793.5	606.5
615	S<565>	3476.5	461.5	665	S<515>	2626.5	461.5	715	S<465>	1776.5	461.5
616	S<564>	3459.5	606.5	666	S<514>	2609.5	606.5	716	S<464>	1759.5	606.5
617	S<563>	3442.5	461.5	667	S<513>	2592.5	461.5	717	S<463>	1742.5	461.5
618	S<562>	3425.5	606.5	668	S<512>	2575.5	606.5	718	S<462>	1725.5	606.5
619	S<561>	3408.5	461.5	669	S<511>	2558.5	461.5	719	S<461>	1708.5	461.5
620	S<560>	3391.5	606.5	670	S<510>	2541.5	606.5	720	S<460>	1691.5	606.5
621	S<559>	3374.5	461.5	671	S<509>	2524.5	461.5	721	S<459>	1674.5	461.5
622	S<558>	3357.5	606.5	672	S<508>	2507.5	606.5	722	S<458>	1657.5	606.5
623	S<557>	3340.5	461.5	673	S<507>	2490.5	461.5	723	S<457>	1640.5	461.5
624	S<556>	3323.5	606.5	674	S<506>	2473.5	606.5	724	S<456>	1623.5	606.5
625	S<555>	3306.5	461.5	675	S<505>	2456.5	461.5	725	S<455>	1606.5	461.5
626	S<554>	3289.5	606.5	676	S<504>	2439.5	606.5	726	S<454>	1589.5	606.5
627	S<553>	3272.5	461.5	677	S<503>	2422.5	461.5	727	S<453>	1572.5	461.5
628	S<552>	3255.5	606.5	678	S<502>	2405.5	606.5	728	S<452>	1555.5	606.5
629	S<551>	3238.5	461.5	679	S<501>	2388.5	461.5	729	S<451>	1538.5	461.5
630	S<550>	3221.5	606.5	680	S<500>	2371.5	606.5	730	S<450>	1521.5	606.5
631	S<549>	3204.5	461.5	681	S<499>	2354.5	461.5	731	S<449>	1504.5	461.5
632	S<548>	3187.5	606.5	682	S<498>	2337.5	606.5	732	S<448>	1487.5	606.5
633	S<547>	3170.5	461.5	683	S<497>	2320.5	461.5	733	S<447>	1470.5	461.5
634	S<546>	3153.5	606.5	684	S<496>	2303.5	606.5	734	S<446>	1453.5	606.5
635	S<545>	3136.5	461.5	685	S<495>	2286.5	461.5	735	S<445>	1436.5	461.5
636	S<544>	3119.5	606.5	686	S<494>	2269.5	606.5	736	S<444>	1419.5	606.5
637	S<543>	3102.5	461.5	687	S<493>	2252.5	461.5	737	S<443>	1402.5	461.5
638	S<542>	3085.5	606.5	688	S<492>	2235.5	606.5	738	S<442>	1385.5	606.5
639	S<541>	3068.5	461.5	689	S<491>	2218.5	461.5	739	S<441>	1368.5	461.5
640	S<540>	3051.5	606.5	690	S<490>	2201.5	606.5	740	S<440>	1351.5	606.5
641	S<539>	3034.5	461.5	691	S<489>	2184.5	461.5	741	S<439>	1334.5	461.5
642	S<538>	3017.5	606.5	692	S<488>	2167.5	606.5	742	S<438>	1317.5	606.5
643	S<537>	3000.5	461.5	693	S<487>	2150.5	461.5	743	S<437>	1300.5	461.5
644	S<536>	2983.5	606.5	694	S<486>	2133.5	606.5	744	S<436>	1283.5	606.5
645	S<535>	2966.5	461.5	695	S<485>	2116.5	461.5	745	S<435>	1266.5	461.5
646	S<534>	2949.5	606.5	696	S<484>	2099.5	606.5	746	S<434>	1249.5	606.5
647	S<533>	2932.5	461.5	697	S<483>	2082.5	461.5	747	S<433>	1232.5	461.5
648	S<532>	2915.5	606.5	698	S<482>	2065.5	606.5	748	S<432>	1215.5	606.5
649	S<531>	2898.5	461.5	699	S<481>	2048.5	461.5	749	S<431>	1198.5	461.5
650	S<530>	2881.5	606.5	700	S<480>	2031.5	606.5	750	S<430>	1181.5	606.5

Preliminary

Table 7. Pad Center Coordinates (continued)

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
751	S<429>	1164.5	461.5	801	S<379>	314.5	461.5	851	S<329>	−535.5	461.5
752	S<428>	1147.5	606.5	802	S<378>	297.5	606.5	852	S<328>	−552.5	606.5
753	S<427>	1130.5	461.5	803	S<377>	280.5	461.5	853	S<327>	−569.5	461.5
754	S<426>	1113.5	606.5	804	S<376>	263.5	606.5	854	S<326>	−586.5	606.5
755	S<425>	1096.5	461.5	805	S<375>	246.5	461.5	855	S<325>	−603.5	461.5
756	S<424>	1079.5	606.5	806	S<374>	229.5	606.5	856	S<324>	−620.5	606.5
757	S<423>	1062.5	461.5	807	S<373>	212.5	461.5	857	S<323>	−637.5	461.5
758	S<422>	1045.5	606.5	808	S<372>	195.5	606.5	858	S<322>	−654.5	606.5
759	S<421>	1028.5	461.5	809	S<371>	178.5	461.5	859	S<321>	−671.5	461.5
760	S<420>	1011.5	606.5	810	S<370>	161.5	606.5	860	S<320>	−688.5	606.5
761	S<419>	994.5	461.5	811	S<369>	144.5	461.5	861	S<319>	−705.5	461.5
762	S<418>	977.5	606.5	812	S<368>	127.5	606.5	862	S<318>	−722.5	606.5
763	S<417>	960.5	461.5	813	S<367>	110.5	461.5	863	S<317>	−739.5	461.5
764	S<416>	943.5	606.5	814	S<366>	93.5	606.5	864	S<316>	−756.5	606.5
765	S<415>	926.5	461.5	815	S<365>	76.5	461.5	865	S<315>	−773.5	461.5
766	S<414>	909.5	606.5	816	S<364>	59.5	606.5	866	S<314>	−790.5	606.5
767	S<413>	892.5	461.5	817	S<363>	42.5	461.5	867	S<313>	−807.5	461.5
768	S<412>	875.5	606.5	818	S<362>	25.5	606.5	868	S<312>	−824.5	606.5
769	S<411>	858.5	461.5	819	S<361>	8.5	461.5	869	S<311>	−841.5	461.5
770	S<410>	841.5	606.5	820	S<360>	−8.5	606.5	870	S<310>	−858.5	606.5
771	S<409>	824.5	461.5	821	S<359>	−25.5	461.5	871	S<309>	−875.5	461.5
772	S<408>	807.5	606.5	822	S<358>	−42.5	606.5	872	S<308>	−892.5	606.5
773	S<407>	790.5	461.5	823	S<357>	−59.5	461.5	873	S<307>	−909.5	461.5
774	S<406>	773.5	606.5	824	S<356>	−76.5	606.5	874	S<306>	−926.5	606.5
775	S<405>	756.5	461.5	825	S<355>	−93.5	461.5	875	S<305>	−943.5	461.5
776	S<404>	739.5	606.5	826	S<354>	−110.5	606.5	876	S<304>	−960.5	606.5
777	S<403>	722.5	461.5	827	S<353>	−127.5	461.5	877	S<303>	−977.5	461.5
778	S<402>	705.5	606.5	828	S<352>	−144.5	606.5	878	S<302>	−994.5	606.5
779	S<401>	688.5	461.5	829	S<351>	−161.5	461.5	879	S<301>	−1011.5	461.5
780	S<400>	671.5	606.5	830	S<350>	−178.5	606.5	880	S<300>	−1028.5	606.5
781	S<399>	654.5	461.5	831	S<349>	−195.5	461.5	881	S<299>	−1045.5	461.5
782	S<398>	637.5	606.5	832	S<348>	−212.5	606.5	882	S<298>	−1062.5	606.5
783	S<397>	620.5	461.5	833	S<347>	−229.5	461.5	883	S<297>	−1079.5	461.5
784	S<396>	603.5	606.5	834	S<346>	−246.5	606.5	884	S<296>	−1096.5	606.5
785	S<395>	586.5	461.5	835	S<345>	−263.5	461.5	885	S<295>	−1113.5	461.5
786	S<394>	569.5	606.5	836	S<344>	−280.5	606.5	886	S<294>	−1130.5	606.5
787	S<393>	552.5	461.5	837	S<343>	−297.5	461.5	887	S<293>	−1147.5	461.5
788	S<392>	535.5	606.5	838	S<342>	−314.5	606.5	888	S<292>	−1164.5	606.5
789	S<391>	518.5	461.5	839	S<341>	−331.5	461.5	889	S<291>	−1181.5	461.5
790	S<390>	501.5	606.5	840	S<340>	−348.5	606.5	890	S<290>	−1198.5	606.5
791	S<389>	484.5	461.5	841	S<339>	−365.5	461.5	891	S<289>	−1215.5	461.5
792	S<388>	467.5	606.5	842	S<338>	−382.5	606.5	892	S<288>	−1232.5	606.5
793	S<387>	450.5	461.5	843	S<337>	−399.5	461.5	893	S<287>	−1249.5	461.5
794	S<386>	433.5	606.5	844	S<336>	−416.5	606.5	894	S<286>	−1266.5	606.5
795	S<385>	416.5	461.5	845	S<335>	−433.5	461.5	895	S<285>	−1283.5	461.5
796	S<384>	399.5	606.5	846	S<334>	−450.5	606.5	896	S<284>	−1300.5	606.5
797	S<383>	382.5	461.5	847	S<333>	−467.5	461.5	897	S<283>	−1317.5	461.5
798	S<382>	365.5	606.5	848	S<332>	−484.5	606.5	898	S<282>	−1334.5	606.5
799	S<381>	348.5	461.5	849	S<331>	−501.5	461.5	899	S<281>	−1351.5	461.5
800	S<380>	331.5	606.5	850	S<330>	−518.5	606.5	900	S<280>	−1368.5	606.5

Preliminary

Table 8. Pad Center Coordinates (continued)

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
901	S<279>	−1385.5	461.5	951	S<229>	−2235.5	461.5	1001	S<179>	−3085.5	461.5
902	S<278>	−1402.5	606.5	952	S<228>	−2252.5	606.5	1002	S<178>	−3102.5	606.5
903	S<277>	−1419.5	461.5	953	S<227>	−2269.5	461.5	1003	S<177>	−3119.5	461.5
904	S<276>	−1436.5	606.5	954	S<226>	−2286.5	606.5	1004	S<176>	−3136.5	606.5
905	S<275>	−1453.5	461.5	955	S<225>	−2303.5	461.5	1005	S<175>	−3153.5	461.5
906	S<274>	−1470.5	606.5	956	S<224>	−2320.5	606.5	1006	S<174>	−3170.5	606.5
907	S<273>	−1487.5	461.5	957	S<223>	−2337.5	461.5	1007	S<173>	−3187.5	461.5
908	S<272>	−1504.5	606.5	958	S<222>	−2354.5	606.5	1008	S<172>	−3204.5	606.5
909	S<271>	−1521.5	461.5	959	S<221>	−2371.5	461.5	1009	S<171>	−3221.5	461.5
910	S<270>	−1538.5	606.5	960	S<220>	−2388.5	606.5	1010	S<170>	−3238.5	606.5
911	S<269>	−1555.5	461.5	961	S<219>	−2405.5	461.5	1011	S<169>	−3255.5	461.5
912	S<268>	−1572.5	606.5	962	S<218>	−2422.5	606.5	1012	S<168>	−3272.5	606.5
913	S<267>	−1589.5	461.5	963	S<217>	−2439.5	461.5	1013	S<167>	−3289.5	461.5
914	S<266>	−1606.5	606.5	964	S<216>	−2456.5	606.5	1014	S<166>	−3306.5	606.5
915	S<265>	−1623.5	461.5	965	S<215>	−2473.5	461.5	1015	S<165>	−3323.5	461.5
916	S<264>	−1640.5	606.5	966	S<214>	−2490.5	606.5	1016	S<164>	−3340.5	606.5
917	S<263>	−1657.5	461.5	967	S<213>	−2507.5	461.5	1017	S<163>	−3357.5	461.5
918	S<262>	−1674.5	606.5	968	S<212>	−2524.5	606.5	1018	S<162>	−3374.5	606.5
919	S<261>	−1691.5	461.5	969	S<211>	−2541.5	461.5	1019	S<161>	−3391.5	461.5
920	S<260>	−1708.5	606.5	970	S<210>	−2558.5	606.5	1020	S<160>	−3408.5	606.5
921	S<259>	−1725.5	461.5	971	S<209>	−2575.5	461.5	1021	S<159>	−3425.5	461.5
922	S<258>	−1742.5	606.5	972	S<208>	−2592.5	606.5	1022	S<158>	−3442.5	606.5
923	S<257>	−1759.5	461.5	973	S<207>	−2609.5	461.5	1023	S<157>	−3459.5	461.5
924	S<256>	−1776.5	606.5	974	S<206>	−2626.5	606.5	1024	S<156>	−3476.5	606.5
925	S<255>	−1793.5	461.5	975	S<205>	−2643.5	461.5	1025	S<155>	−3493.5	461.5
926	S<254>	−1810.5	606.5	976	S<204>	−2660.5	606.5	1026	S<154>	−3510.5	606.5
927	S<253>	−1827.5	461.5	977	S<203>	−2677.5	461.5	1027	S<153>	−3527.5	461.5
928	S<252>	−1844.5	606.5	978	S<202>	−2694.5	606.5	1028	S<152>	−3544.5	606.5
929	S<251>	−1861.5	461.5	979	S<201>	−2711.5	461.5	1029	S<151>	−3561.5	461.5
930	S<250>	−1878.5	606.5	980	S<200>	−2728.5	606.5	1030	S<150>	−3578.5	606.5
931	S<249>	−1895.5	461.5	981	S<199>	−2745.5	461.5	1031	S<149>	−3595.5	461.5
932	S<248>	−1912.5	606.5	982	S<198>	−2762.5	606.5	1032	S<148>	−3612.5	606.5
933	S<247>	−1929.5	461.5	983	S<197>	−2779.5	461.5	1033	S<147>	−3629.5	461.5
934	S<246>	−1946.5	606.5	984	S<196>	−2796.5	606.5	1034	S<146>	−3646.5	606.5
935	S<245>	−1963.5	461.5	985	S<195>	−2813.5	461.5	1035	S<145>	−3663.5	461.5
936	S<244>	−1980.5	606.5	986	S<194>	−2830.5	606.5	1036	S<144>	−3680.5	606.5
937	S<243>	−1997.5	461.5	987	S<193>	−2847.5	461.5	1037	S<143>	−3697.5	461.5
938	S<242>	−2014.5	606.5	988	S<192>	−2864.5	606.5	1038	S<142>	−3714.5	606.5
939	S<241>	−2031.5	461.5	989	S<191>	−2881.5	461.5	1039	S<141>	−3731.5	461.5
940	S<240>	−2048.5	606.5	990	S<190>	−2898.5	606.5	1040	S<140>	−3748.5	606.5
941	S<239>	−2065.5	461.5	991	S<189>	−2915.5	461.5	1041	S<139>	−3765.5	461.5
942	S<238>	−2082.5	606.5	992	S<188>	−2932.5	606.5	1042	S<138>	−3782.5	606.5
943	S<237>	−2099.5	461.5	993	S<187>	−2949.5	461.5	1043	S<137>	−3799.5	461.5
944	S<236>	−2116.5	606.5	994	S<186>	−2966.5	606.5	1044	S<136>	−3816.5	606.5
945	S<235>	−2133.5	461.5	995	S<185>	−2983.5	461.5	1045	S<135>	−3833.5	461.5
946	S<234>	−2150.5	606.5	996	S<184>	−3000.5	606.5	1046	S<134>	−3850.5	606.5
947	S<233>	−2167.5	461.5	997	S<183>	−3017.5	461.5	1047	S<133>	−3867.5	461.5
948	S<232>	−2184.5	606.5	998	S<182>	−3034.5	606.5	1048	S<132>	−3884.5	606.5
949	S<231>	−2201.5	461.5	999	S<181>	−3051.5	461.5	1049	S<131>	−3901.5	461.5
950	S<230>	−2218.5	606.5	1000	S<180>	−3068.5	606.5	1050	S<130>	−3918.5	606.5

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Table 9. Pad Center Coordinates (continued)

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
1051	S<129>	−3935.5	461.5	1101	S<79>	−4785.5	461.5	1151	S<29>	−5635.5	461.5
1052	S<128>	−3952.5	606.5	1102	S<78>	−4802.5	606.5	1152	S<28>	−5652.5	606.5
1053	S<127>	−3969.5	461.5	1103	S<77>	−4819.5	461.5	1153	S<27>	−5669.5	461.5
1054	S<126>	−3986.5	606.5	1104	S<76>	−4836.5	606.5	1154	S<26>	−5686.5	606.5
1055	S<125>	−4003.5	461.5	1105	S<75>	−4853.5	461.5	1155	S<25>	−5703.5	461.5
1056	S<124>	−4020.5	606.5	1106	S<74>	−4870.5	606.5	1156	S<24>	−5720.5	606.5
1057	S<123>	−4037.5	461.5	1107	S<73>	−4887.5	461.5	1157	S<23>	−5737.5	461.5
1058	S<122>	−4054.5	606.5	1108	S<72>	−4904.5	606.5	1158	S<22>	−5754.5	606.5
1059	S<121>	−4071.5	461.5	1109	S<71>	−4921.5	461.5	1159	S<21>	−5771.5	461.5
1060	S<120>	−4088.5	606.5	1110	S<70>	−4938.5	606.5	1160	S<20>	−5788.5	606.5
1061	S<119>	−4105.5	461.5	1111	S<69>	−4955.5	461.5	1161	S<19>	−5805.5	461.5
1062	S<118>	−4122.5	606.5	1112	S<68>	−4972.5	606.5	1162	S<18>	−5822.5	606.5
1063	S<117>	−4139.5	461.5	1113	S<67>	−4989.5	461.5	1163	S<17>	−5839.5	461.5
1064	S<116>	−4156.5	606.5	1114	S<66>	−5006.5	606.5	1164	S<16>	−5856.5	606.5
1065	S<115>	−4173.5	461.5	1115	S<65>	−5023.5	461.5	1165	S<15>	−5873.5	461.5
1066	S<114>	−4190.5	606.5	1116	S<64>	−5040.5	606.5	1166	S<14>	−5890.5	606.5
1067	S<113>	−4207.5	461.5	1117	S<63>	−5057.5	461.5	1167	S<13>	−5907.5	461.5
1068	S<112>	−4224.5	606.5	1118	S<62>	−5074.5	606.5	1168	S<12>	−5924.5	606.5
1069	S<111>	−4241.5	461.5	1119	S<61>	−5091.5	461.5	1169	S<11>	−5941.5	461.5
1070	S<110>	−4258.5	606.5	1120	S<60>	−5108.5	606.5	1170	S<10>	−5958.5	606.5
1071	S<109>	−4275.5	461.5	1121	S<59>	−5125.5	461.5	1171	S<9>	−5975.5	461.5
1072	S<108>	−4292.5	606.5	1122	S<58>	−5142.5	606.5	1172	S<8>	−5992.5	606.5
1073	S<107>	−4309.5	461.5	1123	S<57>	−5159.5	461.5	1173	S<7>	−6009.5	461.5
1074	S<106>	−4326.5	606.5	1124	S<56>	−5176.5	606.5	1174	S<6>	−6026.5	606.5
1075	S<105>	−4343.5	461.5	1125	S<55>	−5193.5	461.5	1175	S<5>	−6043.5	461.5
1076	S<104>	−4360.5	606.5	1126	S<54>	−5210.5	606.5	1176	S<4>	−6060.5	606.5
1077	S<103>	−4377.5	461.5	1127	S<53>	−5227.5	461.5	1177	S<3>	−6077.5	461.5
1078	S<102>	−4394.5	606.5	1128	S<52>	−5244.5	606.5	1178	S<2>	−6094.5	606.5
1079	S<101>	−4411.5	461.5	1129	S<51>	−5261.5	461.5	1179	S<1>	−6111.5	461.5
1080	S<100>	−4428.5	606.5	1130	S<50>	−5278.5	606.5	1180	DUMMY	−6128.5	606.5
1081	S<99>	−4445.5	461.5	1131	S<49>	−5295.5	461.5	1181	DUMMY	−6145.5	461.5
1082	S<98>	−4462.5	606.5	1132	S<48>	−5312.5	606.5	1182	DUMMY	−6162.5	606.5
1083	S<97>	−4479.5	461.5	1133	S<47>	−5329.5	461.5	1183	DUMMY	−6179.5	461.5
1084	S<96>	−4496.5	606.5	1134	S<46>	−5346.5	606.5	1184	DUMMY	−6645.5	606.5
1085	S<95>	−4513.5	461.5	1135	S<45>	−5363.5	461.5	1185	DUMMY	−6662.5	461.5
1086	S<94>	−4530.5	606.5	1136	S<44>	−5380.5	606.5	1186	DUMMY	−6679.5	606.5
1087	S<93>	−4547.5	461.5	1137	S<43>	−5397.5	461.5	1187	DUMMY	−6696.5	461.5
1088	S<92>	−4564.5	606.5	1138	S<42>	−5414.5	606.5	1188	DUMMY	−6713.5	606.5
1089	S<91>	−4581.5	461.5	1139	S<41>	−5431.5	461.5	1189	G<431>	−6730.5	461.5
1090	S<90>	−4598.5	606.5	1140	S<40>	−5448.5	606.5	1190	G<429>	−6747.5	606.5
1091	S<89>	−4615.5	461.5	1141	S<39>	−5465.5	461.5	1191	G<427>	−6764.5	461.5
1092	S<88>	−4632.5	606.5	1142	S<38>	−5482.5	606.5	1192	G<425>	−6781.5	606.5
1093	S<87>	−4649.5	461.5	1143	S<37>	−5499.5	461.5	1193	G<423>	−6798.5	461.5
1094	S<86>	−4666.5	606.5	1144	S<36>	−5516.5	606.5	1194	G<421>	−6815.5	606.5
1095	S<85>	−4683.5	461.5	1145	S<35>	−5533.5	461.5	1195	G<419>	−6832.5	461.5
1096	S<84>	−4700.5	606.5	1146	S<34>	−5550.5	606.5	1196	G<417>	−6849.5	606.5
1097	S<83>	−4717.5	461.5	1147	S<33>	−5567.5	461.5	1197	G<415>	−6866.5	461.5
1098	S<82>	−4734.5	606.5	1148	S<32>	−5584.5	606.5	1198	G<413>	−6883.5	606.5
1099	S<81>	−4751.5	461.5	1149	S<31>	−5601.5	461.5	1199	G<411>	−6900.5	461.5
1100	S<80>	−4768.5	606.5	1150	S<30>	−5618.5	606.5	1200	G<409>	−6917.5	606.5

Preliminary

Table 10. Pad Center Coordinates (continued)

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y	NO	NAME	X	Y
1201	G<407>	-6934.5	461.5	1251	G<307>	-7784.5	461.5	1301	G<207>	-8634.5	461.5
1202	G<405>	-6951.5	606.5	1252	G<305>	-7801.5	606.5	1302	G<205>	-8651.5	606.5
1203	G<403>	-6968.5	461.5	1253	G<303>	-7818.5	461.5	1303	G<203>	-8668.5	461.5
1204	G<401>	-6985.5	606.5	1254	G<301>	-7835.5	606.5	1304	G<201>	-8685.5	606.5
1205	G<399>	-7002.5	461.5	1255	G<299>	-7852.5	461.5	1305	G<199>	-8702.5	461.5
1206	G<397>	-7019.5	606.5	1256	G<297>	-7869.5	606.5	1306	G<197>	-8719.5	606.5
1207	G<395>	-7036.5	461.5	1257	G<295>	-7886.5	461.5	1307	G<195>	-8736.5	461.5
1208	G<393>	-7053.5	606.5	1258	G<293>	-7903.5	606.5	1308	G<193>	-8753.5	606.5
1209	G<391>	-7070.5	461.5	1259	G<291>	-7920.5	461.5	1309	G<191>	-8770.5	461.5
1210	G<389>	-7087.5	606.5	1260	G<289>	-7937.5	606.5	1310	G<189>	-8787.5	606.5
1211	G<387>	-7104.5	461.5	1261	G<287>	-7954.5	461.5	1311	G<187>	-8804.5	461.5
1212	G<385>	-7121.5	606.5	1262	G<285>	-7971.5	606.5	1312	G<185>	-8821.5	606.5
1213	G<383>	-7138.5	461.5	1263	G<283>	-7988.5	461.5	1313	G<183>	-8838.5	461.5
1214	G<381>	-7155.5	606.5	1264	G<281>	-8005.5	606.5	1314	G<181>	-8855.5	606.5
1215	G<379>	-7172.5	461.5	1265	G<279>	-8022.5	461.5	1315	G<179>	-8872.5	461.5
1216	G<377>	-7189.5	606.5	1266	G<277>	-8039.5	606.5	1316	G<177>	-8889.5	606.5
1217	G<375>	-7206.5	461.5	1267	G<275>	-8056.5	461.5	1317	G<175>	-8906.5	461.5
1218	G<373>	-7223.5	606.5	1268	G<273>	-8073.5	606.5	1318	G<173>	-8923.5	606.5
1219	G<371>	-7240.5	461.5	1269	G<271>	-8090.5	461.5	1319	G<171>	-8940.5	461.5
1220	G<369>	-7257.5	606.5	1270	G<269>	-8107.5	606.5	1320	G<169>	-8957.5	606.5
1221	G<367>	-7274.5	461.5	1271	G<267>	-8124.5	461.5	1321	G<167>	-8974.5	461.5
1222	G<365>	-7291.5	606.5	1272	G<265>	-8141.5	606.5	1322	G<165>	-8991.5	606.5
1223	G<363>	-7308.5	461.5	1273	G<263>	-8158.5	461.5	1323	G<163>	-9008.5	461.5
1224	G<361>	-7325.5	606.5	1274	G<261>	-8175.5	606.5	1324	G<161>	-9025.5	606.5
1225	G<359>	-7342.5	461.5	1275	G<259>	-8192.5	461.5	1325	G<159>	-9042.5	461.5
1226	G<357>	-7359.5	606.5	1276	G<257>	-8209.5	606.5	1326	G<157>	-9059.5	606.5
1227	G<355>	-7376.5	461.5	1277	G<255>	-8226.5	461.5	1327	G<155>	-9076.5	461.5
1228	G<353>	-7393.5	606.5	1278	G<253>	-8243.5	606.5	1328	G<153>	-9093.5	606.5
1229	G<351>	-7410.5	461.5	1279	G<251>	-8260.5	461.5	1329	G<151>	-9110.5	461.5
1230	G<349>	-7427.5	606.5	1280	G<249>	-8277.5	606.5	1330	G<149>	-9127.5	606.5
1231	G<347>	-7444.5	461.5	1281	G<247>	-8294.5	461.5	1331	G<147>	-9144.5	461.5
1232	G<345>	-7461.5	606.5	1282	G<245>	-8311.5	606.5	1332	G<145>	-9161.5	606.5
1233	G<343>	-7478.5	461.5	1283	G<243>	-8328.5	461.5	1333	G<143>	-9178.5	461.5
1234	G<341>	-7495.5	606.5	1284	G<241>	-8345.5	606.5	1334	G<141>	-9195.5	606.5
1235	G<339>	-7512.5	461.5	1285	G<239>	-8362.5	461.5	1335	G<139>	-9212.5	461.5
1236	G<337>	-7529.5	606.5	1286	G<237>	-8379.5	606.5	1336	G<137>	-9229.5	606.5
1237	G<335>	-7546.5	461.5	1287	G<235>	-8396.5	461.5	1337	G<135>	-9246.5	461.5
1238	G<333>	-7563.5	606.5	1288	G<233>	-8413.5	606.5	1338	G<133>	-9263.5	606.5
1239	G<331>	-7580.5	461.5	1289	G<231>	-8430.5	461.5	1339	G<131>	-9280.5	461.5
1240	G<329>	-7597.5	606.5	1290	G<229>	-8447.5	606.5	1340	G<129>	-9297.5	606.5
1241	G<327>	-7614.5	461.5	1291	G<227>	-8464.5	461.5	1341	G<127>	-9314.5	461.5
1242	G<325>	-7631.5	606.5	1292	G<225>	-8481.5	606.5	1342	G<125>	-9331.5	606.5
1243	G<323>	-7648.5	461.5	1293	G<223>	-8498.5	461.5	1343	G<123>	-9348.5	461.5
1244	G<321>	-7665.5	606.5	1294	G<221>	-8515.5	606.5	1344	G<121>	-9365.5	606.5
1245	G<319>	-7682.5	461.5	1295	G<219>	-8532.5	461.5	1345	G<119>	-9382.5	461.5
1246	G<317>	-7699.5	606.5	1296	G<217>	-8549.5	606.5	1346	G<117>	-9399.5	606.5
1247	G<315>	-7716.5	461.5	1297	G<215>	-8566.5	461.5	1347	G<115>	-9416.5	461.5
1248	G<313>	-7733.5	606.5	1298	G<213>	-8583.5	606.5	1348	G<113>	-9433.5	606.5
1249	G<311>	-7750.5	461.5	1299	G<211>	-8600.5	461.5	1349	G<111>	-9450.5	461.5
1250	G<309>	-7767.5	606.5	1300	G<209>	-8617.5	606.5	1350	G<109>	-9467.5	606.5

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Table 11. Pad Center Coordinates (continued)

[Unit: um]

NO	NAME	X	Y	NO	NAME	X	Y
1351	G<107>	−9484.5	461.5	1401	G<7>	−10334.5	461.5
1352	G<105>	−9501.5	606.5	1402	G<5>	−10351.5	606.5
1353	G<103>	−9518.5	461.5	1403	G<3>	−10368.5	461.5
1354	G<101>	−9535.5	606.5	1404	G<1>	−10385.5	606.5
1355	G<99>	−9552.5	461.5	1405	DUMMY	−10402.5	461.5
1356	G<97>	−9569.5	606.5	1406	DUMMY	−10419.5	606.5
1357	G<95>	−9586.5	461.5	1407	DUMMY	−10436.5	461.5
1358	G<93>	−9603.5	606.5	1408	DUMMY	−10453.5	606.5
1359	G<91>	−9620.5	461.5				
1360	G<89>	−9637.5	606.5				
1361	G<87>	−9654.5	461.5				
1362	G<85>	−9671.5	606.5				
1363	G<83>	−9688.5	461.5				
1364	G<81>	−9705.5	606.5				
1365	G<79>	−9722.5	461.5				
1366	G<77>	−9739.5	606.5				
1367	G<75>	−9756.5	461.5				
1368	G<73>	−9773.5	606.5				
1369	G<71>	−9790.5	461.5				
1370	G<69>	−9807.5	606.5				
1371	G<67>	−9824.5	461.5				
1372	G<65>	−9841.5	606.5				
1373	G<63>	−9858.5	461.5				
1374	G<61>	−9875.5	606.5				
1375	G<59>	−9892.5	461.5				
1376	G<57>	−9909.5	606.5				
1377	G<55>	−9926.5	461.5				
1378	G<53>	−9943.5	606.5				
1379	G<51>	−9960.5	461.5				
1380	G<49>	−9977.5	606.5				
1381	G<47>	−9994.5	461.5				
1382	G<45>	−10011.5	606.5				
1383	G<43>	−10028.5	461.5				
1384	G<41>	−10045.5	606.5				
1385	G<39>	−10062.5	461.5				
1386	G<37>	−10079.5	606.5				
1387	G<35>	−10096.5	461.5				
1388	G<33>	−10113.5	606.5				
1389	G<31>	−10130.5	461.5				
1390	G<29>	−10147.5	606.5				
1391	G<27>	−10164.5	461.5				
1392	G<25>	−10181.5	606.5				
1393	G<23>	−10198.5	461.5				
1394	G<21>	−10215.5	606.5				
1395	G<19>	−10232.5	461.5				
1396	G<17>	−10249.5	606.5				
1397	G<15>	−10266.5	461.5				
1398	G<13>	−10283.5	606.5				
1399	G<11>	−10300.5	461.5				
1400	G<9>	−10317.5	606.5				

PIN DESCRIPTION

POWER SUPPLY PINS

Table 12. Power supply pin description

Symbol	I/O	Description
VDD	P	Power supply for internal logic and internal RAM. Internally, voltage regulator output is connected to this pin. Connect a capacitor for stabilization. Don't apply any external power to this pin.
VDD3	P	Interface power supply level for logic signals (VDD3: 1.65V ~ 3.3 V)
VDD3O	O	Output port used only near logic pins fixed VDD3 It is connected internally with VDD3 It is not used, leave this pin open
AVDD	O	Generated power output pin for source driver block. AVDD is from step-up1 block. This pin needs to connect a capacitor for storage function. (AVDD: 3.5 ~ 6 V)
GVDD	O	Reference voltage level for grayscale voltage generator. Connect a capacitor for stabilization. When internal GVDD generator is not used, connect an external power supply 5 V.
VCI	P	Power supply for analog circuit blocks (VCI : 2.4 ~ 3.3V)
VSS	P	System ground (0V)
VSS3	P	System ground level for I/O
VSS3O	O	Output port used only near logic pins fixed VSS3 It is connected internally with VSS3 It is not used, leave this pin open
VSSC	P	System ground level for step up circuit block.
AVSS	P	System ground level for source driver block.
VGS	I	Reference voltage input pin for grayscale voltage generator. Connect an external resistor or to system ground.

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Table 13. Power supply pin description (continued)

Symbol	I/O	Description
VCI1	O	Reference input voltage for step-up1 circuit. Connect a capacitor for stabilization. <note> VCI1 cannot exceed 3V
VCL	O	3rd booster output pin used as a power supply for generating VCOML block. Connect a capacitor for stabilization.
VREF	I	Reference voltage for generating GVDD, VCOMH, VCOML
VCOM	O	Power supply pin for the TFT- display counter electrode. The alternating cycle can be set by the M pin. Connect this pin to the TFT-display counter electrode. This output pin is also used for charge-sharing function: When VCIR_MON = “High” period, all outputs of source driver (S1 to S720) are shorted to VCOM level (Hi-z). In case of VCOML < -1V, charge sharing function must not be used. (Set VCIR bit (R0Bh) to be “000” for preventing the abnormal function.)
VCOMR	I/O	Reference voltage input pin for VCOMH. When VCOMH voltage is adjusted externally, halt the internal adjuster of VCOMH by setting the register and insert a variable resistor between GVDD and VSS. When VCOMH is not adjusted externally, leave this pin open and adjust VCOMH by setting the internal register.
VCOMH	O	High level output voltage of VCOM. An internal register can be used to adjust the voltage. Connect this pin to the capacitor for stabilization.
VCOML	O	Low level output voltage of VCOM. An internal register can be used to adjust the voltage. Connect this pin to the capacitor for stabilization.
VGH	O	Positive power output pin for gate level shifter, internal step-up circuits, bias circuits, and operational amplifiers. Connect a capacitor for stabilization.
VGL	O	Negative power output pin for gate level shifter, bias circuits, and operational amplifiers. Connect a capacitor for stabilization. When internal VGL generator is not used, connect an external-voltage power supply higher than –13.75 V. To protect IC against latch up, connect the cathode of the schottky diode to the VSS pad and the anode of the schottky diode to the VGL pad. (refer to application circuit) Connect a capacitor for stabilization.

Table 14. Power supply pin description (continued)

Symbol	I/O	Description
C11M, C11P C12M, C12P	-	Connect the charge-pumping capacitor for generating AVDD level.
C21M, C21P C22M, C22P	-	Connect the charge-pumping capacitor for generating VGH, VGL level.
C31M, C31P	-	Connect the charge-pumping capacitor for generating VCL level.

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SYSTEM / RGB INTERFACE PINS

Table 15. System interface pin description

<table>
<tr><th>Symbol</th><th>I/O</th><th colspan="3">Description</th></tr>
<tr><td>S_PB</td><td>I</td><td colspan="3">Selects the CPU interface mode
"Low" = Parallel Interface, "High" = Serial Interface</td></tr>
<tr><td>ID_MIB</td><td>I</td><td colspan="3">Selects the CPU type
"Low" = Intel 80x-system, "High" = Motolora 68x-system
If S-PB = "High", the pin is used as ID setting bit for a device code.</td></tr>
<tr><td>CSB</td><td>I</td><td colspan="3">Chip select signal input pin.
Low: S6D0170 is selected and can be accessed
High: S6D0170 is not selected and cannot be accessed</td></tr>
<tr><td>RS</td><td>I</td><td colspan="3">Register select pin.
Low: Index/status, High: Instruction parameter, GRAM data
Must be fixed at VSS level when not used.</td></tr>
<tr><td rowspan="4">RW_WRB/
SCL</td><td rowspan="4">I</td><td>Pin function</td><td>CPU type</td><td>Pin description</td></tr>
<tr><td>RW</td><td>68-system</td><td>Read/Write operation selection pin.
Low: Write, High: Read</td></tr>
<tr><td>WRB</td><td>80-system</td><td>Write strobe signal. (Input pin)
Data is fetched at the rising edge.</td></tr>
<tr><td>SCL</td><td>Serial Peripheral Interface (SPI)</td><td>The synchronous clock signal.
(Input pin)</td></tr>
<tr><td rowspan="4">E_RDB</td><td rowspan="4">I</td><td>Pin function</td><td>CPU type</td><td>Pin description</td></tr>
<tr><td>E</td><td>68-system</td><td>Read/Write operation enable pin.</td></tr>
<tr><td>RDB</td><td>80-system</td><td>Read strobe signal. (Input pin)
Read out data at the low level.</td></tr>
<tr><td colspan="3">When SPI mode is selected, fix this pin at VSS level.</td></tr>
<tr><td>SDI</td><td>I</td><td colspan="3">For a serial peripheral interface (SPI), input data is fetched at the rising edge of the SCL signal. Fix this pin at VSS level if the pin is not used.</td></tr>
<tr><td>SDO</td><td>O</td><td colspan="3">For a serial peripheral interface (SPI), serves as the serial data output pin (SDO). Successive bits are output at the falling edge of the SCL signal.
Leave this pin open if the pin is not used.</td></tr>
<tr><td>RESETB</td><td>I</td><td colspan="3">Reset pin Initializes the IC when low. Should be reset after power-on.</td></tr>
<tr><td>DB17-DB0</td><td>I/O</td><td colspan="3">Bi-directional data bus.
When CPU I/F,
18-bit interface : DB 17-0
16-bit interface : DB 17-10, DB 8-1
9-bit interface : DB 17-9
8-bit interface : DB 17-10
When RGB I/F,
18-bit interface : DB 17-0
16-bit interface : DB 17-13, DB 11-1
6-bit interface : DB 17-12
Fix unused pin to the VDD3 or VSS level.</td></tr>
</table>

Table 16. System interface pin description (Continued)

<table>
<tr><th>Symbol</th><th>I/O</th><th colspan="4">Description</th></tr>
<tr><td rowspan="7">ENABLE</td><td rowspan="7">I</td><td colspan="4">Data enable signal pin for RGB interface.
EPL="0": Only in case of ENABLE="Low", the IC can be access via RGB interface.
EPL="1": Only in case of ENABLE="High", the IC can be access via RGB interface</td></tr>
<tr><td>EPL</td><td>ENABLE</td><td>GRAM write</td><td>GRAM address</td></tr>
<tr><td>0</td><td>0</td><td>Valid</td><td>Updated</td></tr>
<tr><td>0</td><td>1</td><td>Invalid</td><td>Held</td></tr>
<tr><td>1</td><td>0</td><td>Invalid</td><td>Held</td></tr>
<tr><td>1</td><td>1</td><td>Valid</td><td>Updated</td></tr>
<tr><td colspan="4">Fix ENABLE pin at VDD3 or VSS level if the pin is not used.</td></tr>
<tr><td>VSYNC</td><td>I</td><td colspan="4">Frame-synchronizing signal.
VSPL= "0": Low active, VSPL="1": High active
Fix this pin at VDD3 or VSS level if the pin is not used.</td></tr>
<tr><td>HSYNC</td><td>I</td><td colspan="4">Line-synchronizing signal.
HSPL="0": Low active, HSPL="1": High active
Fix this pin at VDD3 or VSS level if the pin is not used.</td></tr>
<tr><td>DOTCLK</td><td>I</td><td colspan="4">Input pin for clock signal of external interface: dot clock.
DPL="0": Display data is fetched at DOTCLK's rising edge
DPL="1": Display data is fetched at DOTCLK's falling edge
Fix this pin at VDD3 or VSS level if the pin is not used.</td></tr>
</table>

DISPLAY PINS

Table 17. Display pin description

Symbol	I/O	Description
S1 – S720	O	Source driver output pins. The SS bit can change the shift direction of the source signal. For example, if SS = 0, gray data of <S1/S2/S3> is read from RAM address 0000h. If SS = 1, contents of is RAM address 0000h is out from <S526/S527/S720>. S1, S4, S7, ... S(3n-1) : display Red (R) (BGR = 0) S2, S5, S8, ... S(3n-2) : display Green (G) (BGR = 0) S3, S6, S9, ... S(3n) : display Blue (B) (BGR = 0)
G1 – G432	O	Gate driver output pins. The output of driving circuit is whether VGH or VGL. VGH : Gate-ON level VGL : Gate-OFF level

MISCELLANEOUS PINS

Table 18. Oscillator and internal power regulator pin description

Symbol	I/O	Description
TEST_MODE2-0 TEST_MUX TEST_GRAY TEST_RVDDOFF TEST_IN4-0	I	Input pins used only for test purpose at vendor-side. In normal operation, connect this pin to VSS.
TI_FIX	I	Input pin for preventing abnormal function by set ESD. Fixed values of test register In normal operation, connect this pin to VSS.
M CL1 FLM VCIR_MON GATE_MON	O	Output pins used only for test purpose at vendor-side. In normal operation, leave this pin open.
DUMMY	-	Dummy pins. Leave this pin should be open.
CONTACT1 CONTACT2	-	Pass-through pins Identical pins at input and output part connected without any circuitry.
MTPG	I	Power supply pin for MTP cell writing operation.
MTPD	I	Power supply pin for MTP cell writing operation.
EN_EXCLK	I	Enable external clock input Fix this pin at VSS level if the pin is not used.
EXCLK	I	External clock input pin. Fix this pin at VSS level if the pin is not used.

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FUNCTIONAL DESCRIPTION

SYSTEM INTERFACE

The S6D0170 has ten high-speed system interfaces: an 80-system 18-/16-/9-/8-bit bus, a 68-system 18-/16-/9-/8-bit and two type serial interface (SPI: Serial Peripheral Interface).
The S6D0170 has three 18-bit registers: an index register (IR), a write data register (WDR), and a read data register (RDR). The IR stores index information for control register and GRAM. The WDR temporarily stores data to be written into control register and GRAM. The RDR temporarily stores data read from GRAM. Data written into the GRAM from CPU is initially written to the WDR and then written to the GRAM automatically. Data is read through the RDR when reading from the GRAM, and the first read data is invalid and the second and the following data are valid.
Execution time for instruction, except oscillation start, is 0-clock cycle so that instructions can be written in succession.

Table 19. Register Selection (18-/16-/9-/8- Parallel Interface)

SYSTEM	RW_WRB	E_RDB	RS	Operations
68	0	1	0	Write index to IR
	1	1	0	Read internal status
	0	1	1	Write to control register and GRAM through WDR
	1	1	1	Read from GRAM through RDR
80	0	1	0	Write index to IR
	1	0	0	Read internal status
	0	1	1	Write to control register and GRAM through WDR
	1	0	1	Read from GRAM through RDR

Table 20. CSB signal (GRAM update control)

CSB	Operation
0	Data is written to GRAM, GRAM address is updated
1	Data is not written to GRAM, GRAM address is not updated

Table 21. Register Selection (Serial Peripheral Interface)

R/W bit	RS bit	Operation
0	0	Write index to IR
1	0	Read internal status
0	1	Write data to control register and GRAM through WDR
1	1	Read data from GRAM through RDR

EXTERNAL INTERFACE (RGB-I/F)

The S6D0170 incorporates RGB interface as external interface for motion picture display.
When the RGB interface is selected, the synchronization signals (VSYNC, HSYNC, and DOTCLK) are available for display. The RGB data for display (DB17-0) are written according to enable signal (ENABLE) in synchronization with VSYNC, HSYNC, and DOTCLK signal. This allows flicker-free updating of the screen. See the section on the EXTERNAL DISPLAY INTERFACE.

ADDRESS COUNTER (AC)

The address counter (AC) assigns address to GRAM. When an address-set-instruction is written to the IR, the address information is sent from IR to AC. After writing and reading to the GRAM, the address value of AC is automatically increased/ decreased by 1 according to ID1-0 bit of control register. A window address function allows data to be written only to a window area specified by GRAM.

GRAPHICS RAM (GRAM)

The graphics RAM (GRAM) has 18-bits/pixel and stores the bit-pattern data for 240-RGB x 432-dot display.

GRAYSCALE VOLTAGE GENERATOR

The grayscale voltage circuit generates a certain voltage level that is specified by the grayscale ϒ-adjusting resistor for LCD driver circuit. S6D0170 has three grayscale voltage circuits and ϒ-adjusting resistors for separated RGB gamma adjust. By use of the generator, 262,144 colors can be displayed at the same time. For details, see the GAMMA-ADJUSTING RESISTOR section.

TIMING GENERATOR

The timing generator generates timing signals for the operation of internal circuits such as GRAM.
The GRAM read timing for display and the internal operation timing for CPU access is generated separately to avoid interference with one another. Several important timing signals can be monitored via signal monitoring pin (M, FLM, CL1, VCIR_MON, GATE_MON).

OSCILLATION CIRCUIT (OSC)

The S6D0170 can provide R-C oscillation simply through the internal oscillation-resistor. The appropriate oscillation frequency for operating voltage, display size, and frame frequency can be obtained by adjusting the internal register. Clock pulse can also be supplied externally by EN_EXCLK and EXCLK pins. Since R-C oscillation stops during the standby mode, current consumption can be reduced. For details, see the OSCILLATION CIRCUIT section.

SOURCE DRIVER CIRCUIT

The Liquid-Crystal-Display source driver circuit consists of 720 drivers (S1 to S720).
Display pattern data is latched when 720-channel data has arrived. The latched data then enables the source drivers to generate drive waveform outputs. The SS bit can change the shift direction of 720-channel data by selecting an appropriate direction for the device-mounted configuration.

GATE DRIVER CIRCUIT

The liquid crystal display gate driver circuit consists of 432 gate drivers (G1 to G432).
The VGH or VGL level is output by the signal from the gate control circuit.

GRAM ADDRESS MAP

The image data stored in GRAM corresponds to pixel data on display as shown below:

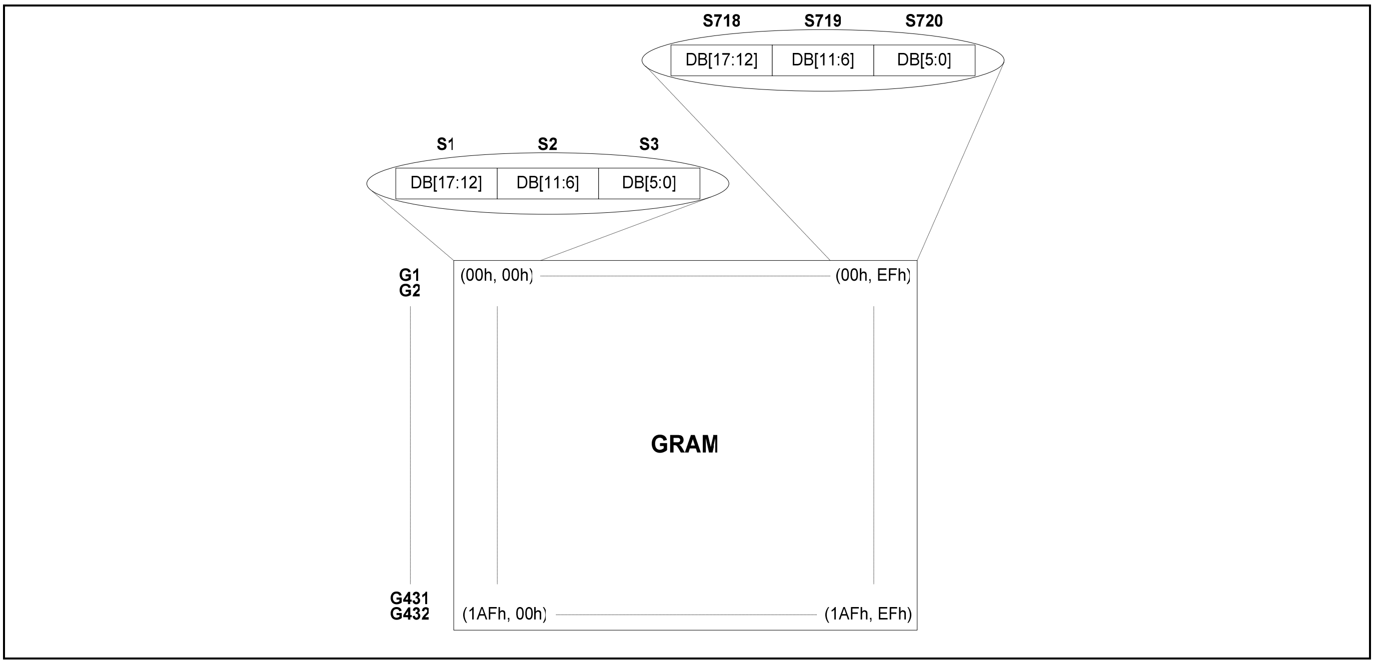

Figure 6. GRAM address (SS="0" and BGR="0")

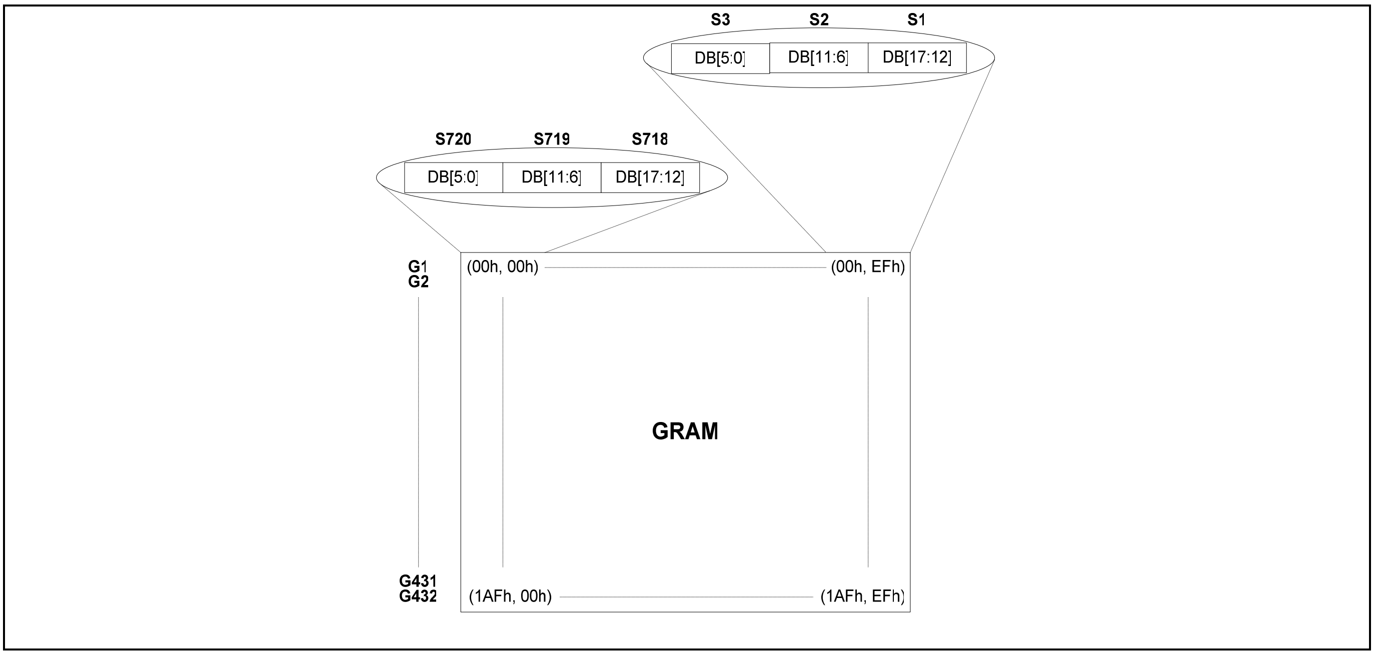

Figure 7. GRAM address (SS="1" and BGR="1")

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INSTRUCTIONS

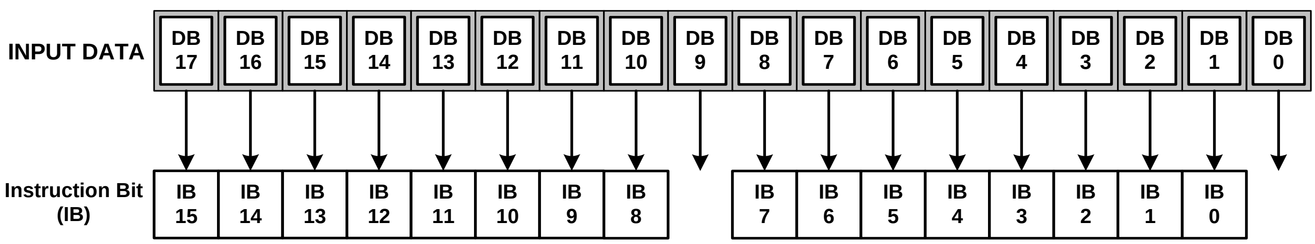
The S6D0170 uses the 18-bit bus architecture. Before the internal operation of the S6D0170 starts, control information is stored temporarily in the registers described below to allow high-speed interfacing with a high-performance microcomputer. The internal operation of the S6D0170 is determined by signals sent from the microcomputer. These signals, which include the register selection signal (RS), the read/write signal (R/W), and the data bus signals (DB17 to DB0), make up the S6D0170 instructions.
There are seven categories of instructions that:

- Specify the index
- Control the display
- Control power management
- Set internal GRAM addresses
- Transfer data to and from the internal GRAM
- Set grayscale level for the internal grayscale palette table
- Interface with the gate driver and power supply IC

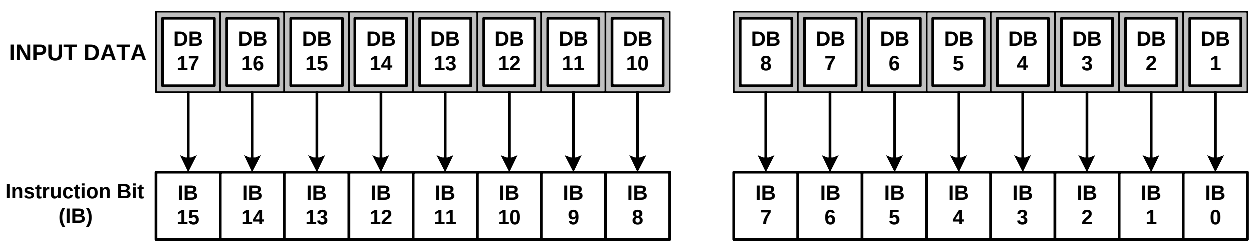
Normally, instructions that write data are used the most. However, an auto-update of internal GRAM addresses after each data write can lighten the microcomputer program load. As instructions are executed in 0 cycles, they can be written in succession.

The 16-bit instruction assignment differs from interface-setup (18-/16-/9-/8-/SPI), so instructions should be fetched according to the data format shown below:

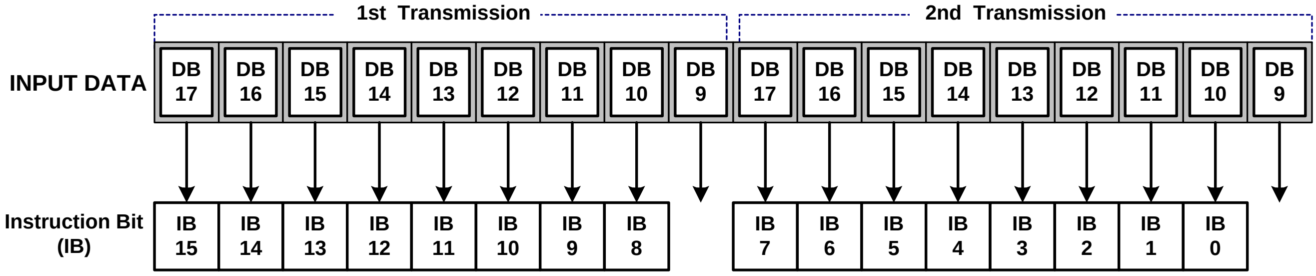
68/80-system 18-bit Interface



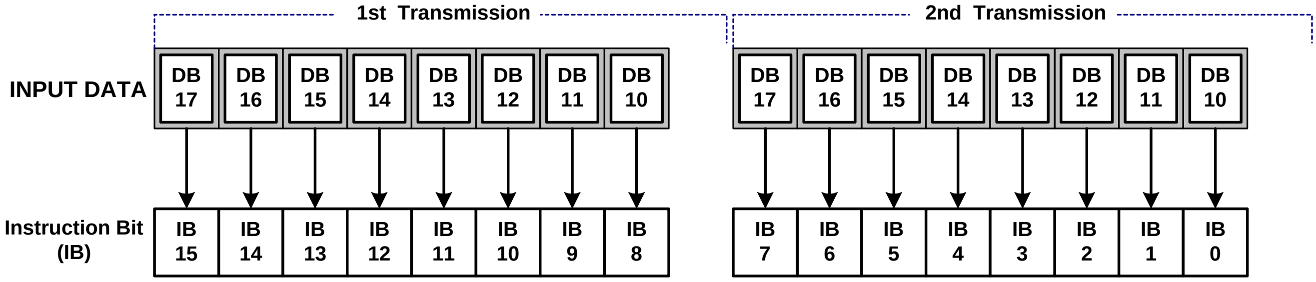
68/80-system 16-bit Interface/SPI (Serial Peripheral Interface)



68/80-system 9-bit Interface



68/80-system 8-bit Interface



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INSTRUCTION TABLE

Table 22. Instruction Table

Reg. No	R/W	RS	IB 15	IB 14	IB 13	IB 12	IB 11	IB 10	IB 9	IB 8	IB 7	IB 6	IB 5	IB 4	IB 3	IB 2	IB 1	IB 0	Register Name / Description
IR	W	0	X	X	X	X	X	X	X	X	X	ID6	ID5	ID4	ID3	ID2	ID1	ID0	**Index /** Sets the index register value
SR	R	0	X	X	X	X	X	X	X	L8	L7	L6	L5	L4	L3	L2	L1	L0	**Status read /** Reads the internal status of the S6D0170
R00h	W	0	No operation																**No Operation**
R01h	W	1	VSPL	HSPL	DPL	EPL	X	SM	GS	SS	X	X	NL5	NL4	NL3	NL2	NL1	NL0	**Driver output control(R01H) /** VSPL: set polarity of VSYNC pin. HSPL: set polarity of HSYNC pin. DPL: set polarity of DOTCLK pin. EPL: set polarity of ENABLE pin. SM: gate driver division drive control GS: gate driver shift direction SS: source driver shift direction. NL5-0: number of driving lines.
R02h	W	1	X	X	X	X	X	X	INV1	INV0	X	X	X	X	X	X	X	X	**LCD-Driving-waveform control (R02H)/** INV1-0: LCD drive AC waveform
R03h	W	1	CLS	MDT1	MDT0	BGR	X	X	X	X	X	AM	I/D1	I/D0	X	X	X	X	**Entry mode(R03H) /** **CLS : Select Color Format** MDT1-0 : Multiple Data Transfer BGR: RGB swap control I/D1-0: address counter Increment / Decrement control AM: horizontal / vertical RAM update
R05h	R	1	0	0	0	0	0	0	0	1	0	1	1	1	0	0	0	0	**Device code read /** Read 0170H
R07h	W	1	X	X	X	TEMON	X	X	X	S_GAM	X	X	X	GON	CL	REV	D1	D0	**Display control (R07H) /** TEMON : Enable Frame Flag Output(FLM) S_GAM: Single gamma enable GON: gate on/off control CL: 8-color display mode enable REV: display area inversion drive D1-0: source output control
R08h	W	1	X	X	X	X	X	X	X	X	FP3	FP2	FP1	FP0	BP3	BP2	BP1	BP0	**Blank period control (R08H)/** FP3-0: Front porch setting BP3-0: Back porch setting
R0Bh	W	1	NO3	NO2	NO1	NO0	SDT3	SDT2	SDT1	SDT0	X	X	X	DIV	RTN3	RTN2	RTN1	RTN0	**Frame cycle control (R0BH)/** NO3-0: specify the amount of non-overlap SDT3-0: set amount of source delay DIV : Division Ratio of internal clock RTN3-0: set the 1-H period
R0Ch	W	1	X	X	X	X	X	X	X	RM	X	X	X	DM	X	X	RIM1	RIM0	**External interface control(R0CH) /** RM: specify the interface for RAM access DM: specify display operation mode RIM1-0: specify RGB-I/F mode
R0Fh	W	1	X	X	X	FOSC 4	FOSC 3	FOSC 2	FOSC 1	FOSC 0	X	X	X	X	X	X	X	OSC ON	**Start oscillation(R00H) /** **FOSC4-0:Adjust frequency of oscillator** OSCON : oscillator ON
R10h	W	1	X	X	X	X	X	SAP2	SAP1	SAP0	X	X	X	X	X	DSTB _1F	DSTB	STB	**Power control 1 (R10H) /** SAP2-0:adjust amount of current for source driver amp DSTB_1F:internal 1F DSTB after STB DSTB: deep standby mode control STB: standby mode control
R11h	W	1	X	X	X	APON	PON3	PON2	PON1	PON	X	X	AON	VCI1_ EN	VC3	VC2	VC1	VC0	**Power control 2 (R11H)/** APON: auto power on control PON3 : VCL booster control PON2 : VGL booster control PON1: VGH booster circuit control PON: AVDD booster circuit control AON: operation start bit for the amplifier. VCI1_EN: VCI1 amplifier control VC3-0:set VCI1 voltage
R12h	W	1	X	BT2	BT1	BT0	X	X	DC11	DC10	X	X	DC21	DC20	X	X	DC31	DC30	Power control 3 (R12H)/ BT2-0:Adjust scale factor DC11-0:Adjust the frequency in stepup1 DC21-0: Adjust the frequency in stepup2 DC31-0: Adjust the frequency in stepup3
R13h	W	1	X	X	RTR_ AMP_ EN	DCR_ EX	X	DCR2	DCR1	DCR0	X	GVD6	GVD5	GVD4	GVD3	GVD2	GVD1	GVD0	**Power control 4 (R13H)/** RTR_AMP_EN: Select AB class amp DCR_EX: Input signal selection. DCR2-0: Set clock cycle for step-up circuit. GVD6-0:set GVDD voltage
R14h	W	1	VCO MG	VCM6	VCM5	VCM4	VCM3	VCM2	VCM1	VCM0	VCM R	VML6	VML5	VML4	VML3	VML2	VML1	VML0	**Power control 5 (R14H)/** VCOMG : VCOML -> GND VCMR: VCOMH control VCM6-0:set the VCOMH voltage VML6-0:set the VCOM Amplitude
R15h	W	1	X	X	X	X	X	X	X	X	X	VCIR 2	VCIR 1	VCIR 0	X	X	LVCM _VCIR	VCIR_ VSSB	**VCI Recycling (R15H)/** VCIR2-0: VCI Recycling period setting LVCM_VCIR: VCIR with VSS VCIR_VSSB : VCIR with VCI and VSS
R1Fh	W	1	X	X	X	X	X	X	X	X	1	0	1	0	1	0	1	0	**Software RESET(R1FH)** AAh : Software RESET

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Table 23. Instruction Table (Continued)

Reg. No	R/W	RS	IB 15	IB 14	IB 13	IB 12	IB 11	IB 10	IB 9	IB 8	IB 7	IB 6	IB 5	IB 4	IB 3	IB 2	IB 1	IB 0	Register Name / Description
R22h	W	1	**WD17-0 : Pin assignment varies according to the interface method**																**Write data to GRAM (R22H)/** WD17-0:Input data for GRAM
	R	1	**RD17-0 : Pin assignment varies according to the interface method**																**Read data from GRAM (R22H)/** RD17-0:Read data from GRAM
R23h	W	0	Select 18/16-Bit Data Bus Interface																**Select Data Bus for Interface (R23H,R24H)**
R24h	W	0	Select 9/8-Bit Data Bus Interface																
R40h	W	1	X	X	X	X	X	X	X	X	X	X	SCN5	SCN4	SCN3	SCN2	SCN1	SCN0	**Gate scan position (R40H)/** SCN5-0 : Scan starting position of gate
R41h	W	1	X	X	X	X	X	X	X	SEA8	SEA7	SEA6	SEA5	SEA4	SEA3	SEA2	SEA1	SEA0	**Vertical scroll control 1 (R41H,R42H)/** SEA7-0 : Scroll End Address SSA7-0 : Scroll Start Address
R42h	W	1	X	X	X	X	X	X	X	SSA8	SSA7	SSA6	SSA5	SSA4	SSA3	SSA2	SSA1	SSA0	
R43h	W	1	X	X	X	X	X	X	X	SST8	SST7	SST6	SST5	SST4	SST3	SST2	SST1	SST0	**Vertical scroll control 2(R43H)/** SST7-0 : Scroll Start and Step
R44h	W	1	X	X	X	X	X	X	X	SE18	SE17	SE16	SE15	SE14	SE13	SE12	SE11	SE10	**Partial screen driving position (R44H,R45H)/** SS18-10: screen start position SE18-10: screen end position
R45h	W	1	X	X	X	X	X	X	X	SS18	SS17	SS16	SS15	SS14	SS13	SS12	SS11	SS10	
R46h	W	1	X	X	X	X	X	X	X	HEA8	HEA7	HEA6	HEA5	HEA4	HEA3	HEA2	HEA1	HEA0	**Horizontal window address (R45H)/** HEA8-0: Horizontal window address end position
R47h	W	1	X	X	X	X	X	X	X	HSA8	HSA7	HSA6	HSA5	HSA4	HSA3	HSA2	HSA1	HSA0	HSA8-0: Horizontal window address start position
R48h	W	1	X	X	X	X	X	X	X	VEA8	VEA7	VEA6	VEA5	VEA4	VEA3	VEA2	VEA1	VEA0	**Vertical window Address (R46H,R47H)/** VEA8-0: Vertical window address end position
R49h	W	1	X	X	X	X	X	X	X	VSA8	VSA7	VSA6	VSA5	VSA4	VSA3	VSA2	VSA1	VSA0	VSA8-0: Vertical window address start position
R50h	W	1	X	X	X	R_VRN04	R_VRN03	R_VRN02	R_VRN01	R_VRN00	X	X	X	R_VRP04	R_VRP03	R_VRP02	R_VRP01	R_VRP00	**Gamma control R1 (R50H)/** Adjust Amplitude voltage for R
R51h	W	1	X	X	X	R_VRN14	R_VRN13	R_VRN12	R_VRN11	R_VRN10	X	X	X	R_VRP14	R_VRP13	R_VRP12	R_VRP11	R_VRP10	**Gamma control R2 (R51H)/** Adjust Amplitude voltage for R
R52h	W	1	X	X	X	X	R_PRN03	R_PRN02	R_PRN01	R_PRN00	X	X	X	X	R_PRP03	R_PRP02	R_PRP01	R_PRP00	**Gamma control R3 (R52H)/** Adjust Gradient for R
R53h	W	1	X	X	X	X	R_PRN13	R_PRN12	R_PRN11	R_PRN10	X	X	X	X	R_PRP13	R_PRP12	R_PRP11	R_PRP10	**Gamma control R4 (R53H)/** Adjust Gradient for R
R54h	W	1	X	X	X	X	R_PKN03	R_PKN02	R_PKN01	R_PKN00	X	X	X	X	R_PKP03	R_PKP02	R_PKP01	R_PKP00	**Gamma control R5 (R54H)/** Adjust Gamma voltage for R
R55h	W	1	X	X	X	X	R_PKN13	R_PKN12	R_PKN11	R_PKN10	X	X	X	X	R_PKP13	R_PKP12	R_PKP11	R_PKP10	**Gamma control R6 (R55H)/** Adjust Gamma voltage for R
R56h	W	1	X	X	X	X	R_PKN23	R_PKN22	R_PKN21	R_PKN20	X	X	X	X	R_PKP23	R_PKP22	R_PKP21	R_PKP20	**Gamma control R7 (R56H)/** Adjust Gamma voltage for R
R57h	W	1	X	X	X	X	R_PKN33	R_PKN32	R_PKN31	R_PKN30	X	X	X	X	R_PKP33	R_PKP32	R_PKP31	R_PKP30	**Gamma control R8 (R57H)/** Adjust Gamma voltage for R
R58h	W	1	X	X	X	X	R_PKN43	R_PKN42	R_PKN41	R_PKN40	X	X	X	X	R_PKP43	R_PKP42	R_PKP41	R_PKP40	**Gamma control R9 (R58H)/** Adjust Gamma voltage for R
R59h	W	1	X	X	X	X	R_PKN53	R_PKN52	R_PKN51	R_PKN50	X	X	X	X	R_PKP53	R_PKP52	R_PKP51	R_PKP50	**Gamma control R10 (R59H)/** Adjust Gamma voltage for R
R60h	W	1	X	X	X	G_VRN04	G_VRN03	G_VRN02	G_VRN01	G_VRN00	X	X	X	G_VRP04	G_VRP03	G_VRP02	G_VRP01	G_VRP00	**Gamma control G1 (R60H)/** Adjust Amplitude voltage for G
R61h	W	1	X	X	X	G_VRN14	G_VRN13	G_VRN12	G_VRN11	G_VRN10	X	X	X	G_VRP14	G_VRP13	G_VRP12	G_VRP11	G_VRP10	**Gamma control G2 (R61H)/** Adjust Amplitude voltage for G
R62h	W	1	X	X	X	X	G_PRN03	G_PRN02	G_PRN01	G_PRN00	X	X	X	X	G_PRP03	G_PRP02	G_PRP01	G_PRP00	**Gamma control G3 (R62H)/** Adjust Gradient for G
R63h	W	1	X	X	X	X	G_PRN13	G_PRN12	G_PRN11	G_PRN10	X	X	X	X	G_PRP13	G_PRP12	G_PRP11	G_PRP10	**Gamma control G4 (R63H)/** Adjust Gradient for G
R64h	W	1	X	X	X	X	G_PKN03	G_PKN02	G_PKN01	G_PKN00	X	X	X	X	G_PKP03	G_PKP02	G_PKP01	G_PKP00	**Gamma control G5 (R64H)/** Adjust Gamma voltage for G
R65h	W	1	X	X	X	X	G_PKN13	G_PKN12	G_PKN11	G_PKN10	X	X	X	X	G_PKP13	G_PKP12	G_PKP11	G_PKP10	**Gamma control G6 (R65H)/** Adjust Gamma voltage for G

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Table 24. Instruction Table (Continued)

Reg. No	R/W	RS	IB 15	IB 14	IB 13	IB 12	IB 11	IB 10	IB 9	IB 8	IB 7	IB 6	IB 5	IB 4	IB 3	IB 2	IB 1	IB 0	Register Name / Description
R66h	W	1	X	X	X	X	G_PKN23	G_PKN22	G_PKN21	G_PKN20	X	X	X	X	G_PKP23	G_PKP22	G_PKP21	G_PKP20	**Gamma control G7 (R66H)/** Adjust Gamma voltage for G
R67h	W	1	X	X	X	X	G_PKN33	G_PKN32	G_PKN31	G_PKN30	X	X	X	X	G_PKP33	G_PKP32	G_PKP31	G_PKP30	**Gamma control G8 (R67H)/** Adjust Gamma voltage for G
R68h	W	1	X	X	X	X	G_PKN43	G_PKN42	G_PKN41	G_PKN40	X	X	X	X	G_PKP43	G_PKP42	G_PKP41	G_PKP40	**Gamma control G9 (R68H)/** Adjust Gamma voltage for G
R69h	W	1	X	X	X	X	G_PKN53	G_PKN52	G_PKN51	G_PKN50	X	X	X	X	G_PKP53	G_PKP52	G_PKP51	G_PKP50	**Gamma control G10 (R69H)/** Adjust Gamma voltage for G
R70h	W	1	X	X	X	B_VRN04	B_VRN03	B_VRN02	B_VRN01	B_VRN00	X	X	X	B_VRP04	B_VRP03	B_VRP02	B_VRP01	B_VRP00	**Gamma control B1 (R70H)/ Adjust Amplitude voltage for B**
R71h	W	1	X	X	X	B_VRN14	B_VRN13	B_VRN12	B_VRN11	B_VRN10	X	X	X	B_VRP14	B_VRP13	B_VRP12	B_VRP11	B_VRP10	**Gamma control B2 (R71H)/ Adjust Amplitude voltage for B**
R72h	W	1	X	X	X	X	B_PRN03	B_PRN02	B_PRN01	B_PRN00	X	X	X	X	B_PRP03	B_PRP02	B_PRP01	B_PRP00	**Gamma control B3 (R72H)/ Adjust Gradiendt for B**
R73h	W	1	X	X	X	X	B_PRN13	B_PRN12	B_PRN11	B_PRN10	X	X	X	X	B_PRP13	B_PRP12	B_PRP11	B_PRP10	**Gamma control B4 (R73H)/ Adjust Gradient for B**
R74h	W	1	X	X	X	X	B_PKN03	B_PKN02	B_PKN01	B_PKN00	X	X	X	X	B_PKP03	B_PKP02	B_PKP01	B_PKP00	**Gamma control B5 (R74H)/ Adjust Gamma voltage for B**
R75h	W	1	X	X	X	X	B_PKN13	B_PKN12	B_PKN11	B_PKN10	X	X	X	X	B_PKP13	B_PKP12	B_PKP11	B_PKP10	**Gamma control B6 (R75H)/ Adjust Gamma voltage for B**
R76h	W	1	X	X	X	X	B_PKN23	B_PKN22	B_PKN21	B_PKN20	X	X	X	X	B_PKP23	B_PKP22	B_PKP21	B_PKP20	**Gamma control B7 (R76H)/ Adjust Gamma voltage for B**
R77h	W	1	X	X	X	X	B_PKN33	B_PKN32	B_PKN31	B_PKN30	X	X	X	X	B_PKP33	B_PKP32	B_PKP31	B_PKP30	**Gamma control B8 (R77H)/ Adjust Gamma voltage for B**
R78h	W	1	X	X	X	X	B_PKN43	B_PKN42	B_PKN41	B_PKN40	X	X	X	X	B_PKP43	B_PKP42	B_PKP41	B_PKP40	**Gamma control B9 (R78H)/ Adjust Gamma voltage for B**
R79h	W	1	X	X	X	X	B_PKN53	B_PKN52	B_PKN51	B_PKN50	X	X	X	X	B_PKP53	B_PKP52	B_PKP51	B_PKP50	**Gamma control B10 (R79H)/ Adjust Gamma voltage for B**
R80h	W	1	X	X	X	MTP_LOAD	X	X	X	MTP_WRB	X	X	X	MTP_SEL	X	X	X	MTP_ERB	**MTP control (R80H) :**
R81h	W	1	X	X	X	X	X	X	X	X	1	0	0	0	1	1	0	0	**Test Key Command (R81H) : 8Ch**
R82h	R	1	X	X	X	X	X	X	X	X	MTP_DOUT7	MTP_DOUT6	MTP_DOUT5	MTP_DOUT4	MTP_DOUT3	MTP_DOUT2	MTP_DOUT1	MTP_DOUT0	**MTP Read Data (R82H) :**
R90h	W	1	Test command1																**Don't use this command**
R91h	W	1	Test command2																**Don't use this command**
R92h	W	1	Test command3																**Don't use this command**
R93h	W	1	Test command4																**Don't use this command**

Note : Commands R01h(SM, GS), R07h(D[1:0], GON, CL, REV), R0Bh(DIV,RTN[3:0]), R43h(SST[8:0]), R44h(SE1[8:0]), and R45h(SS1[8:0]) are updated during Frame Flag to avoid abnormal visual effects. If the other commands are updated during display operating, abnormal visual effects can appear.

INSTRUCTION DESCRIPTIONS

INDEX

The index instruction specifies the RAM control indexes (R00h to R93h). It sets the register number in the range of 00000000 to 11111111 in binary form. However, R90h to R93h are disabled as they are test registers.

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	0	X	X	X	X	X	X	X	X	ID7	ID6	ID5	ID4	ID3	ID2	ID1	ID0

Note) When the register is written, X means “0”. And when the register is read, X means “0” or “1”.

STATUS READ

The status read instruction reads out the internal status of the IC.

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
R	0	0	0	0	0	0	0	0	L8	L7	L6	L5	L4	L3	L2	L1	L0

L8–0: Indicate the driving raster-row position in scan address of GRAM where the liquid crystal display is being driven.

NO OPERATION (R00H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	0	No Operation															

This command does not have any effect on the display module. However it can be used to terminate Memory Write and Read in 22h command. Where, Write and Read Address stayed, when 00h or another command except for 22h is written.

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DRIVER OUTPUT CONTROL (R01H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	VSPL	HSPL	DPL	EPL	X	SM	GS	SS	X	X	NL5	NL4	NL3	NL2	NL1	NL0

*Initial Value = 0000_X000_XX11_0110
Note) When the register is written, X means “0”. And when the register is read, X means “0” or “1”.

VSPL: reverses the polarity of the VSYNC signal.
VSPL= “0”: VSYNC is low active.
VSPL= “1”: VSYNC is high active.

HSPL: reverses the polarity of the HSYNC signal.
HSPL= “0”: HSYNC is low active.
HSPL= “1”: HSYNC is high active.

DPL: reverses the polarity of the DOTCLK signal.
DPL= “0”: Display data is fetched at DOTCLK’s rising edge.
DPL= “1”: Display data is fetched at DOTCLK’s falling edge.

EPL: Set the polarity of ENABLE pin while using RGB interface.

EPL = “0”: ENABLE = ”Low” / write data of DB17-0
ENABLE = ”High” / don’t write data of DB17-0
EPL = “1”: ENABLE = ”High” / write data of DB17-0
ENABLE = ”Low” / don’t write data of DB17-0

Table 25. Relationship between EPL, ENABLE and RAM access

EPL	ENABLE	RAM write	RAM address
0	0	Valid	Updated
0	1	Invalid	Held
1	1	Valid	Updated
1	0	Invalid	Held

SM: Select the division drive method of the gate driver. When SM = 0, even/odd division is selected; SM = 1, upper/lower division drive is selected by NL5-0(Upper NL5-0/2 and Lower NL5-0/2). Various connections between TFT panel and the IC can be supported with the combination of SM and GS bit.

GS: Set the order of Gate Clock generation. When GS = 0, G1 is output first and G160 is finally output. When GS = 1, G432 is output first and G273 is finally output (NL = 6'b010100). But in case of NL = 6'b000001, when GS = 0, G1 is output first and G8 is finally output, and when GS = 1, G432 is output first and G425 is finally output

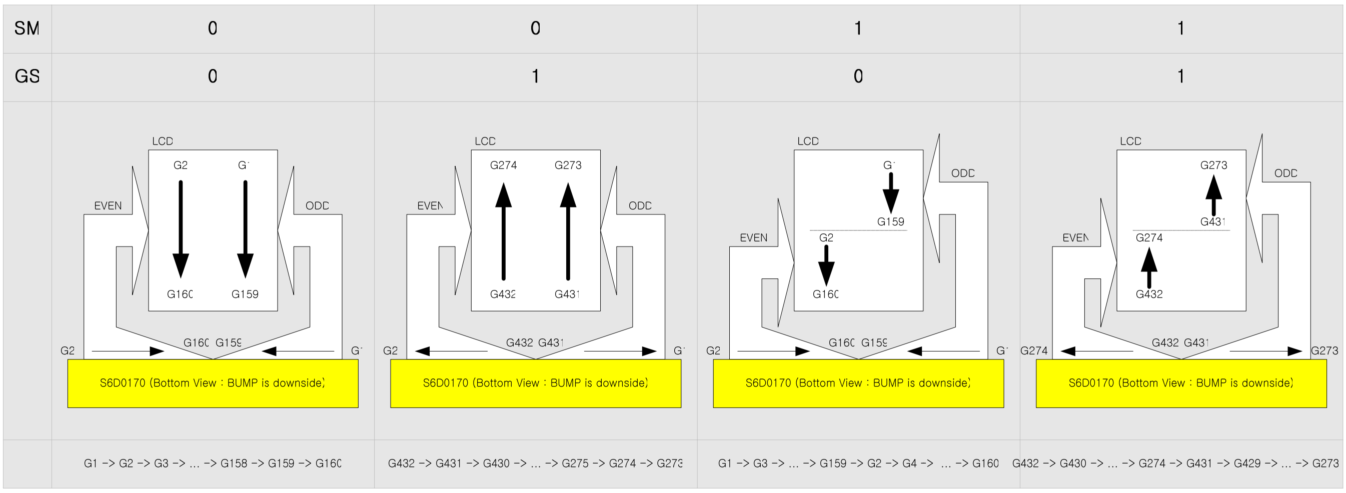

Figure 8. Gate Clock Generation order selection using GS and SM (NL = 6'b010100, SCN = 6'b000000)

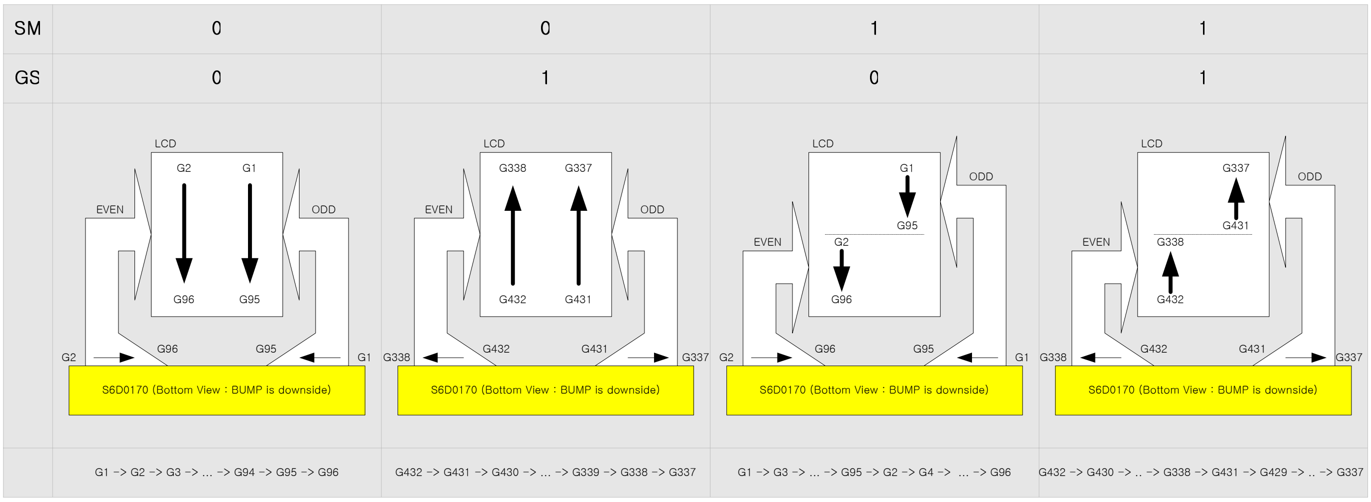

Figure 9. Gate Clock Generation order selection using GS and SM (NL = 6'b001100, SCN = 6'b000000)

SS: Selects the output shift direction of the source driver.

When SS = 0, S1 shifts to S720.

When SS = 1, S720 shifts to S1.

In addition, SS and BGR bits should be specified in case of any change in the RGB order.

When SS = 0 and BGR = 0, <R><G><B> are assigned in order from S1 pin.

When SS = 1 and BGR = 1, <R><G><B> are assigned in order from S720.

Re-write data to GRAM whenever SS and BGR bit are changed.

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NL5–0: Specify the number of raster-rows to be driven. The number of raster-row can be adjusted in units of eight. GRAM address mapping is independent of this setting. The set value should be higher than the panel size.

Table 26. NL bit and Drive Duty

NL5	NL4	NL3	NL2	NL1	NL0	Display size	LCD Raster row	Gate lines used
0	0	0	0	0	0	Setting disabled	Setting disabled	Setting disabled
0	0	0	0	0	1	720 X 8 dots	8	G1 to G8
0	0	0	0	1	0	720 X 16 dots	16	G1 to G24
0	0	0	0	1	1	720 X 24 dots	24	G1 to G24
0	0	0	1	0	0	720 X 32 dots	32	G1 to G32
0	0	0	1	0	1	720 X 40 dots	40	G1 to G40
0	0	0	1	1	0	720 X 48 dots	48	G1 to G48
0	0	0	1	1	1	720 X 56 dots	56	G1 to G56
0	0	1	0	0	0	720 X 64 dots	64	G1 to G64
0	0	1	0	0	1	720 X 72 dots	72	G1 to G72
0	0	1	0	1	0	720 X 80 dots	80	G1 to G80
0	0	1	0	1	1	720 X 88 dots	88	G1 to G88
0	0	1	1	0	0	720 X 96 dots	96	G1 to G96
0	0	1	1	0	1	720 X 104 dots	104	G1 to G104
0	0	1	1	1	0	720 X 112 dots	112	G1 to G112
0	0	1	1	1	1	720 X 120 dots	120	G1 to G120
0	1	0	0	0	0	720 X 128 dots	128	G1 to G128
0	1	0	0	0	1	720 X 136 dots	136	G1 to G136
0	1	0	0	1	0	720 X 144 dots	144	G1 to G144
0	1	0	0	1	1	720 X 152 dots	152	G1 to G152
0	1	0	1	0	0	720 X 160 dots	160	G1 to G160
0	1	0	1	0	1	720 X 168 dots	168	G1 to G168
0	1	0	1	1	0	720 X 176 dots	176	G1 to G176
0	1	0	1	1	1	720 X 184 dots	184	G1 to G184
0	1	1	0	0	0	720 X 192 dots	192	G1 to G192
0	1	1	0	0	1	720 X 200 dots	200	G1 to G200
0	1	1	0	1	0	720 X 208 dots	208	G1 to G208
0	1	1	0	1	1	720 X 216 dots	216	G1 to G216
0	1	1	1	0	0	720 X 224 dots	224	G1 to G224
0	1	1	1	0	1	720 X 232 dots	232	G1 to G232
0	1	1	1	1	0	720 X 240 dots	240	G1 to G240
0	1	1	1	1	1	720 X 248 dots	248	G1 to G248
1	0	0	0	0	0	720 X 256 dots	256	G1 to G256
1	0	0	0	0	1	720 X 264 dots	264	G1 to G264
1	0	0	0	1	0	720 X 272 dots	272	G1 to G272
1	0	0	0	1	1	720 X 280 dots	280	G1 to G280
1	0	0	1	0	0	720 X 288 dots	288	G1 to G288
1	0	0	1	0	1	720 X 296 dots	296	G1 to G296
1	0	0	1	1	0	720 X 304 dots	304	G1 to G304
1	0	0	1	1	1	720 X 312 dots	312	G1 to G312
1	0	1	0	0	0	720 X 320 dots	320	G1 to G320

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Table 27. NL bit and Drive Duty (continued)

NL5	NL4	NL3	NL2	NL1	NL0	Display size	LCD Raster row	Gate lines used
1	0	1	0	0	1	720 X 328 dots	328	G1 to G328
1	0	1	0	1	0	720 X 336 dots	336	G1 to G336
1	0	1	0	1	1	720 X 344 dots	344	G1 to G344
1	0	1	1	0	0	720 X 352 dots	352	G1 to G352
1	0	1	1	0	1	720 X 360 dots	360	G1 to G360
1	0	1	1	1	0	720 X 368 dots	368	G1 to G368
1	0	1	1	1	1	720 X 376 dots	376	G1 to G376
1	1	0	0	0	0	720 X 384 dots	384	G1 to G384
1	1	0	0	0	1	720 X 392 dots	392	G1 to G392
1	1	0	0	1	0	720 X 400 dots	400	G1 to G400
1	1	0	0	1	1	720 X 408 dots	408	G1 to G408
1	1	0	1	0	0	720 X 416 dots	416	G1 to G416
1	1	0	1	0	1	720 X 424 dots	424	G1 to G424
1	1	0	1	1	0	720 X 432 dots	432	G1 to G432

NOTE: A FP (front porch) and BP (back porch) period will be inserted as blanking period (All gates output VGL level) before / after the driver scan through all of the scans.

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LCD-DRIVING-WAVEFORM CONTROL (R02H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	X	X	X	INV1	INV0	X	X	X	X	X	X	X	X

*Initial Value = XXXX_XX00_XXXX_XXXX

INV1-0: Set LCD drive AC waveform.

Table 28. INV Bits Setting

INV1	INV0	Inversion
0	0	1-Line Inversion with Frame Inversion
0	1	Frame Inversion
1	0	Fixed VCOM=L for Gamma Adjustment
1	1	Fixed VCOM=H for Gamma Adjustment

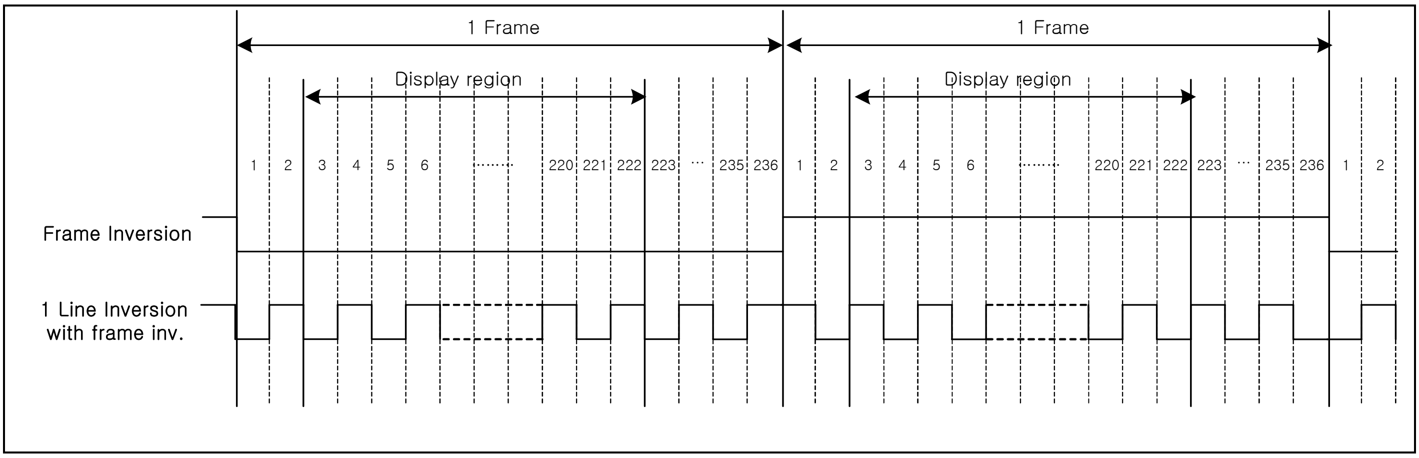

Figure 10. Frame / Line Inversion Timing

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ENTRY MODE (R03H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	CLS	MDT 1	MDT 0	BGR	x	x	x	x	x	AM	I/D1	I/D0	x	x	x	x

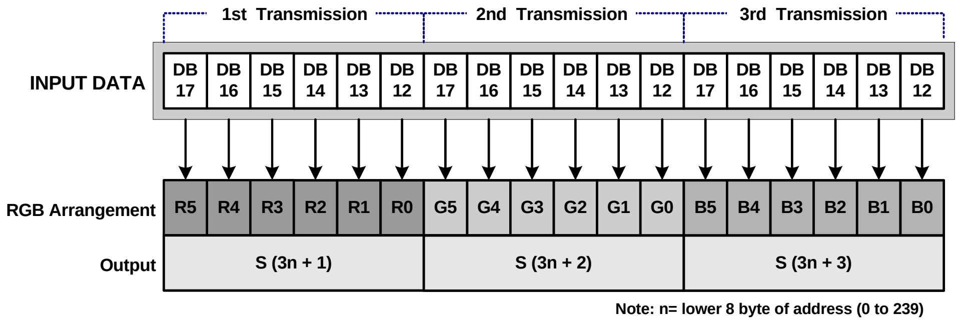
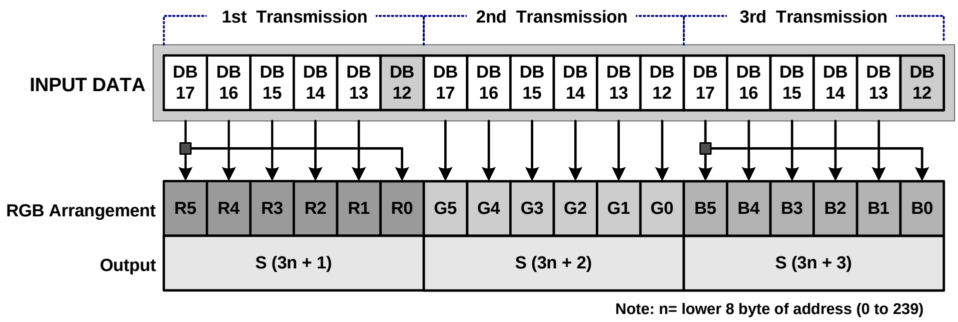
*Initial Value = 0000_XXXX_X000_XXXX

CLS: This bit is used to define the color and interface bus format, When MDT0-1 = 00
CLS = 0: 65K-color mode through 8-bit(Index address 24h) or 16-bit bus(Index address 23h)
CLS = 1: 262K-color mode through 9-bit(Index address 24h) or 18-bit bus(Index address 23h)

MDT1: This bit is active on the 80-system of 8-bit bus, and the data for 1-pixel is transported to the memory for 3 write cycles. This bit is on the 80-system of 16-bit bus, and the data for 1-pixel is transported to the memory for 2 write cycles. When the 80-system interface mode is not set in the 8-bit or16-bit mode, set MDT1 bit to be “0”.

MDT0: When 8-bit or16-bit 80 interface mode and MDT1 bit =1, MDT0 defines color depth for the IC.
8-bit (80-system), MDT0 = 0: 262k-color mode (3 times of 6-bit data transfer to GRAM)
8-bit (80-system), MDT0 = 1: 65k-color mode (5-bit, 6-bit, 5-bit data transfer to GRAM)

16-bit (80-system), MDT0 = 0: 262k-color mode (16-bit, 2-bit data transfer to GRAM)
16-bit (80-system), MDT0 = 1: 262k-color mode (2-bit, 16-bit data transfer to GRAM)

<table>
<tr><th>Interface Mode</th><th>MDT1</th><th>MDT0</th><th>Write data to GRAM</th></tr>
<tr><td>*</td><td>0</td><td>0</td><td>Default value. Multiple Data Transfer (MDT1-0) function is not available. Data Transfer is controlled by interface mode. (Depends on S_PB, ID_MIB pins, CLS register and Index address 23h or 24h)</td></tr>
<tr><td rowspan="3">80-system 8-bit</td><td>0</td><td>1</td><td>Multiple Data Transfer (MDT1-0) function is not available.</td></tr>
<tr><td rowspan="2">1</td><td>0</td><td>
</td></tr>
<tr><td>1</td><td>
</td></tr>
</table>

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Interface Mode	MDT1	MDT0	Data Assignment
80-system 16-bit	0	1	1st / 2nd Transmission; 2nd / 3rd Transmission
	1	0	1st / 2nd Transmission
		1	1st / 2nd Transmission

80-system 16-bit, MDT1 = 0, MDT0 = 1

1st Transmission (DB15–DB10, DB7–DB2), 2nd Transmission (DB15–DB10)

INPUT DATA	DB 15	DB 14	DB 13	DB 12	DB 11	DB 10	DB 7	DB 6	DB 5	DB 4	DB 3	DB 2	DB 15	DB 14	DB 13	DB 12	DB 11	DB 10
RGB Arrangement	R1 5	R1 4	R1 3	R1 2	R1 1	R1 0	G1 5	G1 4	G1 3	G1 2	G1 1	G1 0	B1 5	B1 4	B1 3	B1 2	B1 1	B1 0
Output	S (3n + 1)						S (3n + 2)						S (3n + 3)					

Note: n= lower 8 byte of address (0 to 239)

2nd Transmission (DB7–DB2), 3rd Transmission (DB15–DB10, DB7–DB2)

INPUT DATA	DB 7	DB 6	DB 5	DB 4	DB 3	DB 2	DB 15	DB 14	DB 13	DB 12	DB 11	DB 10	DB 7	DB 6	DB 5	DB 4	DB 3	DB 2
RGB Arrangement	R2 5	R2 4	R2 3	R2 2	R2 1	R2 0	G2 5	G2 4	G2 3	G2 2	G2 1	G2 0	B2 5	B2 4	B2 3	B2 2	B2 1	B2 0
Output	S (3n + 4)						S (3n + 5)						S (3n + 6)					

Note: n= lower 8 byte of address (0 to 239)

MDT1 = 1, MDT0 = 0

1st Transmission (DB17–DB10, DB8–DB1), 2nd Transmission (DB17, DB16)

INPUT DATA	DB 17	DB 16	DB 15	DB 14	DB 13	DB 12	DB 11	DB 10	DB 8	DB 7	DB 6	DB 5	DB 4	DB 3	DB 2	DB 1	DB 17	DB 16
RGB Arrangement	R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0
Output	S (3n + 1)						S (3n + 2)						S (3n + 3)					

Note: n= lower 8 byte of address (0 to 239)

MDT1 = 1, MDT0 = 1

1st Transmission (DB2, DB1), 2nd Transmission (DB17–DB10, DB8–DB1)

INPUT DATA	DB 2	DB 1	DB 17	DB 16	DB 15	DB 14	DB 13	DB 12	DB 11	DB 10	DB 8	DB 7	DB 6	DB 5	DB 4	DB 3	DB 2	DB 1
RGB Arrangement	R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0
Output	S (3n + 1)						S (3n + 2)						S (3n + 3)					

Note: n= lower 8 byte of address (0 to 239)

BGR: About writing 18-bit data to GRAM, it is changed <R><G><B> into <B><G><R>.

- BGR = 0; {DB[17:12], DB[11:6], DB[5:0]} is assigned to {R, G, B}. Actually the analog value that corresponds to DB[17:12] is output firstly at source output
- BGR = 1; {DB[17:12], DB[11:6], DB[5:0]} is assigned to {B, G, R}. Actually the analog value that corresponds to DB[5:0] is output firstly at source output.

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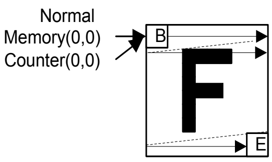
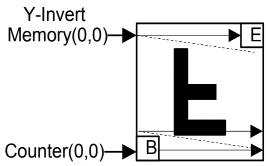
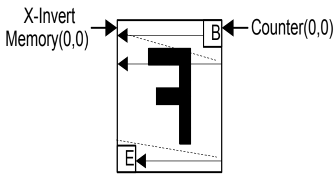
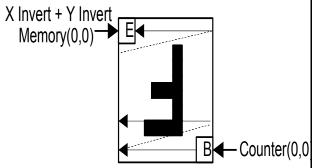
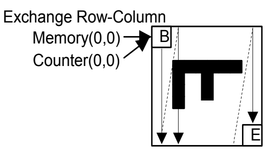
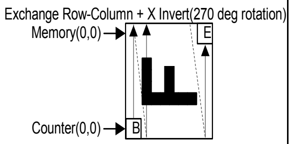
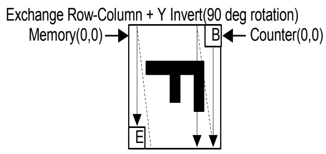
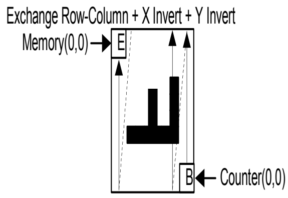
I/D1-0: When I/D1-0 = 0, the address counter (AC) is automatically increased by 1 after the data is written to the GRAM. When I/D1-0 = 1, the AC is automatically decreased by 1 after the data is written to the GRAM. Automatic address counter updating is performed when reading data from GRAM.

AM: Set the automatic update method of the AC after the data is written to GRAM. When AM = "0", the data is continuously written in horizontally. When AM = "1", the data is continuously written vertically. When window addresses are specified, the GRAM in the window range can be written to according to the I/D1-0 and AM.

Table 29. Address Direction Setting

	I/D1-0="00" H: increment V: increment	I/D1-0="01" H: increment V: decrement	I/D1-0="10" H: decrement V: increment	I/D1-0="11" H: decrement V: decrement
AM=0 Horizontal				
AM=1 Vertical				

Note) There is not possible to avoid the tearing effect in normal display if AM = 1.

DEVICE CODE READ (R05H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
R	1	0	0	0	0	0	0	0	1	0	1	1	1	0	0	0	0

If this register is read forcibly, *0170h is read.

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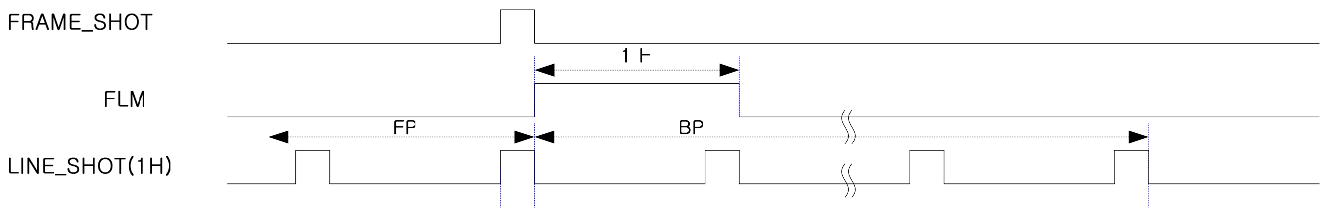
DISPLAY CONTROL (R07H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	TEM ON	X	X	X	S_GA M	X	X	X	GON	CL	REV	D1	D0

*Initial Value = XXX0_XXX0_XXX0_0000

TEMON:
TEMON = 1, Enable the Frame flag output signal from the FLM signal line for preventing Tearing Effect.
TEMON = 0, Disable the Frame flag output signal from the FLM signal line for preventing Tearing Effect.



S_GAM: Single gamma adjust enable signal. When S_GAM = 1, single gamma adjustment mode is selected and gamma curve adjusted by Gamma control R register (R50h~R59h). In this case, Gamma control G register (R60h~R69h) and Gamma control B register (R70h~R79h) register values are ignored.

GON: Gate on/off control signal. All gate outputs are set to be gate off level when GON = 0.
When GON = 1, gate driver is working: G1 to G432 output is either VGH or VGL level. See the Instruction set up flow for further description on the display on/off flow.

D 1-0	GON	Gate Output
00	X	All gates off (AVSS)
01/10/11	0	All gates off (VGL)
	1	Gate on(VGH / VGL)

CL: CL = 1 selects 8-color display mode. For details, see the section on 8-COLOR DISPLAY MODE.

CL	Number of display colors
0	262,144 colors
1	8 colors

REV: Displays all character and graphics display sections with reversal when REV = 1. For details, see the Reversed Display Function section. Since the grayscale level can be reversed, display of the same data is enabled on normally white and normally black panels.

REV	GRAM Data	Display Area	
		Positive	Negative
0	6'b000000 : 6'b111111	V63 : V0	V0 : V63
1	6'b000000 : 6'b111111	V0 : V63	V63 : V0

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D1–0: Display is on when D1 = 1 and off when D1 = 0. When off, the display data remains in the GRAM, and can be re-displayed instantly by setting D1 = 1. When D1 is 0, the display is off with the entire source outputs set to the VSS level. Because of this, the S6D0170 can control the charging current for the LCD with AC driving. Control the display on/off while control GON. For details, see the Instruction set up flow.

When D1–0 = 01, the internal display of the S6D0170 is performed although the display is off. When D1-0 = 00, the internal display operation halts and the display is off.

D1	D0	GON	Source output	Gate Output	VCOM Output	display
0	0	X	AVSS	AVSS	AVSS	off
0	1	0	AVSS	VGL	AVSS	off
		1	AVSS	Operate	AVSS	on
1	0	0	White on Normally White Panel Black on Normally Black Panel	VGL	Operate	off
		1	White on Normally White Panel Black on Normally Black Panel	Operate	Operate	on
1	1	0	Normal Display	VGL	Operate	off
		1	Normal Display	Operate	Operate	on

Notes:

1. Writing from MCU to GRAM is independent of D1–0.
2. In standby mode, D1–0 = 00. However, the register contents of D1–0 are not modified.
3. When source output is the same phase with VCOM, white screen is displayed at normally white LCD panel

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BLANK PERIOD CONTROL (R08H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	x	x	x	x	x	x	x	x	FP3	FP2	FP1	FP0	BP3	BP2	BP1	BP0

*Initial Value = XXXX_XXXX_1000_1000

The blanking period in the front and end of the display area can be defined using this register.
When N-raster-row is driving, a blank period is inserted after all screens are drawn. Front and Back porch can be adjusted using FP3-0 and BP3-0 bits (R08h).

FP3-0/BP3-0: Set the blanking periods (the front and back porch) which are placed at the beginning and the end of the data in. FP3-0 is for a front porch and BP3-0 is for a back porch. When front and back porches are set, the settings should meet the following conditions.

BP + FP ≤ 16 raster-rows
FP ≥ 2 raster-rows
BP ≥ 2 raster-rows

When the external display interface is in use, the front porch (FP) will start on the falling edge of the VSYNC signal and display operation commences at the end of the front-porch period. The back porch (BP) will start when data for the number of raster-rows specified by the NL bits has been displayed. During the period between the completion of the back-porch period and the next VSYNC signal, the display will remain blank.

Do not change the value of BP and FP during display on (D=11).
If you change the value at that time, display will be off for a maximum one frame period.

Table 30. Front/Back Porch

FP3 BP3	FP2 BP2	FP1 BP1	FP0 BP0	# of Raster Periods In the Front Porch # of Raster Periods In the Back Porch
0	0	0	0	Setting Disabled
0	0	0	1	Setting Disabled
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
. . .	. . .	. . .	. . .	. . .
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	Setting Disabled

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FRAME CYCLE CONTROL (R0BH)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	NO3	NO2	NO1	NO0	SDT3	SDT2	SDT1	SDT0	x	x	x	DIV	RTN3	RTN2	RTN1	RTN0

*Initial Value = 0001_0001_XXX0_0000

DIV: Set the division ratio of clocks for internal operation. Internal operations are driven by clocks, which are frequency divided according to the value of this register. Frame frequency can be adjusted with this. When changing number of the drive cycle, adjust the frame frequency.

Table 31 : Frame Frequency Control

DIV	Division Ratio	Internal operation clock frequency(INCLK)
0	1	fosc/1
1	2	fosc/2

[NOTE] fosc = R-C oscillation frequency. The clock which is divided by DIV is called as INCLK below

NO3-0: Set amount of non-overlay for the gate output.

NO3	NO2	NO1	NO0	Amount of non-overlap		
				Internal Operation (synchronized with internal clock)	RGB I/F Operation (synchronized with DOTCLK)	
					18-bit RGB	6-bit RGB
0	0	0	0	Setting disable	Setting disable	Setting disable
0	0	0	1	1 INCLK	8 dot clock	8*3 dot clock
0	0	1	0	2 INCLK	16 dot clock	16*3 dot clock
0	0	1	1	3 INCLK	24 dot clock	24*3 dot clock
0	1	0	0	4 INCLK	32 dot clock	32*3 dot clock
0	1	0	1	5 INCLK	40 dot clock	40*3 dot clock
0	1	1	0	6 INCLK	48 dot clock	48*3 dot clock
0	1	1	1	7 INCLK	56 dot clock	56*3 dot clock
1	0	0	0	8 INCLK	64 dot clock	64*3 dot clock
1	0	0	1	9 INCLK	72 dot clock	72*3 dot clock
1	0	1	0	10 INCLK	80 dot clock	80*3 dot clock
1	0	1	1	Setting disable	88 dot clock	88*3 dot clock
1	1	0	0	Setting disable	96 dot clock	96*3 dot clock
1	1	0	1	Setting disable	104 dot clock	104*3 dot clock
1	1	1	0	Setting disable	112dot clock	112*3 dot clock
1	1	1	1	Setting disable	120dot clock	120*3 dot clock

Note: The amount of non-overlap time is defined from starting time of 1H.

SDT3-0: Set delay amount from gate edge (end) to source output.

SDT3	SDT2	SDT1	SDT0	Delay amount of the source output		
				Internal Operation (synchronized with internal clock)	RGB I/F Operation (synchronized with DOTCLK)	
					18-bit RGB	6-bit RGB
0	0	0	0	Setting disable	Setting disable	Setting disable
0	0	0	1	1 INCLK	8 dot clock	8*3 dot clock
0	0	1	0	2 INCLK	16 dot clock	16*3 dot clock
0	0	1	1	3 INCLK	24 dot clock	24*3 dot clock
0	1	0	0	4 INCLK	32 dot clock	32*3 dot clock
0	1	0	1	5 INCLK	40 dot clock	40*3 dot clock
0	1	1	0	6 INCLK	48 dot clock	48*3 dot clock
0	1	1	1	7 INCLK	56 dot clock	56*3 dot clock
1	0	0	0	8 INCLK	64 dot clock	64*3 dot clock
1	0	0	1	9 INCLK	72 dot clock	72*3 dot clock
1	0	1	0	10 INCLK	80 dot clock	80*3 dot clock
1	0	1	1	Setting disable	88 dot clock	88*3 dot clock
1	1	0	0	Setting disable	96 dot clock	96*3 dot clock
1	1	0	1	Setting disable	104 dot clock	104*3 dot clock
1	1	1	0	Setting disable	112dot clock	112*3 dot clock
1	1	1	1	Setting disable	120dot clock	120*3 dot clock

RTN3-0: Set the 1H period (1 raster-row).

RTN3	RTN2	RTN1	RTN0	1 Horizontal clock cycle (CL1)
0	0	0	0	16 INCLK
0	0	0	1	17 INCLK
0	0	1	0	18 INCLK
0	0	1	1	19 INCLK
0	1	0	0	20 INCLK
0	1	0	1	21 INCLK
0	1	1	0	22 INCLK
0	1	1	1	23 INCLK
1	0	0	0	24 INCLK
1	0	0	1	25 INCLK
1	0	1	0	26 INCLK
1	0	1	1	27 INCLK
1	1	0	0	28 INCLK
1	1	0	1	29 INCLK
1	1	1	0	30 INCLK
1	1	1	1	31 INCLK

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EXTERNAL DISPLAY INTERFACE CONTROL (R0CH)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	x	x	x	x	x	x	x	RM	x	x	x	DM	x	x	RIM1	RIM0

*Initial Value = XXXX_XXX0_XXX0_XX00

RM: Specifies the interface for RAM accesses. RAM accesses can be performed through the interface specified by the bits of RIM1-0. When the display data is written via the RGB interface, 1 should be set. This bit and the DM bits can be set independently. The display data can be written via the system interface by clearing this bit while the RGB interface is used.

RM	Interface for RAM Access
0	System interface
1	RGB interface

DM: Specify the display operation mode. The interface can be set based on the bits of DM. This setting enables switching interface between internal operation and the external display interface.
Switching between two external display interfaces (RGB interface) should not be done.

DM	Display Operation Mode
0	Internal clock operation
1	RGB interface

RIM1-0: Specify the RGB interface mode when the RGB interface is used. Specifically, this setting specifies the mode when the bits of DM and RM are set to RGB interface. These bits should be set before display operation through the RGB interface and should not be set during operation.

RIM1	RIM0	RGB Interface Mode
0	0	18-bit RGB interface (one transfer/pixel)
0	1	16-bit RGB interface (one transfer /pixel)
1	0	6-bit RGB interface (three transfers /pixel)
1	1	Setting disabled

You should notice that some display functions, which will be described later, cannot be used according to the display mode shown below.

Table 32. Display Functions and Display Modes

Function	External Clock Operation Mode	Internal Clock Operation Mode
Partial Display	Cannot be used	Can be used
Scroll Function	Cannot be used	Can be used
Rotation	Cannot be used	Can be used
Mirroring	Cannot be used	Can be used
Window Function	Cannot be used	Can be used

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Depending on the external display interface setting, various interfaces can be specified to match the display state. While displaying motion pictures (RGB interface), the data for display can be written in high-speed write mode, which achieves both low power consumption and high-speed access.

Table 33. Display State and Interface

Display State	Operation Mode	RAM Access (RM)	Display Operation Mode (DM)
Still Pictures	Internal Clock	System interface (RM=0)	Internal clock (DM=0)
Motion Pictures	RGB interface (1)	RGB interface (RM=1)	RGB interface (DM=1)
Rewrite still picture area while displaying motion pictures	RGB interface (2)	System interface (RM=0)	RGB interface (DM=1)

NOTE: 1) The instruction register can only be set through the system interface(SPI).
2) The RGB interface mode should not be set during operation.
For the transition flow for each operation mode, see the External Display Interface section.

Internal Clock Mode

All display operation is controlled by signals generated by the internal clock in internal clock mode.
All inputs through the external display interface are invalid. The internal RAM can be accessed only via the system interface.

RGB Interface Mode (1)

The display operations are controlled by the frame synchronization clock (VSYNC), raster-row synchronization signal (HSYNC), and dot clock (DOTCLK) in RGB interface mode. These signals should be supplied during display operation in this mode.

The display data is transferred to the internal RAM via DB17-0 for each pixel. Combining the function of the high-speed write mode and the window address enables display of both the motion picture area and the internal RAM area simultaneously. In this method, data is only transferred when the screen is updated, which reduces the amount of data transferred.

The periods of the front (FP), back (BP) porch, and the display are automatically generated in the S6D0170 by counting the raster-row synchronization signal (HSYNC) based on the frame synchronization signal (VSYNC).

RGB Interface Mode (2)

When RGB interface is in use, data can be written to RAM via the system interface. This write operation should be performed while data for display is not being transferred via RGB interface (ENABLE = active). Before the next data transfer for display via RGB interface, the setting above should be changed, and then the address and index (R22h) should be set.

START OSCILLATION (R0FH)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	x	x	x	FOSC4	FOSC3	FOSC2	FOSC1	FOSC0	x	x	x	x	x	x	x	OSC ON

*Initial Value = XXX0_0101_XXXX_XXX1

FOSC4-0: Select the oscillation frequency of internal oscillator

Table 34. Display State and Interface

FOSC[4:0]	Oscillation Frequency
00000	x Min multiplier
00001	x multiplier 1
…	…
00101	x 1.000 (Default)
00110	x multiplier 6
…	…
11110	x Max. multiplier
11111	Setting disabled

Note : If the default oscillation frequency is 430 KHz and register setting of FOSC[4:0] is 00000, internal oscillator oscillation frequency is 430KHz × "Min multiplier" = Min. oscillator frequency

OSCON: This instruction starts the oscillator from the Halt State in the standby mode. After this instruction, wait at least 10 ms for oscillation to stabilize before giving the next instruction.
When OSCON = 1: OSC ON, OSCON = 0: OSC OFF

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POWER CONTROL 1 (R10H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	X	X	SAP2	SAP1	SAP0	X	X	X	X	X	DSTB_1F	DSTB	STB

*Initial Value = XXXX_X000_XXXX_XX00

SAP2-0: Adjust the slew-rate of the operational amplifier for the source driver. If higher SAP2-0 is set, LCD panel having higher resolution or higher frame frequency can be driven because the slew-rate of the operational amplifier is increased. But these bits must be set as adequate value because the amount of fixed current of the operational amplifier is also adjusted. During non-display, operational amplifiers are turned off, so current consumption can be reduced.

SAP2	SAP1	SAP0	Slew Rate of Operational Amplifier for Source driver	Amount of Current in Operational Amplifier for Source driver
0	0	0	Slow	Small
0	0	1	Medium Slow1	Medium Low1
0	1	0	Medium Slow2	Medium Low2
0	1	1	Medium Slow3	Medium Low3
1	0	0	Medium	Medium
1	0	1	Medium Fast1	Medium High1
1	1	0	Medium Fast2	Medium High2
1	1	1	Fast	Large

DSTB_1F: When DSTB_1F = 0, internally the S6D0170 enters the deep standby mode after 1-line time. In this case CSB signal must be raised up before 1-line time. If not, the S6D0170 enters wake-up sequence by CSB rising. When DSTB_1F = 1, internally the S6D0170 enters the deep standby mode after 1-frame time. In this case CSB signal must be raised up before 1-frame time. If not, the S6D0170 enters wake-up sequence by CSB rising.

DSTB: When DSTB = 1, the S6D0170 enters the deep standby mode, where the power supply for the internal logic is turned off to save more power than the standby mode. The GRAM data and the instruction sets are prohibited during the deep standby mode and they must be reset after releasing from the deep standby mode. For more information, see the Standby Mode section.

STB: When STB = 1, the S6D0170 enters the standby mode, where display operation completely stops, halting all the internal operations including the internal R-C oscillator. Further, no external clock pulses are supplied. For more information, see the Standby Mode section.

Outputs	Conditions
VCOM	AVSS
Gate	AVSS
Source	AVSS

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POWER CONTROL 2 (R11H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	APON	PON3	PON2	PON1	PON	X	X	AON	VCI1_EN	VC3	VC2	VC1	VC0

*Initial Value = XXX0_0000_XX00_0000

APON: This is an auto-operation-starting bit for the booster circuits. In case of APON=0, the auto booster sequence circuit is stopped, but the booster circuits are independently operated by PON, PON1, PON2 and PON3 bits. In case of APON=1, booster circuits are automatically and sequentially operated. For further information about timing, please refer to the SET UP FLOW OF POWER SUPPLY.

PON3: This is an operation-starting bit for the booster circuit 3(VCL). In case of VCLON = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the VCLON= 1, please refer to the SET UP FLOW OF POWER SUPPLY.

PON2: This is an operation-starting bit for the booster circuit 2(VGL). In case of VGLON = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the VGLON= 1, please refer to the SET UP FLOW OF POWER SUPPLY.

PON1: This is an operation-starting bit for the booster circuit 2(VGH). In case of PON1 = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the PON1= 1, please refer to the SET UP FLOW OF POWER SUPPLY.

PON: This is an operation-starting bit for the booster circuit1. In case of PON = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the PON = 1, please refer to the SET UP FLOW OF POWER SUPPLY.

AON: This is an operation-starting bit for the Amplifier. In case of AON = 0, the circuit is stopped and vice versa. For further information about timing for adjusting to the AON= 1, please refer to the SET UP FLOW OF POWER SUPPLY.

VCI1_EN: Internal VCI1 generation amplifier operation control bit. When VCI1_EN=0, VCI1 voltage is not generated

VC3-0: Set the VCI1 voltage. These bits set the VCI1 voltage up to 3V as the nominal output (upper limit value may depend on VCI voltage)

VC3	VC2	VC1	VC0	VCI1
0	0	0	0	1.35
0	0	0	1	1.75
0	0	1	0	2.07
0	0	1	1	2.16
0	1	0	0	2.25
0	1	0	1	2.34
0	1	1	0	2.43
0	1	1	1	2.52
1	0	0	0	2.58
1	0	0	1	2.64
1	0	1	0	2.70
1	0	1	1	2.76
1	1	0	0	2.82
1	1	0	1	2.88
1	1	1	0	2.94
1	1	1	1	3

Note: Don't set any higher VCI1 level than VCI

POWER CONTROL 3 (R12H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	BT2	BT1	BT0	X	X	DC11	DC10	X	X	DC21	DC20	X	X	DC31	DC30

*Initial Value = X000_XX00_XX00_XX00

BT2–0: The output factor of step-up is switched. Adjust scale factor of the step-up circuit by the voltage used. When the step-up operating frequency is high, the driving ability of the step-up circuit and the display quality become high, but the current consumption is increased. Adjust the frequency considering the display quality and the current consumption.

BT2	BT1	BT0	VGH	VGL	Notes*	
0	0	0	5 X Vci1	-3X Vci1	13.75V	-8.25V
0	0	1	5 X Vci1	-4X Vci1	13.75V	-11V
0	1	0	6 X Vci1	-3X Vci1	16.5V	-8.25V
0	1	1	6 X Vci1	-4X Vci1	16.5V	-11V
1	0	0	6 X Vci1	-5X Vci1	16.5V	-13.75V
1	0	1	7 X Vci1	-4X Vci1	19.25V	-11V
1	1	0	Setting disabled			
1	1	1	Setting disabled			

Note: The values in table above are upper-limit by register setting. (when VCI1=2.75V)

DC11-0: The operating frequency in the step-up circuit1 is selected. When the step-up operating frequency is high, the driving ability of the step-up circuit and the display quality become high, but the current consumption is increased. Adjust the frequency considering the display quality and the current consumption.

DC11	DC10	Internal Operation (synchronized with internal clock)	RGB I/F Operation (synchronized with DOTCLK)
		f(CL1) : f(DCCLK1)	f(DCCLK):f(DCCLK1)
0	0	1:4	1:1
0	1	1:2	1:0.5
1	0	1:1	1:0.25
1	1	Setting disabled	Setting disabled

Note: DCCLK1 is pumping clock for step-up circuit1
f(1H) is horizontal frequency (1 raster-row)

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DC21-0: The operating frequency in the step-up circuit 2 is selected**.**

DC21	DC20	Internal Operation (synchronized with internal clock) f(CL1) : f(DCCLK2)	RGB I/F Operation (synchronized with DOTCLK) f(DCCLK) : f(DCCLK2)
0	0	1:2	1:0.5
0	1	1:1	1:0.25
1	0	1:0.5	1:0.125
1	1	1:0.25	1:0.0625

Note: DCCLK2 is pumping clock for step-up circuit2

DC31-0: The operating frequency in the step-up circuit 3 is selected**.**

DC31	DC30	Internal Operation (synchronized with internal clock) f(CL1) : f(DCCLK3)	RGB I/F Operation (synchronized with DOTCLK) f(DCCLK) : f(DCCLK3)
0	0	1:4	1:1
0	1	1:2	1:0.5
1	0	1:1	1:0.25
1	1	Setting disabled	Setting disabled

Note: DCCLK3 is pumping clock for step-up circuit3

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POWER CONTROL 4 (R13H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	RTR_AMP_EN	DCR_EX	X	DCR2	DCR1	DCR0	X	GVD6	GVD5	GVD4	GVD3	GVD2	GVD1	GVD0

*Initial Value = XXX0_X000_X000_0000

RTR_AMP_EN: VCOMH and VCOML amp select.

RTR_AMP_EN	Operational Amplifier Type of VCOMH and VCOML	Amount of Static Current in Operational Amplifier for VCOM driver
0	A Class	Small
1	AB Class	Large

DCR_EX: Input signal selection register for external interface mode. (0: internal operation clock, 1: DOTCLK)
Set DCR_EX bit to 1 for DOTCLK to be DCCLK (clock cycle for step-up circuit) source when external interface mode is in use (DM=1).

DCR 2-0: Set clock cycle for step-up circuit in external interface mode. Please set DCR_EX bit to "1" and DCR1-0 value when external interface is in use. In this case, DOTCLK must be input periodically and continuously.

DCR2	DCR1	DCR0	Clock cycle for step-up circuits (DCCLK) in external interface mode	
			18-bit RGB	6-bit RGB
0	0	0	6 dot clock	18 dot clock
0	0	1	8 dot clock	24 dot clock
0	1	0	12 dot clock	36 dot clock
0	1	1	16 dot clock	48 dot clock
1	0	0	24 dot clock	72 dot clock
1	0	1	32 dot clock	96 dot clock
1	1	0	48 dot clock	144 dot clock
1	1	1	64 dot clock	192 dot clock

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GVD6-0: Set the amplified factor of the GVDD voltage (the voltage for the Gamma voltage). It allows to amplified from 2.5v to 5.0v

GVD6	GVD5	GVD4	GVD3	GVD 2	GVD 1	GVD 0	GVDD Voltage	GVD6	GVD5	GVD4	GVD 3	GVD 2	GVD 1	GVD 0	GVDD Voltage
0	0	0	0	0	0	0	2.500	0	1	0	0	0	0	0	3.130
0	0	0	0	0	0	1	2.520	0	1	0	0	0	0	1	3.150
0	0	0	0	0	1	0	2.539	0	1	0	0	0	1	0	3.169
0	0	0	0	0	1	1	2.559	0	1	0	0	0	1	1	3.189
0	0	0	0	1	0	0	2.579	0	1	0	0	1	0	0	3.209
0	0	0	0	1	0	1	2.598	0	1	0	0	1	0	1	3.228
0	0	0	0	1	1	0	2.618	0	1	0	0	1	1	0	3.248
0	0	0	0	1	1	1	2.638	0	1	0	0	1	1	1	3.268
0	0	0	1	0	0	0	2.657	0	1	0	1	0	0	0	3.287
0	0	0	1	0	0	1	2.677	0	1	0	1	0	0	1	3.307
0	0	0	1	0	1	0	2.697	0	1	0	1	0	1	0	3.327
0	0	0	1	0	1	1	2.717	0	1	0	1	0	1	1	3.346
0	0	0	1	1	0	0	2.736	0	1	0	1	1	0	0	3.366
0	0	0	1	1	0	1	2.756	0	1	0	1	1	0	1	3.386
0	0	0	1	1	1	0	2.776	0	1	0	1	1	1	0	3.406
0	0	0	1	1	1	1	2.795	0	1	0	1	1	1	1	3.425
0	0	1	0	0	0	0	2.815	0	1	1	0	0	0	0	3.445
0	0	1	0	0	0	1	2.835	0	1	1	0	0	0	1	3.465
0	0	1	0	0	1	0	2.854	0	1	1	0	0	1	0	3.484
0	0	1	0	0	1	1	2.874	0	1	1	0	0	1	1	3.504
0	0	1	0	1	0	0	2.894	0	1	1	0	1	0	0	3.524
0	0	1	0	1	0	1	2.913	0	1	1	0	1	0	1	3.543
0	0	1	0	1	1	0	2.933	0	1	1	0	1	1	0	3.563
0	0	1	0	1	1	1	2.953	0	1	1	0	1	1	1	3.583
0	0	1	1	0	0	0	2.972	0	1	1	1	0	0	0	3.602
0	0	1	1	0	0	1	2.992	0	1	1	1	0	0	1	3.622
0	0	1	1	0	1	0	3.012	0	1	1	1	0	1	0	3.642
0	0	1	1	0	1	1	3.031	0	1	1	1	0	1	1	3.661
0	0	1	1	1	0	0	3.051	0	1	1	1	1	0	0	3.681
0	0	1	1	1	0	1	3.071	0	1	1	1	1	0	1	3.701
0	0	1	1	1	1	0	3.091	0	1	1	1	1	1	0	3.720
0	0	1	1	1	1	1	3.110	0	1	1	1	1	1	1	3.740

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GVD6	GVD5	GVD4	GVD3	GVD 2	GVD 1	GVD 0	GVDD Voltage	GVD6	GVD5	GVD4	GVD 3	GVD 2	GVD 1	GVD 0	GVDD Voltage
1	0	0	0	0	0	0	3.760	1	1	0	0	0	0	0	4.390
1	0	0	0	0	0	1	3.780	1	1	0	0	0	0	1	4.409
1	0	0	0	0	1	0	3.799	1	1	0	0	0	1	0	4.429
1	0	0	0	0	1	1	3.819	1	1	0	0	0	1	1	4.449
1	0	0	0	1	0	0	3.839	1	1	0	0	1	0	0	4.469
1	0	0	0	1	0	1	3.858	1	1	0	0	1	0	1	4.488
1	0	0	0	1	1	0	3.878	1	1	0	0	1	1	0	4.508
1	0	0	0	1	1	1	3.898	1	1	0	0	1	1	1	4.528
1	0	0	1	0	0	0	3.917	1	1	0	1	0	0	0	4.547
1	0	0	1	0	0	1	3.937	1	1	0	1	0	0	1	4.567
1	0	0	1	0	1	0	3.957	1	1	0	1	0	1	0	4.587
1	0	0	1	0	1	1	3.976	1	1	0	1	0	1	1	4.606
1	0	0	1	1	0	0	3.996	1	1	0	1	1	0	0	4.626
1	0	0	1	1	0	1	4.016	1	1	0	1	1	0	1	4.646
1	0	0	1	1	1	0	4.035	1	1	0	1	1	1	0	4.665
1	0	0	1	1	1	1	4.055	1	1	0	1	1	1	1	4.685
1	0	1	0	0	0	0	4.075	1	1	1	0	0	0	0	4.705
1	0	1	0	0	0	1	4.094	1	1	1	0	0	0	1	4.724
1	0	1	0	0	1	0	4.114	1	1	1	0	0	1	0	4.744
1	0	1	0	0	1	1	4.134	1	1	1	0	0	1	1	4.764
1	0	1	0	1	0	0	4.154	1	1	1	0	1	0	0	4.783
1	0	1	0	1	0	1	4.173	1	1	1	0	1	0	1	4.803
1	0	1	0	1	1	0	4.193	1	1	1	0	1	1	0	4.823
1	0	1	0	1	1	1	4.213	1	1	1	0	1	1	1	4.843
1	0	1	1	0	0	0	4.232	1	1	1	1	0	0	0	4.862
1	0	1	1	0	0	1	4.252	1	1	1	1	0	0	1	4.882
1	0	1	1	0	1	0	4.272	1	1	1	1	0	1	0	4.902
1	0	1	1	0	1	1	4.291	1	1	1	1	0	1	1	4.921
1	0	1	1	1	0	0	4.311	1	1	1	1	1	0	0	4.941
1	0	1	1	1	0	1	4.331	1	1	1	1	1	0	1	4.961
1	0	1	1	1	1	0	4.350	1	1	1	1	1	1	0	4.980
1	0	1	1	1	1	1	4.370	1	1	1	1	1	1	1	5.000

Note: Don't set any higher GVDD level than AVDD-0.5V

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POWER CONTROL 5 (R14H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	VCOMG	VCM6	VCM5	VCM4	VCM3	VCM2	VCM1	VCM0	VCMR	VML6	VML5	VML4	VML3	VML2	VML1	VML0

*Initial Value =1000_0000_0000_0000

VCOMG: When VCOMG = 1, low level of VCOM signal is to be fixed at AVSS. Therefore, the amplitude of VCOM signal is determined as |VCOMH – AVSS| regardless of VML setting. In this case, VCOML pin can be open or connected to GND, because VCOML amp is off and VCOML output is floated.
When VCOMG=0, the amplitude of VCOM signal is determined as |VCOMH – VCOML|.

VCMR: In case of VCMR is LOW, VCOMH is adjusted by VCM6-0 Register and VCOMR pin is used to monitor the input current of the AMP which output the VCOMH voltage.
In case of VCMR is HIGH, VCM6-0 register is ignored and VCOMH voltage is adjusted by VCOMR voltage. VCOMR voltage is externally supplied. The relationship between VCOMH and VCOMR is given as VCOMH=2.5 X VCOMR.

VCM6-0: Set the upper level of the VCOM (VCOMH).

VCM6	VCM5	VCM4	VCM3	VCM2	VCM1	VCM0	VCOMH Voltage	VCM6	VCM5	VCM4	VCM3	VCM2	VCM1	VCM0	VCOMH Voltage
0	0	0	0	0	0	0	2.500	0	1	0	0	0	0	0	3.130
0	0	0	0	0	0	1	2.520	0	1	0	0	0	0	1	3.150
0	0	0	0	0	1	0	2.539	0	1	0	0	0	1	0	3.169
0	0	0	0	0	1	1	2.559	0	1	0	0	0	1	1	3.189
0	0	0	0	1	0	0	2.579	0	1	0	0	1	0	0	3.209
0	0	0	0	1	0	1	2.598	0	1	0	0	1	0	1	3.228
0	0	0	0	1	1	0	2.618	0	1	0	0	1	1	0	3.248
0	0	0	0	1	1	1	2.638	0	1	0	0	1	1	1	3.268
0	0	0	1	0	0	0	2.657	0	1	0	1	0	0	0	3.287
0	0	0	1	0	0	1	2.677	0	1	0	1	0	0	1	3.307
0	0	0	1	0	1	0	2.697	0	1	0	1	0	1	0	3.327
0	0	0	1	0	1	1	2.717	0	1	0	1	0	1	1	3.346
0	0	0	1	1	0	0	2.736	0	1	0	1	1	0	0	3.366
0	0	0	1	1	0	1	2.756	0	1	0	1	1	0	1	3.386
0	0	0	1	1	1	0	2.776	0	1	0	1	1	1	0	3.406
0	0	0	1	1	1	1	2.795	0	1	0	1	1	1	1	3.425
0	0	1	0	0	0	0	2.815	0	1	1	0	0	0	0	3.445
0	0	1	0	0	0	1	2.835	0	1	1	0	0	0	1	3.465
0	0	1	0	0	1	0	2.854	0	1	1	0	0	1	0	3.484
0	0	1	0	0	1	1	2.874	0	1	1	0	0	1	1	3.504
0	0	1	0	1	0	0	2.894	0	1	1	0	1	0	0	3.524
0	0	1	0	1	0	1	2.913	0	1	1	0	1	0	1	3.543
0	0	1	0	1	1	0	2.933	0	1	1	0	1	1	0	3.563
0	0	1	0	1	1	1	2.953	0	1	1	0	1	1	1	3.583
0	0	1	1	0	0	0	2.972	0	1	1	1	0	0	0	3.602
0	0	1	1	0	0	1	2.992	0	1	1	1	0	0	1	3.622
0	0	1	1	0	1	0	3.012	0	1	1	1	0	1	0	3.642
0	0	1	1	0	1	1	3.031	0	1	1	1	0	1	1	3.661
0	0	1	1	1	0	0	3.051	0	1	1	1	1	0	0	3.681
0	0	1	1	1	0	1	3.071	0	1	1	1	1	0	1	3.701
0	0	1	1	1	1	0	3.091	0	1	1	1	1	1	0	3.720
0	0	1	1	1	1	1	3.110	0	1	1	1	1	1	1	3.740

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VCM6	VCM5	VCM4	VCM3	VCM2	VCM1	VCM0	VCOMH Voltage	VCM6	VCM5	VCM4	VCM3	VCM2	VCM1	VCM0	VCOMH Voltage
1	0	0	0	0	0	0	3.760	1	1	0	0	0	0	0	4.390
1	0	0	0	0	0	1	3.780	1	1	0	0	0	0	1	4.409
1	0	0	0	0	1	0	3.799	1	1	0	0	0	1	0	4.429
1	0	0	0	0	1	1	3.819	1	1	0	0	0	1	1	4.449
1	0	0	0	1	0	0	3.839	1	1	0	0	1	0	0	4.469
1	0	0	0	1	0	1	3.858	1	1	0	0	1	0	1	4.488
1	0	0	0	1	1	0	3.878	1	1	0	0	1	1	0	4.508
1	0	0	0	1	1	1	3.898	1	1	0	0	1	1	1	4.528
1	0	0	1	0	0	0	3.917	1	1	0	1	0	0	0	4.547
1	0	0	1	0	0	1	3.937	1	1	0	1	0	0	1	4.567
1	0	0	1	0	1	0	3.957	1	1	0	1	0	1	0	4.587
1	0	0	1	0	1	1	3.976	1	1	0	1	0	1	1	4.606
1	0	0	1	1	0	0	3.996	1	1	0	1	1	0	0	4.626
1	0	0	1	1	0	1	4.016	1	1	0	1	1	0	1	4.646
1	0	0	1	1	1	0	4.035	1	1	0	1	1	1	0	4.665
1	0	0	1	1	1	1	4.055	1	1	0	1	1	1	1	4.685
1	0	1	0	0	0	0	4.075	1	1	1	0	0	0	0	4.705
1	0	1	0	0	0	1	4.094	1	1	1	0	0	0	1	4.724
1	0	1	0	0	1	0	4.114	1	1	1	0	0	1	0	4.744
1	0	1	0	0	1	1	4.134	1	1	1	0	0	1	1	4.764
1	0	1	0	1	0	0	4.154	1	1	1	0	1	0	0	4.783
1	0	1	0	1	0	1	4.173	1	1	1	0	1	0	1	4.803
1	0	1	0	1	1	0	4.193	1	1	1	0	1	1	0	4.823
1	0	1	0	1	1	1	4.213	1	1	1	0	1	1	1	4.843
1	0	1	1	0	0	0	4.232	1	1	1	1	0	0	0	4.862
1	0	1	1	0	0	1	4.252	1	1	1	1	0	0	1	4.882
1	0	1	1	0	1	0	4.272	1	1	1	1	0	1	0	4.902
1	0	1	1	0	1	1	4.291	1	1	1	1	0	1	1	4.921
1	0	1	1	1	0	0	4.311	1	1	1	1	1	0	0	4.941
1	0	1	1	1	0	1	4.331	1	1	1	1	1	0	1	4.961
1	0	1	1	1	1	0	4.350	1	1	1	1	1	1	0	4.980
1	0	1	1	1	1	1	4.370	1	1	1	1	1	1	1	5.000

Note: Don't set any higher GVDD level than AVDD-0.5V

Preliminary

VML6-0: Set the Amplitude of the VCOM voltage. VCOML is adjusted automatically by setting the Amplitude of VCOM voltage.

VML6	VML5	VML4	VML3	VML2	VML1	VML0	Amplitude Voltage	VML6	VML5	VML4	VML3	VML2	VML1	VML0	Amplitude Voltage
0	0	0	0	0	0	0	3.000	0	1	0	0	0	0	0	3.756
0	0	0	0	0	0	1	3.024	0	1	0	0	0	0	1	3.780
0	0	0	0	0	1	0	3.047	0	1	0	0	0	1	0	3.803
0	0	0	0	0	1	1	3.071	0	1	0	0	0	1	1	3.827
0	0	0	0	1	0	0	3.094	0	1	0	0	1	0	0	3.850
0	0	0	0	1	0	1	3.118	0	1	0	0	1	0	1	3.874
0	0	0	0	1	1	0	3.142	0	1	0	0	1	1	0	3.898
0	0	0	0	1	1	1	3.165	0	1	0	0	1	1	1	3.921
0	0	0	1	0	0	0	3.189	0	1	0	1	0	0	0	3.945
0	0	0	1	0	0	1	3.213	0	1	0	1	0	0	1	3.969
0	0	0	1	0	1	0	3.236	0	1	0	1	0	1	0	3.992
0	0	0	1	0	1	1	3.260	0	1	0	1	0	1	1	4.016
0	0	0	1	1	0	0	3.283	0	1	0	1	1	0	0	4.039
0	0	0	1	1	0	1	3.307	0	1	0	1	1	0	1	4.063
0	0	0	1	1	1	0	3.331	0	1	0	1	1	1	0	4.087
0	0	0	1	1	1	1	3.354	0	1	0	1	1	1	1	4.110
0	0	1	0	0	0	0	3.378	0	1	1	0	0	0	0	4.134
0	0	1	0	0	0	1	3.402	0	1	1	0	0	0	1	4.157
0	0	1	0	0	1	0	3.425	0	1	1	0	0	1	0	4.181
0	0	1	0	0	1	1	3.449	0	1	1	0	0	1	1	4.205
0	0	1	0	1	0	0	3.472	0	1	1	0	1	0	0	4.228
0	0	1	0	1	0	1	3.496	0	1	1	0	1	0	1	4.252
0	0	1	0	1	1	0	3.520	0	1	1	0	1	1	0	4.276
0	0	1	0	1	1	1	3.543	0	1	1	0	1	1	1	4.299
0	0	1	1	0	0	0	3.567	0	1	1	1	0	0	0	4.323
0	0	1	1	0	0	1	3.591	0	1	1	1	0	0	1	4.346
0	0	1	1	0	1	0	3.614	0	1	1	1	0	1	0	4.370
0	0	1	1	0	1	1	3.638	0	1	1	1	0	1	1	4.394
0	0	1	1	1	0	0	3.661	0	1	1	1	1	0	0	4.417
0	0	1	1	1	0	1	3.685	0	1	1	1	1	0	1	4.441
0	0	1	1	1	1	0	3.709	0	1	1	1	1	1	0	4.465
0	0	1	1	1	1	1	3.732	0	1	1	1	1	1	1	4.488

VML6	VML5	VML4	VML3	VML2	VML1	VML0	Amplitude Voltage	VML6	VML5	VML4	VML3	VML2	VML1	VML0	Amplitude Voltage
1	0	0	0	0	0	0	4.512	1	1	0	0	0	0	0	5.268
1	0	0	0	0	0	1	4.535	1	1	0	0	0	0	1	5.291
1	0	0	0	0	1	0	4.559	1	1	0	0	0	1	0	5.315
1	0	0	0	0	1	1	4.583	1	1	0	0	0	1	1	5.339
1	0	0	0	1	0	0	4.606	1	1	0	0	1	0	0	5.362
1	0	0	0	1	0	1	4.630	1	1	0	0	1	0	1	5.386
1	0	0	0	1	1	0	4.654	1	1	0	0	1	1	0	5.409
1	0	0	0	1	1	1	4.677	1	1	0	0	1	1	1	5.433
1	0	0	1	0	0	0	4.701	1	1	0	1	0	0	0	5.457
1	0	0	1	0	0	1	4.724	1	1	0	1	0	0	1	5.480
1	0	0	1	0	1	0	4.748	1	1	0	1	0	1	0	5.504
1	0	0	1	0	1	1	4.772	1	1	0	1	0	1	1	5.528
1	0	0	1	1	0	0	4.795	1	1	0	1	1	0	0	5.551
1	0	0	1	1	0	1	4.819	1	1	0	1	1	0	1	5.575
1	0	0	1	1	1	0	4.843	1	1	0	1	1	1	0	5.598
1	0	0	1	1	1	1	4.866	1	1	0	1	1	1	1	5.622
1	0	1	0	0	0	0	4.890	1	1	1	0	0	0	0	5.646
1	0	1	0	0	0	1	4.913	1	1	1	0	0	0	1	5.669
1	0	1	0	0	1	0	4.937	1	1	1	0	0	1	0	5.693
1	0	1	0	0	1	1	4.961	1	1	1	0	0	1	1	5.717
1	0	1	0	1	0	0	4.984	1	1	1	0	1	0	0	5.740
1	0	1	0	1	0	1	5.008	1	1	1	0	1	0	1	5.764
1	0	1	0	1	1	0	5.031	1	1	1	0	1	1	0	5.787
1	0	1	0	1	1	1	5.055	1	1	1	0	1	1	1	5.811
1	0	1	1	0	0	0	5.079	1	1	1	1	0	0	0	5.835
1	0	1	1	0	0	1	5.102	1	1	1	1	0	0	1	5.858
1	0	1	1	0	1	0	5.126	1	1	1	1	0	1	0	5.882
1	0	1	1	0	1	1	5.150	1	1	1	1	0	1	1	5.906
1	0	1	1	1	0	0	5.173	1	1	1	1	1	0	0	5.929
1	0	1	1	1	0	1	5.197	1	1	1	1	1	0	1	5.953
1	0	1	1	1	1	0	5.220	1	1	1	1	1	1	0	5.976
1	0	1	1	1	1	1	5.244	1	1	1	1	1	1	1	6.000

NOTES: Available setting range of VCOML is from VCL+0.5V to 0V. The Amplitude of VCOM cannot exceed 6V.

Preliminary

VCI RECYCLING (R15H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	x	x	x	x	x	x	x	x	x	VCIR 2	VCIR 1	VCIR 0	x	x	LVCM_ VCIR	VCIR_ VSSB

*Initial Value = XXXX_XXXX_X000_XXX0

VCIR2-0: VCI recycling period is sustained for the number of clock cycle which is set on VCIR2-0. When VcomL<-1, set these bits as "000" for preventing the abnormal function.

VCIR 2	VCIR 1	VCIR 0	Amount of non-overlap		
			Internal Operation (synchronized with internal clock)	RGB I/F Operation (synchronized with DOTCLK)	
				18-bit RGB	6-bit RGB
0	0	0	No VCIR	No VCIR	No VCIR
0	0	1	2 INCLK	16 dot clock	16*3 dot clock
0	1	0	4 INCLK	32 dot clock	32*3 dot clock
0	1	1	6 INCLK	48 dot clock	48*3 dot clock
1	0	0	8 INCLK	64 dot clock	64*3 dot clock
1	0	1	10 INCLK	80 dot clock	80*3 dot clock
1	1	0	12 INCLK	96 dot clock	96*3 dot clock
1	1	1	14 INCLK	112 dot clock	112*3 dot clock

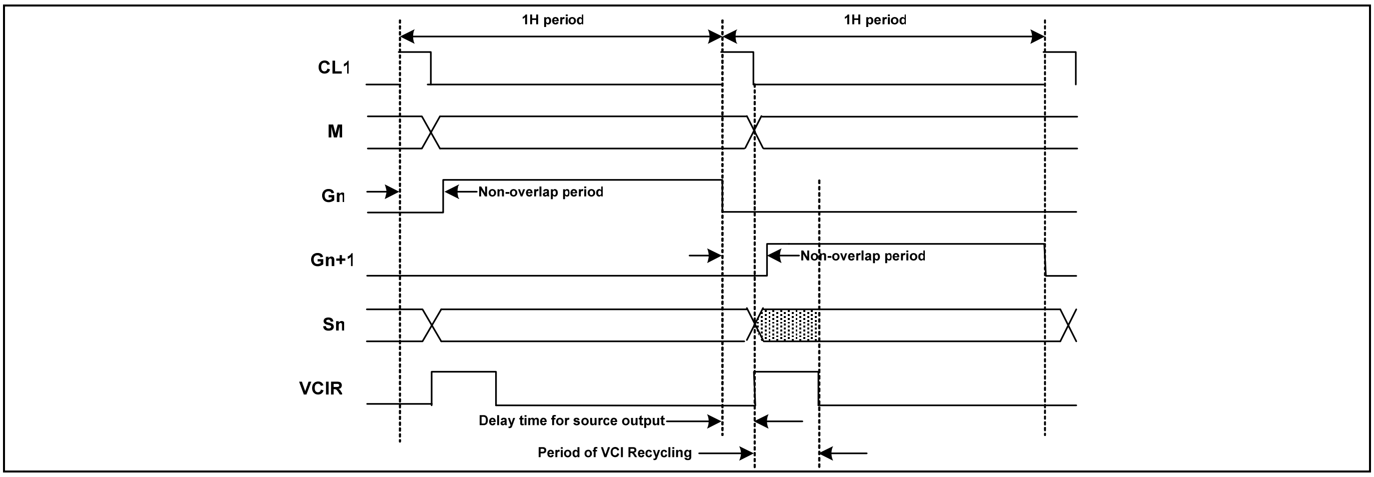


Figure 11. Set Delay From Gate Output To Source Output And VCIR Signal

Note: The values specified by the bits of VCIR2-0, SDT3-0 and NO3-0 vary in a reference clock for each interface mode.
Internal operation mode: Internal R-C oscillation clock
RGB-I/F mode : DOTCLK

LVCM_VCIR: Control bit for charge-recycling mode for VCOMH<VCI case.
VCIR_VSSB: Control bit for charge-recycling mode for VCOM driver. Default setting is '0'.

LVCM_VSSB	VCIR_VSSB	Charge Recycling Mode for VCOM Driver
0	0	Charge recycling using VCI and VSS
0	1	Charge recycling using only VCI
1	X	Charge recycling using only VSS

Note 1: when you set VCOMG=1(R14h), then set VCIR_VSSB=1
Note 2: when you set VCOMH<VCI, then set LVCM_VCIR=1

SOFTWARE RESET (R1FH)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	X	X	X	X	X	1	0	1	0	1	0	1	0

When Software Reset parameter is AAh, It cause a software reset. This register automatically set to Zero after a Software Reset. After Software Reset, wait at least 10 ms for oscillation to stabilization before giving the next instruction.

WRITE DATA TO GRAM (R22H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	RAM write data (WD17-0): Pin assignment varies according to the interface method. (see the following figure for more information)															
W	1	WD15	WD14	WD13	WD12	WD11	WD10	WD9	WD8	WD7	WD6	WD5	WD4	WD3	WD2	WD1	WD0
		When RGB-interface															

WD17-0: Input data for GRAM can be expanded to 18 bits. The expansion format varies according to the interface method. The input data selects the grayscale level. After a write, the address is automatically updated according to I/D bit settings. The GRAM can be accessed in standby mode. When 16- or 8-bit interface is in use, the write data is expanded to 18 bits by writing the MSB of the <R><B> data to its LSB.

When data is written to RAM used by RGB interface via the system interface, please make sure that write data conflicts do not occur.

When the 18-bit RGB interface is in use, 18-bit data is written to RAM via DB17-0 and 262,144-colors are available. When the 16-bit RGB interface is in use, the MSB is written to its LSB and 65,536-colors are available.

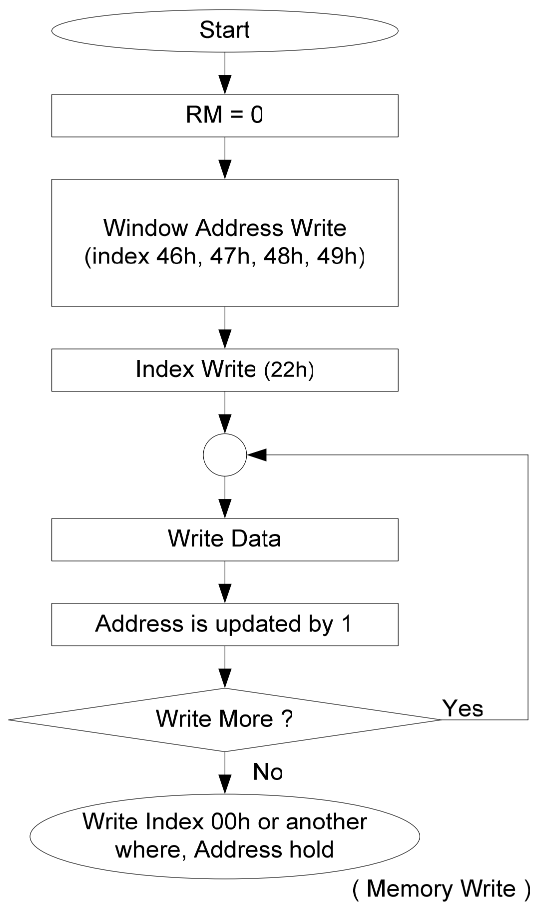


Figure 12. Memory Data Write Sequence

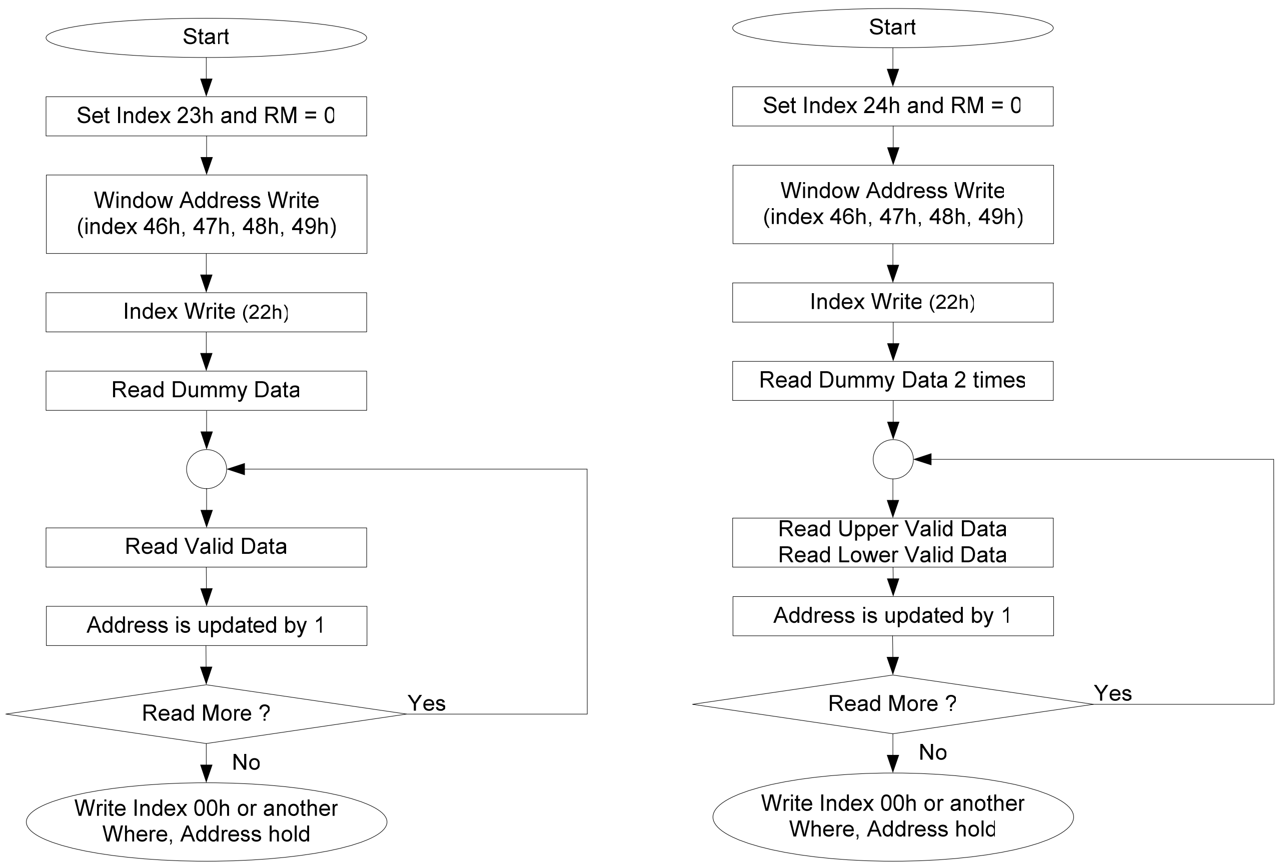
Preliminary

READ DATA FROM GRAM (R22H)

R/W	RS	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
R	1	RAM Read data (RD17-0): Pin assignment varies according to the interface method. (see the following figure for more information)																	

RD17–0: Read 18-bit data from the GRAM. When the data is read to the CPU, the first-word read immediately after the GRAM address setting is latched from the GRAM to the internal read-data latch. The data on the data bus (DB15–0) becomes invalid and the second-word read is normal.

In case of 16-/8-bit interface, the LSB of <R><B> color data will not be read.
This function is not available in RGB interface mode.



18-/16- System Interface **9-/8- System Interface**

Figure 13. Memory Data Read Sequence

SELECT DATA BUS 1 (R23H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	0	Select 18-/16-bit Data Bus Interface															

SELECT DATA BUS 2 (R24H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	0	Select 9-/8-bit Data Bus Interface															

We can select system interface mode by pins and instruction as following.

Pins		Registers				
S_PB	ID_MIB	Index Address	Command (CLS)	Command MDT[1]	Command MDT[0]	Description
0 (Parallel)	0 (80 mode)	default & 24h (9/8bit)	0 (80 8bit)	0	x	80-system 8-bit bus interface
				1	0	80-system 8-bit 260k bus interface
					1	80-system 8-bit 65k bus interface
			1(80 9bit)	x	x	80-system 9-bit bus interface
		Index 23 h (18/16bit)	0 (80 16bit)	0	0	80-system 16-bit bus interface
					1	80-system 16-bit bus interface
				1	x	80-system 16-bit 260k bus interface
			1(80 18bit)	x	x	80-system 18-bit bus interface
	1 (68 mode)	default & 24h (9/8bit)	0 (68 8bit)	0	x	68-system 8-bit bus interface
				1	0	68-system 8-bit 260k bus interface
					1	68-system 8-bit 65k bus interface
			1 (68 9bit)	x	x	68-system 9-bit bus interface
		Index 23 h (18/16bit)	0 (68 16bit)	0	0	68-system 16-bit bus interface
					1	68-system 16-bit bus interface
				1	x	68-system 16-bit 260k bus interface
			1 (68 18bit)	x	x	68-system 18-bit bus interface
1 (Serial)	ID	x	x	x	x	Serial peripheral interface (SPI)

For details, see the SYSTEM INTERFACE.

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GATE SCAN POSITION (R40H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	0	0	0	0	0	0	0	0	0	0	SCN5	SCN4	SCN3	SCN2	SCN1	SCN0

*Initial Value = XXXX_XXXX_XX00_0000

SCN5-0: Set the scanning starting position of the gate driver.

SCN5	SCN4	SCN3	SCN2	SCN1	SCN0	Scanning start position	
						GS=0	GS=1
0	0	0	0	0	0	G1	G432
0	0	0	0	0	1	G9	G424
0	0	0	0	1	0	G17	G416
:	:	:	:	:	:	:	:
1	1	0	0	1	0	G401	G32
1	1	0	0	1	1	G409	G24
1	1	0	1	0	0	G417	G16

[NOTE] Ensure that gate start position (SCN) + the number of LCD driver lines (NL) ≤ 432 when GS = 0, and that gate start position (SCN) – the number of LCD driver lines (NL) ≥ 0 when GS = 1

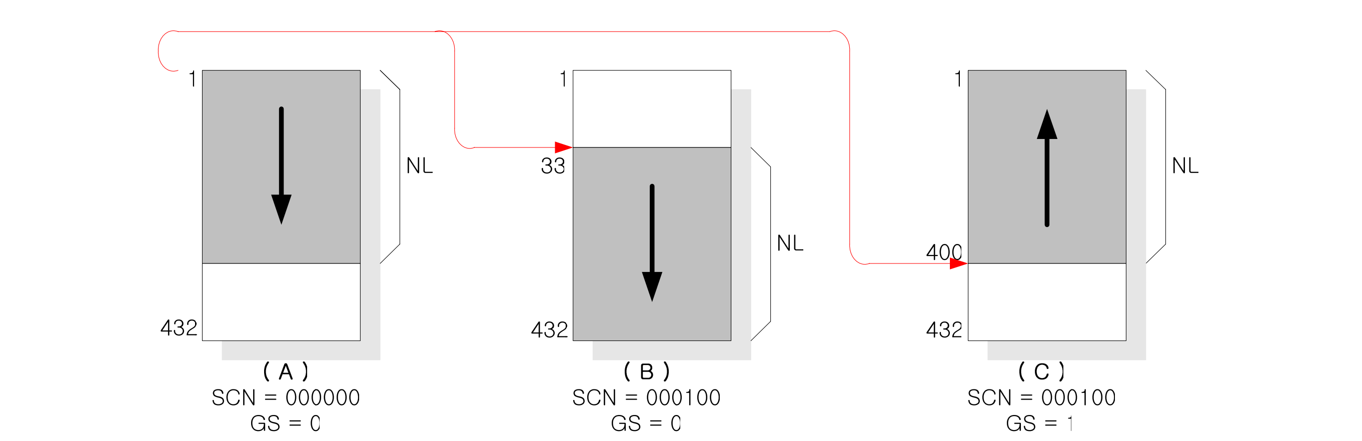

Figure 14. Relationship between NL and SCN set up value

VERTICAL SCROLL CONTROL 1 (R41H, R42H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	X	X	X	X	SEA 8	SEA 7	SEA 6	SEA 5	SEA 4	SEA 3	SEA 2	SEA 1	SEA 0
W	1	X	X	X	X	X	X	X	SSA 8	SSA 7	SSA 6	SSA 5	SSA 4	SSA 3	SSA 2	SSA 1	SSA 0

*41h Initial Value = XXXX_XXX1_1010_1111
*42h Initial Value = XXXX_XXX0_0000_0000

SSA8-0: Specify scroll start address at the scroll display for vertical smooth scrolling.

SSA8	SSA7	SSA6	SSA5	SSA4	SSA3	SSA2	SSA1	SSA0	Scroll Start Address
0	0	0	0	0	0	0	0	0	0 raster-row
0	0	0	0	0	0	0	0	1	1 raster-row
0	0	0	0	0	0	0	1	0	2 raster-row
0	0	0	0	0	0	0	1	1	3 raster-row
0	0	0	0	0	0	1	0	0	4 raster-row
0	0	0	0	0	0	1	0	1	5 raster-row
:	:	:	:	:	:	:			:
1	1	0	1	0	1	1	0	0	428 raster-row
1	1	0	1	0	1	1	0	1	429 raster-row
1	1	0	1	0	1	1	1	0	430 raster-row
1	1	0	1	0	1	1	1	1	431 raster-row

SEA8-0: Specify scroll end address at the scroll display for vertical smooth scrolling.

SEA8	SEA7	SEA6	SEA5	SEA4	SEA3	SEA2	SEA1	SEA0	Scroll End Address
0	0	0	0	0	0	0	0	0	0 raster-row
0	0	0	0	0	0	0	0	1	1 raster-row
0	0	0	0	0	0	0	1	0	2 raster-row
0	0	0	0	0	0	0	1	1	3 raster-row
0	0	0	0	0	0	1	0	0	4 raster-row
0	0	0	0	0	0	1	0	1	5 raster-row
:	:	:	:	:	:	:			:
1	1	0	1	0	1	1	0	0	428 raster-row
1	1	0	1	0	1	1	0	1	429 raster-row
1	1	0	1	0	1	1	1	0	430 raster-row
1	1	0	1	0	1	1	1	1	431 raster-row

Note: Don't set any higher raster-row than 431 (“1AF”H)
Set SS18-10 ≦ SSA8-0, if set out of range, SSA8-0 = SS18-10.
Set SE18-10 ≧ SEA8-0, if set out of range, SEA8-0 = SE18-10

Preliminary

VERTICAL SCROLL CONTROL 2 (R43H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	X	X	X	X	SST 8	SST 7	SST 6	SST 5	SST 4	SST 3	SST 2	SST 1	SST 0

*Initial Value = XXXX_XXX0_0000_0000

SST8-0: Specify scroll start and step at the scroll display for vertical smooth scrolling. Any raster-row from the 1st to 432th can be scrolled for the number of the raster-row. After 431th raster-row is displayed, the display restarts from the first raster-row. When SST8-0 = 00000000, Vertical Scroll Function is disabled.

SST8	SST7	SST6	SST5	SST4	SST3	SST2	SST1	SST0	Scroll Step
0	0	0	0	0	0	0	0	0	Scroll Disabled
0	0	0	0	0	0	0	0	1	1 raster-row
0	0	0	0	0	0	0	1	0	2 raster-row
0	0	0	0	0	0	0	1	1	3 raster-row
0	0	0	0	0	0	1	0	0	4 raster-row
0	0	0	0	0	0	1	0	1	5 raster-row
:	:	:	:	:	:	:			:
1	1	0	1	0	1	1	0	0	428 raster-row
1	1	0	1	0	1	1	0	1	429 raster-row
1	1	0	1	0	1	1	1	0	430 raster-row
1	1	0	1	0	1	1	1	1	431 raster-row

Note: Don't set any higher raster-row than 431 ("1AF"H)
Set SS18-10 < SSA8-0 + SST8-0 ≦ SEA8-0 ≦ SE18-10, if set out of range, Scroll function is disabled

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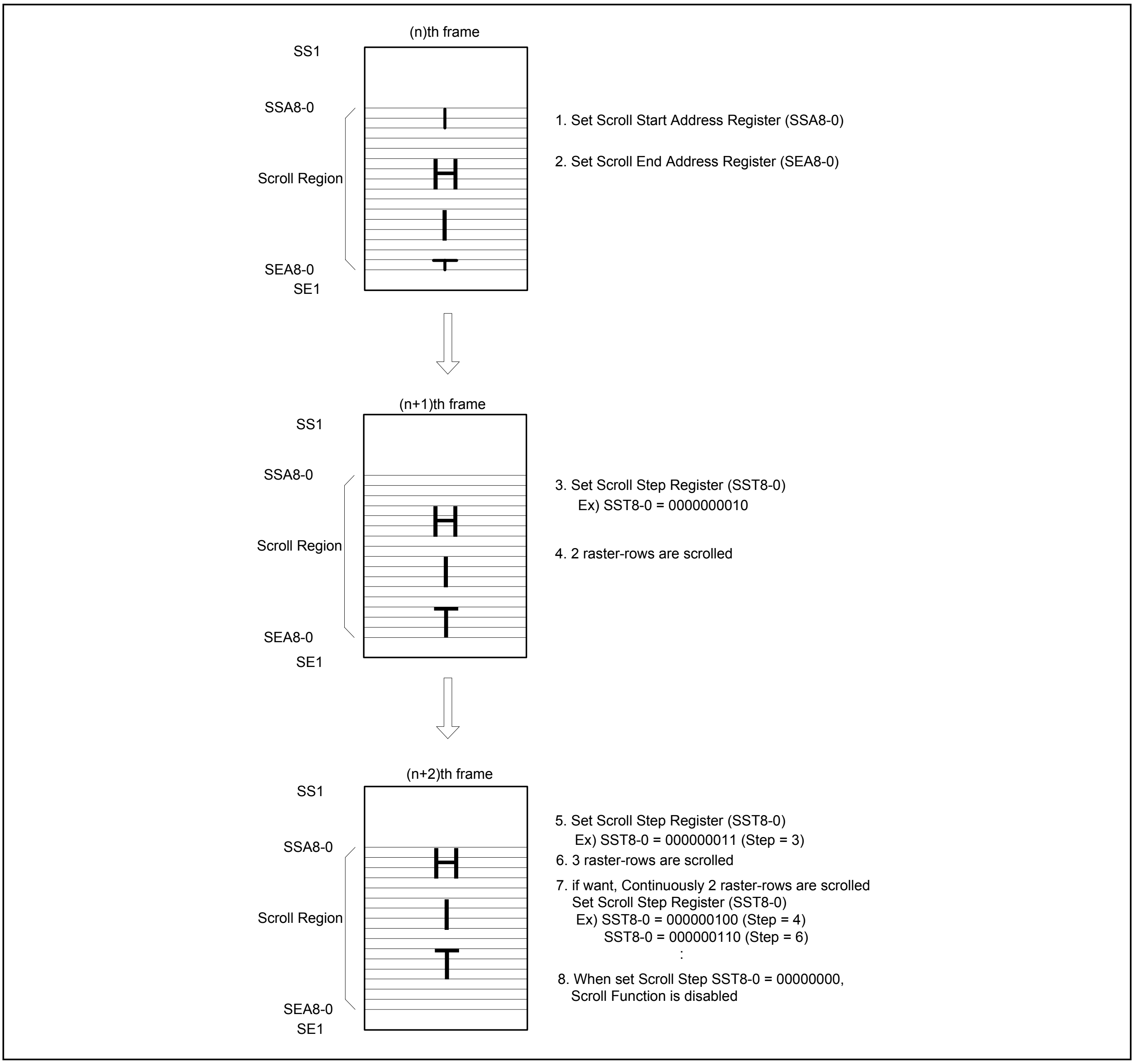


Figure 15. Vertical Scroll Display

PARTIAL SCREEN DRIVING POSITION (R44H, R45H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	X	X	X	X	SE18	SE17	SE16	SE15	SE14	SE13	SE12	SE11	SE10
W	1	X	X	X	X	X	X	X	SS18	SS17	SS16	SS15	SS14	SS13	SS12	SS11	SS10

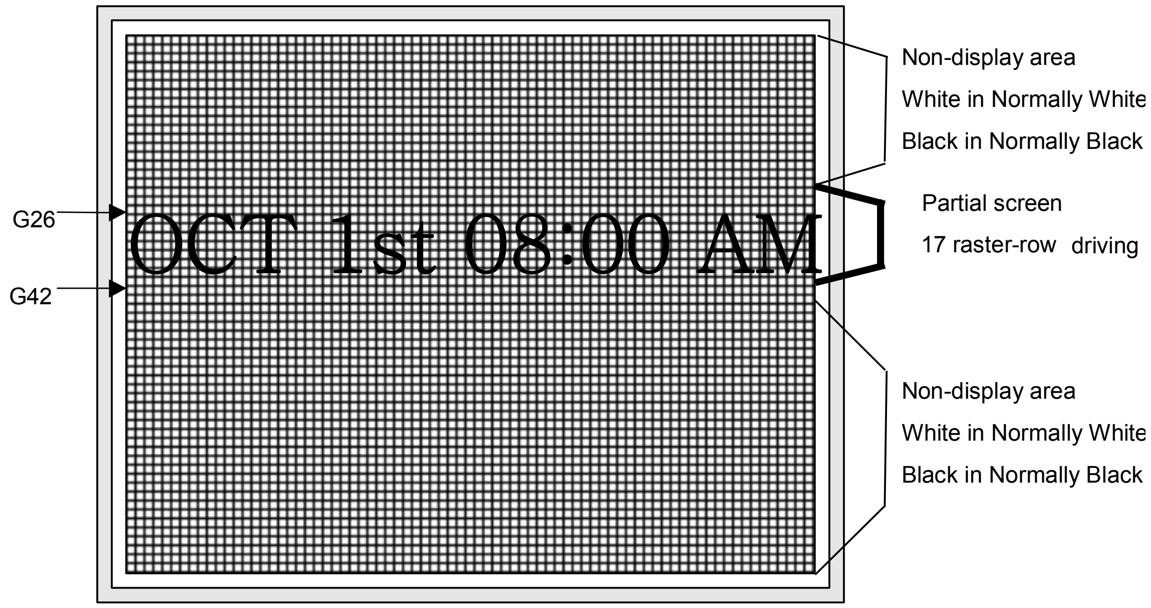
*44h Initial Value = XXXX_XXX1_1010_1111
*45h Initial Value = XXXX_XXX0_0000_0000

SE18–10: Specify the driving end position for the screen in a line unit. The LCD driving is performed to the 'set value + 1' gate driver. For instance, when SS18–10 = 019h and SE18–10 = 029h are set, the LCD driving is performed from G26 to G42, and non-display driving is performed for G1 to G25, G43, and others. Ensure that SS18–10 ≤ SE18–10 ≤1Afh.

SS18–10: Specify the drive starting position for the first screen in a line unit. The LCD driving starts from the 'set value +1' gate driver.

Note: DO NOT set the partial setting when the operation is in the normal display condition. Set this register only when in the partial display condition.
Ex) SS18-0=007h and SE18-0=010h are performed from G8 to G17.

The S6D0170 can select and drive partial screens at any position with the screen-driving position registers (R44H, R45H). Any partial screens required for display are selectively driven and reducing LCD-driving voltage and power consumption.



- Driving raster-row: NL5-0 = 110110 (432 lines)
- Partial screen setting: SS18-10 = 019H, SE18-10 = 029H

Figure 16. Driving On Partial Screen

Preliminary

HORIZONTAL RAM ADDRESS POSITION (R46H,R47H)

VERTICAL RAM ADDRESS POSITION (R48H,R49H)

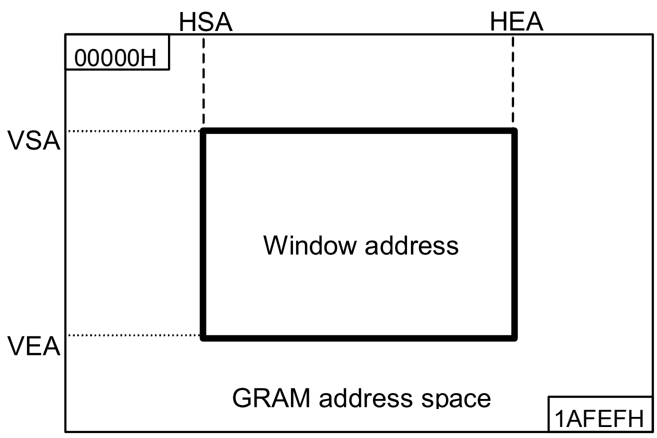
R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	X	X	X	X	HEA 8	HEA 7	HEA 6	HEA 5	HEA 4	HEA 3	HEA 2	HEA 1	HEA 0
W	1	X	X	X	X	X	X	X	HSA 8	HSA 7	HSA 6	HSA 5	HSA 4	HSA 3	HSA 2	HSA 1	HSA 0
W	1	X	X	X	X	X	X	X	VEA 8	VEA 7	VEA 6	VEA 5	VEA 4	VEA 3	VEA 2	VEA 1	VEA 0
W	1	X	X	X	X	X	X	X	VSA 8	VSA 7	VSA 6	VSA 5	VSA 4	VSA 3	VSA 2	VSA 1	VSA 0

*46h Initial Value = XXXX_XXX0_1110_1111
*47h Initial Value = XXXX_XXX0_0000_0000
*48h Initial Value = XXXX_XXX1_1010_1111
*49h Initial Value = XXXX_XXX0_0000_0000

HSA8-0/HEA8-0: Specify the horizontal start/end positions of a window for access in memory. Data can be written to the GRAM from the address specified by HSA8-0 to the address specified by HEA 8-0. Note that an address must be set before RAM is written..

VSA8-0/VEA8-0: Specify the vertical start/end positions of a window for access in memory. Data can be written to the GRAM from the address specified by VSA8-0 to the address specified by VEA8-0. Note that an address must be set before RAM is written.

When AM=0, 000h ≤ HSA8-0 ≤ HEA8-0 ≤ 0EFh and 000h ≤ VSA8-0 ≤ VEA8-0 ≤ 1AFh
AM=1, 000h ≤ HSA8-0 ≤ HEA8-0 ≤ 1AFh and 000h ≤ VSA8-0 ≤ VEA8-0 ≤ 0EFh



Window address setting range

“000”h ≤ HSA8-0≤ HEA8-0≤ “0EF”h
“000”h ≤ VSA8-0≤ VEA8-0≤ “1AF”h

NOTE: 1. Ensure that the window address area is within the GRAM address space

Figure 17. Window Address Setting Range(AM=0)

Preliminary

GAMMA CONTROL R (R50H TO R59H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	R_ VRN 04	R_ VRN 03	R_ VRN 02	R_ VRN 01	R_ VRN 00	X	X	X	R_ VRP 04	R_ VRP 03	R_ VRP 02	R_ VRP 01	R_ VRP 00
W	1	X	X	X	R_ VRN 14	R_ VRN 13	R_ VRN 12	R_ VRN 11	R_ VRN 10	X	X	X	R_ VRP 14	R_ VRP 13	R_ VRP 12	R_ VRP 11	R_ VRP 10
W	1	X	X	X	X	R_ PRN 03	R_ PRN 02	R_ PRN 01	R_ PRN 00	X	X	X	X	R_ PRP 03	R_ PRP 02	R_ PRP 01	R_ PRP 00
W	1	X	X	X	X	R_ PRN 13	R_ PRN 12	R_ PRN 11	R_ PRN 10	X	X	X	X	R_ PRP 13	R_ PRP 12	R_ PRP 11	R_ PRP 10
W	1	X	X	X	X	R_ PKN 03	R_ PKN 02	R_ PKN 01	R_ PKN 00	X	X	X	X	R_ PKP 03	R_ PKP 02	R_ PKP 01	R_ PKP 00
W	1	X	X	X	X	R_ PKN 13	R_ PKN 12	R_ PKN 11	R_ PKN 10	X	X	X	X	R_ PKP 13	R_ PKP 12	R_ PKP 11	R_ PKP 10
W	1	X	X	X	X	R_ PKN 23	R_ PKN 22	R_ PKN 21	R_ PKN 20	X	X	X	X	R_ PKP 23	R_ PKP 22	R_ PKP 21	R_ PKP 20
W	1	X	X	X	X	R_ PKN 33	R_ PKN 32	R_ PKN 31	R_ PKN 30	X	X	X	X	R_ PKP 33	R_ PKP 32	R_ PKP 31	R_ PKP 30
W	1	X	X	X	X	R_ PKN 43	R_ PKN 42	R_ PKN 41	R_ PKN 40	X	X	X	X	R_ PKP 43	R_ PKP 42	R_ PKP 41	R_ PKP 40
W	1	X	X	X	X	R_ PKN 53	R_ PKN 52	R_ PKN 51	R_ PKN 50	X	X	X	X	R_ PKP 53	R_ PKP 52	R_ PKP 51	R_ PKP 50

*Initial Value = All “0”

R_VRN14-00: The amplitude adjust register for the negative polarity output
R_VRP14-00: The amplitude adjust register for the positive polarity output

R_PRN13-00: The gradient adjust register for the negative polarity output
R_PRP13-00: The gradient adjust register for the positive polarity output

R_PKN53-00: The gamma fine adjust register for the negative polarity output
R_PKP53–00: The gamma fine adjust register for the positive polarity output

For details, see the GAMMA ADJUSTMENT FUNCTION.

Preliminary

GAMMA CONTROL G (R60H TO R69H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	G_VRN04	G_VRN03	G_VRN02	G_VRN01	G_VRN00	X	X	X	G_VRP04	G_VRP03	G_VRP02	G_VRP01	G_VRP00
W	1	X	X	X	G_VRN14	G_VRN13	G_VRN12	G_VRN11	G_VRN10	X	X	X	G_VRP14	G_VRP13	G_VRP12	G_VRP11	G_VRP10
W	1	X	X	X	X	G_PRN03	G_PRN02	G_PRN01	G_PRN00	X	X	X	X	G_PRP03	G_PRP02	G_PRP01	G_PRP00
W	1	X	X	X	X	G_PRN13	G_PRN12	G_PRN11	G_PRN10	X	X	X	X	G_PRP13	G_PRP12	G_PRP11	G_PRP10
W	1	X	X	X	X	G_PKN03	G_PKN02	G_PKN01	G_PKN00	X	X	X	X	G_PKP03	G_PKP02	G_PKP01	G_PKP00
W	1	X	X	X	X	G_PKN13	G_PKN12	G_PKN11	G_PKN10	X	X	X	X	G_PKP13	G_PKP12	G_PKP11	G_PKP10
W	1	X	X	X	X	G_PKN23	G_PKN22	G_PKN21	G_PKN20	X	X	X	X	G_PKP23	G_PKP22	G_PKP21	G_PKP20
W	1	X	X	X	X	G_PKN33	G_PKN32	G_PKN31	G_PKN30	X	X	X	X	G_PKP33	G_PKP32	G_PKP31	G_PKP30
W	1	X	X	X	X	G_PKN43	G_PKN42	G_PKN41	G_PKN40	X	X	X	X	G_PKP43	G_PKP42	G_PKP41	G_PKP40
W	1	X	X	X	X	G_PKN53	G_PKN52	G_PKN51	G_PKN50	X	X	X	X	G_PKP53	G_PKP52	G_PKP51	G_PKP50

*Initial Value = All "0"

G_VRN14-00: The amplitude adjust register for the negative polarity output
G_VRP14-00: The amplitude adjust register for the positive polarity output

G_PRN13-00: The gradient adjust register for the negative polarity output
G_PRP13-00: The gradient adjust register for the positive polarity output

G_PKN53-00: The gamma fine adjust register for the negative polarity output
G_PKP53–00: The gamma fine adjust register for the positive polarity output

For details, see the GAMMA ADJUSTMENT FUNCTION.

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GAMMA CONTROL B (R70H TO R79H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	B_VRN04	B_VRN03	B_VRN02	B_VRN01	B_VRN00	X	X	X	B_VRP04	B_VRP03	B_VRP02	B_VRP01	B_VRP00
W	1	X	X	X	B_VRN14	B_VRN13	B_VRN12	B_VRN11	B_VRN10	X	X	X	B_VRP14	B_VRP13	B_VRP12	B_VRP11	B_VRP10
W	1	X	X	X	X	B_PRN03	B_PRN02	B_PRN01	B_PRN00	X	X	X	X	B_PRP03	B_PRP02	B_PRP01	B_PRP00
W	1	X	X	X	X	B_PRN13	B_PRN12	B_PRN11	B_PRN10	X	X	X	X	B_PRP13	B_PRP12	B_PRP11	B_PRP10
W	1	X	X	X	X	B_PKN03	B_PKN02	B_PKN01	B_PKN00	X	X	X	X	B_PKP03	B_PKP02	B_PKP01	B_PKP00
W	1	X	X	X	X	B_PKN13	B_PKN12	B_PKN11	B_PKN10	X	X	X	X	B_PKP13	B_PKP12	B_PKP11	B_PKP10
W	1	X	X	X	X	B_PKN23	B_PKN22	B_PKN21	B_PKN20	X	X	X	X	B_PKP23	B_PKP22	B_PKP21	B_PKP20
W	1	X	X	X	X	B_PKN33	B_PKN32	B_PKN31	B_PKN30	X	X	X	X	B_PKP33	B_PKP32	B_PKP31	B_PKP30
W	1	X	X	X	X	B_PKN43	B_PKN42	B_PKN41	B_PKN40	X	X	X	X	B_PKP43	B_PKP42	B_PKP41	B_PKP40
W	1	X	X	X	X	B_PKN53	B_PKN52	B_PKN51	B_PKN50	X	X	X	X	B_PKP53	B_PKP52	B_PKP51	B_PKP50

*Initial Value = All "0"

B_VRN14-00: The amplitude adjust register for the negative polarity output
B_VRP14-00: The amplitude adjust register for the positive polarity output

B_PRN13-00: The gradient adjust register for the negative polarity output
B_PRP13-00: The gradient adjust register for the positive polarity output

B_PKN53-00: The gamma fine adjust register for the negative polarity output
B_PKP53–00: The gamma fine adjust register for the positive polarity output

For details, see the GAMMA ADJUSTMENT FUNCTION.

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MTP CONTROL (R80H)

TEST KEY COMMAND (R81H)

MTP READ DATA (R82H)

R/W	RS	IB15	IB14	IB13	IB12	IB11	IB10	IB9	IB8	IB7	IB6	IB5	IB4	IB3	IB2	IB1	IB0
W	1	X	X	X	MTP_ LOAD	X	X	X	MTP_ WRB	X	X	X	MTP_ SEL	X	X	X	MTP_ ERB
W	1	X	X	X	X	X	X	X	X	1	0	0	0	1	1	0	0
R	1	X	X	X	X	X	X	X	X	MTP_ DOUT 7	MTP_ DOUT 6	MTP_ DOUT 5	MTP_ DOUT 4	MTP_ DOUT 3	MTP_ DOUT 2	MTP_ DOUT 1	MTP_ DOUT 0

*80h Initial Value = XXX0_XXX1_XXX1_XXX1

MTP_LOAD: When MTP_LOAD is High, MTP data is loaded into internal register
MTP_WRB: MTP Write enable signal. If you want to write data to MTP cell, set MTP_WRB = 0
MTP_SEL: Select the VCOMH voltage setting register between MTP data and VCM5-0 itself.

MTP_SEL	VCOMH Control Data
0	VCM[6:0] Register
1	MTP data

MTP_ERB: Enable for MTP initial or erase. When MTP_ERB = 0, MTP initialization or erase is enabled.

Test Key Command: Protection command. When Test Key Command =8Ch, MTP_WRB and MTP_ERB are valid

MTP_DOUT7-0: Data of MTP Output

Preliminary

MTP SEQUENCE FLOW

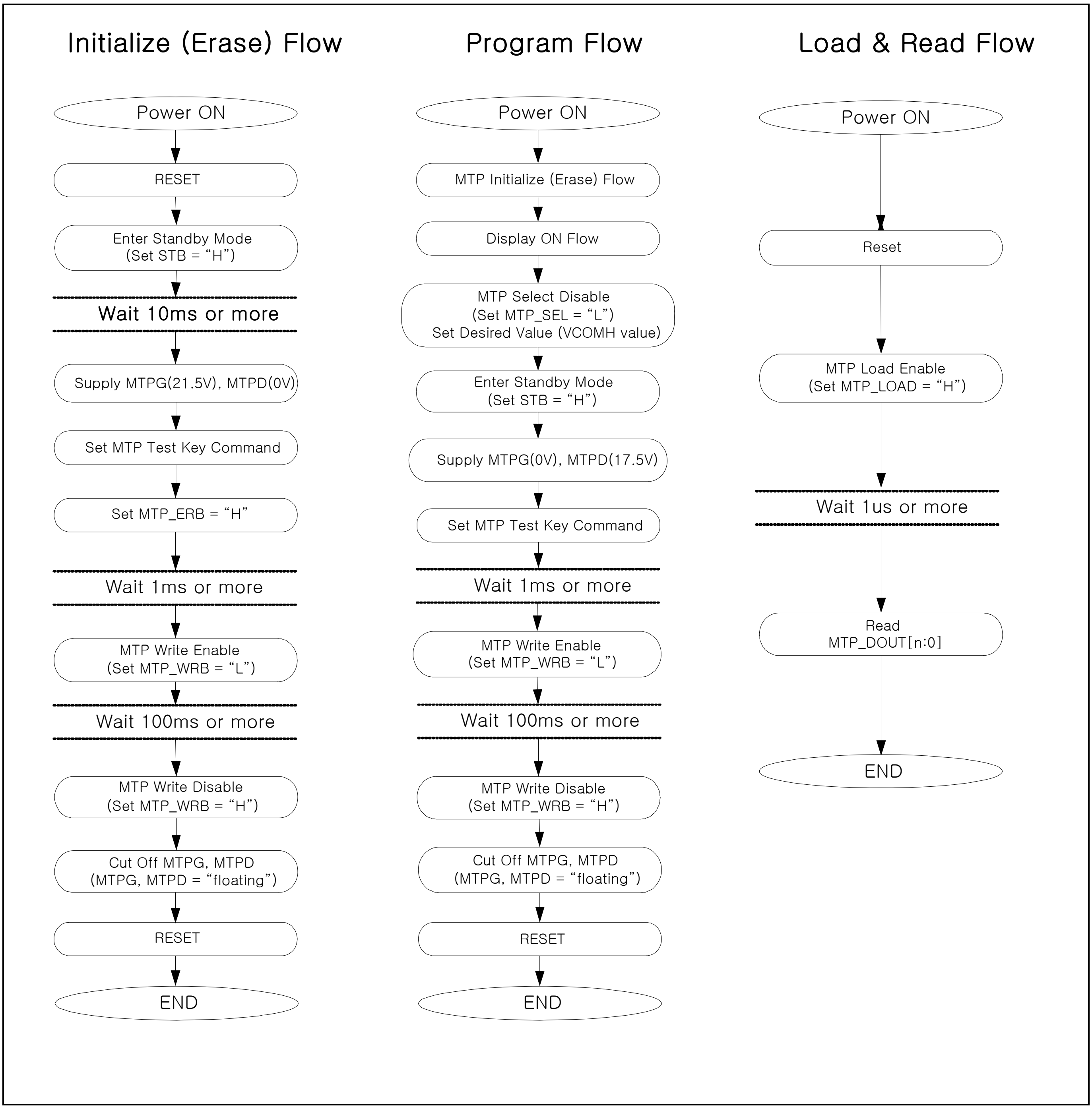

Figure 18. MTP Sequence Flow

Preliminary

MTP CHARACTERISTICS

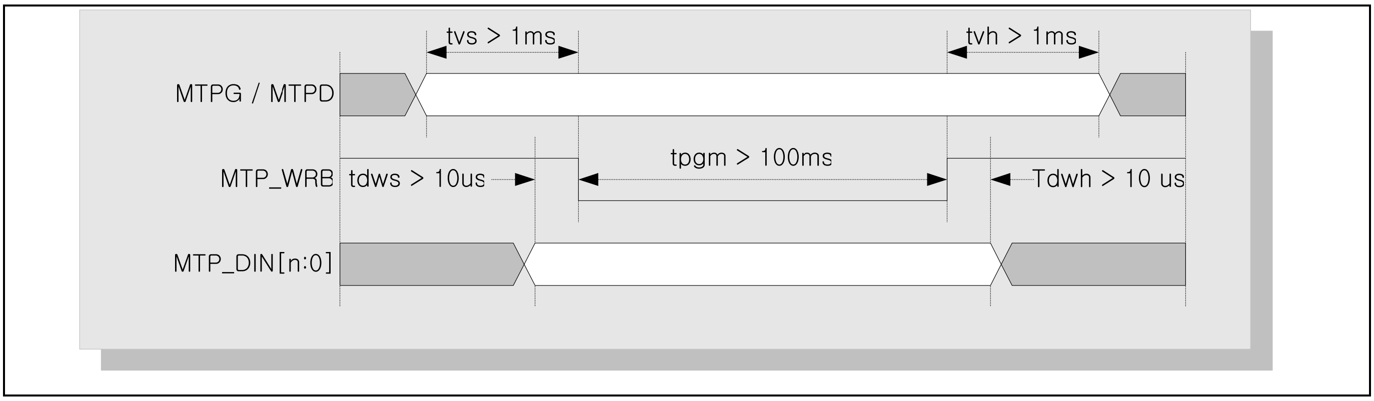

Figure 19. Voltages and waveforms for MTP programming

Table 35. MTP writing time

Timing	Min	Max	Unit
tvs	1	-	ms
tvh	1	-	ms
tdws	10	-	us
tdwh	10	-	us
tpgm	100	200	ms

Table 36. MTPG/D Voltage Tolerance

Item	Pgm	Min	Typ	Max	Unit
Tolerance of MTPG	Erase	21.0	21.5	22.0	V
	Write		0		
Tolerance of MTPD	Erase		0		V
	Write	17.0	17.5	18.0	

Table 37. Current consumption during setting MTP

Item	Symbol	condition	Min	Typ	Max	Unit
Current consumption during setting MTP	I_{MTPG}	MTPG = 21.5V	-	-	0.6	mA
	I_{MTPD}	MTPD = 17.5V	-	-	0.6	

* Note: simulation result, with common power condition VDD3=2.8V, VDD=1.5V, VCI=2.8V

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RESTRICTION ON PARTIAL DISPLAY AREA SETTING

The following restrictions must be satisfied when setting the start line (SS18 to 10) and end line (SE18 to 10) of the partial screen driving position register (R44H, R45H) for the S6D0170. Note that incorrect display may occur if the restrictions are not satisfied.

Table 38. Restrictions on the partial Screen Driving Position Register Setting

Register setting	Display operation
(SE18 to 10) – (SS18 to 10) = NL	Full screen display Normally displays (SS18 to 10) to (SE18 to 10)
(SE18 to 10) – (SS18 to 10) < NL	Partial display Normally displays (SS18 to 10) to (SE18 to 10) White display for all other times (RAM data is not related at all)
(SE18 to 10) – (SS18 to 10) > NL	Setting disabled

NOTE: 000h ≤ SS18 to 10 ≤ SE18 to 10 ≤ 1Afh

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WINDOW ADDRESS FUNCTION

When data is written to the on-chip GRAM, a window address-range that is specified by the horizontal address register (start: HSA7-0, end: HEA 7-0) and vertical address register (start: VSA8-0, end: VEA8-0) can be updated consecutively. Data is written to addresses in the direction specified by the I/D1-0bit. When image data is being written, data can be written consecutively without thinking a data wrap by doing this. The window must be specified to be within the GRAM address area described as following example. Addresses must be set within the window address.

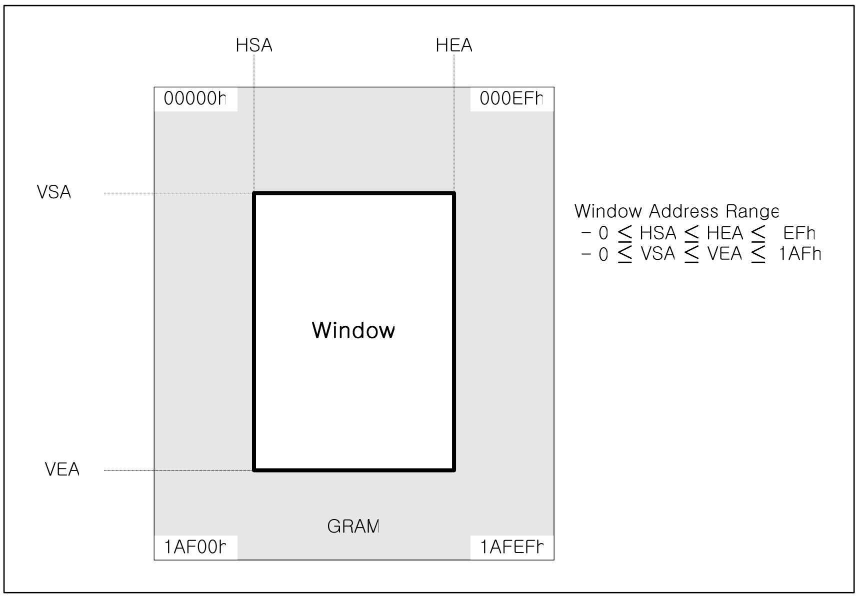

Figure 20. Example of address operation in the window address function

RESET FUNCTION

The S6D0170 is initialized internally by RESET input. The reset input should be held 'L' for at least 10us. Do not access the GRAM nor initially set the instructions until the R-C oscillation frequency is stable after power has been supplied (10 ms).

GRAM DATA INITIALIZATION

GRAM is not automatically initialized by reset input but must be initialized by software while display is off (D1-0 = 00).

OUTPUT PIN INITIALIZATION

1. LCD driver output pins (Source output) : Output AVSS level
(Gate output) : Output AVSS level

Preliminary

POWER SUPPLY

POWER SUPPLY CIRCUIT

The following figure shows a configuration of the voltage generation circuit of S6D0170. The step-up circuits consist of step-up circuits 1 to 3. Step-up circuit1 doubles input voltage supplied from VCI1 for AVDD level. Step-up circuit2 make 2.5, 3 or 3.5 times AVDD level for VGH level, and make -1.5, -2 or -2.5 times AVDD level for VGL level. Step-up circuit3 reverses the VCI1 level with respect to VSS to generate VCL level. These step-up circuits generate power supplies AVDD, VGH, VGL, and VCL. Reference voltages such as GVDD, VCOMH and VCOML are generated with VREF from the voltage adjustment circuit. Connect VCOM to the TFT panel.

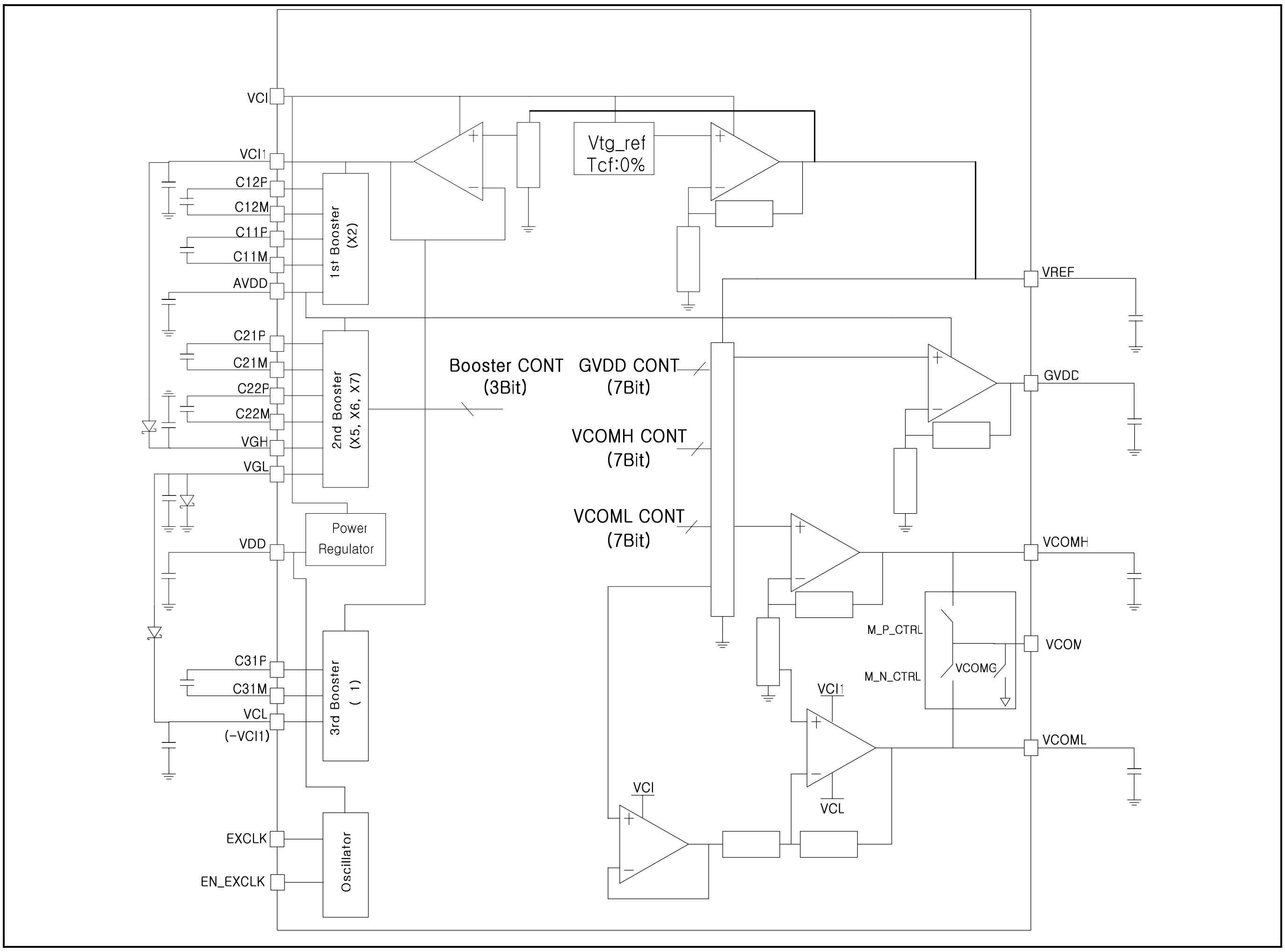

Figure 21. Configuration of the Internal Power-Supply Circuit

Notes: Use the 1uF capacitor.
Schottky diode between VGL and VSS is positively necessary for latch-up free.
Schottky diodes between VCL and VGL and between VGH and VCI1 may be necessary when latch-up occurs even if latch-up free power supply set-up flow is applied.
The Capacitor between VREF and VSS may be used by case when VCOM swing level is fluctuation

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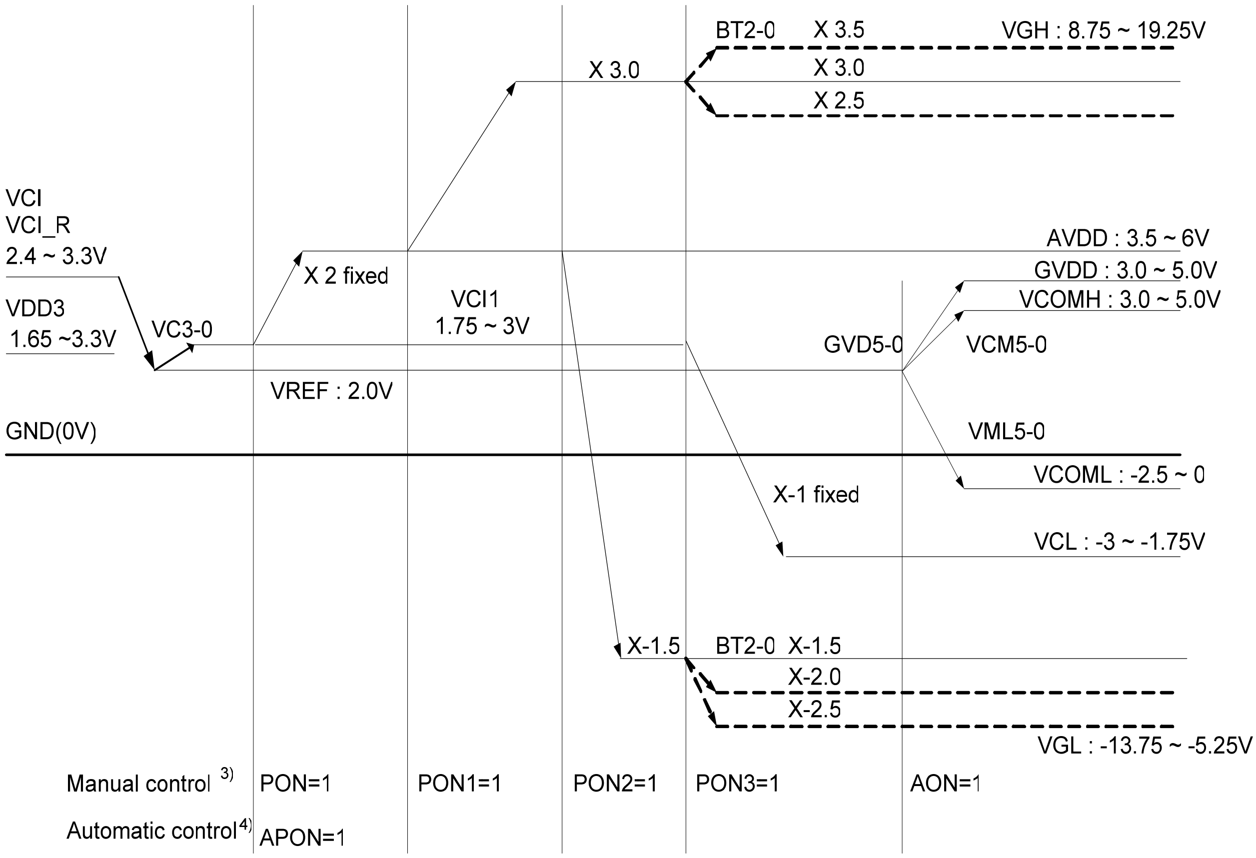
PATTERN DIAGRAMS FOR VOLTAGE SETTING

The following figure shows a pattern diagram for the voltage setting and an example of waveforms.



Note: 1) Set the conditions of AVDD - GVDD > 0.5V with loads because they differ depending on the display load to be driven.
2) PON, PON1, PON2, PON3, AON instructions are separately offered for flexibility of power-up sequence time control.
3) APON instruction is an auto power-up sequence operating switch. This operating takes more than 9 frame time.

cf> Pattern diagram above shows not only a relationship of each generated levels but practical power-up sequence.

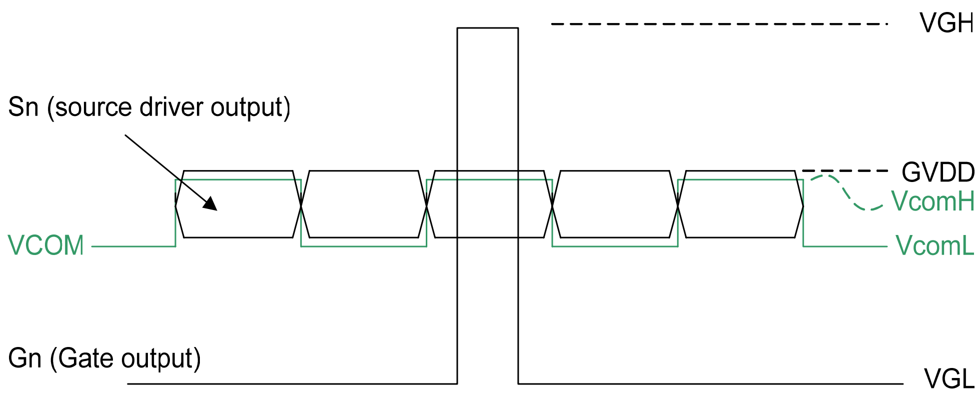


Figure 22. Power-Up Pattern Diagram & An Example Of Source/VCOM Waveforms

SAMSUNG ELECTRONICS

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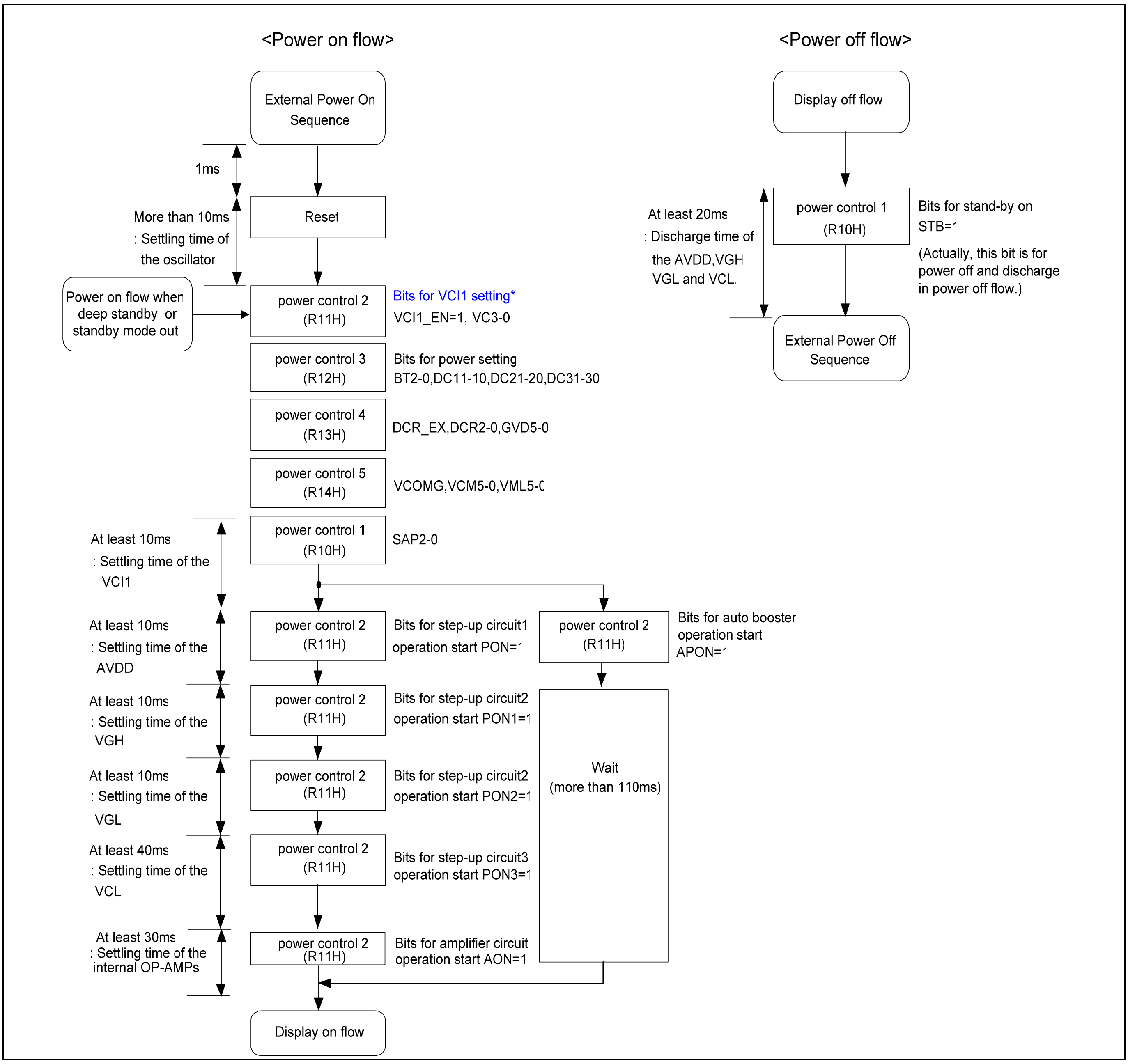
SET UP FLOW OF POWER

Apply the power in a sequential way as shown in the following figure. The settling time of the oscillation circuit, step-up1/2/3 circuits, and operational amplifier depend on the external resistance or capacitance value.



Note: The VCI1 voltage level is set to an user setting value when a PON3 value is high

1) When PON1 = 0 : VCI1 = 1.35V
2) When PON1= 1 : VCI1 = 2.0V
3) When PON1= 1 and PON3=1 : VCI1 = User setting value (VC3-0)
4) Wait more than 80ms, When Reference Frame Frequency is 60Hz

Figure 23. Setup Flow of Power

VOLTAGE REGULATION FUNCTION

The S6D0170 has the internal voltage regulator. By the use of this function, unexpected damages on internal logic circuit can be avoided. Furthermore, power consumption can also be obtained. Detailed function description and application configuration is described in the following diagram.

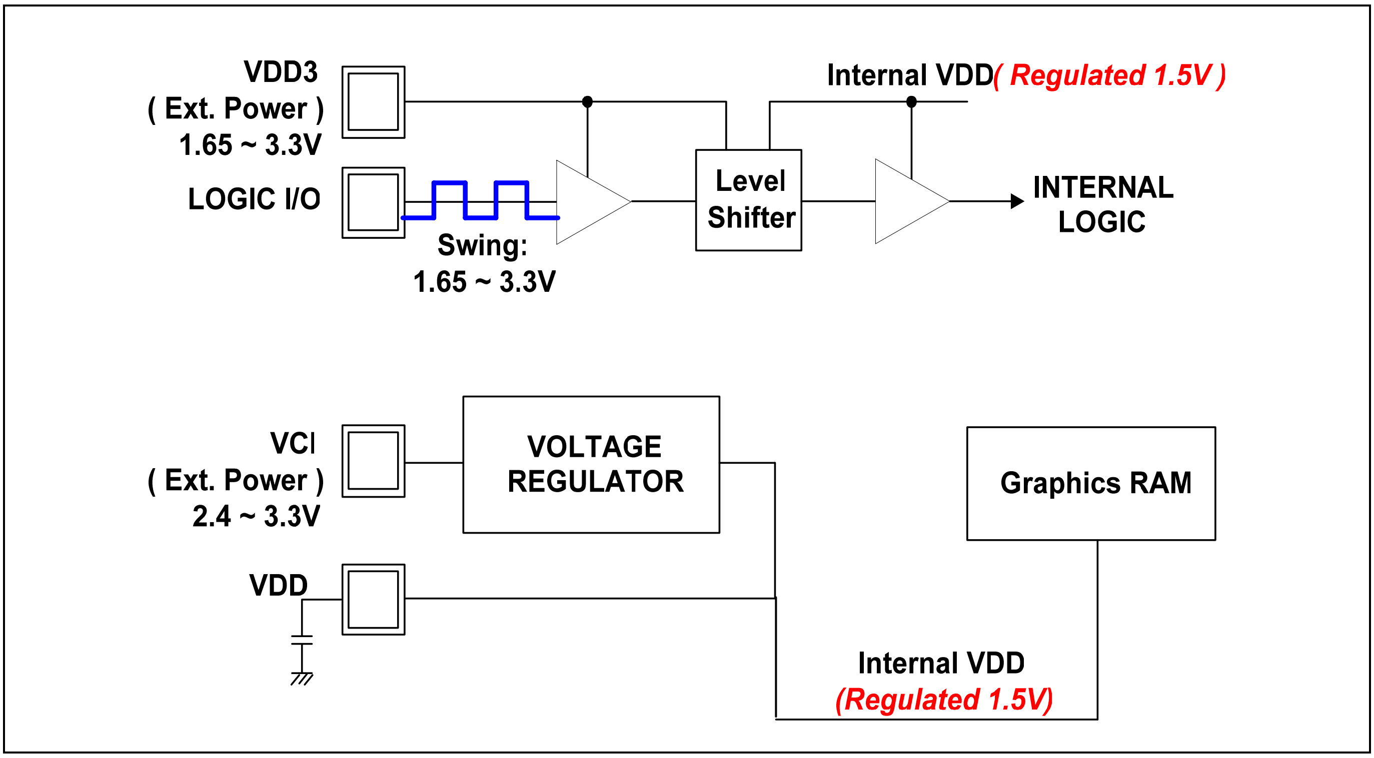


Figure 24. Voltage Regulation Function

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INTERFACE SPECIFICATION

The S6D0170 incorporates a system interface, which is used to set instructions, and an external display interface, which is used to display motion pictures. Selecting these interfaces to match the screen data (motion picture or still picture) enables efficient transfer of data for display.

The external display interface includes RGB interface. This allows flicker-free screen update.
When RGB interface is selected, the synchronization signals (VSYNC, HSYNC, and DOTCLK) are available for use in operating the display. The display data (DB17-0) is written according to the values of the data enable signal (ENABLE) in synchronization with the VSYNC, HSYNC, and DOTCLK signals. In addition, using the window address function enables rewriting only to the internal RAM area to display motion pictures. Using this function also enables simultaneously display of the motion picture area and the RAM data that was written.

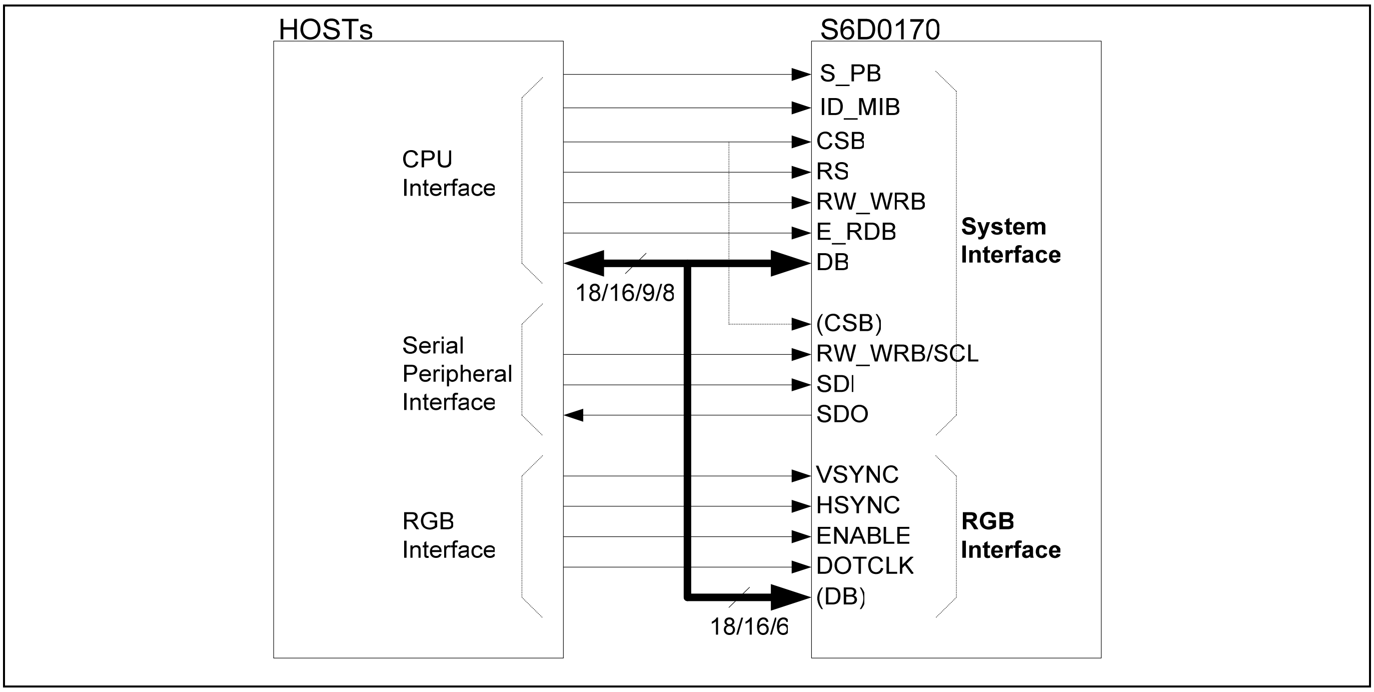

Figure 25. System Interface and RGB Interface

Preliminary

SYSTEM INTERFACE

S6D0170 is enabling to set instruction and access to RAM by selecting S_PB,ID_MIB pins and Instruction in the system interface mode.

Table 39. System Interface mode

Pins		Registers				
S_PB	ID_MIB	Index Address	Command (CLS)	Command MDT[1]	Command MDT[0]	Description
0 (Parallel)	0 (80 mode)	default & 24h (9/8bit)	0 (80 8bit)	0	x	80-system 8-bit bus interface
				1	0	80-system 8-bit 260k bus interface
					1	80-system 8-bit 65k bus interface
			1(80 9bit)	x	x	80-system 9-bit bus interface
		Index 23 h (18/16bit)	0 (80 16bit)	0	0	80-system 16-bit bus interface
					1	80-system 16-bit bus interface
				1	x	80-system 16-bit 260k bus interface
			1(80 18bit)	x	x	80-system 18-bit bus interface
	1 (68 mode)	default & 24h (9/8bit)	0 (68 8bit)	0	x	68-system 8-bit bus interface
				1	0	68-system 8-bit 260k bus interface
					1	68-system 8-bit 65k bus interface
			1 (68 9bit)	x	x	68-system 9-bit bus interface
		Index 23 h (18/16bit)	0 (68 16bit)	0	0	68-system 16-bit bus interface
					1	68-system 16-bit bus interface
				1	x	68-system 16-bit 260k bus interface
			1 (68 18bit)	x	x	68-system 18-bit bus interface
1 (Serial)	ID	x	x	x	x	Serial peripheral interface (SPI)

We can select system interface mode by pins and instruction, don't care 8-/16-bit bus system after power on.

1. In case of 8/9-bit bus system.

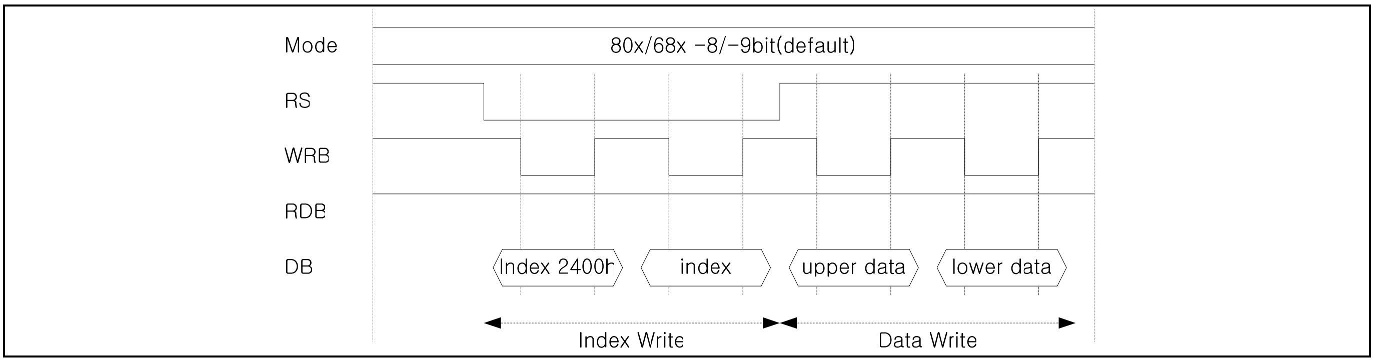

Figure 26. 8/9-bit bus system

2. In case of 18/16-bit bus system

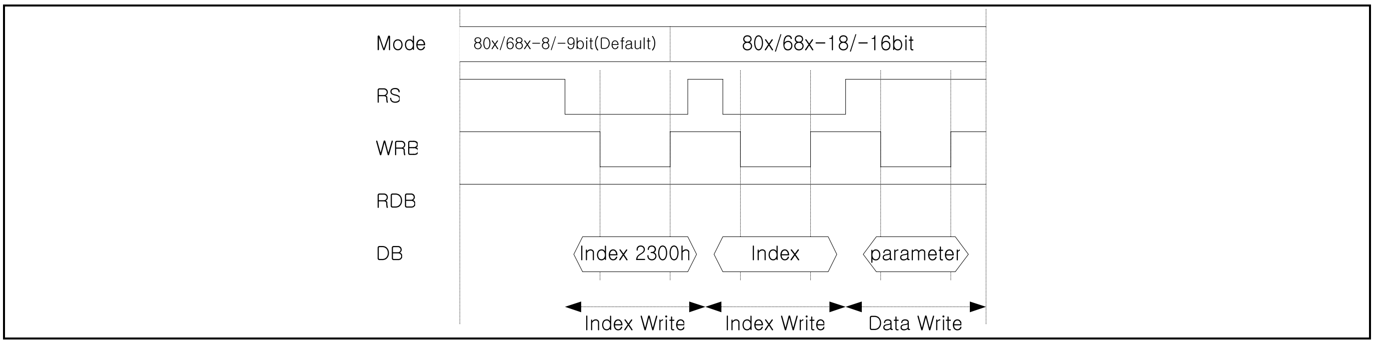

Figure 27. 18/16-bit bus system

Preliminary

68-SYSTEM 18-BIT BUS INTERFACE

Bit Assignment

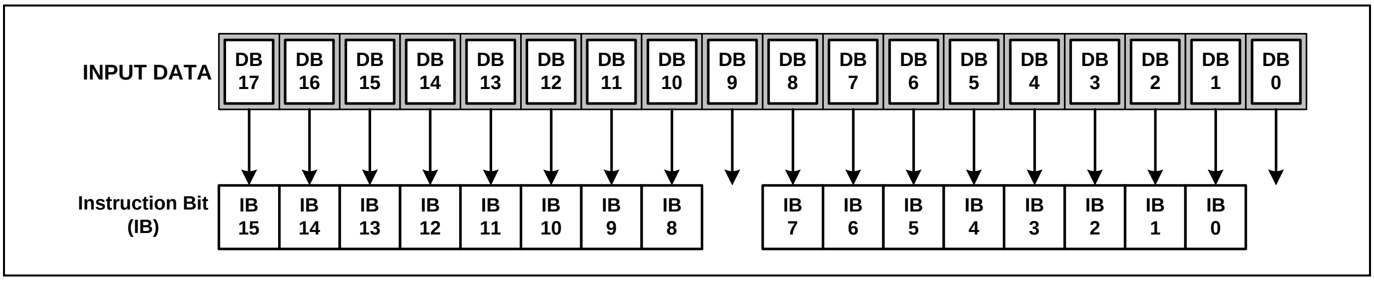

Figure 28. Instruction Format For 18-Bit Interface

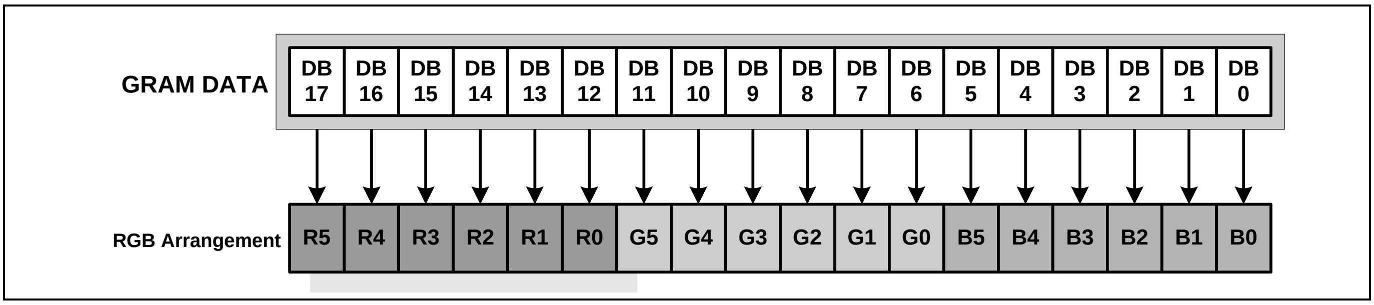

Figure 29. RAM Data Write Format For 18-Bit Interface

Timing Diagram

There are 4 timing conditions for 68-system 18-bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

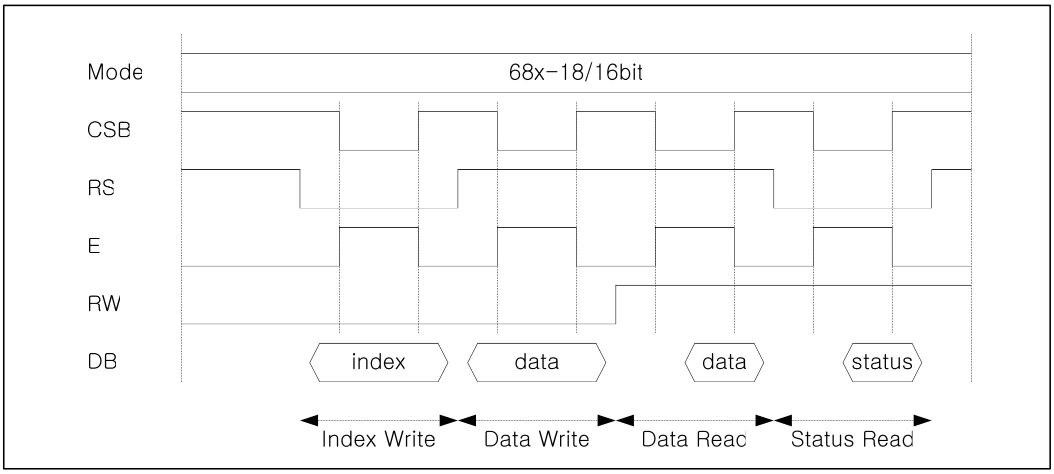

Figure 30. Timing Diagram of 68-System 18-Bit bus interface

SAMSUNG ELECTRONICS

68-SYSTEM 16-BIT BUS INTERFACE

Bit Assignment

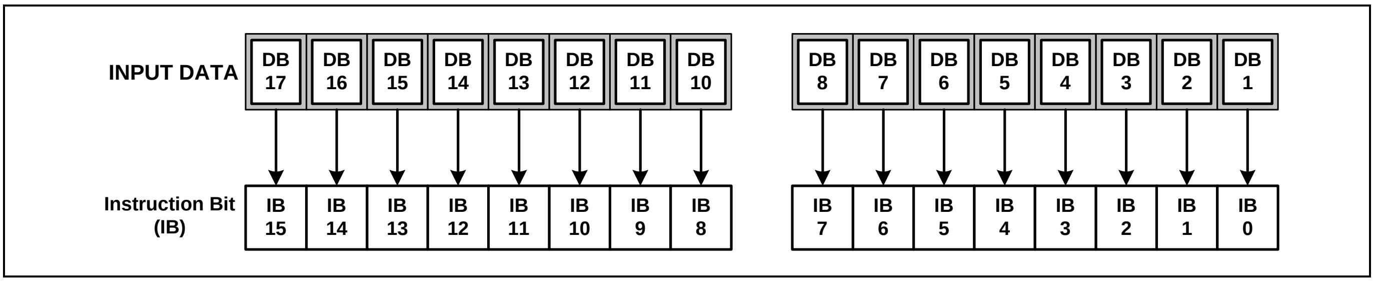

Figure 31. Instruction Format For 16-Bit Interface

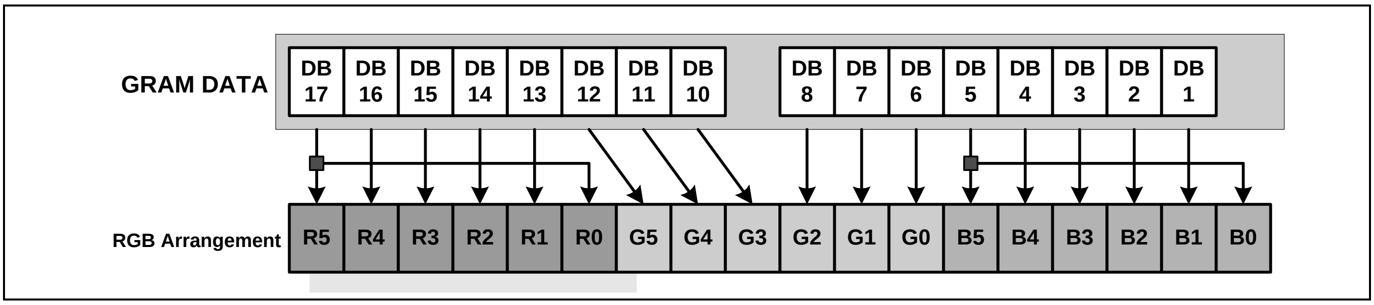

Figure 32. RAM Data Write Format For 16-Bit Interface

Timing Diagram

There are 4 timing conditions for 68-system 16-bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

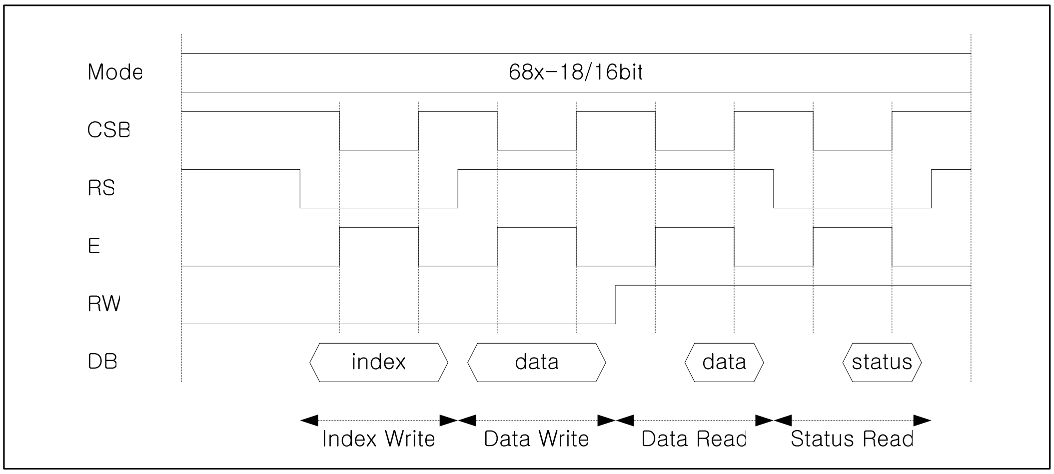

Figure 33. Timing Diagram of 68-System 16-Bit bus interface

Preliminary

68-SYSTEM 9-BIT BUS INTERFACE

Bit Assignment

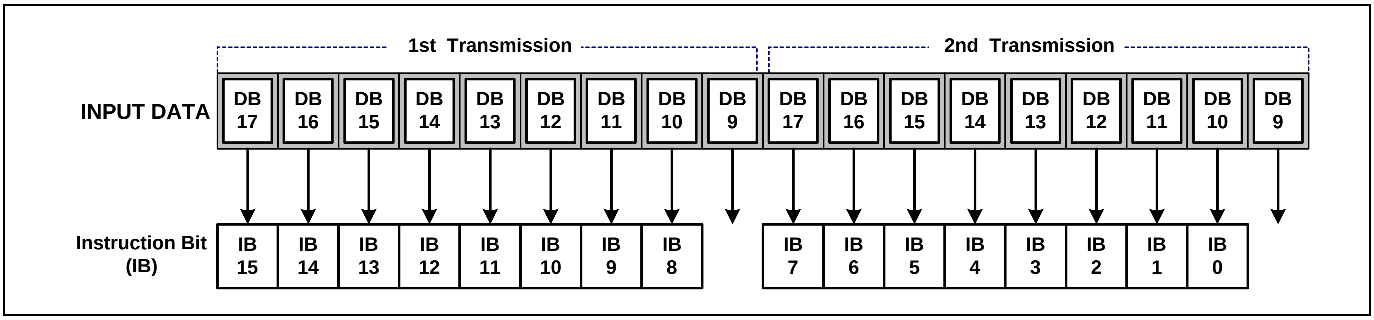

Figure 34. Instruction Format For 9-Bit Interface

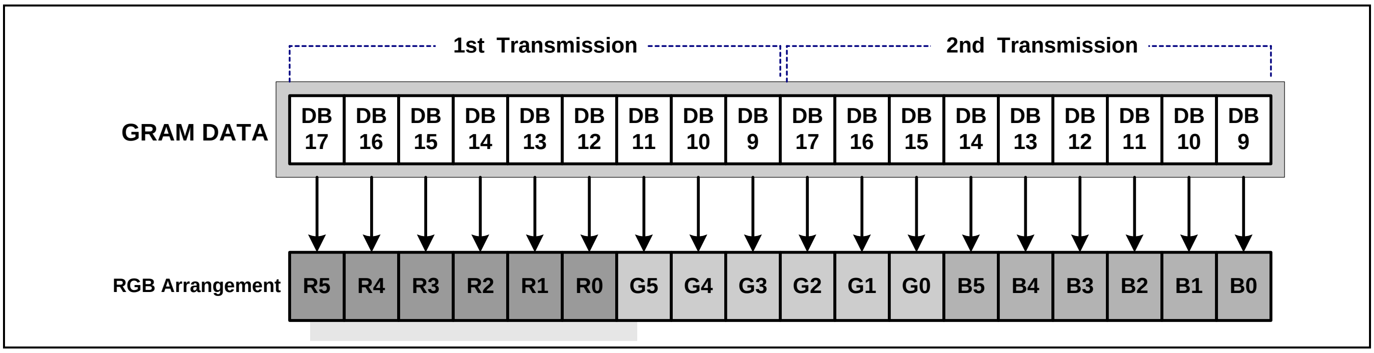

Figure 35. RAM Data Write Format For 9-Bit Interface

Timing Diagram

There are 4 timing conditions for 68-system 9-bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.
In this mode, 16-bit instructions and GRAM data are divided into two half words and the transfer starts from the upper half word. Note that the upper byte must also be written when the index register is written.

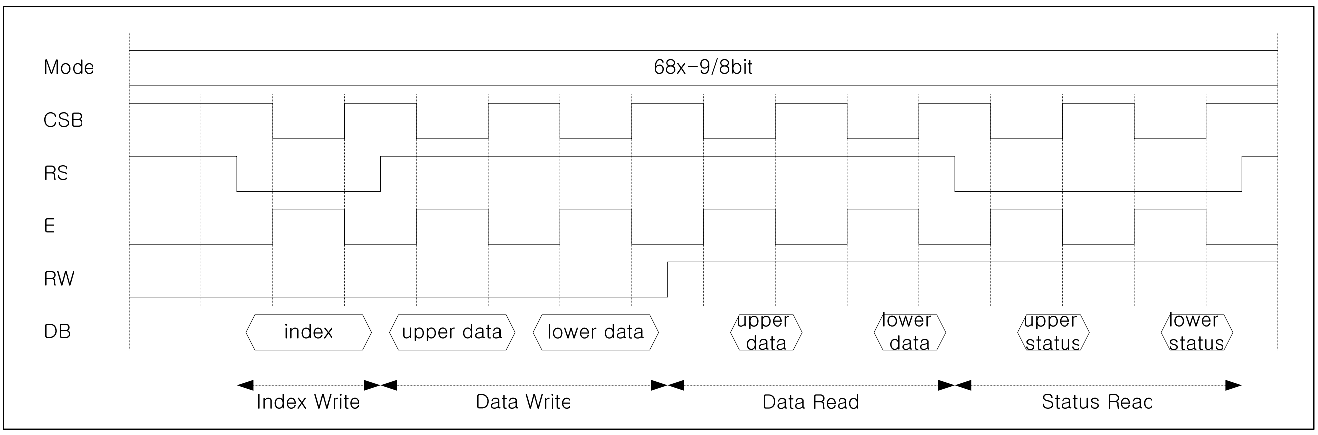

Figure 36. Timing Diagram of 68-System 9-Bit bus interface

SAMSUNG ELECTRONICS

Preliminary

68-SYSTEM 8-BIT BUS INTERFACE

Bit Assignment

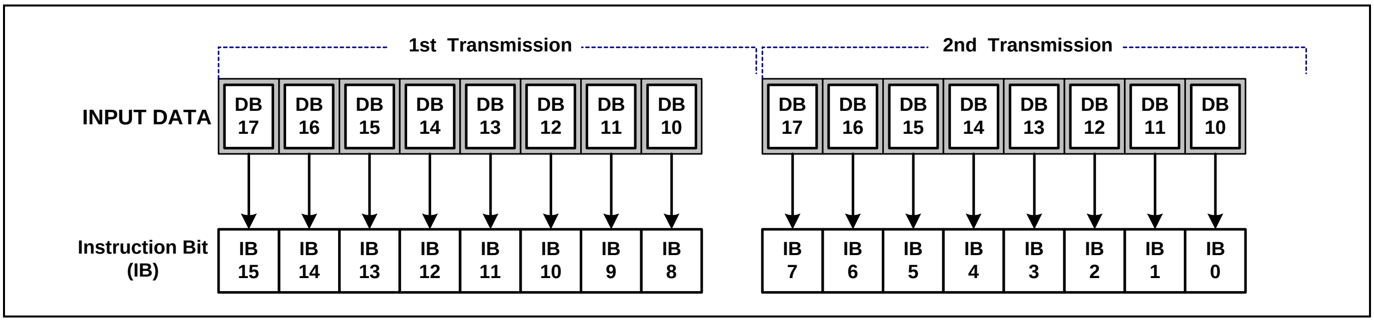

Figure 37. Instruction Format For 8-Bit Interface

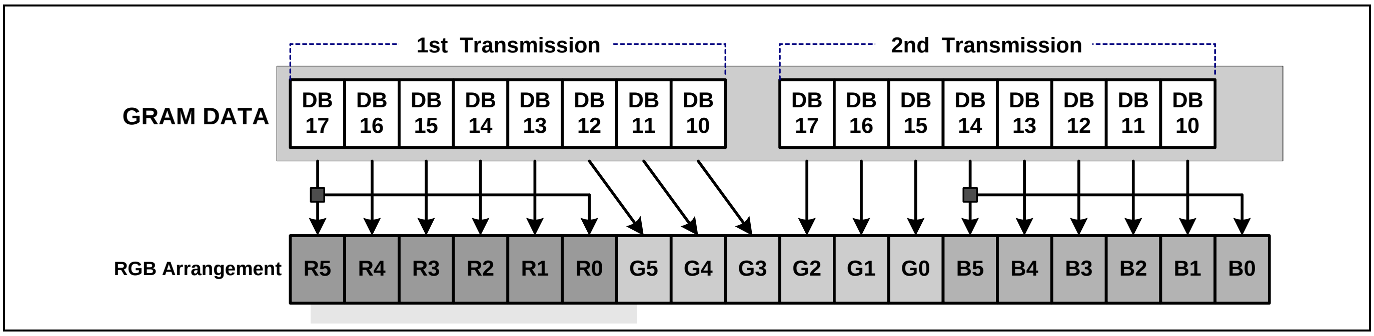

Figure 38. RAM Data Write Format For 8-Bit Interface

Timing Diagram

There are 4 timing conditions for 68-system 8-bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.
In this mode, 16-bit instructions and GRAM data are divided into two half words and the transfer starts from the upper half word. Note that the upper byte must also be written when the index register is written.

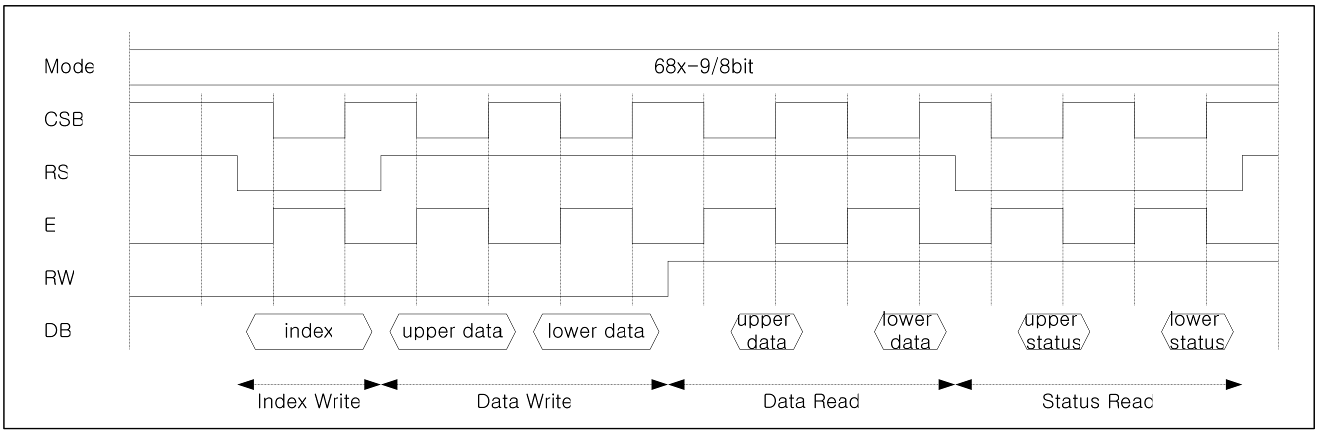

Figure 39. Timing Diagram of 68-System 8-Bit bus interface

Preliminary

80-SYSTEM 18-BIT BUS INTERFACE

Bit Assignment

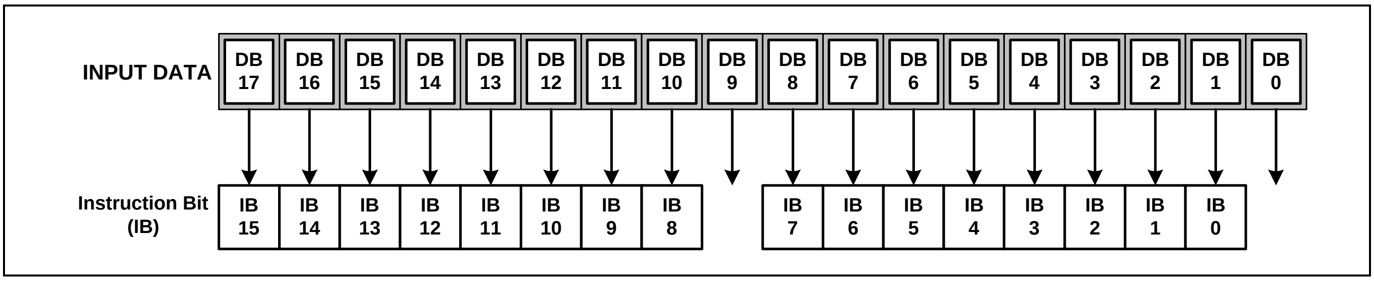

Figure 40. Instruction Format For 18-Bit Interface

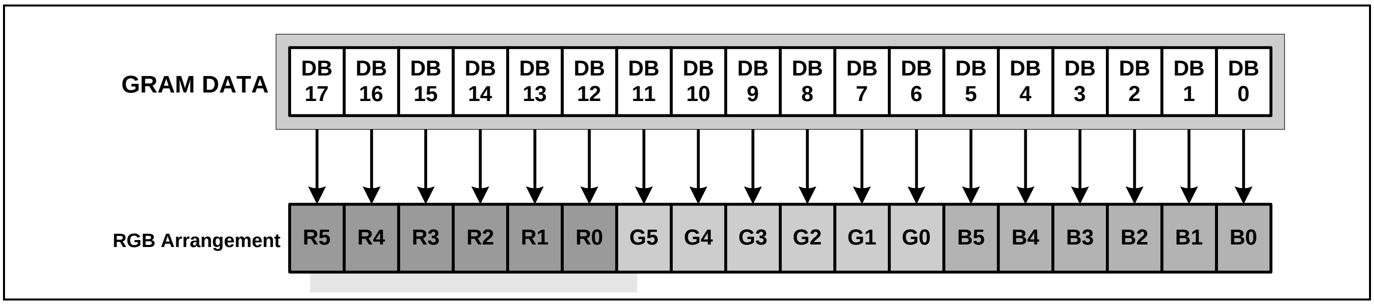

Figure 41. RAM Data Write Format For 18-Bit Interface

Timing Diagram

There are 4 timing conditions for 80-system 18-bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

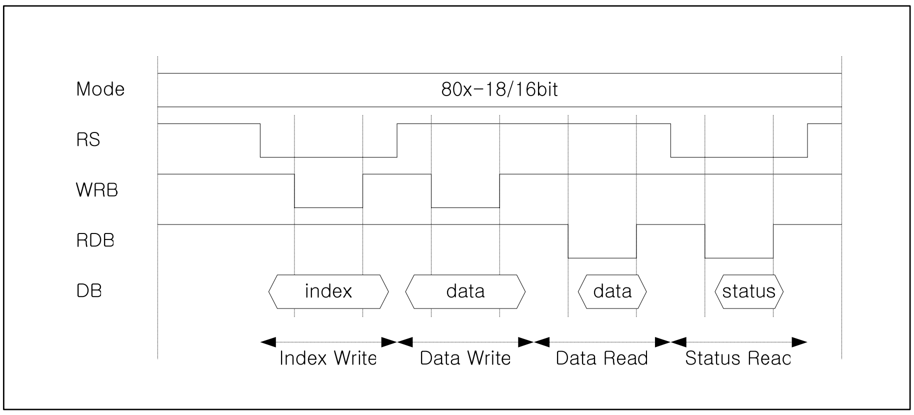

Figure 42. Timing Diagram of 80-System 18-Bit bus interface

80-SYSTEM 16-BIT BUS INTERFACE

Bit Assignment

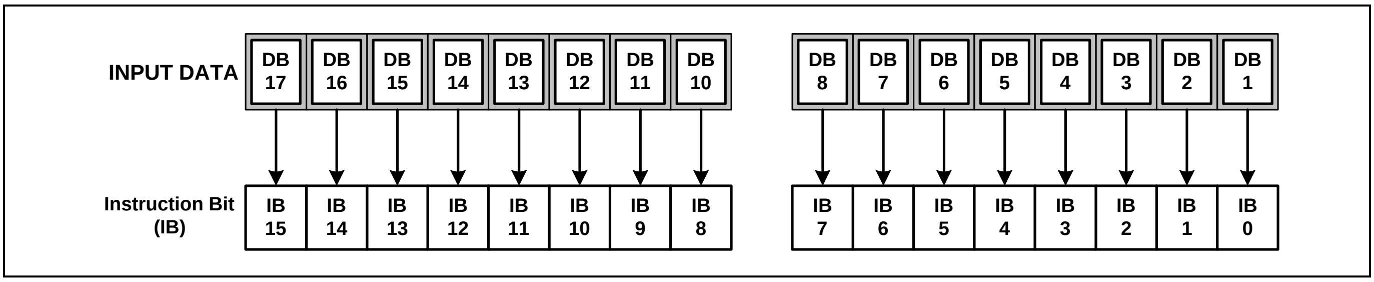

Figure 43. Instruction Format For 16-Bit Interface

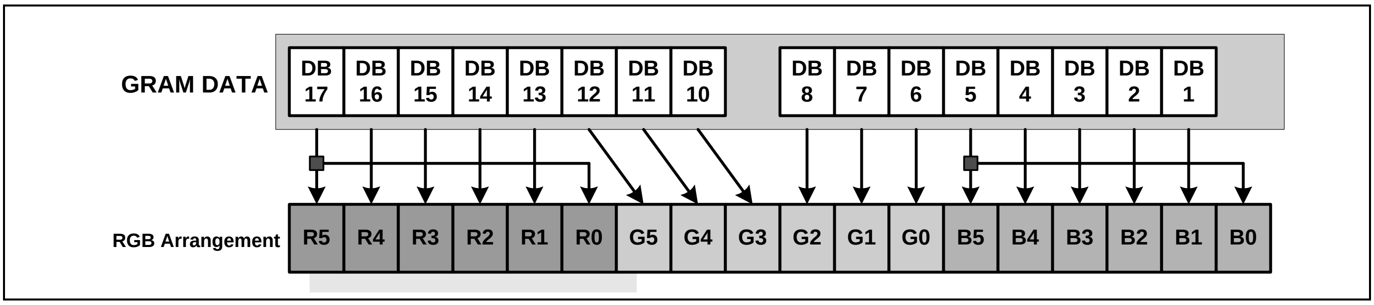

Figure 44. RAM Data Write Format For 16-Bit Interface

Timing Diagram

There are 4 timing conditions for 80-system 16-bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.

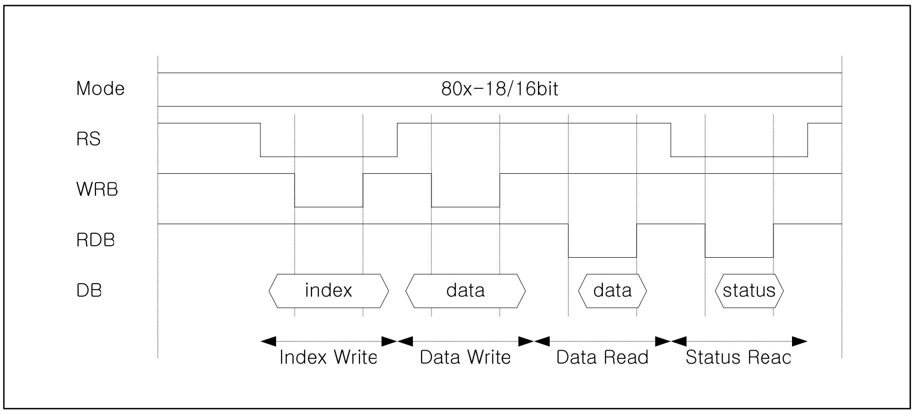

Figure 45. Timing Diagram of 80-System 16-Bit bus interface

Preliminary

80-SYSTEM 9-BIT BUS INTERFACE

Bit Assignment

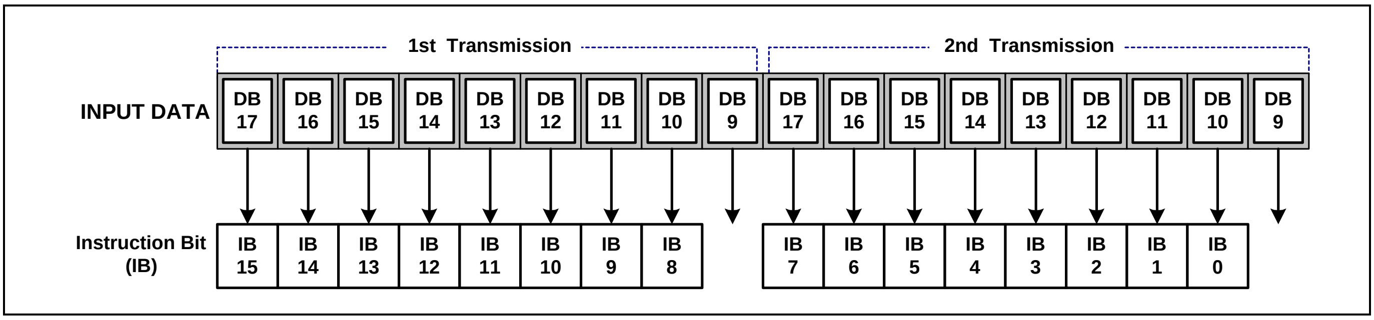

Figure 46. Instruction Format For 9-Bit Interface

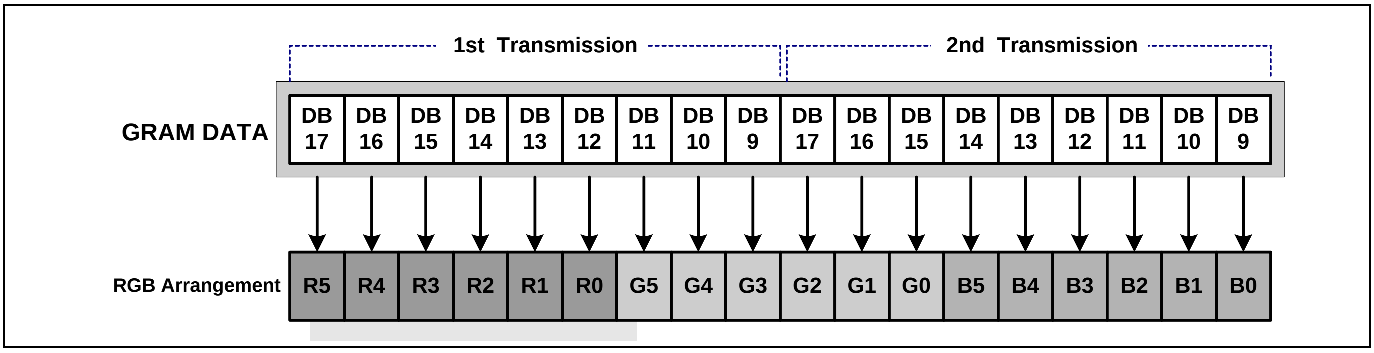

Figure 47. RAM Data Write Format For 9-Bit Interface

Timing Diagram

There are 4 timing conditions for 80-system 9-bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.
In this mode, 16-bit instructions and GRAM data are divided into two half words and the transfer starts from the upper half word. Note that the upper byte must also be written when the index register is written.

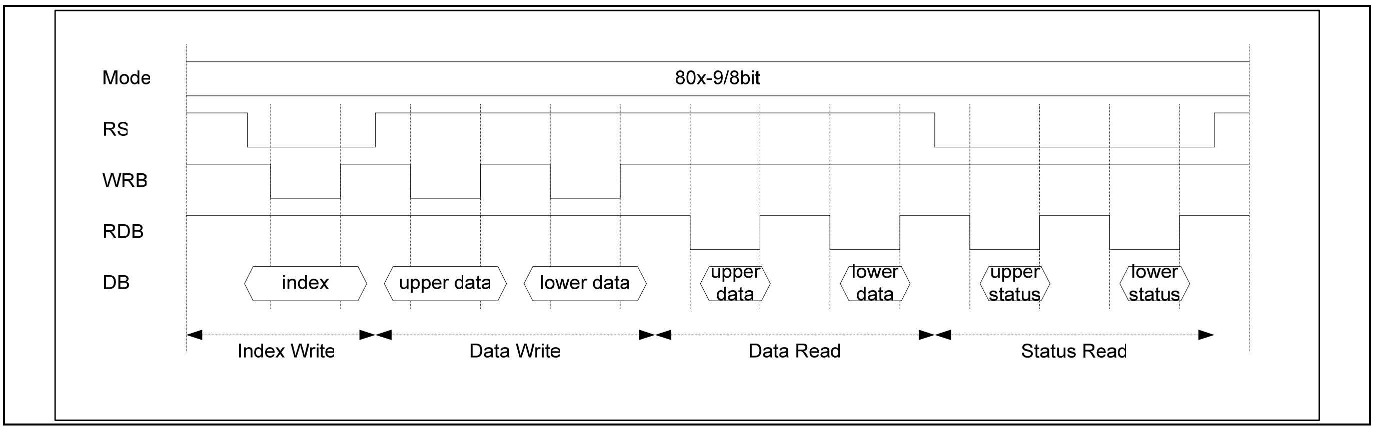

Figure 48. Timing Diagram of 80-System 9-Bit bus interface

SAMSUNG ELECTRONICS

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80-SYSTEM 8-BIT BUS INTERFACE

Bit Assignment

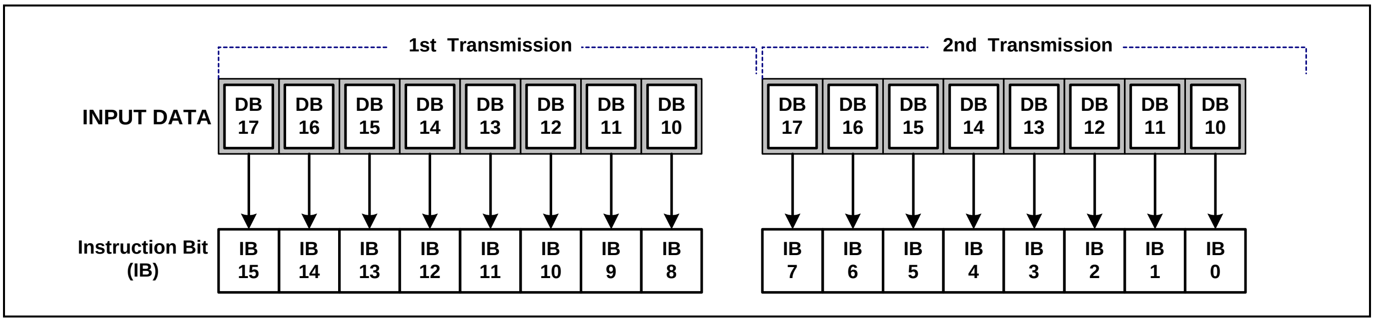

Figure 49. Instruction Format For 8-Bit Interface

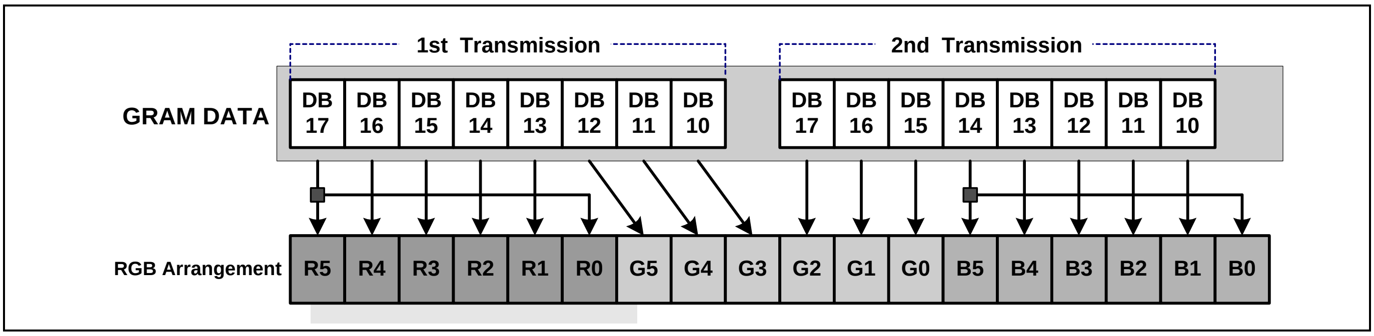

Figure 50. RAM Data Write Format For 8-Bit Interface

Timing Diagram

There are 4 timing conditions for 80-system 8-bit CPU interface, which are index write timing condition, data write timing condition, data read timing condition and status read timing condition.
In this mode, 16-bit instructions and GRAM data are divided into two half words and the transfer starts from the upper half word. Note that the upper byte must also be written when the index register is written.

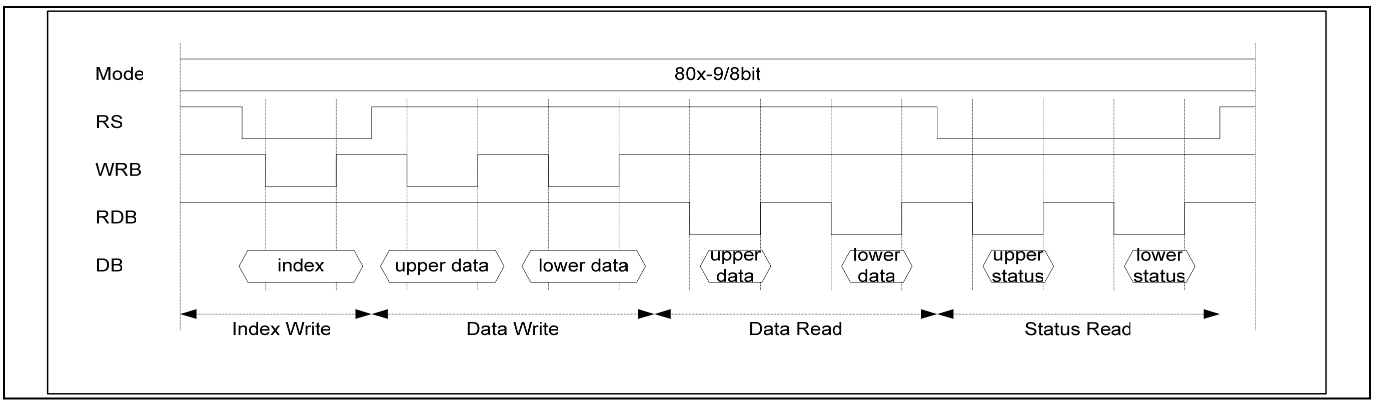

Figure 51. Timing Diagram of 80-System 8-Bit bus interface

68-/80-SYSTEM 8-/9-BIT INTERFACE SYNCHRONIZATION FUNCTION

The S6D0170 supports the transfer synchronization function, which resets the upper/lower counter to count upper/lower 8-/9-bit data transfer in the 8-/9-bit bus interface. Noise causing transfer mismatch between the upper and lower bits can be corrected by a reset triggered by writing a “22h” instruction. The next transfer starts from the upper bits. Executing synchronization function periodically can recover any runaway in the display system.

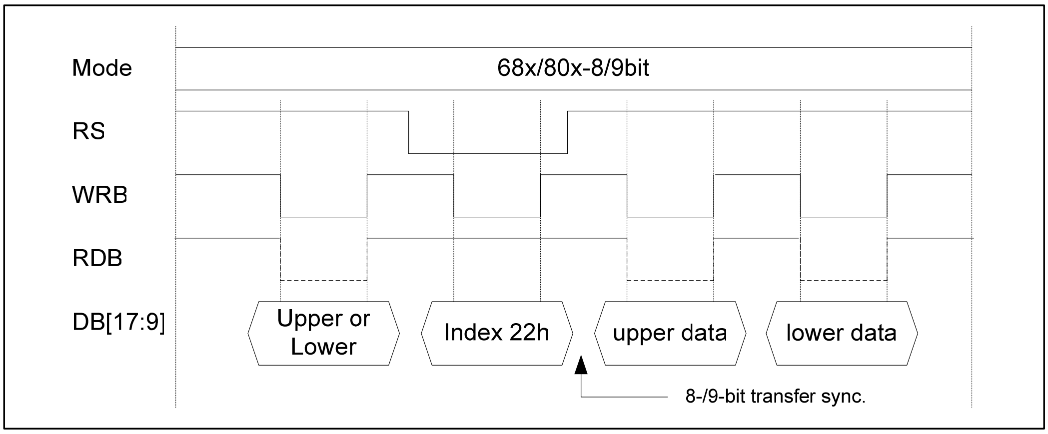

Figure 52. 8-/9-bit Interface Transfer Synchronization

Preliminary

SERIAL PERIPHERAL INTERFACE

Setting the S_PB pin to the VDD3 level allows serial peripheral interface (SPI) transfer, using the chip select line (CS*), serial transfer clock line (SCL), serial input data (SDI), and serial output data (SDO). For a serial interface, the ID_MIB pin function uses an ID pin. If the chip is set up for serial interface, the DB17-0 pins are used only data bus of RGB Interface.
The S6D0170 initiates serial data transfer by transferring the start byte at the falling edge of CSB input. It ends serial data transfer at the rising edge of CSB input.
The S6D0170 is selected when the 6-bit chip address in the start byte matches the 6-bit device identification code that is assigned to the S6D0170. When selected, the S6D0170 receives the subsequent data string. The least significant bit (LSB) of the identification code can be determined by the ID pin. The five upper bits must be 01110. Two different chip addresses must be assigned to a single S6D0170 because the seventh bit of the start byte is used as a register select bit (RS). That is, when RS = 0, data can be written to the index register or status can be read, and when RS = 1, an instruction can be issued or data can be written to or read from RAM. Read or write operation is selected according to the eighth bit of the start byte (R/W bit). The data is received when the R/W bit is 0, and is transmitted when the R/W bit is 1.
After receiving the start byte, the S6D0170 receives or transmits the subsequent data byte-by-byte. The data is transferred with the MSB first. All S6D0170 instructions are 16 bits. Two bytes are received with the MSB first (DB17 to DB0), then the instructions are internally executed. After the start byte has been received, the first byte is fetched as the upper eight bits of the instruction and the second byte is fetched as the lower eight bits of the instruction.
Four bytes of RAM read data after the start byte are invalid. The S6D0170 starts to read correct RAM data from the fifth byte.

Table 40. Start Byte Format

Transfer bit	S	1	2	3	4	5	6	7	8
Start byte format	Transfer start	Device ID code						RS	R/W
		0	1	1	1	0	ID		

NOTE: ID bit is selected by the ID_MIB pin.

Table 41. RS and R/W Bit Function

RS	R/W	Function
0	0	Set index register
0	1	Read status
1	0	Writes instruction or RAM data
1	1	Reads instruction or RAM data

Bit Assignment

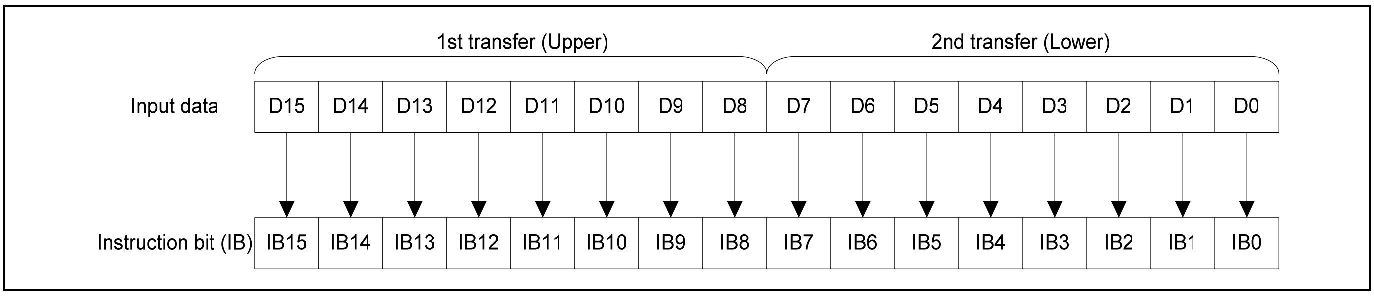

Figure 53. Bit Assignment of Instructions on SPI

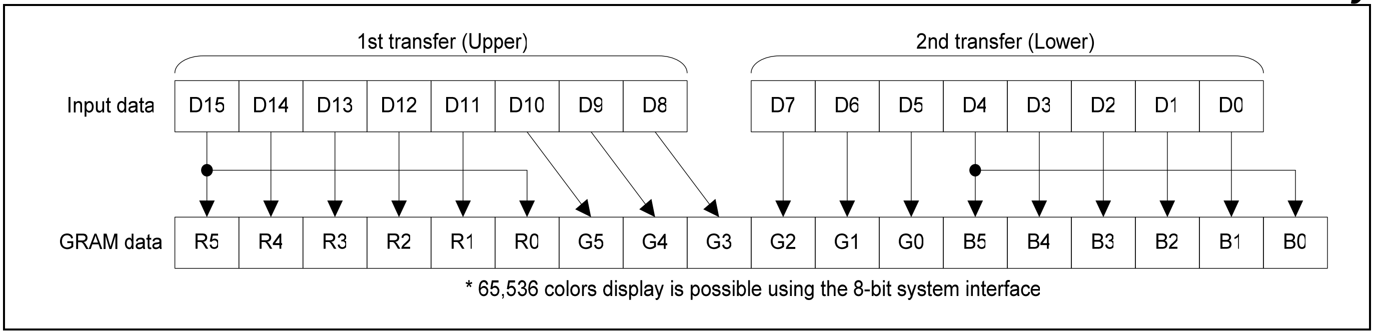

Figure 54. Bit Assignment of GRAM Data on SPI

Timing Diagram

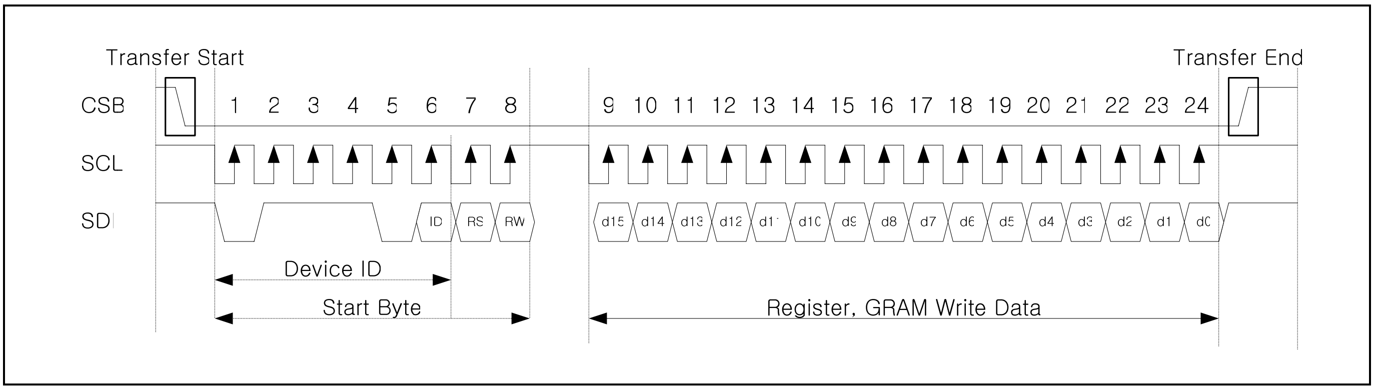

Figure 55. Basic Timing Diagram of Data Transfer through SPI

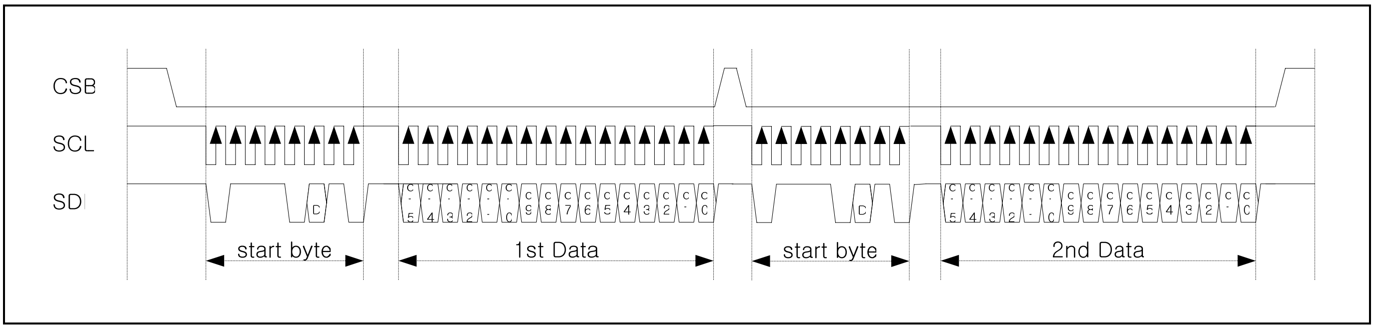

Figure 56. Timing Diagram of Consecutive Register Data-Write through SPI

Preliminary

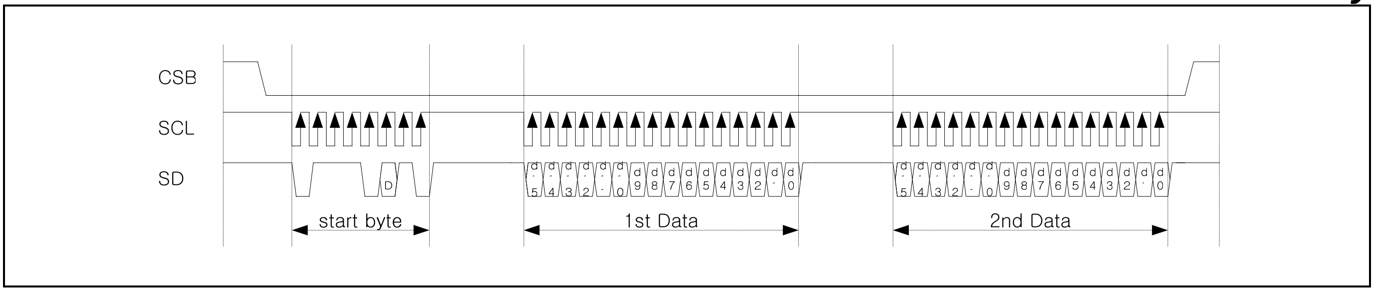

Figure 57. Timing Diagram of Consecutive GRAM Data-Write through SPI

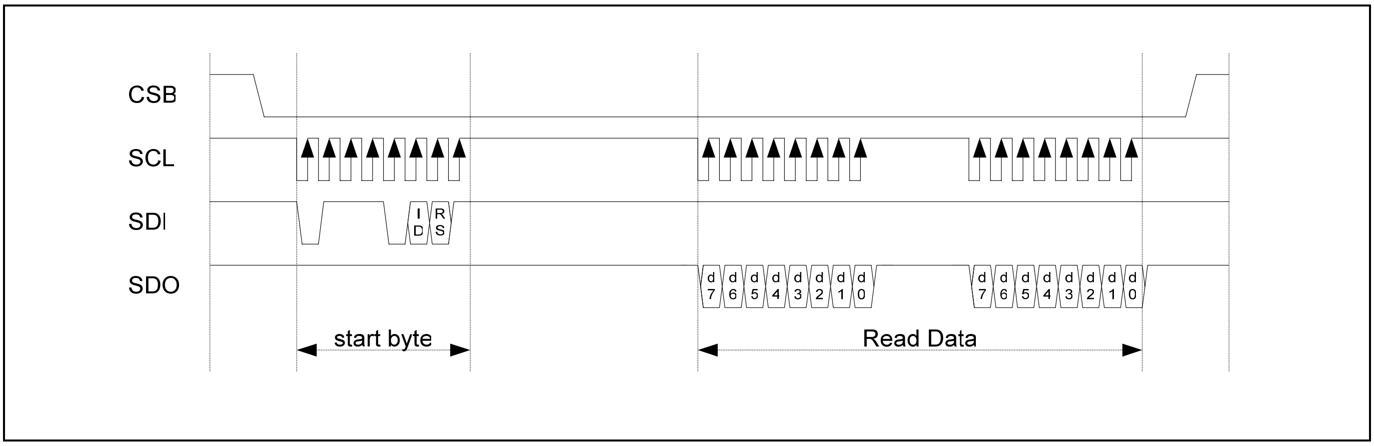

Figure 58. Timing Diagram of Register Read through SPI

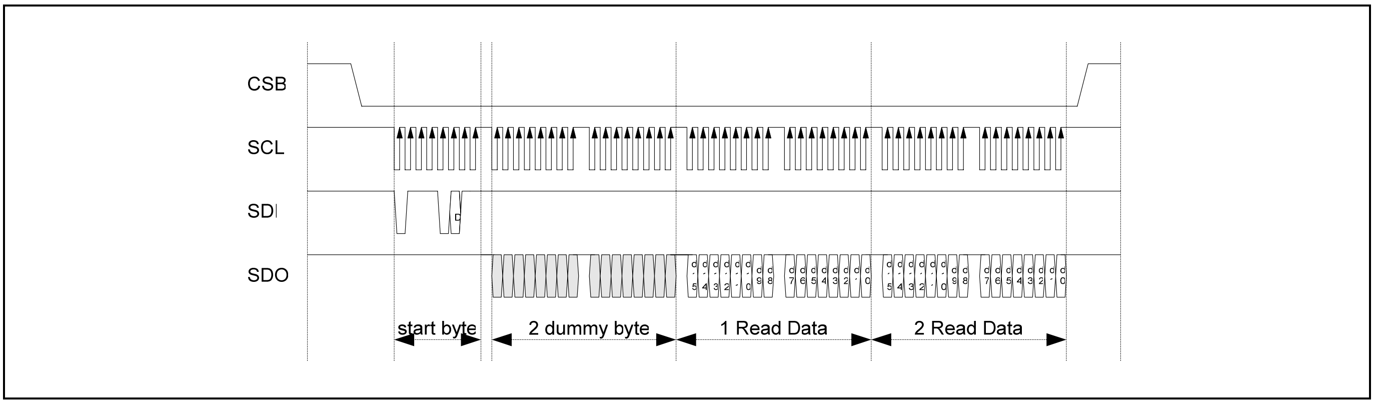

Figure 59. Timing Diagram of GRAM-Data Read through SPI

Preliminary

INDEX AND PARAMETER RECOGNITION

If more parameter command is being sent, exceed parameters are ignored.

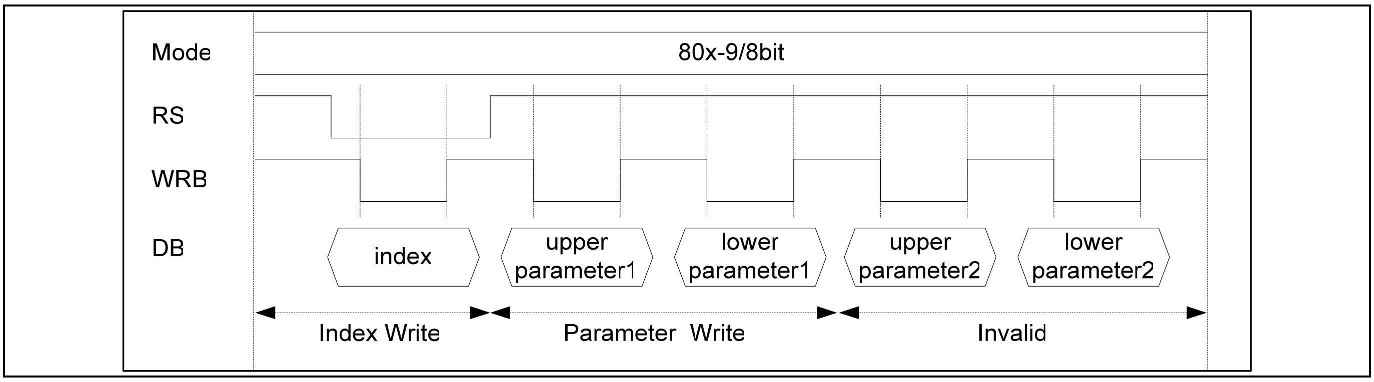

Figure 60. Index and parameter recognition with 8-/9-bit interface

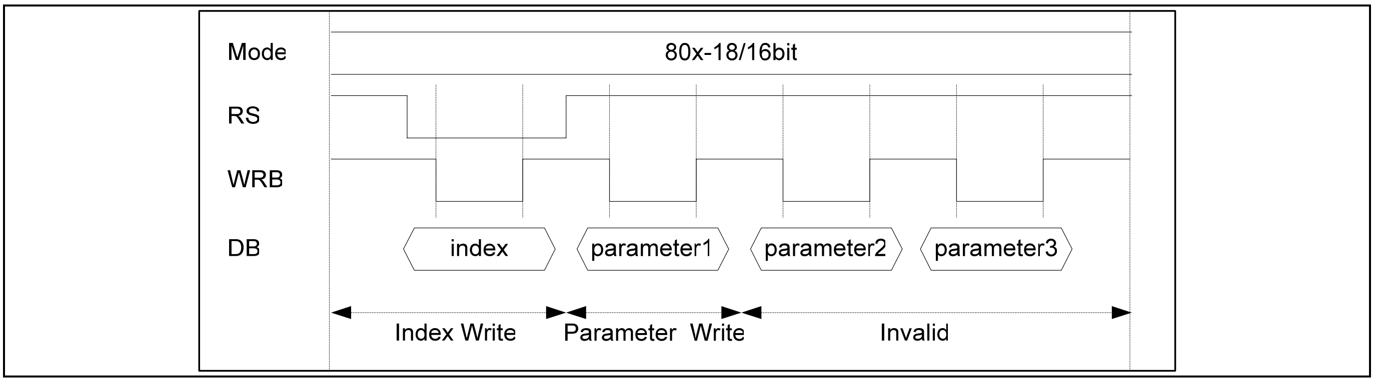

Figure 61. Index and parameter recognition with 18-/16-bit interface

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EXTERNAL DISPLAY INTERFACE

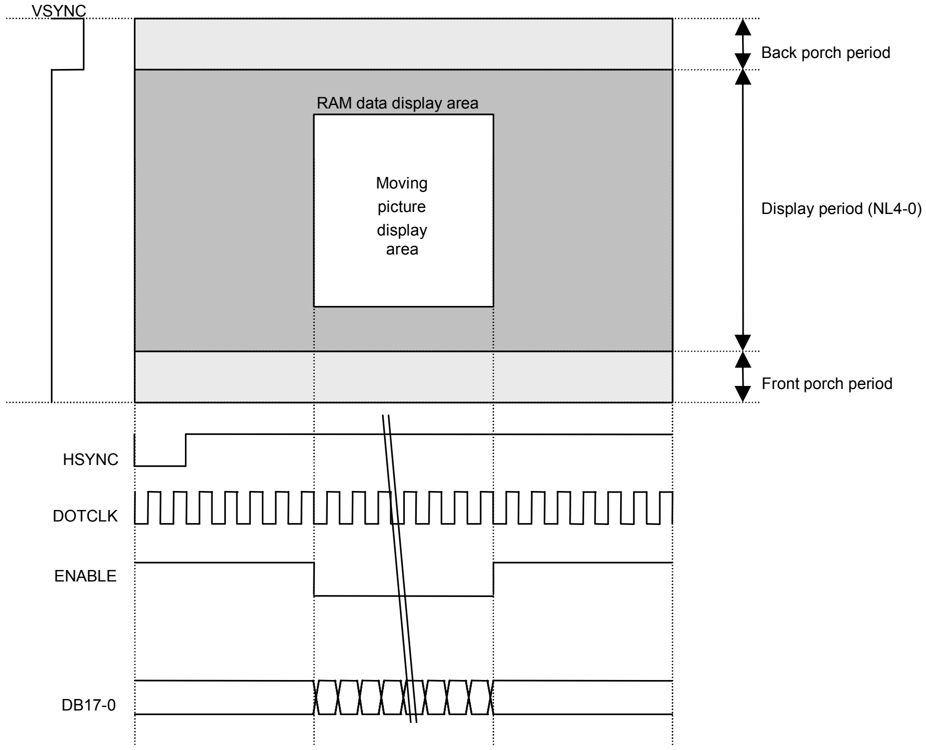
The following interfaces are available as external display interface. It is determined by bit setting of RIM1-0. RAM accesses can be performed via the RGB interface.

Table 42. RIM Bits

RIM1	RIM0	RGB Interface	DB Pin
0	0	18-bit RGB interface	DB17 to 0
0	1	16-bit RGB interface	DB17 to13, 11 to 1
1	0	6-bit RGB interface	DB17 to12
1	1	Setting disabled	

RGB INTERFACE

The RGB interface is performed in synchronization with VSYNC, HSYNC, and DOTCLK. Combining the function of the high-speed write mode and the window address enables transfer only the screen to be updated and reduce the power consumption.



VSYNC : Frame synchronized signal
HSYNC : Line synchronized signal
DOTCLK : Dot clock
ENABLE : Data enable signal
DB17-0 : RGB(6:6:6) display data

Back porch period(BPP) : 14H ≥ BP3-0 ≥2H
Front porch period(FPP) : 14H ≥ FP3-0 ≥2H
FPP+BPP=16H
Display period(DP) : NL4-0≤220H
A number of line of 1-frame : FPP+DP+BPP

Figure 62. RGB Interface (In Case Of EPL = 0, DPL = 0, VSPL = 0, HSPL = 0)

ENABLE SIGNAL

The relationship between EPL and ENABLE signals is shown below. When ENABLE is not active, the address is not updates. When ENABLE is active, the address is updated.

Table 43. Relationship between EPL and ENABLE

EPL	ENABLE	RAM WRITE	RAM ADDRESS
0	0	Valid	Updated
0	1	Invalid	Hold
1	0	Invalid	Hold
1	1	Valid	Update

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18-BIT RGB INTERFACE

Bit Assignment

Input data	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
GRAM data	R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0

* 262,144 colors display is possible using the 18-bit RGB interface

Figure 63. Bit Assignment of GRAM Data on 18bit RGB Interface

Timing Diagram

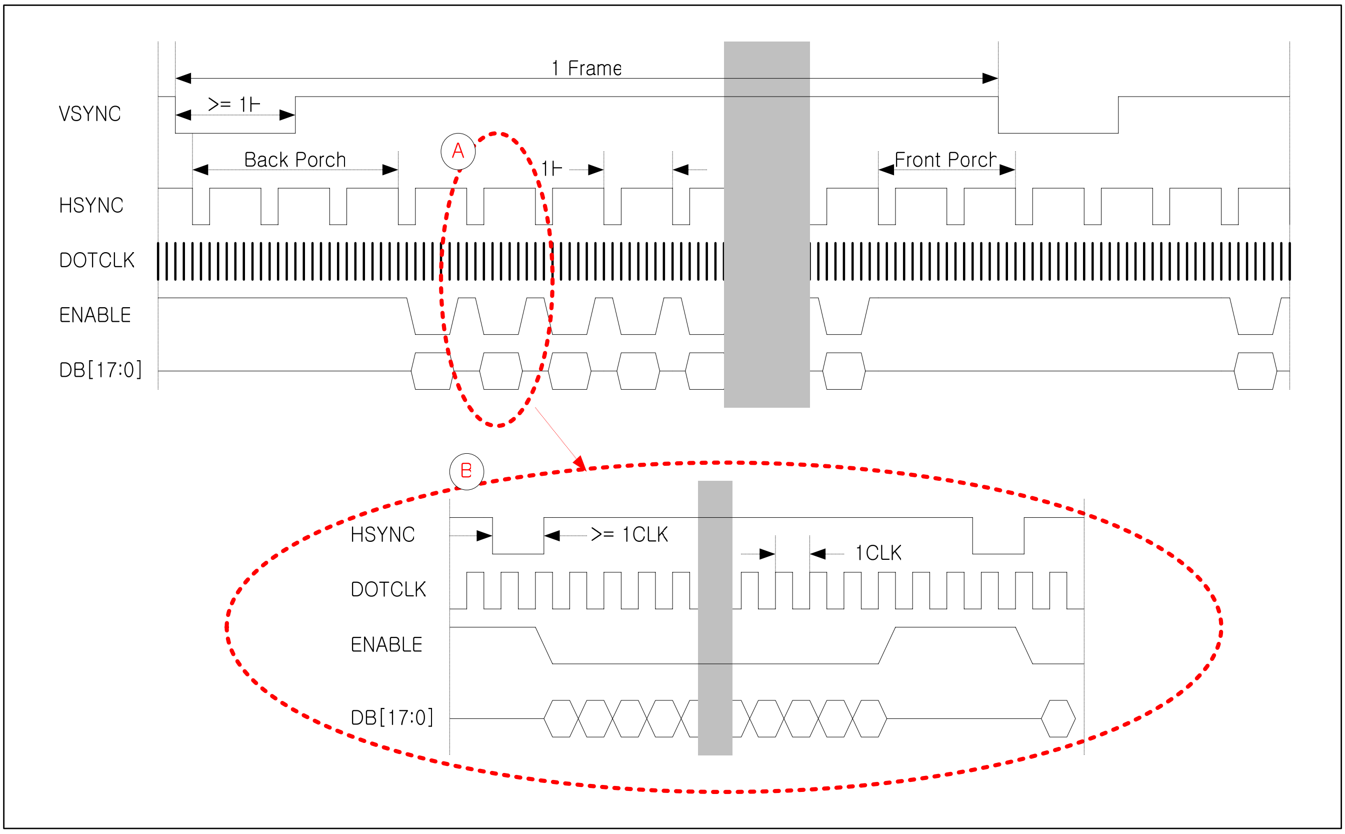

Figure 64. Timing Diagram of 18/16bit RGB Interface

Preliminary

16-BIT RGB INTERFACE

Bit Assignment

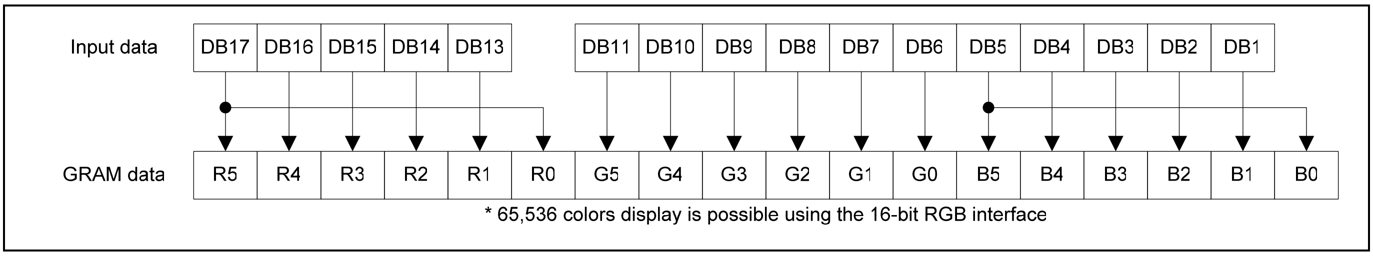

Figure 65. Bit Assignment of GRAM Data on 16bit RGB Interface

Timing Diagram

There are two timing conditions for RGB Interface that is determined according to RIM.

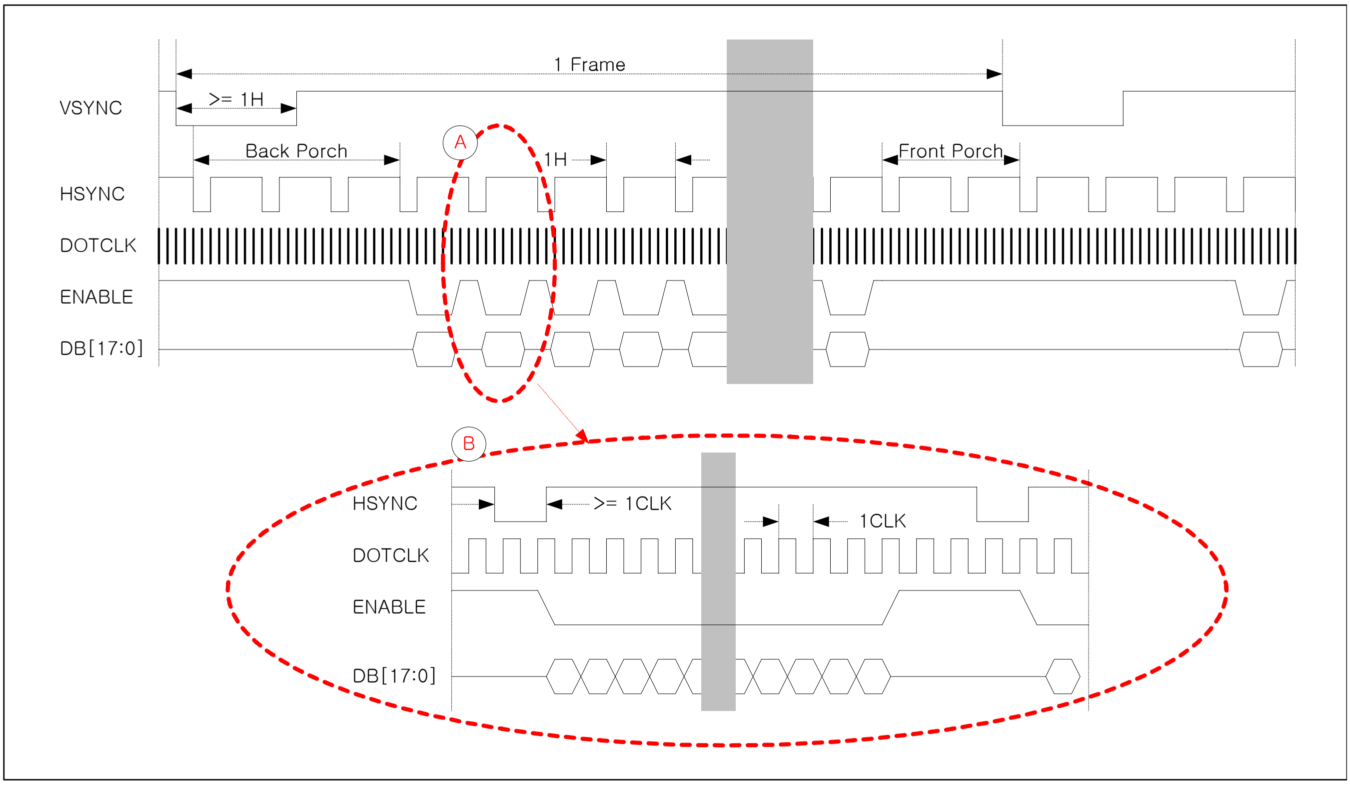

Figure 66. Timing Diagram of 18/16bit RGB Interface

Preliminary

6-BIT RGB INTERFACE

In order to transfer data on 6bit RGB Interface there should be three transfers.

Bit Assignment

	1st transfer						2nd transfer						3rd transfer					
Input data	DB17	DB16	DB15	DB14	DB13	DB12	DB17	DB16	DB15	DB14	DB13	DB12	DB17	DB16	DB15	DB14	DB13	DB12
GRAM data	R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0

* 262,144 colors display is possible using the 6-bit RGB interface

Figure 67. Bit Assignment of GRAM Data on 6bit RGB Interface

Timing Diagram

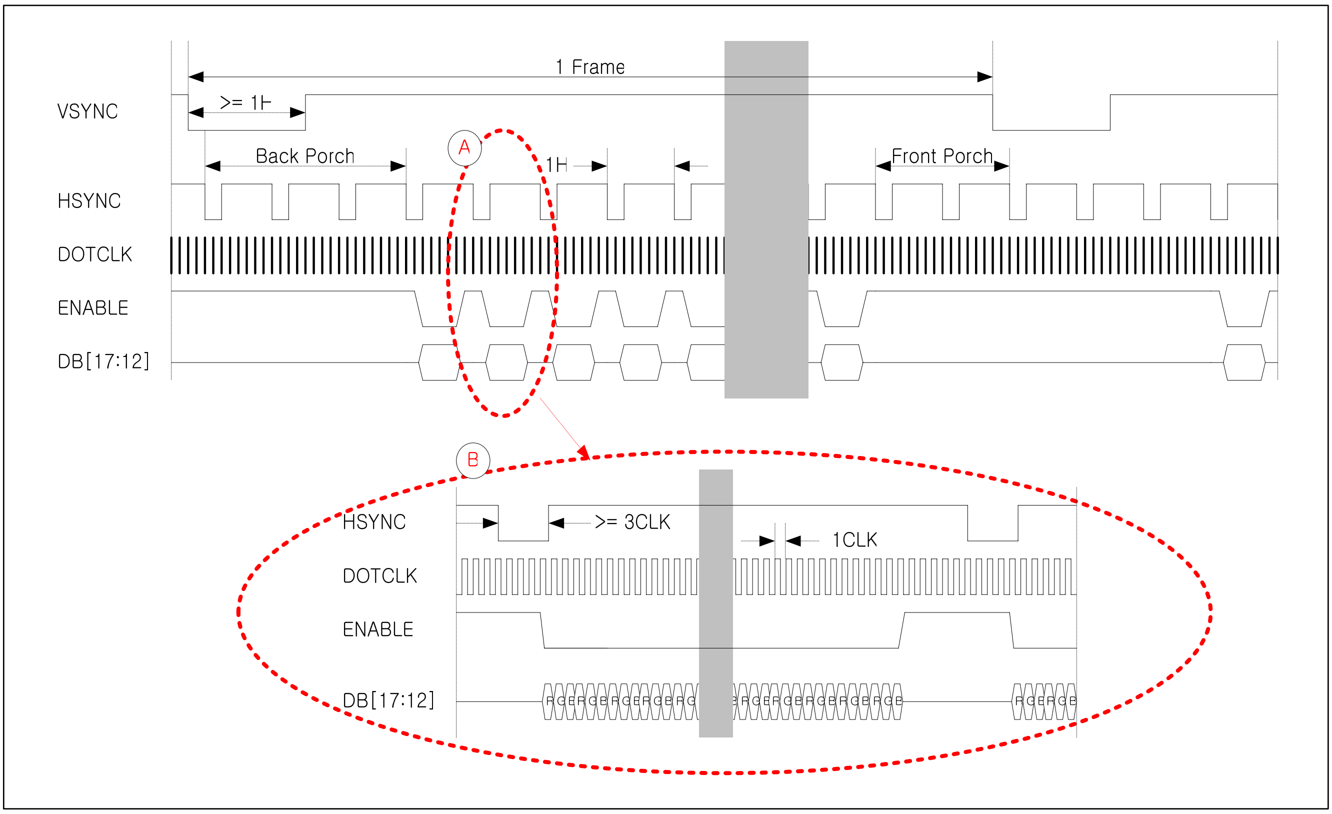

Figure 68. Timing Diagram of 6bit RGB Interface

[NOTES] 1. Three clocks are regarded as one clock for transfer when data is transferred in 6-bit interface.
2006.┌. VSYNC, HSYNC, ENABLE, DOTCLK, and DB[17:12] should be transferred in units of three clocks..

Transfer Synchronization

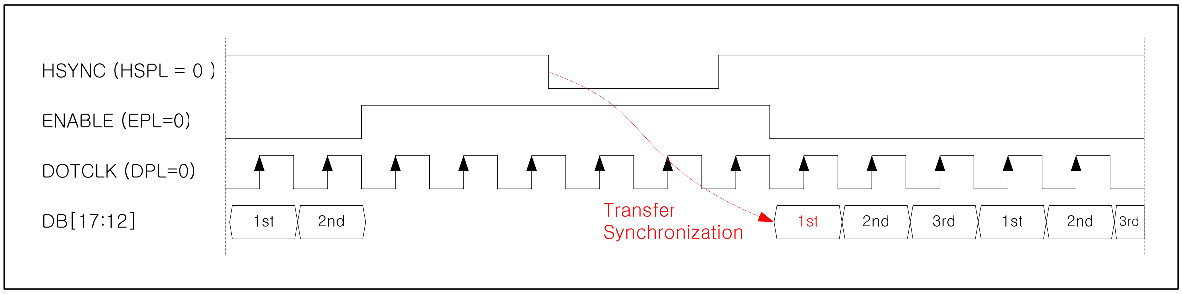

Figure 69. Transfer Synchronization Function in 6-bit RGB Interface mode

NOTE: The figure shows Transfer Synchronization function for 6bit RGB Interface. S6D0170 has a transfer counter to count 1st, 2nd and 3rd data transfer of 6bit RGB Interface. The transfer counter is reset on the falling edge of HSYNC and enters the 1st data transmission state. Transfer mismatch can be corrected at every new transfer restarts with HSYNC signal. In this method, when data is consecutively transferred in such a way as displaying motion pictures, the effect of transfer mismatch will be reduced and recovered by normal operation.

NOTE: The internal display is operated in units of three DOTCLKs. When DOTCLK is not input in units of pixels, clock mismatch occurs and the frame, which is operated, and the next frame are not displayed correctly.

Time chart for RGB interface is shown below. (In case of EPL = 0)

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MOTION PICTURE DISPLAY

The S6D0170 incorporates RGB interface to display motion pictures and RAM to store data for display.
For displaying motion pictures, the S6D0170 has the following features.
- Motion picture area can only be transferred by the window address function.
- Motion picture area to be rewritten can only be transferred.
- Reducing the amount of data transferred enables reduce the power consumption to the whole system.
- Still picture area, such as an icon, can be updated while displaying motion pictures combining with the system interface.

RAM ACCESS VIA RGB INTERFACE AND SPI

RAM can be accessed via the system interface when RGB interface is in use. When data is written to RAM during RGB interface mode, the ENABLE bit should be low to stop data writing via RGB interface, because RAM writing is always performed in synchronization with the DOTCLK input when ENABLE is high. After this RAM access via the system interface, a waiting time is needed for a write/read bus cycle before the next RAM access starts via RGB interface. When a RAM write conflict occurs, data writing is not guaranteed.
Example of display motion picture via RGB-I/F and updating still picture via the system interface are shown below.

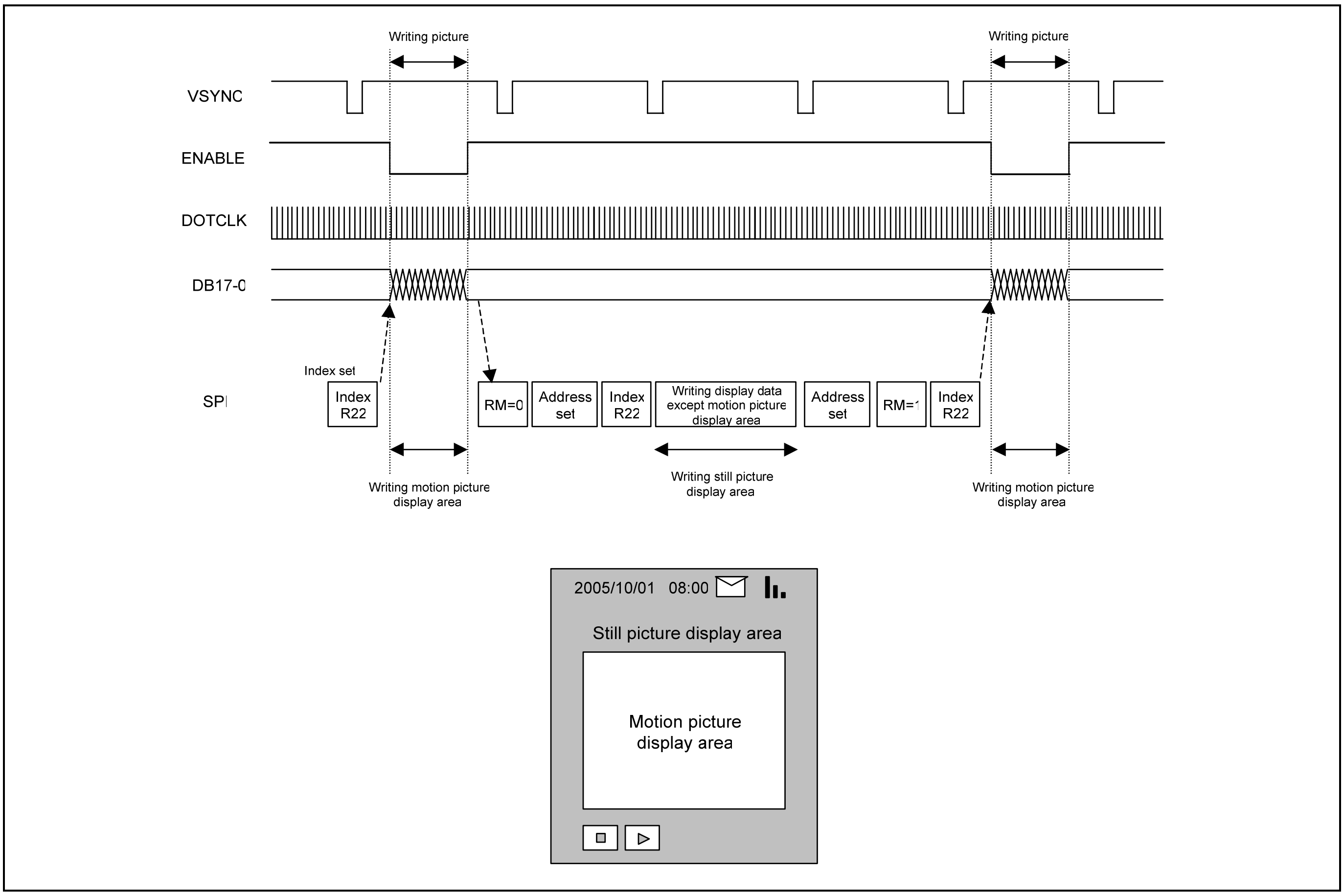

Figure 70. Example Of Updating Still Picture Area During Displaying Motion Picture (In Case Of EPL = 0, VSPL = 0)

USAGE ON EXTERNAL DISPLAY INTERFACE

1. When external display interface is in use, the following functions are not available.

Table 44. External Display Interface and Internal Display Operation

Function	External Display Interface	Internal Display Operation
Partial Display	Not Available	Available
Scroll Function	Not Available	Available
Rotation	Not Available	Available
Mirroring	Not Available	Available
Window Function	Not Available	Available

2. VSYNC, HSYNC, and DOTCLK signals should be supplied during display operation via RGB interface.

3. Please make sure that when setting bits of NO3-0, SDT3-0, and VCIR2-0 in RGB interface, the clock on which operations are based changes from the internal operating clock to DOTCLK.

4. RGB data are transferred for three clock cycles in 6-bit RGB interface. Data transferred, therefore, should be transferred in units of RGB.

5. Interface signals, VSYNC, HSYNC, DOTCLK, ENABLE and DB17-0 should be set in units of RGB (pixels) to match RGB transfer.

6. Transitions between internal operation mode and external display interface should follow the mode transition sequence shown below.

7. During the period between the completion of displaying one frame data and the next VSYNC signal, the display will remain front porch period.

8. An address set is done on the falling edge of VSYNC every frame in RGB interface.

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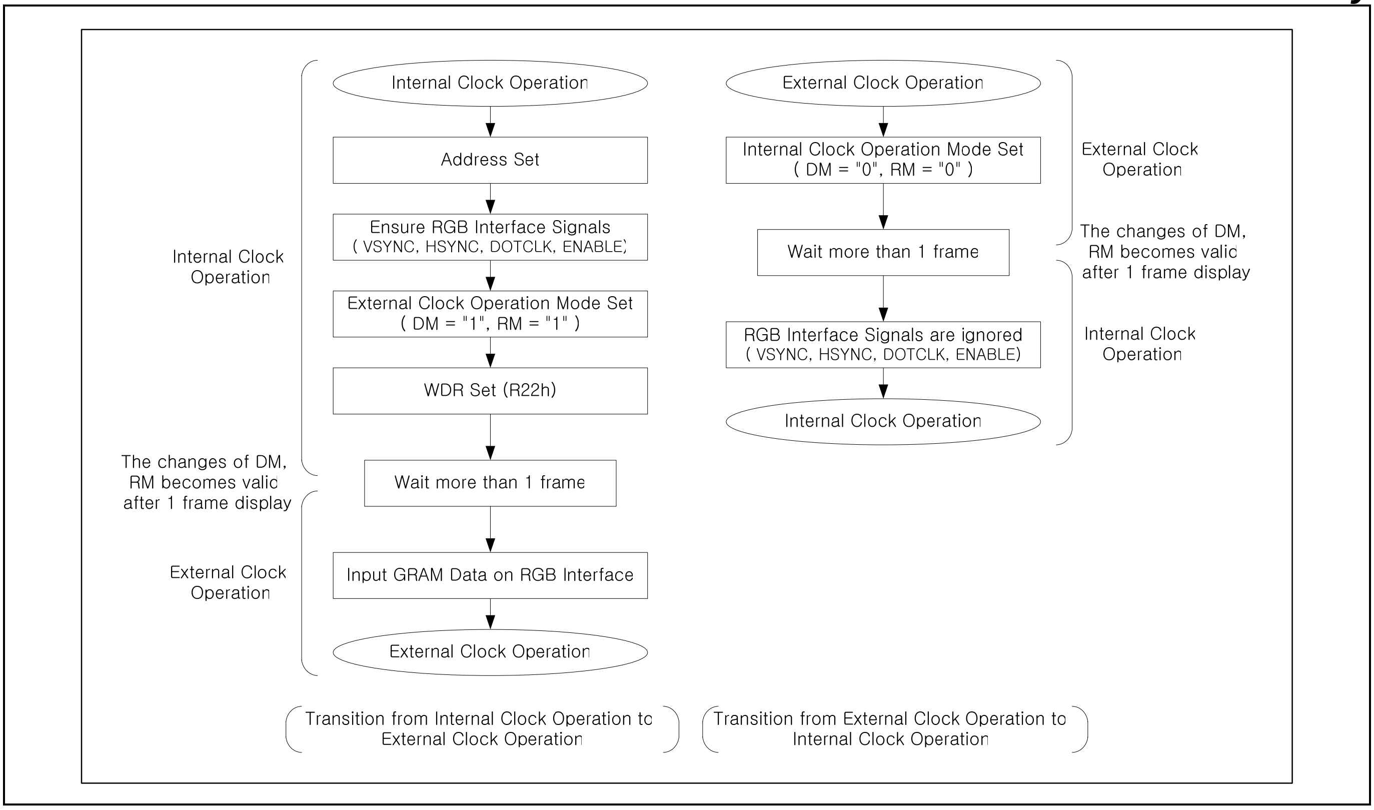

Figure 71. Transition between the Internal Operating Clock Mode And RGB Interface Mode

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DISPLAY SIGNAL TIMING

DISPLAY TIMING IN 1 FRAME (RGB I/F)

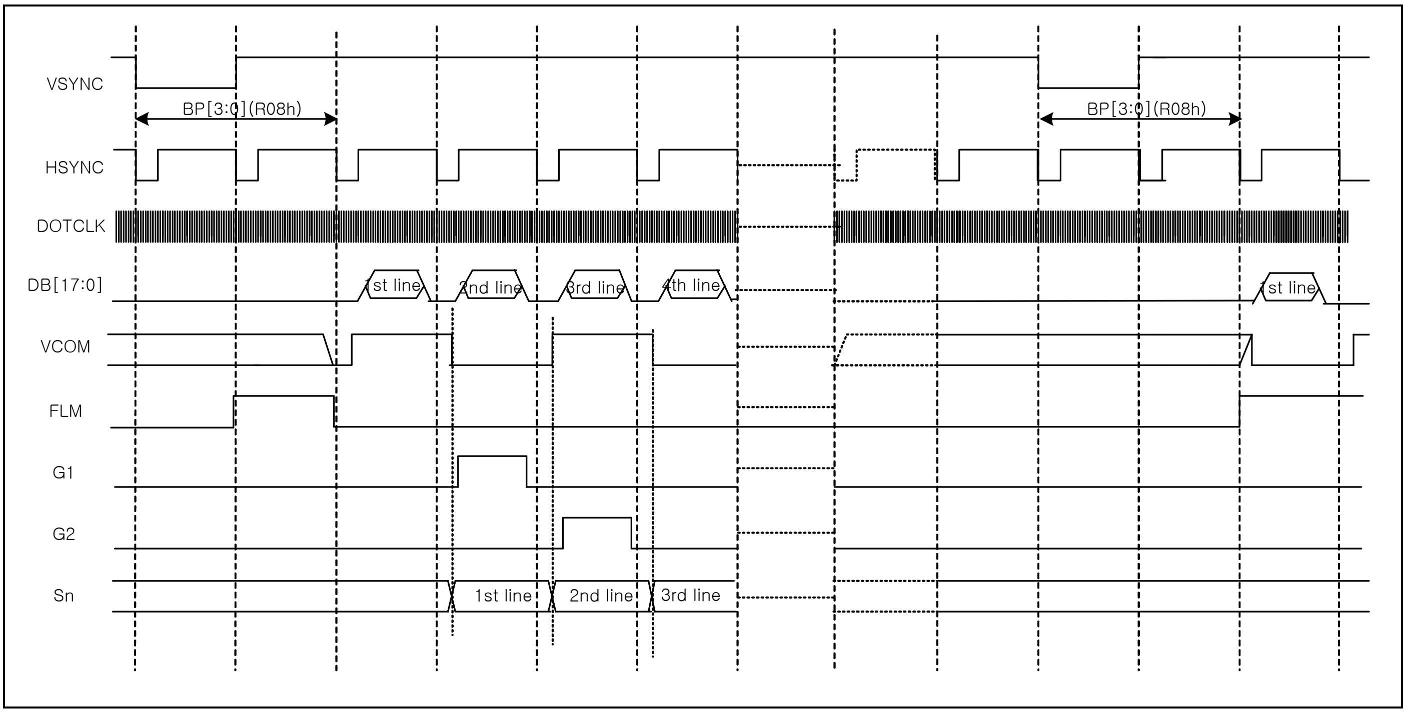

Figure 72. Display Timing in 1 Frame

Preliminary

DISPLAY TIMING IN 1 HORIZONTAL LINE (RGB I/F)

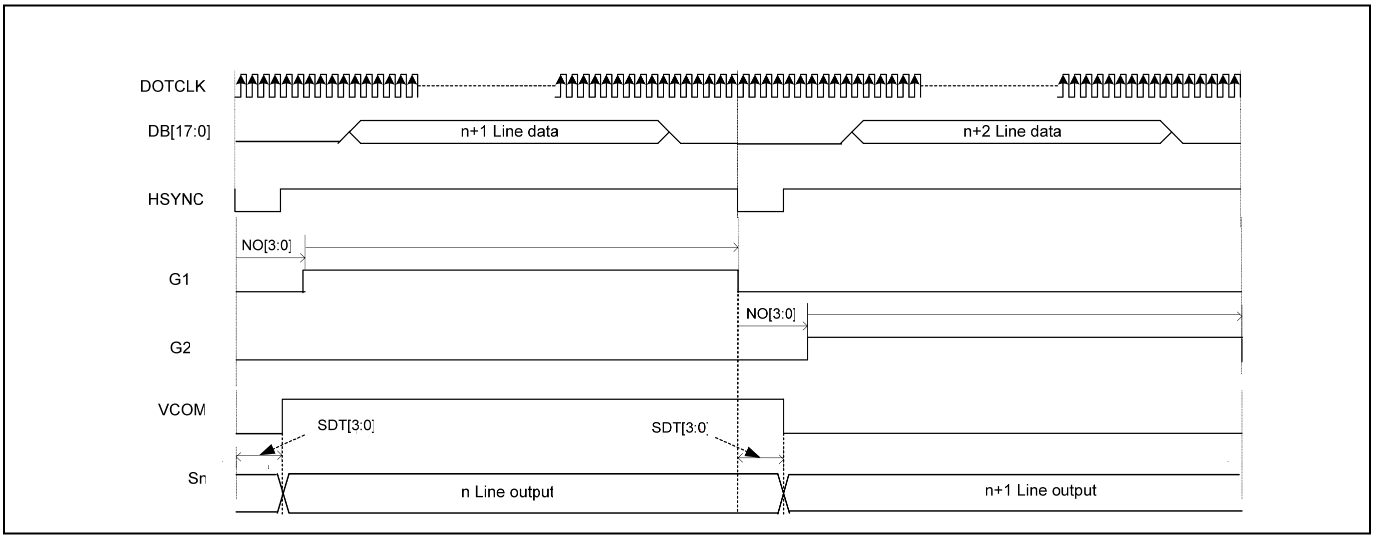

Figure 73. Display Timing in 1 Horizontal Line

Preliminary

GAMMA ADJUSTMENT FUNCTION

S6D0170 provides the gamma adjustment function to display 262,144 colors simultaneously. The gamma adjustment is executed by the gradient adjustment register, the reference adjustment register, the amplitude adjustment resister and the micro-adjustment register that determine the 8 grayscale levels. Furthermore, since these registers have the positive polarities and negative polarities, adjust them to match LCD panel respectively.

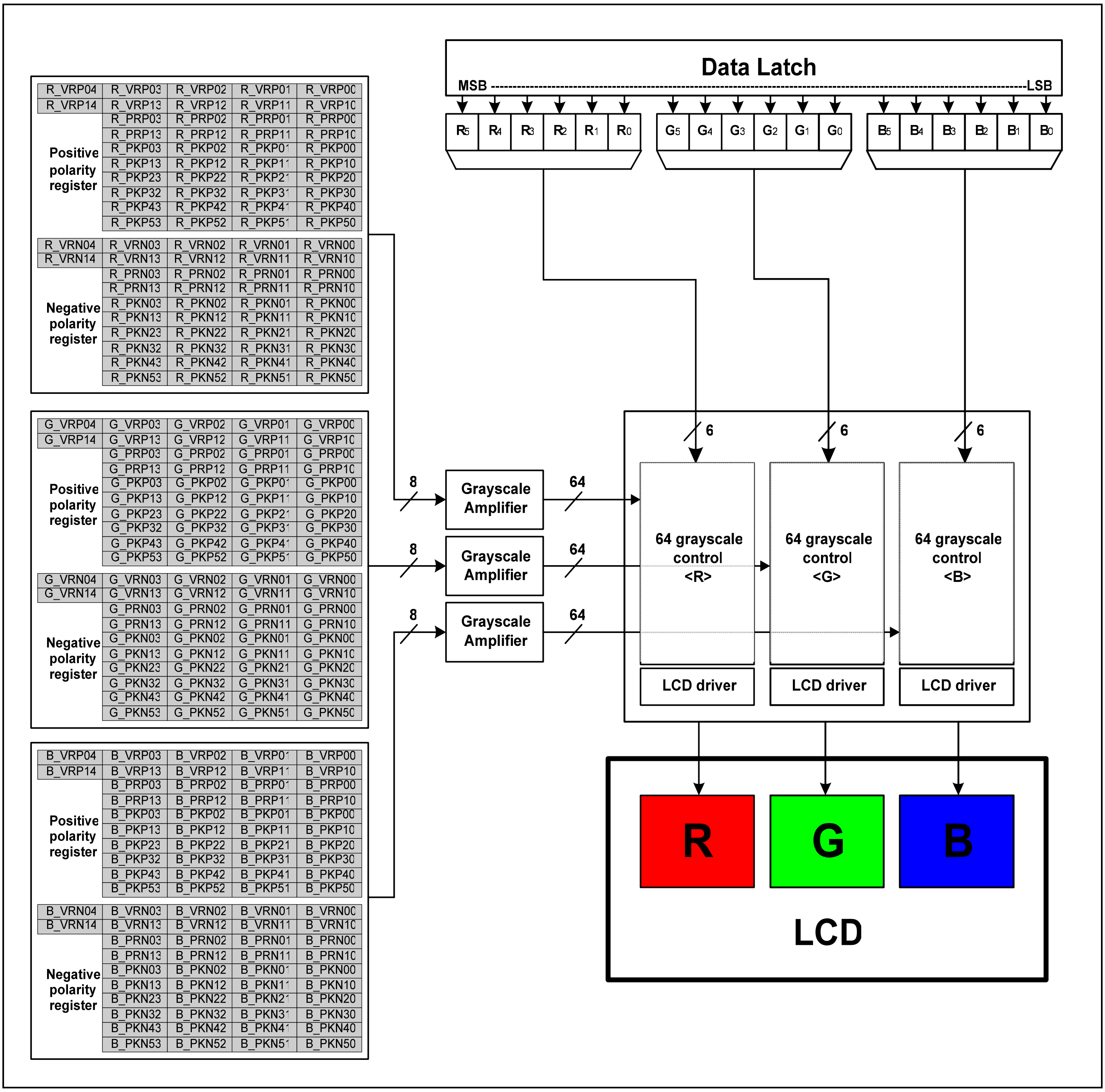

Figure 74. Grayscale Control

STRUCTURE OF GRAYSCALE AMPLIFIER

The structure of grayscale amplifier is shown as below. The 8 voltage levels (VIN0-VIN7) between GVDD and VGS are determined by the gradient adjustment register, the reference adjustment register, the amplitude adjustment resister and the micro-adjustment register. Each level is split into 8 levels again by the internal ladder resistor network. As a result, grayscale amplifier generates 64 voltage levels ranging from V0 to V63.

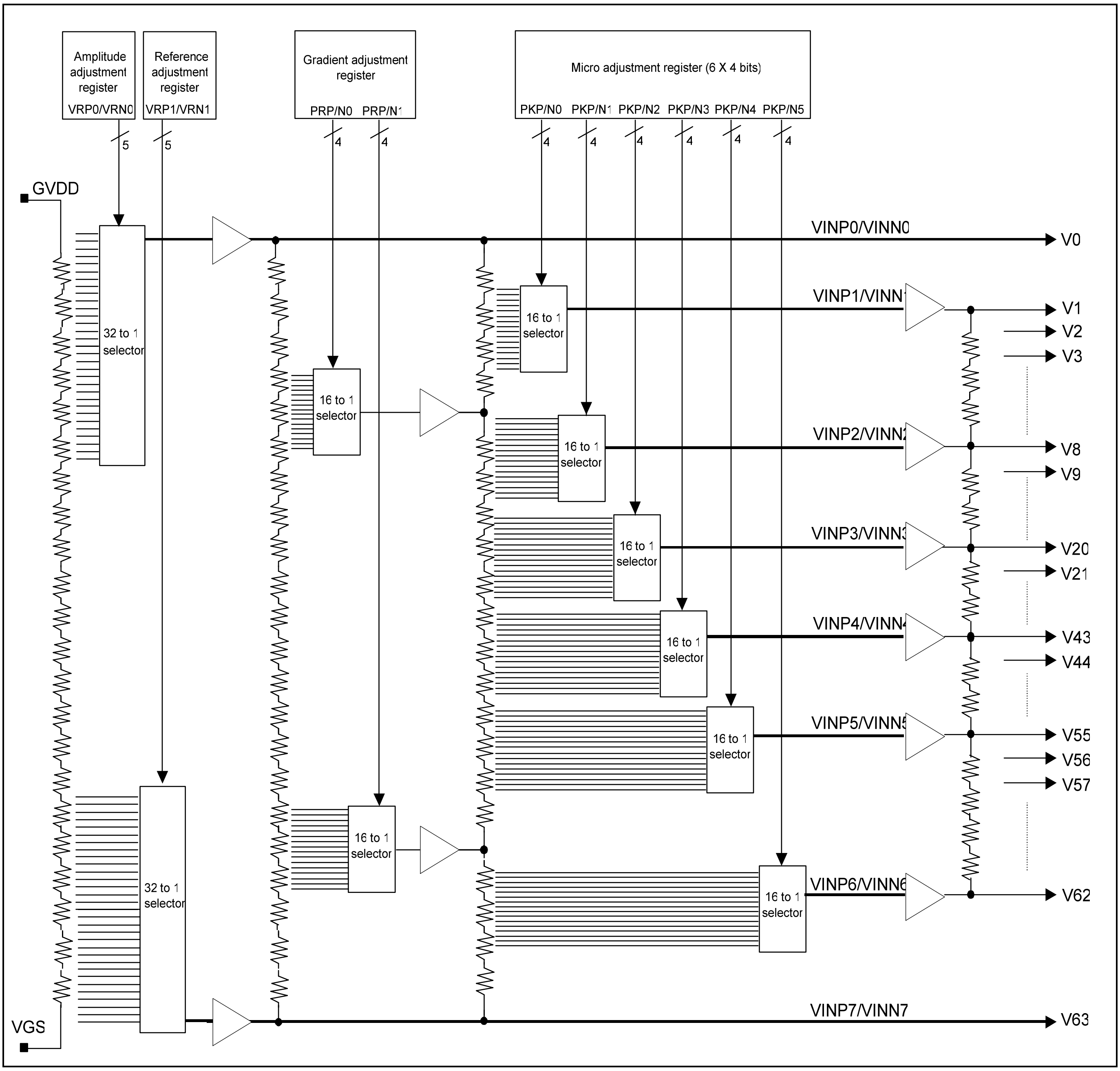

Figure 75. Structure of Grayscale Amplifier

Preliminary

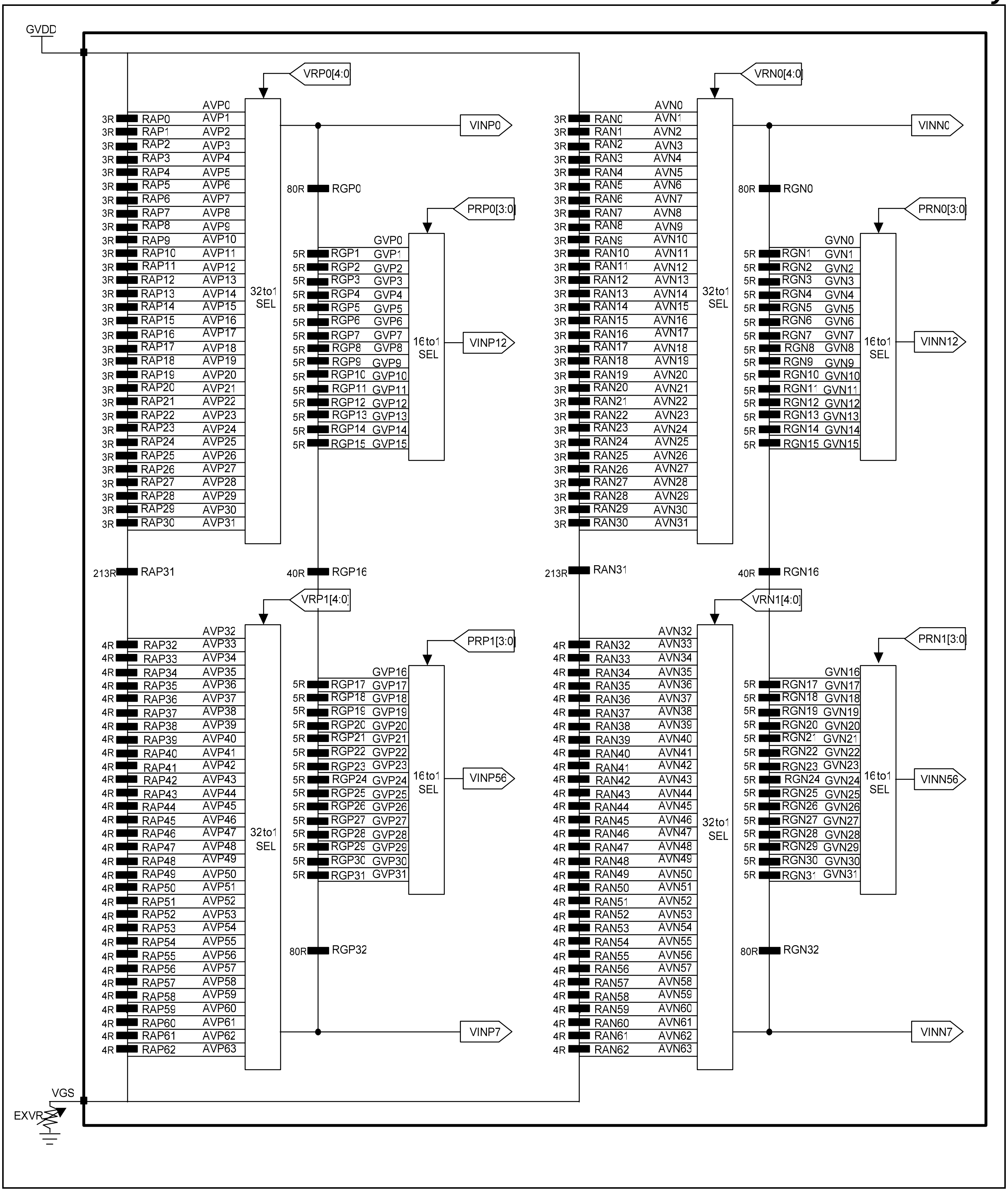

Figure 76. Structure of Resistor Ladder Network 1

Preliminary

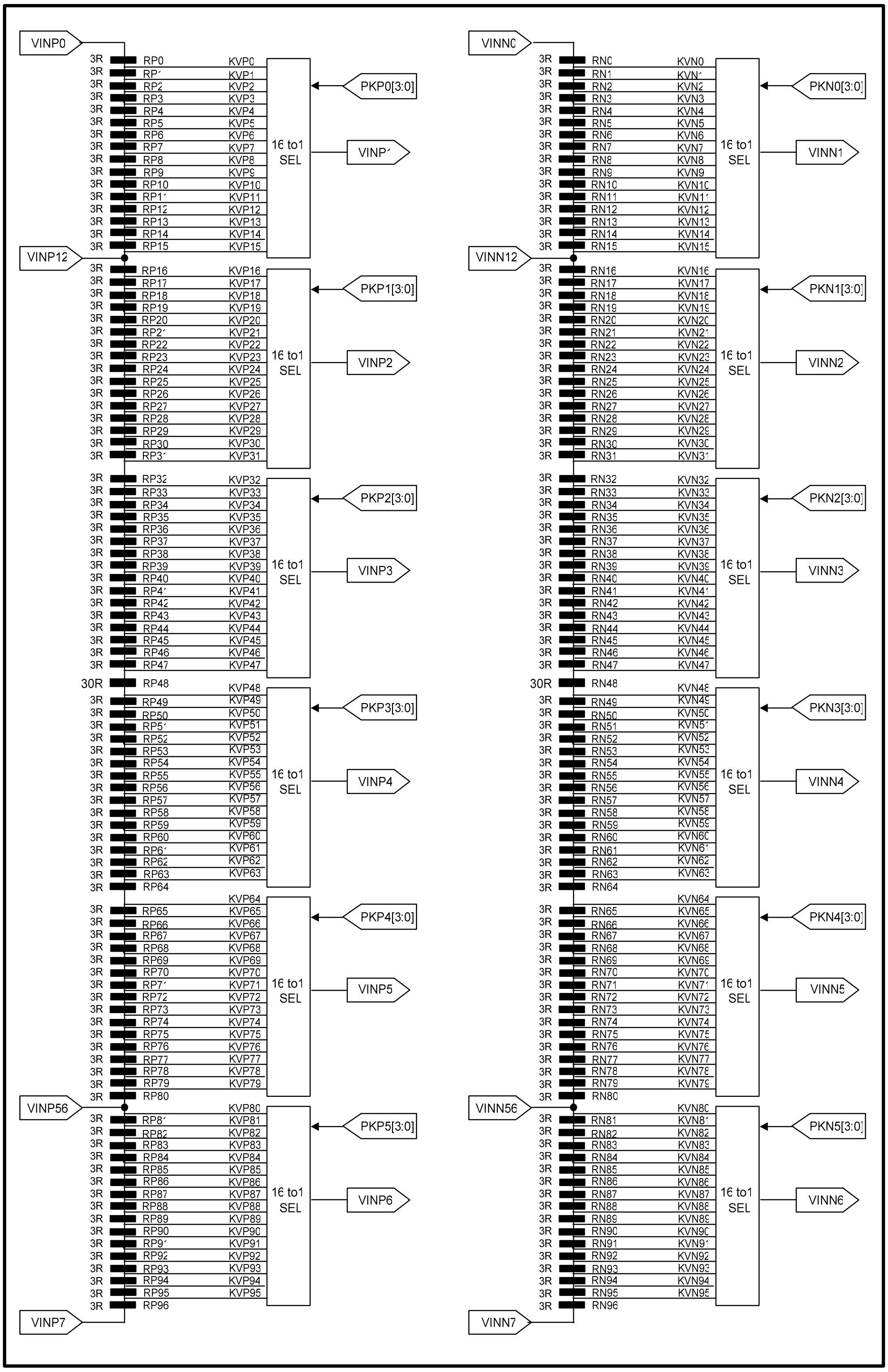

Figure 77. Structure of Resistor Ladder Network 2

GAMMA ADJUSTMENT REGISTER

This block has registers to set up the grayscale voltage adjusting to the gamma specification of the LCD panel. These registers can independently set up to positive/negative polarities and there are 4 types of register groups to adjust gradient and amplitude on number of the grayscale, characteristics of the grayscale voltage. (Average <R><G><B> are common.) The following figure indicates the operation of each adjustment register.

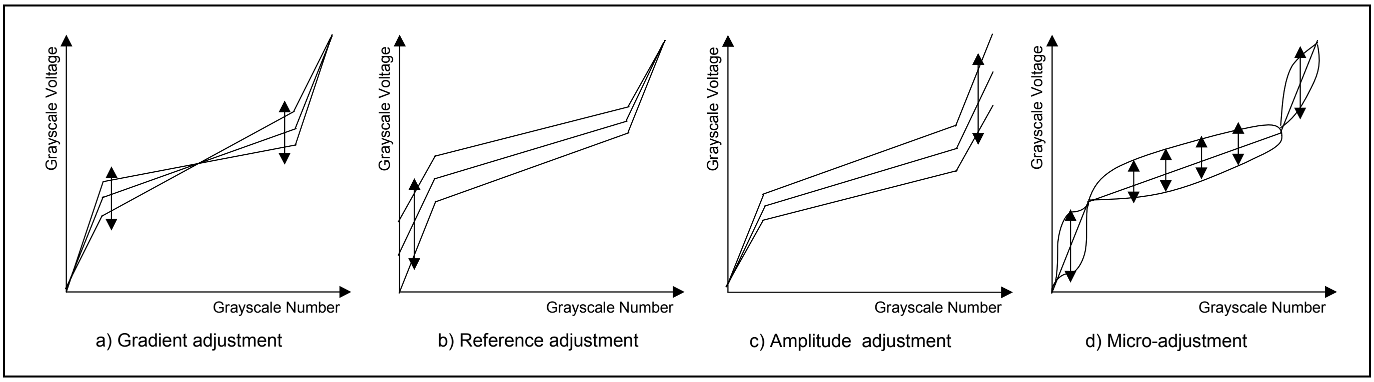

Figure 78. The Operation Of Adjusting Register

Gradient adjusting resistor

The gradient adjustment register is to adjust the gradient in the middle of the grayscale characteristics for the voltage without changing the dynamic range. To accomplish the adjustment, it controls the VINP12/VINN12 and VINP56/VINN56 voltage level by the 16 to 1 selector towards the 16-leveled reference voltage generated from the resistor ladder between VINP0/VINN0 and VINP7/VINN7. Also, there is an independent register on the positive/negative polarities in order for corresponding to asymmetry drive.

Reference adjusting resistor

The Reference adjustment register is to adjust the reference of the grayscale voltage. To accomplish the adjustment, it controls the VINP7/VINN7 voltage level by 32 to 1 selector towards the 32-leveled reference voltage generated from the resistor ladder between GVDD and VGS.

Amplitude adjusting resistor

The Amplitude adjustment register is to adjust the amplitude of the grayscale voltage. To accomplish the adjustment, it controls the VINP0/VINN0 voltage level by 32 to 1 selector towards the 32-leveled reference voltage generated from the resistor ladder between GVDD and VGS.

Micro-adjusting resistor

The Micro adjustment register is to make subtle adjustment of the grayscale voltage level. To accomplish the adjustment, it controls the each reference voltage level by the 16 to 1 selector towards the 16-leveled reference voltage generated from the resistor ladder. Also, there is an independent register on the positive/negative polarities as well as other adjustment registers.

Table 45. Gamma Adjustment Register

Register	Positive polarity	Negative polarity	Set-up contents
Gradient adjustment	R_PRP0[3:0] G_PRP0[3:0] B_PRP0[3:0]	R_PRN0[3:0] G_PRN0[3:0] B_PRN0[3:0]	The voltage of VINP12/VINN12 is selected by the 16 to 1 selector
	R_PRP1[3:0] G_PRP1[3:0] B_PRP2[3:0]	R_PRN1[3:0] G_PRN1[3:0] B_PRN1[3:0]	The voltage of VINP56/VINN56 is selected by the 16 to 1 selector
Reference adjustment	R_VRP1[4:0] G_VRP1[4:0] B_VPR1[4:0]	R_VRN1[4:0] G_VRN1[4:0] B_VRN1[4:0]	The voltage of VINP7/VINN7 is selected by the 32 to 1 selector
Amplitude adjustment	R_VRP0[4:0] G_VRP0[4:0] B_VRP0[4:0]	R_VRN0[4:0] G_VRN0[4:0] B_VRN0[4:0]	The voltage of VINP0/VINN0 is selected by the 32 to 1 selector
Micro adjustment	R_PKP0[3:0] G_PKP0[3:0] B_PKP0[3:0]	R_PKN0[3:0] G_PKN0[3:0] B_PKN0[3:0]	The voltage of grayscale number 1 is selected by the 16 to 1 selector
	R_PKP1[3:0] G_PKP1[3:0] B_PKP1[3:0]	R_PKN1[3:0] G_PKN1[3:0] B_PKN_[3:0]	The voltage of grayscale number 8 is selected by the 16 to 1 selector
	R_PKP2[3:0] G_PKP2[3:0] B_PKP2[3:0]	R_PKN2[3:0] G_PKN2[3:0] B_PKN2[3:0]	The voltage of grayscale number 20 is selected by the 16 to 1 selector
	R_PKP3[3:0] G_PKP3[3:0] B_PKP3[3:0]	R_PKN3[3:0] G_PKN3[3:0] B_PIN3[3:0]	The voltage of grayscale number 43 is selected by the 16 to 1 selector
	R_PKP4[3:0] G_PKP4[3:0] B_PKP4[3:0]	R_PKN4[3:0] G_PKN4[3:0] B_PKN4[3:0]	The voltage of grayscale number 55 is selected by the 16 to 1 selector
	R_PKP5[3:0] G_PKP5[3:0] B_PKP5[3:0]	R_PKN5[3:0] G_PKN5[3:0] B_PKN5[3:0]	The voltage of grayscale number 62 is selected by the 16 to 1 selector

RESISTOR LADDER NETWORK / SELECTOR

This block outputs the reference voltage of the grayscale voltage. There are four ladder resistors including the 8 to 1 selector selecting voltage generated by the ladder resistance voltage. Also, there are pins that connect to the external volume resistor. In addition, it allows compensating the dispersion of length between one panel and another.

RESISTOR LADDER NETWORK 1 / SELECTOR

There are 4 adjustments that are for the gradient adjustment (VRHP(N) / VRLP(N)) and for the reference / amplitude adjustment (VRP(N)1 / VRP(N)0). The voltage level is set by the gradient adjustment register and the reference / amplitude adjustment registers as below.

Table 46. Amplitude Adjustment

Register value VRP(N)0 [4:0]	Selected voltage VINP(N)0	Formula of VINP(N)0
00000	AVP(N)0	(430R/430R) * (GVDD-VGS) + VGS
00001	AVP(N)1	(427R/430R) * (GVDD-VGS) + VGS
00010	AVP(N)2	(424R/430R) * (GVDD-VGS) + VGS
00011	AVP(N)3	(421R/430R) * (GVDD-VGS) + VGS
00100	AVP(N)4	(418R/430R) * (GVDD-VGS) + VGS
00101	AVP(N)5	(415R/430R) * (GVDD-VGS) + VGS
00110	AVP(N)6	(412R/430R) * (GVDD-VGS) + VGS
00111	AVP(N)7	(409R/430R) * (GVDD-VGS) + VGS
01000	AVP(N)8	(406R/430R) * (GVDD-VGS) + VGS
01001	AVP(N)9	(403R/430R) * (GVDD-VGS) + VGS
01010	AVP(N)10	(400R/430R) * (GVDD-VGS) + VGS
01011	AVP(N)11	(397R/430R) * (GVDD-VGS) + VGS
01100	AVP(N)12	(394R/430R) * (GVDD-VGS) + VGS
01101	AVP(N)13	(391R/430R) * (GVDD-VGS) + VGS
01110	AVP(N)14	(388R/430R) * (GVDD-VGS) + VGS
01111	AVP(N)15	(385R/430R) * (GVDD-VGS) + VGS
10000	AVP(N)16	(382R/430R) * (GVDD-VGS) + VGS
10001	AVP(N)17	(379R/430R) * (GVDD-VGS) + VGS
10010	AVP(N)18	(376R/430R) * (GVDD-VGS) + VGS
10011	AVP(N)19	(373R/430R) * (GVDD-VGS) + VGS
10100	AVP(N)20	(370R/430R) * (GVDD-VGS) + VGS
10101	AVP(N)21	(367R/430R) * (GVDD-VGS) + VGS
10110	AVP(N)22	(364R/430R) * (GVDD-VGS) + VGS
10111	AVP(N)23	(361R/430R) * (GVDD-VGS) + VGS
11000	AVP(N)24	(358R/430R) * (GVDD-VGS) + VGS
11001	AVP(N)25	(355R/430R) * (GVDD-VGS) + VGS
11010	AVP(N)26	(352R/430R) * (GVDD-VGS) + VGS
11011	AVP(N)27	(349R/430R) * (GVDD-VGS) + VGS
11100	AVP(N)28	(346R/430R) * (GVDD-VGS) + VGS
11101	AVP(N)29	(343R/430R) * (GVDD-VGS) + VGS
11110	AVP(N)30	(340R/430R) * (GVDD-VGS) + VGS
11111	AVP(N)31	(337R/430R) * (GVDD-VGS) + VGS

Table 47. Reference Adjustment

Register value VRP(N)1 [4:0]	Selected voltage VINP(N)7	Formula of VINP(N)7
00000	AVP(N)63	(0R/430R) * (GVDD-VGS) + VGS
00001	AVP(N)62	(4R/430R) * (GVDD-VGS) + VGS
00010	AVP(N)61	(8R/430R) * (GVDD-VGS) + VGS
00011	AVP(N)60	(12R/430R) * (GVDD-VGS) + VGS
00100	AVP(N)59	(16R/430R) * (GVDD-VGS) + VGS
00101	AVP(N)58	(20R/430R) * (GVDD-VGS) + VGS
00110	AVP(N)57	(24R/430R) * (GVDD-VGS) + VGS
00111	AVP(N)56	(28R/430R) * (GVDD-VGS) + VGS
01000	AVP(N)55	(32R/430R) * (GVDD-VGS) + VGS
01001	AVP(N)54	(36R/430R) * (GVDD-VGS) + VGS
01010	AVP(N)53	(40R/430R) * (GVDD-VGS) + VGS
01011	AVP(N)52	(44R/430R) * (GVDD-VGS) + VGS
01100	AVP(N)51	(48R/430R) * (GVDD-VGS) + VGS
01101	AVP(N)50	(52R/430R) * (GVDD-VGS) + VGS
01110	AVP(N)49	(56R/430R) * (GVDD-VGS) + VGS
01111	AVP(N)48	(60R/430R) * (GVDD-VGS) + VGS
10000	AVP(N)47	(64R/430R) * (GVDD-VGS) + VGS
10001	AVP(N)46	(68R/430R) * (GVDD-VGS) + VGS
10010	AVP(N)45	(72R/430R) * (GVDD-VGS) + VGS
10011	AVP(N)44	(76R/430R) * (GVDD-VGS) + VGS
10100	AVP(N)43	(80R/430R) * (GVDD-VGS) + VGS
10101	AVP(N)42	(84R/430R) * (GVDD-VGS) + VGS
10110	AVP(N)41	(88R/430R) * (GVDD-VGS) + VGS
10111	AVP(N)40	(92R/430R) * (GVDD-VGS) + VGS
11000	AVP(N)39	(96R/430R) * (GVDD-VGS) + VGS
11001	AVP(N)38	(100R/430R) * (GVDD-VGS) + VGS
11010	AVP(N)37	(104R/430R) * (GVDD-VGS) + VGS
11011	AVP(N)36	(108R/430R) * (GVDD-VGS) + VGS
11100	AVP(N)35	(112R/430R) * (GVDD-VGS) + VGS
11101	AVP(N)34	(116R/430R) * (GVDD-VGS) + VGS
11110	AVP(N)33	(120R/430R) * (GVDD-VGS) + VGS
11111	AVP(N)32	(124R/430R) * (GVDD-VGS) + VGS

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Table 48. Gradient Adjustment (1)

Register value PRP(N)0 [2:0]	Selected voltage VINP(N)12	Formula of VINP(N)12
0000	GVP(N)0	(270R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0001	GVP(N)1	(265R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0010	GVP(N)2	(260R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0011	GVP(N)3	(255R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0100	GVP(N)4	(250R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0101	GVP(N)5	(245R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0110	GVP(N)6	(240R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0111	GVP(N)7	(235R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1000	GVP(N)8	(230R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1001	GVP(N)9	(225R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1010	GVP(N)10	(220R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1011	GVP(N)11	(215R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1100	GVP(N)12	(210R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1101	GVP(N)13	(205R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1110	GVP(N)14	(200R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1111	GVP(N)15	(195R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7

Table 49. Gradient Adjustment (2)

Register value PRP(N)1 [2:0]	Selected voltage VINP(N)56	Formula of VINP(N)56
0000	GVP(N)31	(80R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0001	GVP(N)30	(85R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0010	GVP(N)29	(90R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0011	GVP(N)28	(95R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0100	GVP(N)27	(100R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0101	GVP(N)26	(105R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0110	GVP(N)25	(110R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
0111	GVP(N)24	(115R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1000	GVP(N)23	(120R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1001	GVP(N)22	(125R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1010	GVP(N)21	(130R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1011	GVP(N)20	(135R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1100	GVP(N)19	(140R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1101	GVP(N)18	(145R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1110	GVP(N)17	(150R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7
1111	GVP(N)16	(155R/350R) * (VINP(N)0-VINP(N)7) + VINP(N)7

RESISTOR LADDER NETWORK 2 / SELECTOR

In the 16-to-1 selector, the voltage level must be selected by the given ladder resistance and the micro-adjustment register and output the six types of the reference voltage, VIN1 to VIN6.
Following figure explains the relationship between the micro-adjustment register and the selected voltage.

Table 50. Relationship between Micro-adjustment Register and Selected Voltage

Register value PKP(N) [2:0]	Selected voltage					
	VINP(N)1	VINP(N)2	VINP(N)3	VINP(N)4	VINP(N)5	VINP(N)6
0000	KVP(N)0	KVP(N)16	KVP(N)32	KVP(N)63	KVP(N)79	KVP(N)95
0001	KVP(N)1	KVP(N)17	KVP(N)33	KVP(N)62	KVP(N)78	KVP(N)94
0010	KVP(N)2	KVP(N)18	KVP(N)34	KVP(N)61	KVP(N)77	KVP(N)93
0011	KVP(N)3	KVP(N)19	KVP(N)35	KVP(N)60	KVP(N)76	KVP(N)92
0100	KVP(N)4	KVP(N)20	KVP(N)36	KVP(N)59	KVP(N)75	KVP(N)91
0101	KVP(N)5	KVP(N)21	KVP(N)37	KVP(N)58	KVP(N)74	KVP(N)90
0110	KVP(N)6	KVP(N)22	KVP(N)38	KVP(N)57	KVP(N)73	KVP(N)89
0111	KVP(N)7	KVP(N)23	KVP(N)39	KVP(N)56	KVP(N)72	KVP(N)88
1000	KVP(N)8	KVP(N)24	KVP(N)40	KVP(N)55	KVP(N)71	KVP(N)87
1001	KVP(N)9	KVP(N)25	KVP(N)41	KVP(N)54	KVP(N)70	KVP(N)86
1010	KVP(N)10	KVP(N)26	KVP(N)42	KVP(N)53	KVP(N)69	KVP(N)85
1011	KVP(N)11	KVP(N)27	KVP(N)43	KVP(N)52	KVP(N)68	KVP(N)84
1100	KVP(N)12	KVP(N)28	KVP(N)44	KVP(N)51	KVP(N)67	KVP(N)83
1101	KVP(N)13	KVP(N)29	KVP(N)45	KVP(N)50	KVP(N)66	KVP(N)82
1110	KVP(N)14	KVP(N)30	KVP(N)46	KVP(N)49	KVP(N)65	KVP(N)81
1111	KVP(N)15	KVP(N)31	KVP(N)47	KVP(N)48	KVP(N)64	KVP(N)80

The grayscale levels are determined by the following formulas listed in the next pages.

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Table 51. Formulas for Calculating Gamma Adjusting Voltage (Positive Polarity) 1

Pins	Formula	Micro-adjusting register value	Reference voltage
KVP0	(45R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "0000"	VINP1
KVP1	(42R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "0001"	
KVP2	(39R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "0010"	
KVP3	(36R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "0011"	
KVP4	(33R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "0100"	
KVP5	(30R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "0101"	
KVP6	(27R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "0110"	
KVP7	(24R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "0111"	
KVP8	(21R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "1000"	
KVP9	(18R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "1001"	
KVP10	(15R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "1010"	
KVP11	(12R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "1011"	
KVP12	(9R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "1100"	
KVP13	(6R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "1101"	
KVP14	(3R/48R) * (VINP0 – VINP12) + VINP12	PKP0[3:0] = "1110"	
KVP15	VINP12	PKP0[3:0] = "1111"	
KVP16	(219R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "0000"	VINP2
KVP17	(216R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "0001"	
KVP18	(213R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "0010"	
KVP19	(210R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "0011"	
KVP20	(207R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "0100"	
KVP21	(204R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "0101"	
KVP22	(201R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "0110"	
KVP23	(198R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "0111"	
KVP24	(195R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "1000"	
KVP25	(192R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "1001"	
KVP26	(189R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "1010"	
KVP27	(186R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "1011"	
KVP28	(183R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "1100"	
KVP29	(180R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "1101"	
KVP30	(177R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "1110"	
KVP31	(174R/222R) * (VINP12-VINP56) + VINP56	PKP1[3:0] = "1111"	
KVP32	(171R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "0000"	VINP3
KVP33	(168R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "0001"	
KVP34	(165R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "0010"	
KVP35	(162R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "0011"	
KVP36	(159R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "0100"	
KVP37	(156R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "0101"	
KVP38	(153R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "0110"	
KVP39	(150R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "0111"	
KVP40	(147R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "1000"	
KVP41	(144R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "1001"	
KVP42	(141R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "1010"	
KVP43	(138R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "1011"	
KVP44	(135R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "1100"	
KVP45	(132R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "1101"	
KVP46	(129R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "1110"	
KVP47	(126R/222R) * (VINP12-VINP56) + VINP56	PKP2[3:0] = "1111"	

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Pins	Formula	Micro-adjusting register value	Reference voltage
KVP48	(96R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "1111"	
KVP49	(93R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "1110"	
KVP50	(90R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "1101"	
KVP51	(87R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "1100"	
KVP52	(84R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "1011"	
KVP53	(81R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "1010"	
KVP54	(78R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "1001"	
KVP55	(75R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "1000"	VINP4
KVP56	(72R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "0111"	
KVP57	(69R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "0110"	
KVP58	(66R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "0101"	
KVP59	(63R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "0100"	
KVP60	(60R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "0011"	
KVP61	(57R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "0010"	
KVP62	(54R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "0001"	
KVP63	(51R/222R) * (VINP12-VINP56) + VINP56	PKP4[3:0] = "0000"	
KVP64	(48R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "1111"	
KVP65	(45R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "1110"	
KVP66	(42R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "1101"	
KVP67	(39R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "1100"	
KVP68	(36R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "1011"	
KVP69	(33R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "1010"	
KVP70	(30R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "1001"	
KVP71	(27R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "1000"	
KVP72	(24R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "0111"	VINP5
KVP73	(21R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "0110"	
KVP74	(18R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "0101"	
KVP75	(15R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "0100"	
KVP76	(12R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "0011"	
KVP77	(9R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "0010"	
KVP78	(6R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "0001"	
KVP79	(3R/222R) * (VINP12-VINP56) + VINP56	PKP5[3:0] = "0001"	
KVP80	VINP56	PKP6[3:0] = "1111"	
KVP81	(45R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "1110"	
KVP82	(42R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "1101"	
KVP83	(39R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "1100"	
KVP84	(36R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "1011"	
KVP85	(33R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "1010"	
KVP86	(30R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "1001"	
KVP87	(27R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "1000"	
KVP88	(24R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "0111"	VINP6
KVP89	(21R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "0110"	
KVP90	(18R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "0101"	
KVP91	(15R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "0100"	
KVP92	(12R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "0011"	
KVP93	(9R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "0010"	
KVP94	(6R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "0001"	
KVP95	(3R/48R) * (VINP56 – VINP7) + VINP7	PKP6[3:0] = "0000"	

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Table 52. Formulas for Calculating Gamma Adjusting Voltage (Positive Polarity) 2

Grayscale voltage	Formula	Grayscale voltage	Formula
V0	VINP0	V32	V20-(V20-V43)*(12/23)
V1	VINP1	V33	V20-(V20-V43)*(13/23)
V2	V1-(V1-V8)*(28/96)	V34	V20-(V20-V43)*(14/23)
V3	V1-(V1-V8)*(42/96)	V35	V20-(V20-V43)*(15/23)
V4	V1-(V1-V8)*(60/96)	V36	V20-(V20-V43)*(16/23)
V5	V1-(V1-V8)*(69/96)	V37	V20-(V20-V43)*(17/23)
V6	V1-(V1-V8)*(78/96)	V38	V20-(V20-V43)*(18/23)
V7	V1-(V1-V8)*(87/96)	V39	V20-(V20-V43)*(19/23)
V8	VINP2	V40	V20-(V20-V43)*(20/23)
V9	V8-(V8-V20)*(2/24)	V41	V20-(V20-V43)*(21/23)
V10	V8-(V8-V20)*(4/24)	V42	V20-(V20-V43)*(22/23)
V11	V8-(V8-V20)*(6/24)	V43	VINP4
V12	V8-(V8-V20)*(8/24)	V44	V43-(V43-V55)*(2/24)
V13	V8-(V8-V20)*(10/24)	V45	V43-(V43-V55)*(4/24)
V14	V8-(V8-V20)*(12/24)	V46	V43-(V43-V55)*(6/24)
V15	V8-(V8-V20)*(14/24)	V47	V43-(V43-V55)*(8/24)
V16	V8-(V8-V20)*(16/24)	V48	V43-(V43-V55)*(10/24)
V17	V8-(V8-V20)*(18/24)	V49	V43-(V43-V55)*(12/24)
V18	V8-(V8-V20)*(20/24)	V50	V43-(V43-V55)*(14/24)
V19	V8-(V8-V20)*(22/24)	V51	V43-(V43-V55)*(16/24)
V20	VINP3	V52	V43-(V43-V55)*(18/24)
V21	V20-(V20-V43)*(1/23)	V53	V43-(V43-V55)*(20/24)
V22	V20-(V20-V43)*(2/23)	V54	V43-(V43-V55)*(22/24)
V23	V20-(V20-V43)*(3/23)	V55	VINP5
V24	V20-(V20-V43)*(4/23)	V56	V55-(V55-V62)*(9/96)
V25	V20-(V20-V43)*(5/23)	V57	V55-(V55-V62)*(18/96)
V26	V20-(V20-V43)*(6/23)	V58	V55-(V55-V62)*(27/96)
V27	V20-(V20-V43)*(7/23)	V59	V55-(V55-V62)*(36/96)
V28	V20-(V20-V43)*(8/23)	V60	V55-(V55-V62)*(54/96)
V29	V20-(V20-V43)*(9/23)	V61	V55-(V55-V62)*(68/96)
V30	V20-(V20-V43)*(10/23)	V62	VINP6
V31	V20-(V20-V43)*(11/23)	V63	VINP7

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Table 53. Formulas for Calculating Gamma Adjusting Voltage (Negative Polarity) 1

Pins	Formula	Micro-adjusting register value	Reference voltage
KVN0	(45R/48R) * (VINP0 – VINN12) + VINN12	PKP0[3:0] = "0000"	VINN1
KVN1	(42R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "0001"	
KVN2	(39R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "0010"	
KVN3	(36R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "0011"	
KVN4	(33R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "0100"	
KVN5	(30R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "0101"	
KVN6	(27R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "0110"	
KVN7	(24R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "0111"	
KVN8	(21R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "1000"	
KVN9	(18R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "1001"	
KVN10	(15R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "1010"	
KVN11	(12R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "1011"	
KVN12	(9R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "1100"	
KVN13	(6R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "1101"	
KVN14	(3R/48R) * (VINN0 – VINN12) + VINN12	PKN0[3:0] = "1110"	
KVN15	VINN12	PKN0[3:0] = "1111"	
KVN16	(219R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "0000"	VINN2
KVN17	(216R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "0001"	
KVN18	(213R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "0010"	
KVN19	(210R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "0011"	
KVN20	(207R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "0100"	
KVN21	(204R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "0101"	
KVN22	(201R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "0110"	
KVN23	(198R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "0111"	
KVN24	(195R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "1000"	
KVN25	(192R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "1001"	
KVN26	(189R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "1010"	
KVN27	(186R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "1011"	
KVN28	(183R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "1100"	
KVN29	(180R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "1101"	
KVN30	(177R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "1110"	
KVN31	(174R/222R) * (VINN12-VINN56) + VINN56	PKN1[3:0] = "1111"	
KVN32	(171R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "0000"	VINN3
KVN33	(168R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "0001"	
KVN34	(165R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "0010"	
KVN35	(162R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "0011"	
KVN36	(159R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "0100"	
KVN37	(156R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "0101"	
KVN38	(153R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "0110"	
KVN39	(150R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "0111"	
KVN40	(147R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "1000"	
KVN41	(144R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "1001"	
KVN42	(141R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "1010"	
KVN43	(138R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "1011"	
KVN44	(135R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "1100"	
KVN45	(132R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "1101"	
KVN46	(129R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "1110"	
KVN47	(126R/222R) * (VINN12-VINN56) + VINN56	PKN2[3:0] = "1111"	

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Pins	Formula	Micro-adjusting register value	Reference voltage
KVN48	(96R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "1111"	VINN4
KVN49	(93R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "1110"	
KVN50	(90R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "1101"	
KVN51	(87R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "1100"	
KVN52	(84R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "1011"	
KVN53	(81R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "1010"	
KVN54	(78R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "1001"	
KVN55	(75R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "1000"	
KVN56	(72R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "0111"	
KVN57	(69R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "0110"	
KVN58	(66R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "0101"	
KVN59	(63R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "0100"	
KVN60	(60R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "0011"	
KVN61	(57R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "0010"	
KVN62	(54R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "0001"	
KVN63	(51R/222R) * (VINN12-VINN56) + VINN56	PKN4[3:0] = "0000"	
KVN64	(48R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "1111"	VINN5
KVN65	(45R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "1110"	
KVN66	(42R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "1101"	
KVN67	(39R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "1100"	
KVN68	(36R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "1011"	
KVN69	(33R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "1010"	
KVN70	(30R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "1001"	
KVN71	(27R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "1000"	
KVN72	(24R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "0111"	
KVN73	(21R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "0110"	
KVN74	(18R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "0101"	
KVN75	(15R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "0100"	
KVN76	(12R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "0011"	
KVN77	(9R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "0010"	
KVN78	(6R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "0001"	
KVN79	(3R/222R) * (VINN12-VINN56) + VINN56	PKN5[3:0] = "0000"	
KVN80	VINN56	PKN6[3:0] = "1111"	VINN6
KVN81	(45R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "1110"	
KVN82	(42R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "1101"	
KVN83	(39R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "1100"	
KVN84	(36R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "1011"	
KVN85	(33R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "1010"	
KVN86	(30R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "1001"	
KVN87	(27R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "1000"	
KVN88	(24R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "0111"	
KVN89	(21R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "0110"	
KVN90	(18R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "0101"	
KVN91	(15R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "0100"	
KVN92	(12R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "0011"	
KVN93	(9R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "0010"	
KVN94	(6R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "0001"	
KVN95	(3R/48R) * (VINN56 – VINN7) + VINN7	PKN6[3:0] = "0000"	

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Table 54. Formulas for Calculating Gamma Adjusting Voltage (Negative Polarity) 2

Grayscale voltage	Formula	Grayscale voltage	Formula
V0	VINN0	V32	V20-(V20-V43)*(12/23)
V1	VINN1	V33	V20-(V20-V43)*(13/23)
V2	V1-(V1-V8)*(28/96)	V34	V20-(V20-V43)*(14/23)
V3	V1-(V1-V8)*(42/96)	V35	V20-(V20-V43)*(15/23)
V4	V1-(V1-V8)*(60/96)	V36	V20-(V20-V43)*(16/23)
V5	V1-(V1-V8)*(69/96)	V37	V20-(V20-V43)*(17/23)
V6	V1-(V1-V8)*(78/96)	V38	V20-(V20-V43)*(18/23)
V7	V1-(V1-V8)*(87/96)	V39	V20-(V20-V43)*(19/23)
V8	VINN2	V40	V20-(V20-V43)*(20/23)
V9	V8-(V8-V20)*(2/24)	V41	V20-(V20-V43)*(21/23)
V10	V8-(V8-V20)*(4/24)	V42	V20-(V20-V43)*(22/23)
V11	V8-(V8-V20)*(6/24)	V43	VINN4
V12	V8-(V8-V20)*(8/24)	V44	V43-(V43-V55)*(2/24)
V13	V8-(V8-V20)*(10/24)	V45	V43-(V43-V55)*(4/24)
V14	V8-(V8-V20)*(12/24)	V46	V43-(V43-V55)*(6/24)
V15	V8-(V8-V20)*(14/24)	V47	V43-(V43-V55)*(8/24)
V16	V8-(V8-V20)*(16/24)	V48	V43-(V43-V55)*(10/24)
V17	V8-(V8-V20)*(18/24)	V49	V43-(V43-V55)*(12/24)
V18	V8-(V8-V20)*(20/24)	V50	V43-(V43-V55)*(14/24)
V19	V8-(V8-V20)*(22/24)	V51	V43-(V43-V55)*(16/24)
V20	VINN3	V52	V43-(V43-V55)*(18/24)
V21	V20-(V20-V43)*(1/23)	V53	V43-(V43-V55)*(20/24)
V22	V20-(V20-V43)*(2/23)	V54	V43-(V43-V55)*(22/24)
V23	V20-(V20-V43)*(3/23)	V55	VINN5
V24	V20-(V20-V43)*(4/23)	V56	V55-(V55-V62)*(9/96)
V25	V20-(V20-V43)*(5/23)	V57	V55-(V55-V62)*(18/96)
V26	V20-(V20-V43)*(6/23)	V58	V55-(V55-V62)*(27/96)
V27	V20-(V20-V43)*(7/23)	V59	V55-(V55-V62)*(36/96)
V28	V20-(V20-V43)*(8/23)	V60	V55-(V55-V62)*(54/96)
V29	V20-(V20-V43)*(9/23)	V61	V55-(V55-V62)*(68/96)
V30	V20-(V20-V43)*(10/23)	V62	VINN6
V31	V20-(V20-V43)*(11/23)	V63	VINN7

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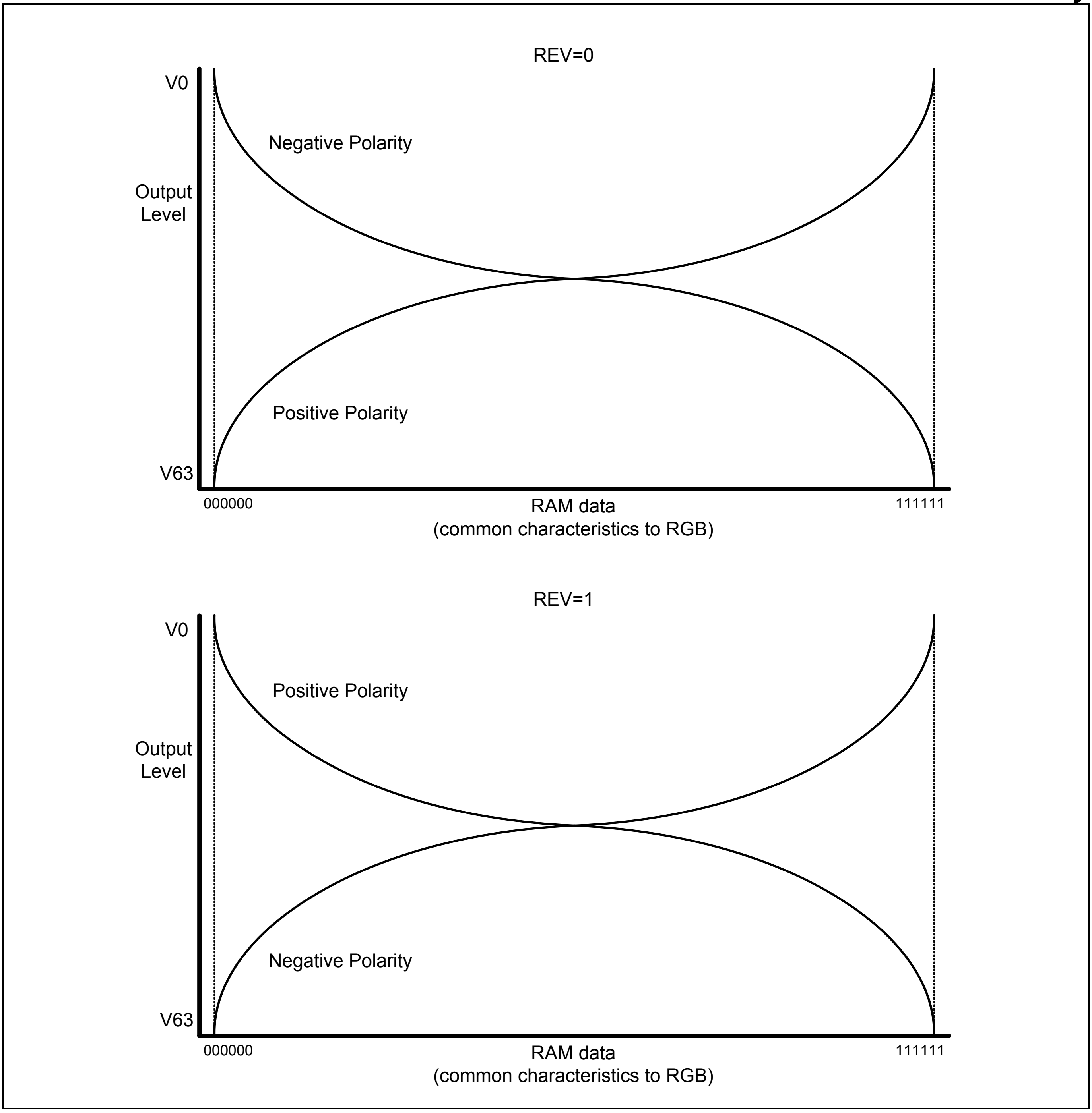

Figure 79. Relationship between RAM Data And Output Voltage

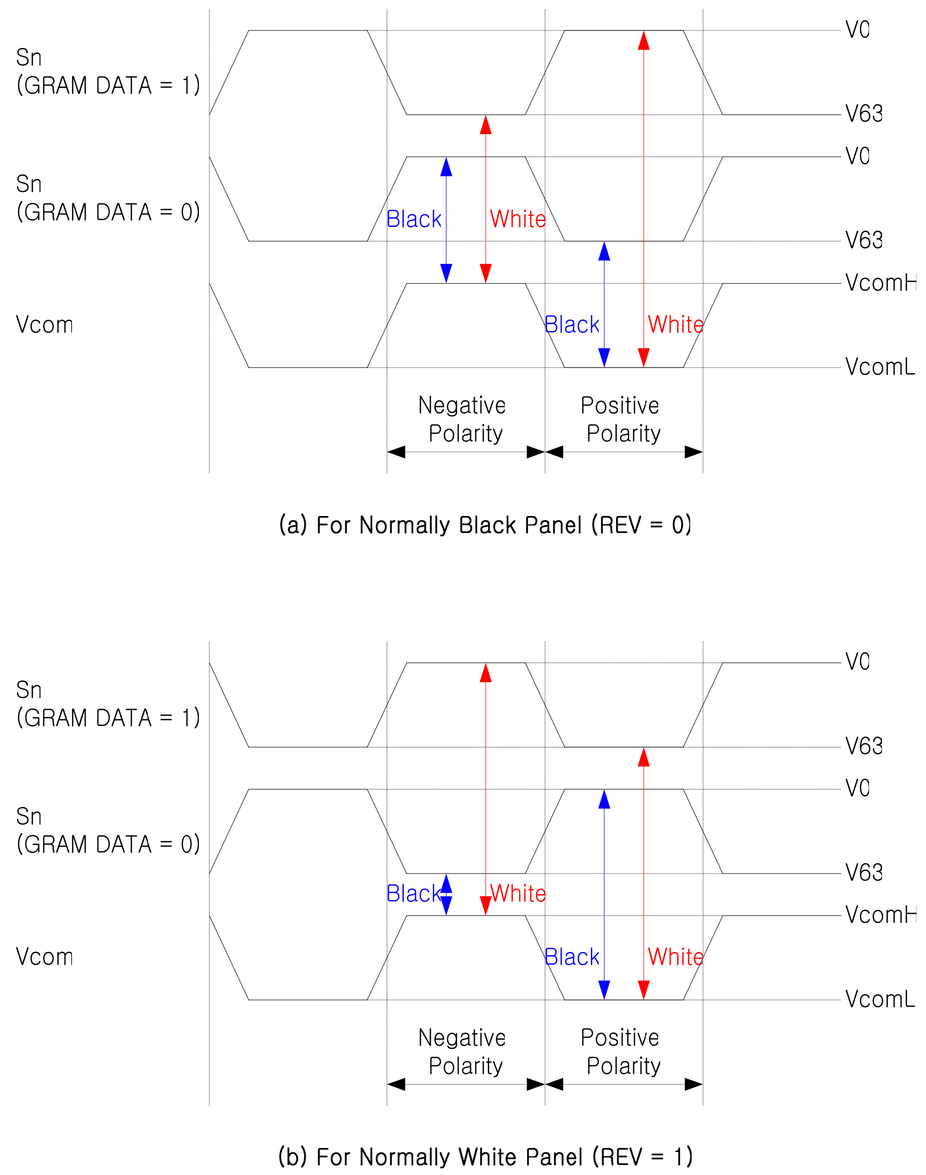

Figure 80. Relationship between Source Output and VCOM

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THE 8-COLOR DISPLAY MODE

The S6D0170 incorporates 8-color display mode. The grayscale levels to be used are V0(GVDD) and V63(AVSS) and all the other levels (V1~V62) are halt. So that it attempts to lower power consumption.
During the 8-color mode, the Gamma micro adjustment register, PKP00-PKP52 and PKN00-PKN52 are invalid. The grayscale levels (V1-V62) are in OFF condition in order to select V0/V63.

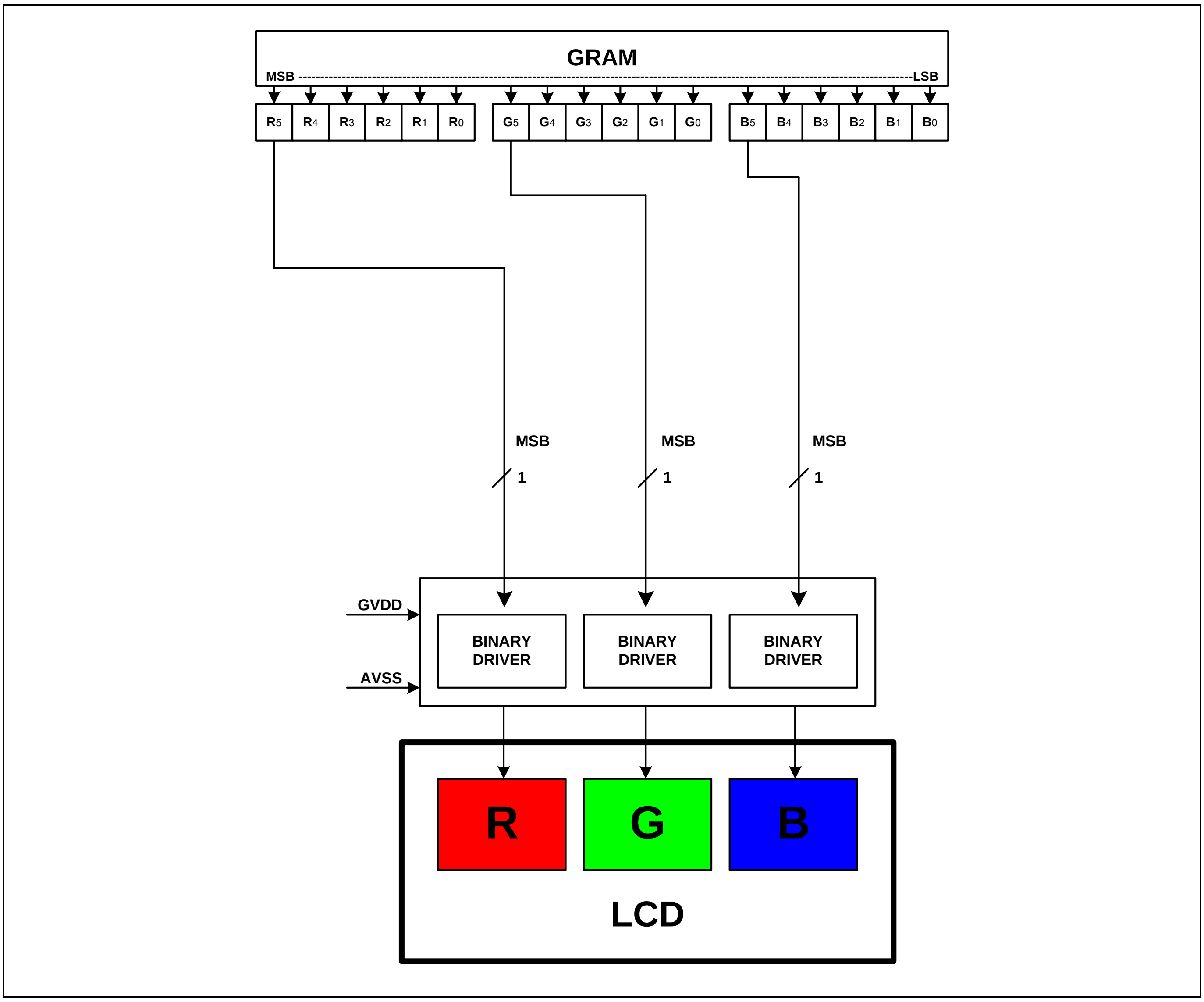

Figure 81. 8-Color Display Control

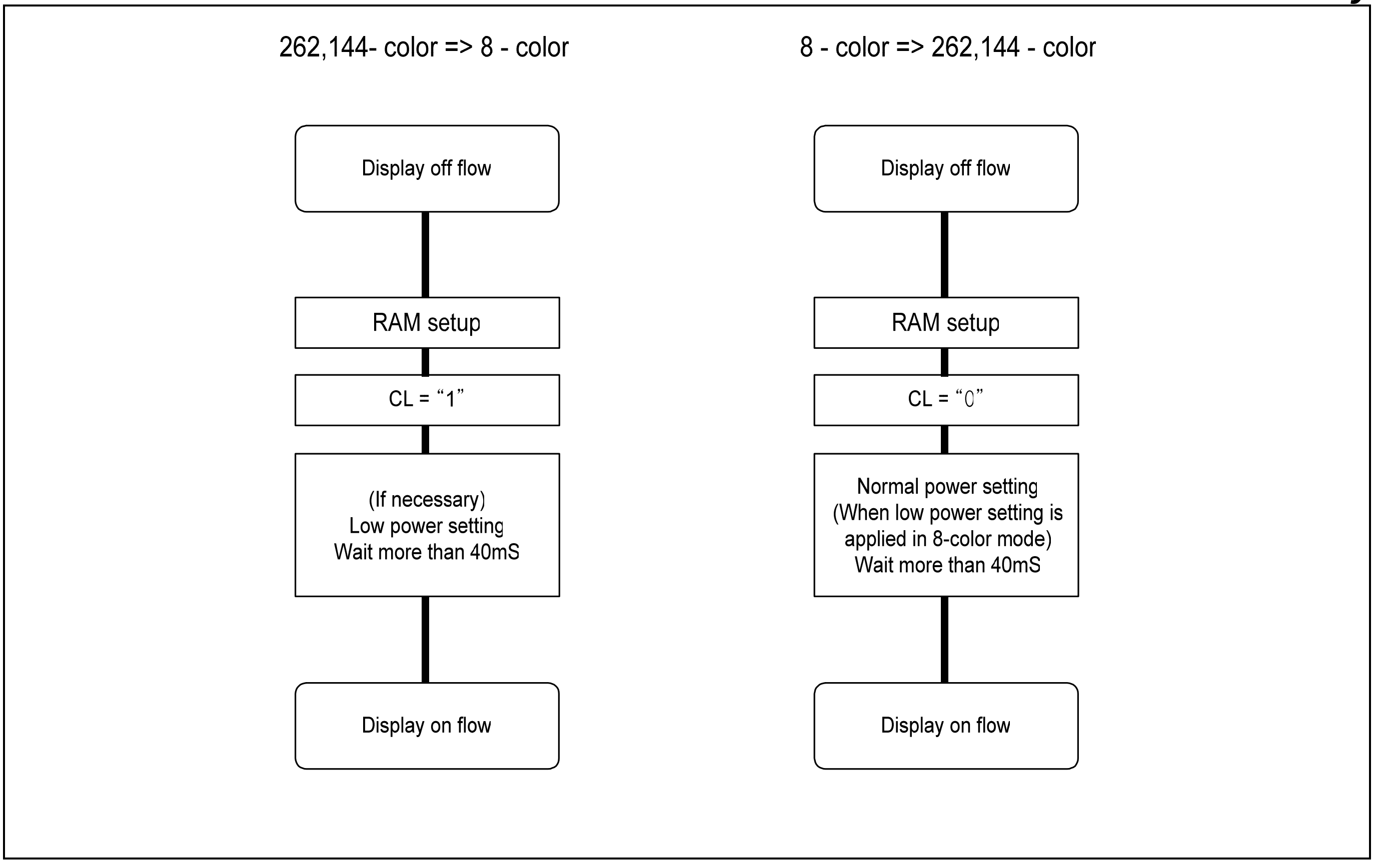

Figure 82. Setup Procedure For The 8-Color Mode

SYSTEM STRUCTURE EXAMPLE

The following figure indicates the system structure, which composes the 240 (width) x 432 (length) dots TFT-LCD panel.

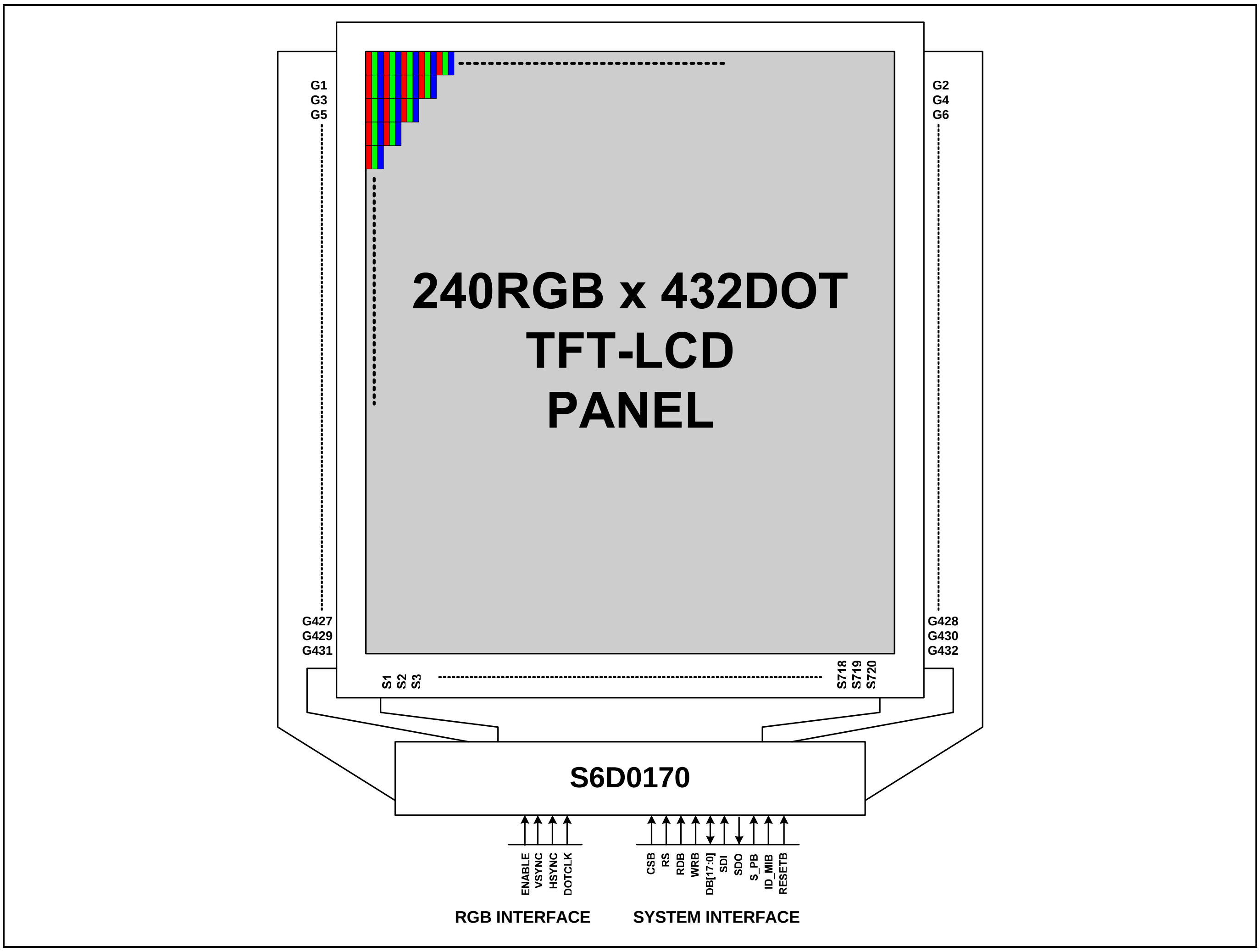

Figure 83. System Structure

Note: Driver IC is "Top view (Bump view)".

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SET UP FLOW OF DISPLAY

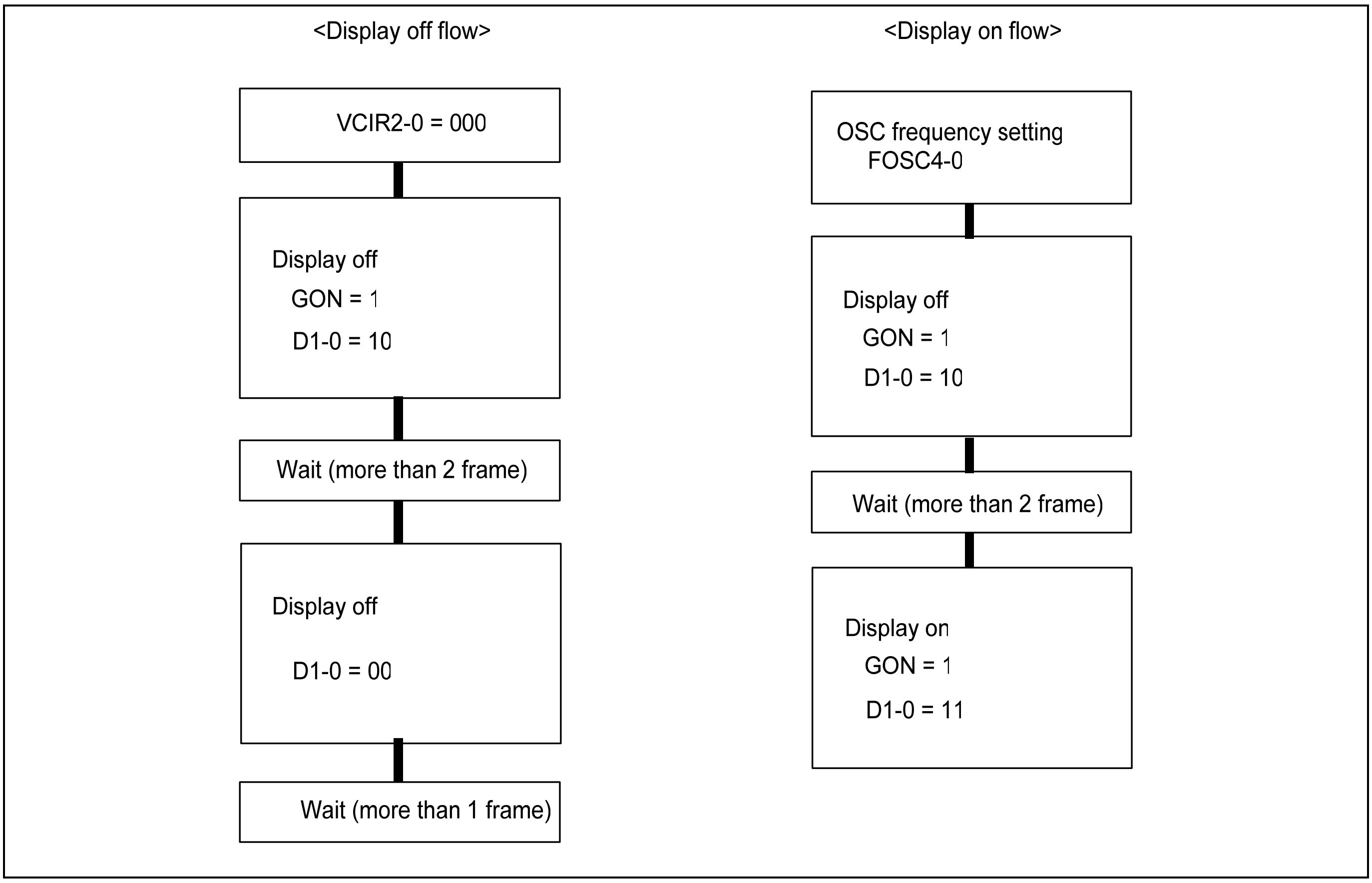

Figure 84. Setup flow of Display

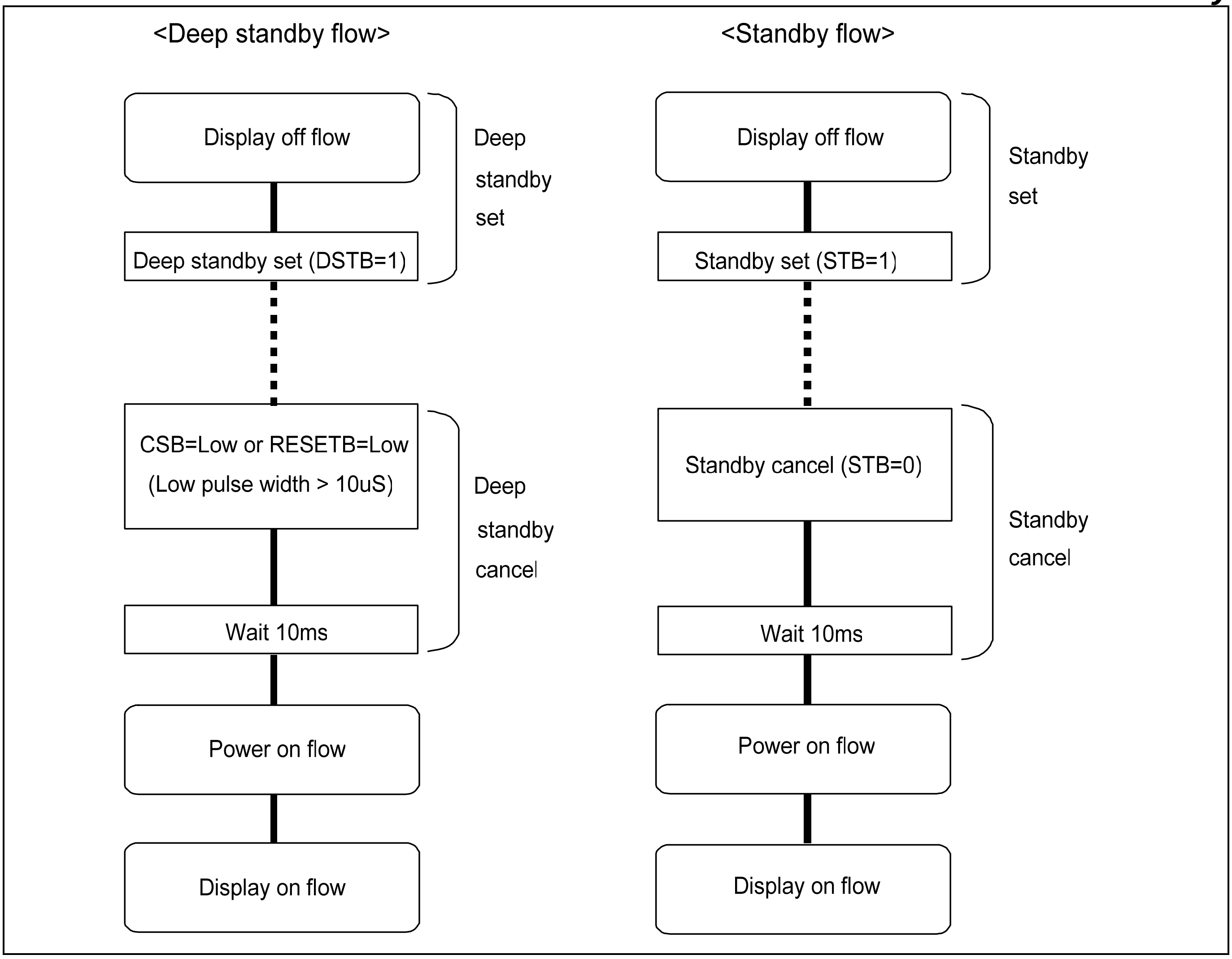

Figure 85. Setup flow of Deep Standby / Standby

Notes:

1. When IC enter deep standby mode, all instruction setting values are initialized.
2. When IC enter standby mode, below instruction setting values are initialized.
   ; R11h (APON, PON3, PON2, PON1, PON, AON, VCI1_EN, VC3-0), R14h (VCOMG)
   ; R07h (D1-0, GON, CL, REV)

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OSCILLATION CIRCUIT

The S6D0170 can provide R-C oscillation. S6D0170 internal oscillator does not need to attach the external resistor. The appropriate oscillation frequency for operating voltage, display size, and frame frequency can be obtained by adjusting the oscillator frequency control register setting. Since R-C oscillation stops during the standby mode, power consumption can be reduced.

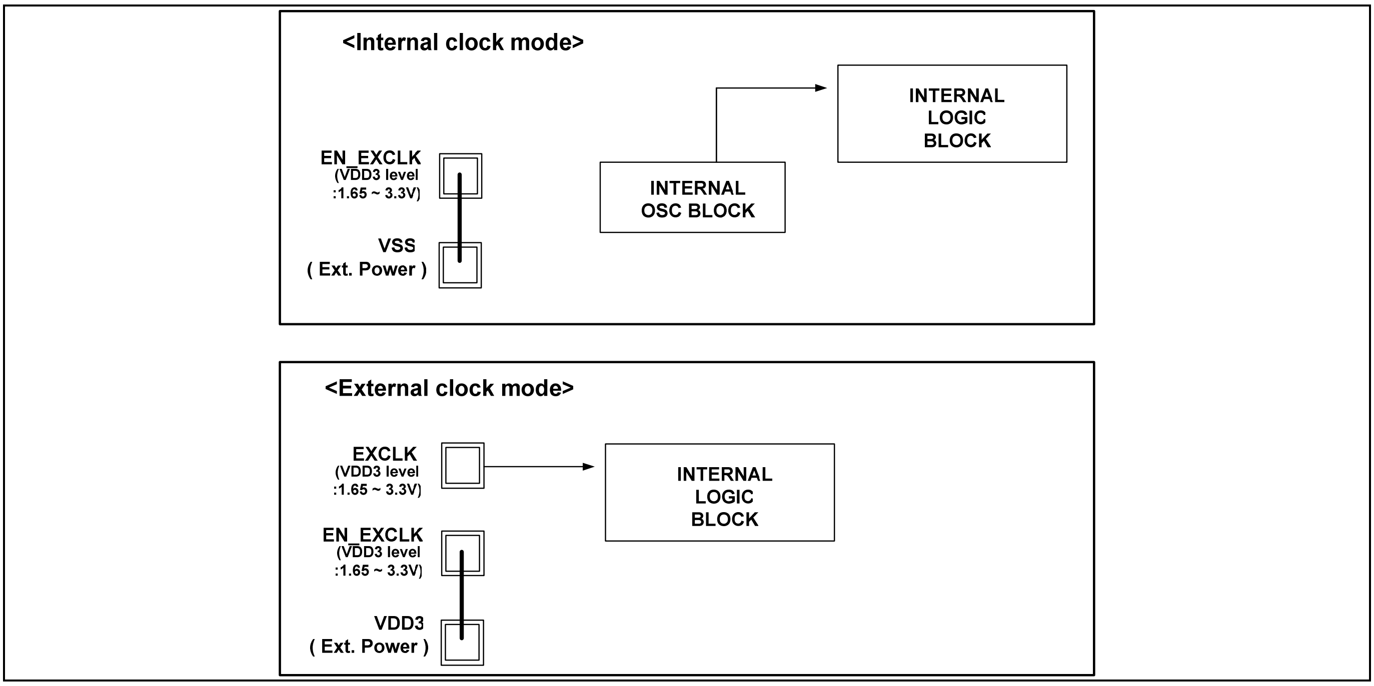

Figure 86. Application Diagram For Oscillation Circuitry

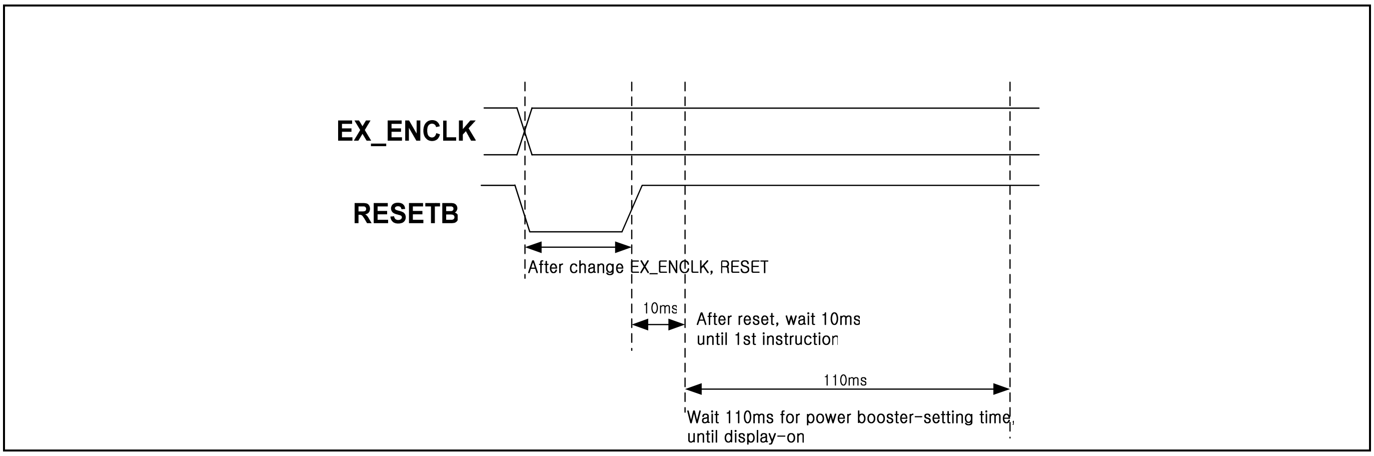

Figure 87. Oscillator sequence

FRAME FREQUENCY ADJUSTING FUNCTION

The S6D0170 has an on-chip frame-frequency adjustment function. The frame frequency can be adjusted by the instruction setting (RTN) during the LCD driver as the oscillation frequency is always same.
If the oscillation frequency is set to high, animation or a static image can be displayed in suitable ways by changing the frame frequency. When a static image is displayed, the frame frequency can be set low and the low-power consumption mode can be entered. When high-speed screen switching for an animated display, etc. is required, the frame frequency can be set high.

Relationship between LCD drive duty and frame frequency

The relationships between the LCD drive duty and the frame frequency is calculated by the following expression. The frame frequency can be adjusted in the 1H period adjusting bit (RTN).

$$\text{Frame Frequency} = \frac{f_{OSC}}{\text{Clock cycles per raster-row x division ratio x (Line+B)}} \text{ [Hz]}$$

f_{OSC}: R-C oscillation frequency
Line: Number of raster-rows (NL bit)
Clock cycles per raster-row: RTN bit
Division ratio: DIV bit
B: Blank period(Back porch + Front Porch)

Figure 88. Formula For The Frame Frequency

Calculation Example:

Driver raster-rows: 432
Division Ratio: 1OSC (DIV=0)
1H period: 16 clock (RTN2 to 0 = 000)
B: Blank period (BP + FP): 16

f_osc = 60Hz x 16 clock x 1OSC x (432 + 16) lines = 430.08 [kHz]

The calculation result says that the required oscillation frequency in the display condition listed above is 430.08 kHz. So the external resistance value of the R-C oscillator should be selected to make oscillation frequency 430.08 kHz.

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APPLICATION CIRCUIT

The following figure indicates a typical application circuit for S6D0170.

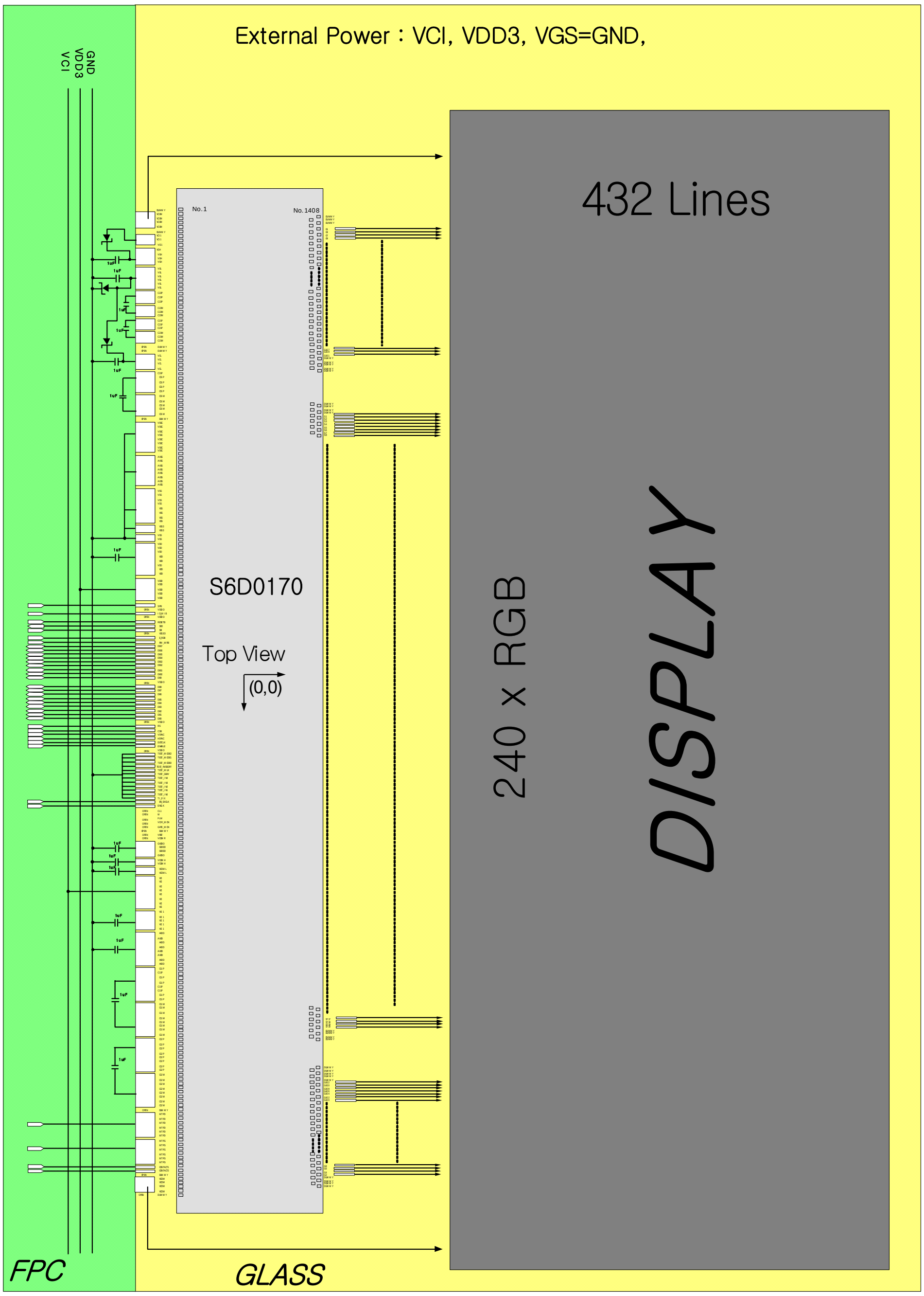

Figure 89. S6D0170 ITO / FPC Application Example

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SPECIFICATIONS

ABSOLUTE MAXIMUM RATINGS

Table 55. Absolute Maximum Rating

(VSS = 0V)

Item	Symbol	Rating	Unit
Supply voltage for logic block	VDD - VSS	-0.3 ~ 3.0	V
Supply voltage for I/O block	VDD3 - VSS	-0.3 ~ 5.0	V
Supply voltage for step-up circuit	VCI - VSS	-0.3 ~ 5.0	V
LCD Supply Voltage range	AVDD – VSS	-0.3 ~ 6.5	V
	VGH - VSS	-0.3 ~ 22.0	V
	VSS – VGL	-0.3 ~ 22.0	V
	VSS - VCL	-0.3 ~ 5.0	V
	|VGH – VGL|	-0.3 ~ 33	V
Input Voltage range	Vin	- 0.3 to VDD3 +0.3	V
Maximum rewritable time of MTP	tmtp	1000	times
Operating temperature	Topr	-40 ~ +85	°C
Storage temperature	Tstg	-55 ~ +110	°C

Notes:

1. Absolute maximum rating is the limit value. When the IC is exposed operating environment beyond this range, the IC do not assure operations and may be damaged permanently, not be able to be recovered.
2. Operating temperature is the range of device-operating temperature. They do not guarantee chip performance.
3. Absolute maximum rating is guaranteed only when our company's package used.

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DC CHARACTERISTICS

Table 56. DC Characteristics

(VSS = 0V)

Characteristic	Symbol	CONDITION	MIN	TYP	MAX	Unit	Note
Operating voltage	VDD3		1.65	-	3.3	V	*1
LCD driving voltage	VGH		8.75	-	19.25	V	
	VGL		-13.5	-	-5.25	V	
	VCL		-3	-	-1.75	V	
	AVDD		3.5	-	6	V	
	GVDD		3.0	-	5.0	V	
Input high voltage	V_{IH}		0.7*VDD3	-	VDD3	V	*2
Input low voltage	V_{IL}		0	-	0.3*VDD3	V	*2
Output high voltage	V_{OH}	I_{OH} = -0.5mA	0.8*VDD3	-	VDD3	V	*3
Output low voltage	V_{OL}	I_{OL} = 0.5mA	0.0	-	0.2*VDD3	V	*3
Input leakage current	I_{IL}	VIN = VSS or VDD3	-1.0		1.0	uA	*2
Output leakage current	I_{OL}	VIN = VSS or VDD3	-3.0		3.0	uA	*3
Operating frequency	Fosc	Frame freq. = 60 Hz Display line = 432 Temp = 25°C VDD=1.5V	T.B.D	430	T.B.D	kHz	*4
Internal reference power supply voltage	VCI		2.4	-	3.3	V	
1st step-up input voltage	VCI1		1.75	-	3	V	
1st step-up output efficiency	AVDD	ILOAD = 4 mA	90	95	100	%	
2nd step-up output efficiency	VGH	ILOAD = 0.1 mA	90	95	100	%	
3rd step-up output efficiency	VGL	ILOAD = 0.1 mA	90	95	100	%	
4th step-up output efficiency	VCL	ILOAD = 0.3 mA	90	95	100	%	

Notes:

1. VSS = 0V.
2. Applied pins; CSB, RDB, WRB, RS, DB0 to DB17, RESETB
3. Applied pins; DB0 to DB17
4. Target frame frequency = 60 Hz, Display line = 432, Back porch = 8, Front porch = 8, RTN2-0 = “000” DIV =”0”, FOSC4-0=”00101” (You can observe fosc at CL1 (fosc / 16)

cf> The Fosc value could be influenced by module condition. (e.g. parasitic capacitance of ITO, FPC, contact pads, etc.) To get expected Fosc value, the module maker has to carry out proper adjustment.

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Table 57. DC Characteristics For LCD Driver Outputs

(VDD = 1.5V, VDD3 = 3.0V, VSS = 0V)

Characteristic	Symbol	CONDITION	MIN	TYP	MAX	Unit	Note
LCD Gate driver output On resistance	Ronvgh	I(sink)=100uA	-	-	2	kΩ	-
	Ronvgl	I(source)=100uA	-	-	2	kΩ	-
LCD source driver output On resistance	Ronp	I(sink)=100uA	-	-	20	kΩ	-
	Ronn	I(source)=100uA	-	-	20	kΩ	-
LCD Binary driver Output On resistance	Ronpb	I(sink)=100uA	-	-	300	kΩ	-
	Ronnb	I(source)=100uA	-	-	300	kΩ	-
Output voltage deviation (Mean value)	ΔV_O	$4.2V \leq V_{SO}$	-	-	±20	mV	*1
		$0.8V < V_{SO} < 4.2V$	-	-	±15	mV	*1
		$0.2 \leq V_{SO} \leq 0.8V$	-	-	±20	mV	*1
		$V_{SO} < 0.2V$	-	-	T.B.D	mV	*1
LCD source driver delay	tSD	SAP = “000”	-	-	60	us	*2
		SAP = “001”	-	-	50	us	*2
		SAP = “010”	-	-	40	us	*2
		SAP = “011”	-	-	30	us	*2
		SAP = “100”	-	-	25	us	*2
		SAP = “101”	-	-	20	us	*2
		SAP = “110”			15	us	*2
		SAP = “111”			12	us	*2

Characteristic	Symbol	CONDITION	MIN	TYP	MAX	Unit	Note
Current consumption during normal operation	IVDD3	No load, Ta = 25 °C VCI=3V Frame(f)=60Hz	-	T.B.D	T.B.D	uA	-
	IVCI		-	T.B.D	T.B.D	mA	-
Current consumption during STBY	IVDD3	Standby mode, Ta = 25 °C (TYP) Ta= 70°C (MAX) VCI=3V,VDD=1.5V	-	T.B.D	T.B.D	uA	-
	IVCI			10	20	uA	
Current consumption during Deep-STBY	IVDD3	Deep standby mode, Ta = 70 °C VCI=3V,VDD=0	-	T.B.D	T.B.D	uA	-
	IVCI			T.B.D	5	uA	

Note:

1. V_{SO} the output voltage of analog output pins S1 to S720

2. t_{SD}: LCD Source driver delay / AVDD = 5.5V, GVDD = 5.0V, Ta = 25 °C

t_{SD} = 1 / [Frame Freq. x {The number of raster-row + The number of blank period (FP+BP)}]

− The marginal time for pixel load charge

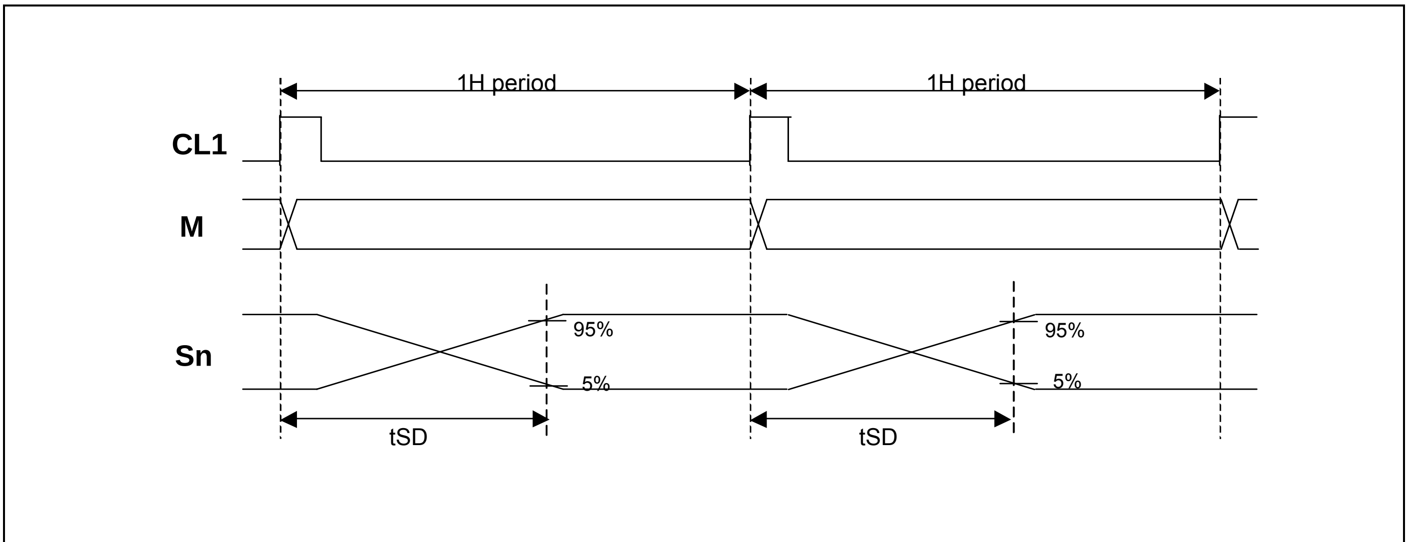

Figure 90. LCD source driver delay

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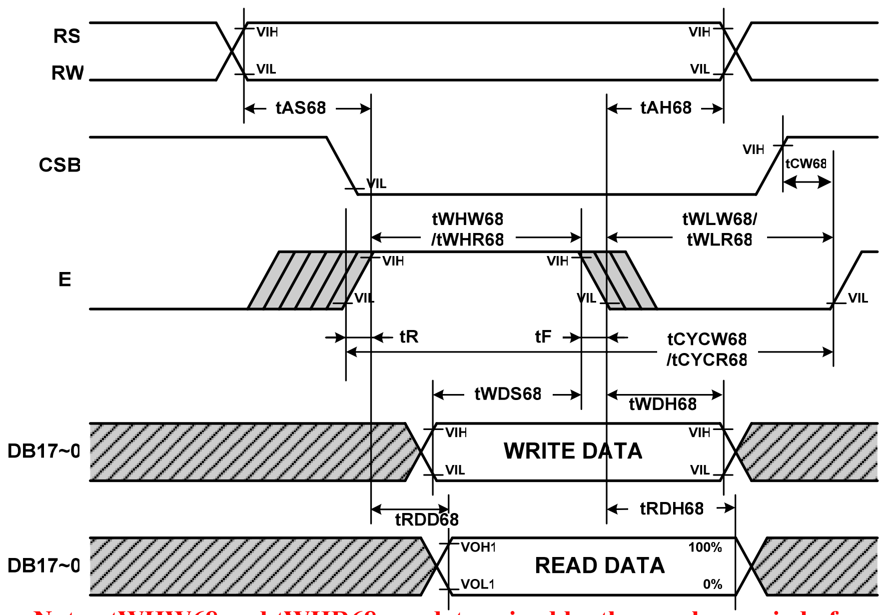
AC CHARACTERISTICS

Table 58. Parallel Write Interface Characteristics (68 Mode)

(VDD = 1.5V, VDD3 = 1.65 to 3.3V, T_A = -40 to +85°C)

Characteristic		Symbol	Specification		Unit
			Min.	Max.	
Cycle time	Write	tCYCW68	62	-	ns
	Read	tCYCR68	500	-	
Pulse rise / fall time		tR, tF	-	10	
Pulse width low	Write	tWHW68	T.B.D	-	
	Read	tWHR68	250	-	
Pulse width high	Write	tWLW68	20	-	
	Read	tWLR68	200	-	
RS,RW to CSB, E setup time		tAS68	(0)	-	
RS,RW to CSB, E hold time		tAH68	(0)	-	
CSB to E time		tCW68	10	-	
Write data setup time		tWDS68	30	-	
Write data hold time		tWDH68	15	-	
Read data delay time		tRDD68	-	200	
Read data hold time		tRDH68	20	60	



Note : tWHW68 and tWHR68 are determined by the overlap period of low CSB and high E

Figure 91. AC Characteristics (68 Mode)

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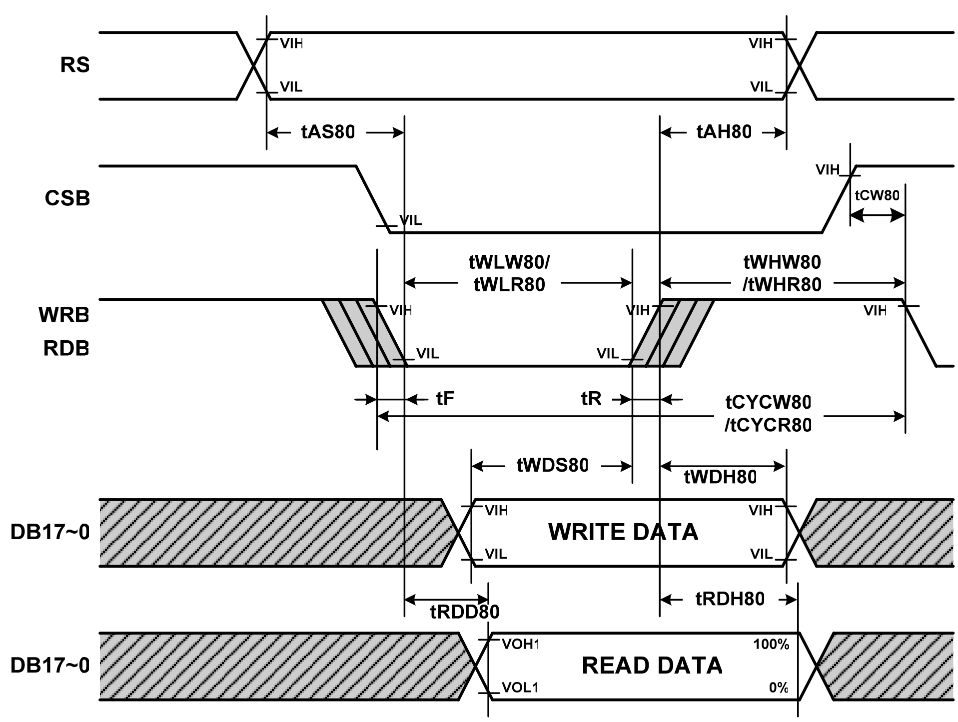
Table 59. Parallel Write Interface Characteristics (80 Mode)

(VDD = 1.5V, VDD3 = 1.65 to 3.3V, T_A = -40 to +85°C)

Characteristic		Symbol	Specification		Unit
			Min.	Max.	
Cycle time	Write	tCYCW80	62	-	
	Read	tCYCR80	500	-	
Pulse rise / fall time		tR, tF	-	10	
Pulse width low	Write	tWLW80	T.B.D	-	
	Read	tWLR80	250	-	
Pulse width high	Write	tWHW80	20	-	
	Read	tWHR80	200	-	
RS to CSB, WRB(RDB) setup time		tAS80	(0)	-	ns
RS to CSB, WRB(RDB) hold time		tAH80	(0)	-	
CSB to WRB(RDB) time		tcw80	10	-	
Write data setup time		tWDS80	30	-	
Write data hold time		tWDH80	15	-	
Read data delay time		tRDD80	-	200	
Read data hold time		tRDH80	20	60	



Note : tWLW80 and tWLR80 are determined by the overlap period of low CSB and low WRB or low CSB and low RDB

Figure 92. AC Characteristics (80 Mode)

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Table 60. Clock Synchronized Serial Write Mode Characteristics

(VDD = 1.5V, VDD3 = 1.65 to 3.3V, T_A = -40 to +85°C)

Characteristic	Symbol	specification		Unit
		Min.	Max.	
Serial clock write cycle time	tscyc	100	-	ns
Serial clock read cycle time	tscyc	500	-	ns
Serial clock rise / fall time	tR, tF	-	15	ns
Pulse width high for write	tSCHW	40	-	ns
Pulse width high for read	tSCHR	230	-	ns
Pulse width low for write	tSCLW	40	-	ns
Pulse width low for read	tSCLR	230	-	ns
Chip Select setup time	tCSS	20	-	ns
Chip Select hold time	tCSH	60	-	ns
Serial input data setup time	tSIDS	30	-	ns
Serial input data hold time	tSIDH	30	-	ns
Serial output data delay time	tSODD	-	130	ns
Serial output data hold time	tSODH	5	-	ns

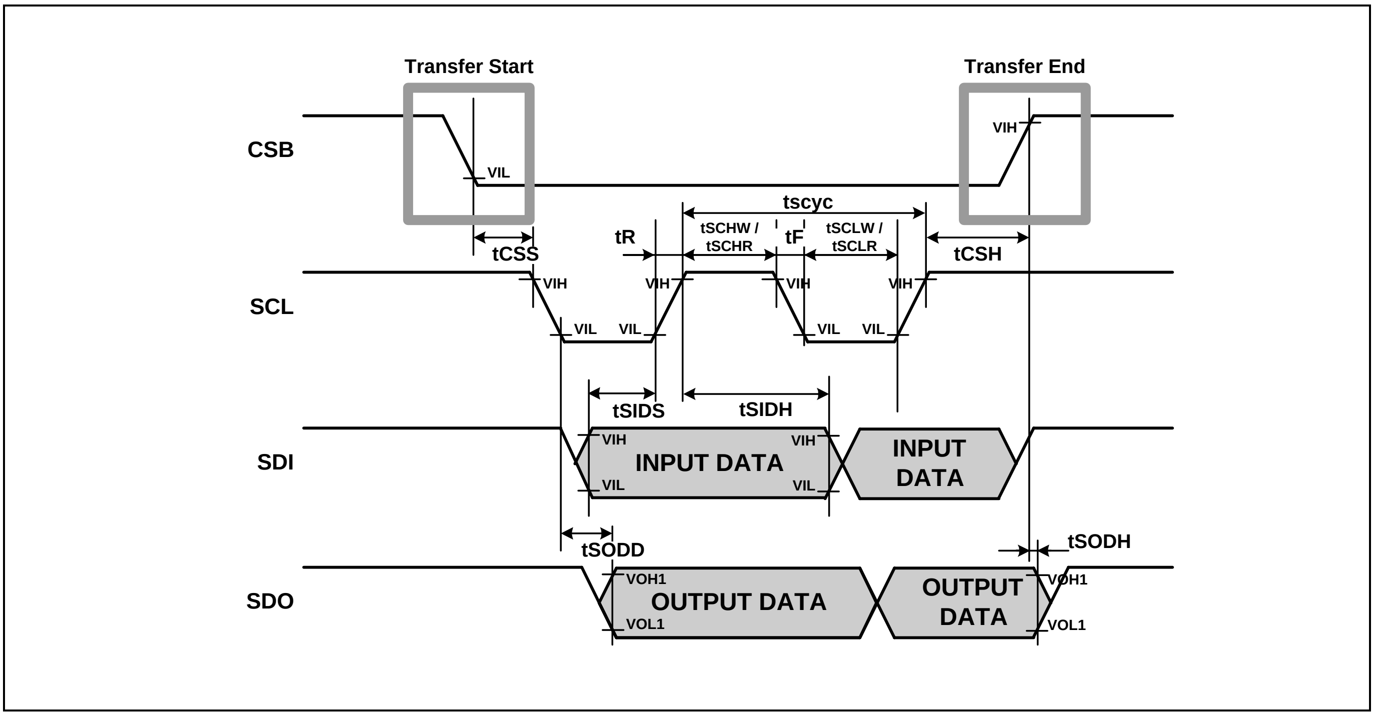

Figure 93. AC Characteristics (SPI Mode)

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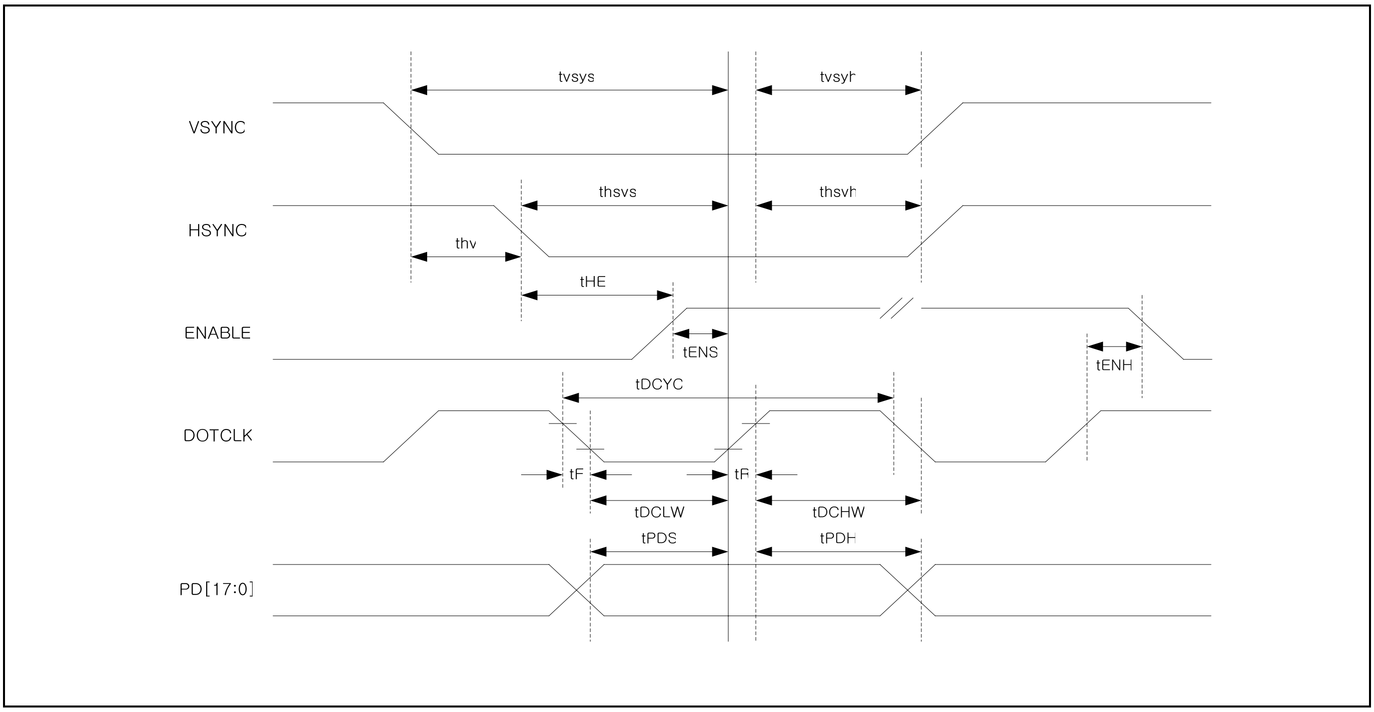
Table 61. RGB Data Interface Characteristics

(VDD = 1.5V, VDD3 = 1.65 to 3.3V, T_A = -40 to +85°C)

Characteristic	Symbol	18/16bit RGB interface		6bit RGB interface		Unit
		Min.	Max.	Min.	Max.	
DOTCLK cycle time	tDCYC	100	-	100	-	ns
DOTCLK rise / fall time	tR, tF	-	15	-	15	
DOTCLK Pulse width high	tDCHW	40	-	40	-	
DOTCLK Pulse width low	tDCLW	40	-	40	-	
Vertical Sync Setup Time	tvsys	30	-	30	-	
Vertical Sync Hold Time	tvsyh	30	-	30	-	
Horizontal Sync Setup Time	thsys	30		30		
Horizontal Sync Hold Time	thsyh	30		30		
ENABLE setup time	tENS	30	-	30	-	
ENABLE hold time	tENH	20	-	20	-	
PD data setup time	tPDS	30	-	30	-	
PD data hold time	tPDH	20	-	20	-	
HSYNC-ENABLE Time	tHE	1	HBP	1	HBP	tDCYC
VSYNC-HSYNC Time	thv	1	175	1	527	

Note: HBP is Horizontal Back-porch.



(When VSPL=0, HSPL=0, DPL=0, EPL=1)

Figure 94. AC Characteristics (RGB Interface Mode)

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DSTB TIMING

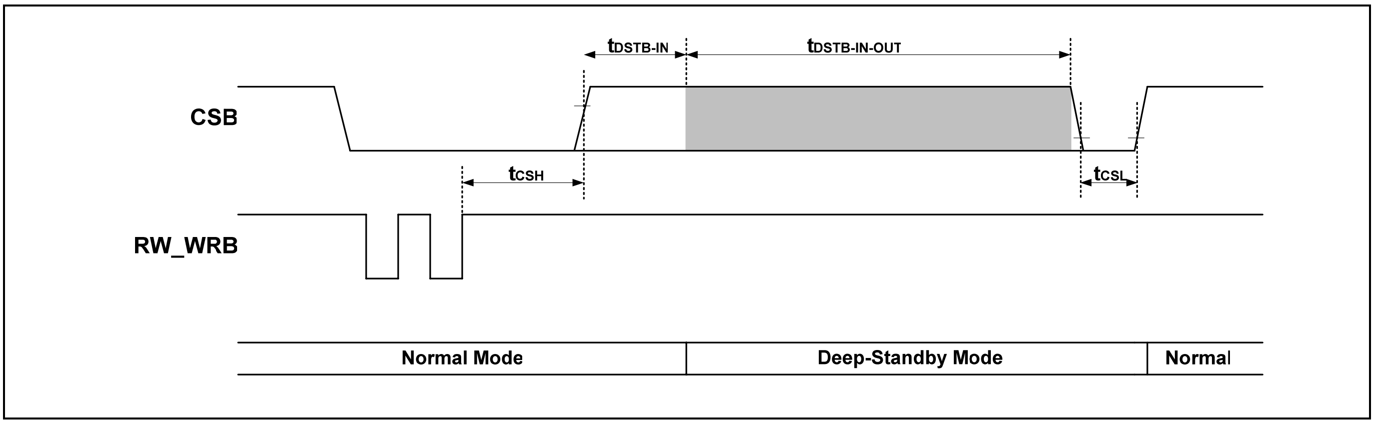

Table 62. Deep-Standby AC Characteristics

Characteristics	Symbol	DSTB_1F	Min	Typ	Max	Unit
WRB to rising edge of CSB	t_{CSH}	0	-	-	< 1	line
		1	-	-	< 1	frame
DSTB-IN	$t_{DSTB\text{-}IN}$	0	1	-	2	line
		1	1	-	2	frame
DSTB-IN to DSTB-OUT	$t_{DSTB\text{-}IN\text{-}OUT}$	-	3	-	-	ms
Low Pulse Width of CSB for Deep-Standby Wake-Up	t_{CSL}	-	2	-	-	Us

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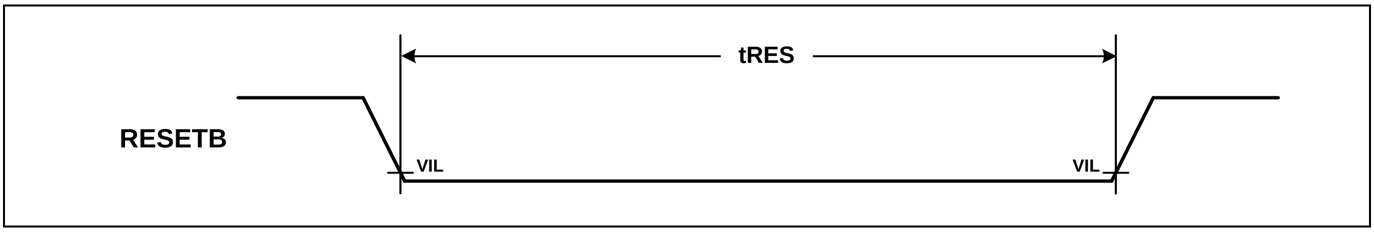
RESET TIMING



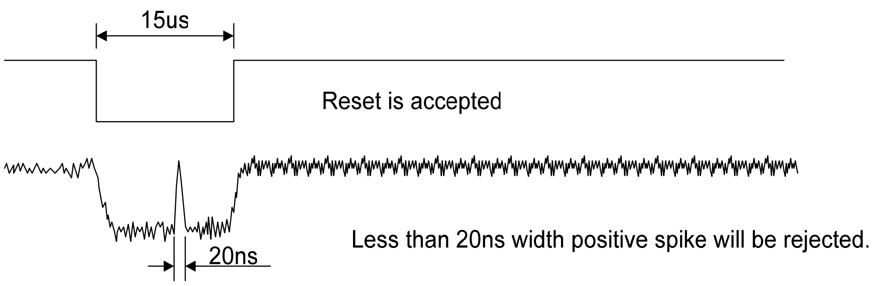
Note: Reset low pulse width shorter than 10us do not make reset. It means undesired short pulse such as glitch, bouncing noise or electrostatic discharge do not cause irregular system reset. Please refer to the table below.

Parameter	Description	Min	Max	Unit
tRES	Reset low pulse width	T.B.D	-	us

Figure 95. AC characteristics (RESET timing)

Table 63. Reset Operation regarding tRES Pulse Width

tRES Pulse	Action
Shorter than 5 us	No reset
Longer than T.B.D us	Reset
Between 5 us and T.B.D us	Not determined



1. User may or may not use RESETB pin. In order to use it, user should satisfy the conditions described in the above tables. But when not wants to use RESETB, user may fix this pin to VDD3 level because internally generated POR (Power-On-Reset) is used.
2. Spike Rejection also applies during a valid reset pulse as shown below:

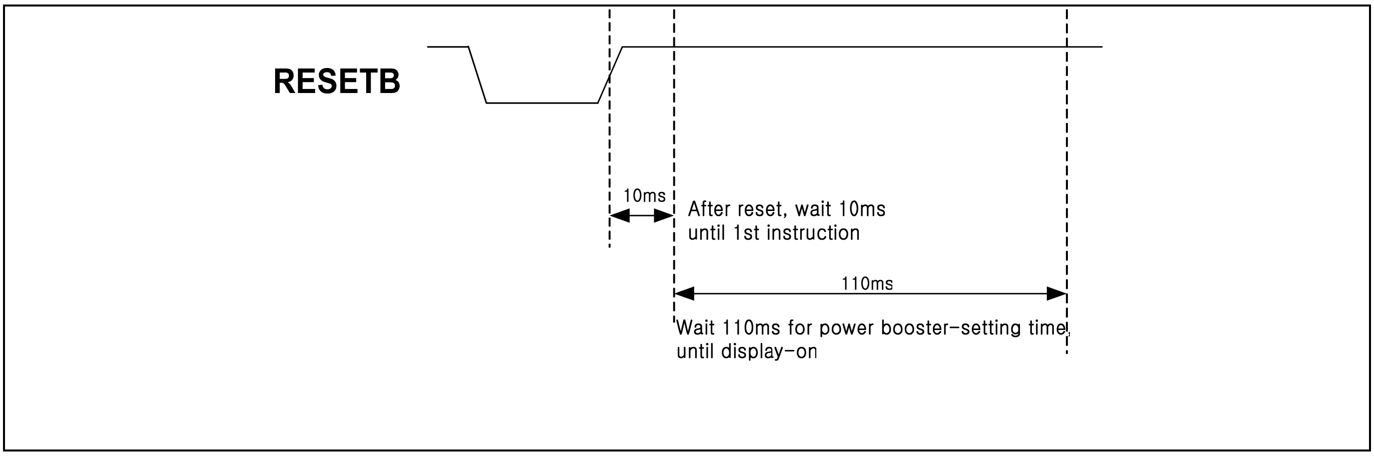


Figure 96. Timing Diagram of RESETB

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EXTERNAL POWER ON / OFF SEQUENCE

VDD3 must be applied earlier than VCI or at least applied simultaneously with VCI. When regulator cap is 1μF, RESETB must be applied after VCI have been applied. The applied time gap between VCI and RESETB is minimum 1ms. As regulator cap becomes larger, this time gap must be increased. Otherwise function is not guaranteed.

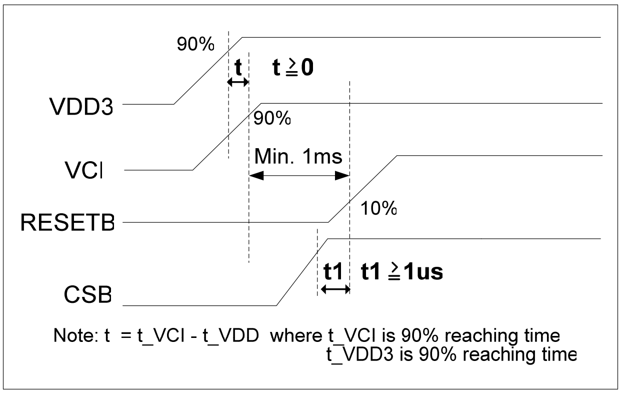

Figure 97. External power on sequence

b) EXTERNAL POWER OFF SEQUENCE

VDD3 must be powered down later than VCI or at least powered down simultaneously with VCI. VCI must be powered down after RESETB have been powered down. The time gap of powered down between RESETB and VCI is minimum 1ms. Otherwise function is not guaranteed.

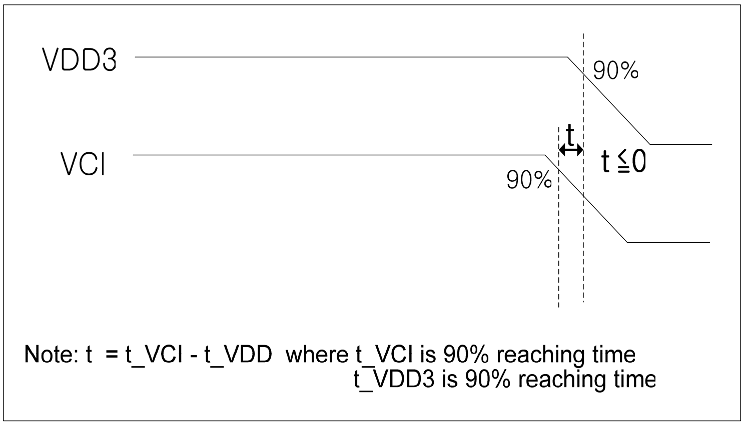

Figure 98. External Power Off sequence

Preliminary

REVISION HISTORY

Version	Content	Author	Date
0.00	New released	J.H.LEE, D.K.KIM	2006-02-27
0.10	Changed selection Interface mode - Table11. Pin description - Added R03h, CLS, MDT[1:0]=01(page40-41) - Modified Interface specification(page88-89) - Added AM(page42) Updated Display control(R07h) – page43 Updated Oscillator specifications (page 50) - Added table 28 - Added OSCEX_EN instruction - OSC7:0 → FOSC5:0 Merged VREFI and VREFO in VREF - Table 9. pin description - Figure 20. (page 81) Modified AVDD maximum voltage 5.5V to 6V - S6D0170 operating voltage (page 6) - VCI1 voltage level (page 53) - Revised Figure 21 - Table 49 DC characteristics Modified external power on/off sequence (page 149) - Figure 3. COG align key - Table 8. Power supply pin description - Table 10. Power supply pin description - Table 14. Oscillator and internal power regulator pin description	J.H.LEE D.K.KIM	2006-04-09
0.20	- Modified Table 1,14,18,19,20 - Fig 71. Structure of grayscale amplifier - VCI1 voltage level (page 53) - GVD 5-0 → GVD 6-0 (page 30, page 57) - VCM 5-0 → VCM 6-0 (page 30, page 60) - VML 5-0 → VML 6-0 (page 30, page 62) - Modified figure 9, 10, 20, 22 - Removed AD[16:0] - Updated MTP control(Figure 16) - Revised Figure 47, 50, 84, 89, 90 - Added Figure 51, 59, 60 - Revised Table 49	J.H.LEE D.K.KIM	2006-04-21

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Version	Content	Author	Date
0.30	- P31/P74 Revised HEA, HAS - P33 Revised Status Read -P35 Added Comments -P67 Revised Figure 10 -P79 Revised Figure 16 -P101 Revised Figure57, 58 - Modified Pin description (added VDD3O, VSS3O) - Modified VCI1 range - Modified Figure 20 - Changed COG Align key Figure 3	D.K.KIM	2006-05-23
0.40	- Modified VCI1 setup flow (modified page 86. note) - Figure 3~5 COG Align Key, Align Key configuration - Table 2~11 Pad Center Coordinates - Figure 87 Oscillator sequence - Table 52 Absolute Maximum rating - Table 53 DC characteristic - Table 54 DC characteristics For LCD driver outputs - Figure 96 Timing Diagram of RESETB - Revised 80 ms => 110 ms : Figure 25 - Added R05h : Device Code - Revised SM,GS : Figure8,9 - Revised Memory Write/Read Address : Figure12,13 - Revised Non-Display Area : Figure 16 - Revised MTP Program Flow : Figure 19 - Revised Document of INDEX - Revised VOH1=>100%, VOL1=>0% : Figure93, 94	J.H.LEE S.J.YOON D.K.KIM	2006-06-15
0.50	- Added Initial Value of Instruction at each instruction table - Removed RESET Function(Instruction Set Initialization) - Added more source driver delay condition in DC Characteristics Table	J.H.LEE S.J.YOON D.K.KIM	2006-06-26
0.60	- Table 2. pad coordinates GVDDO → GVDD - Table 15. Pin description added SDI not used status - Figure 78. Structure of Resistor Ladder Network 1 - Table 63. Amplitude Adjustment - Table 64. Reference Adjustment	J.H.LEE S.J.YOON D.K.KIM	2006-08-24
0.70	- Revised MTP Program Flow - Figure 18. MTP Sequence Flow - Figure 19. Voltages and waveforms for MTP programming - Table 35. MTP writing time - Table 36. MTPG/D Voltage Tolerance - Table 37. Current consumption during setting MTP	Y.H.KIM	2006-09-20

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Version	Content	Author	Date
0.80	- Added instruction description of DSTB_1F(R10h) - Added instruction description of RTR_AMP_EN(R13h) - Added instruction description of LVCM_VCIR(R15h) - Added note for BP/FP - Figure 8, 9 Gate Direction - Figure 15 Scroll Function - Figure 23. Setup flow of power - Figure 26, 27 Set Interface mode - Figure 83. System structure - Figure 85. Setup flow of Deep Standby / Standby : Add notes - Figure 98. External power off sequence - Table 22. Instruction table : DSTB_1F,RTR_AMP_EN,LVCM_VCIR - Table 57. ΔVO (VSO<0.2V) - Table 58,59 tWLWxx : 20 ➔ T.B.D - Added DSTB timing diagram	J.H.LEE Y.H.KIM D.K.KIM	2006-12-04

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NOTICE

Precautions for Light

Light has characteristics to move electrons in the integrated circuitry of semiconductors, therefore may change the characteristics of semiconductor devices when irradiated with light. Consequently, the users of the packages which may expose chips to external light such as COB, COG, TCP and COF must consider effective methods to block out light from reaching the IC on all parts of the surface area, the top, bottom and the sides of the chip. Follow the precautions below when using the products.

1. Consider and verify the protection of penetrating light to the IC at substrate (board or glass) or product design stage.

2. Always test and inspect products under the environment with no penetration of light.